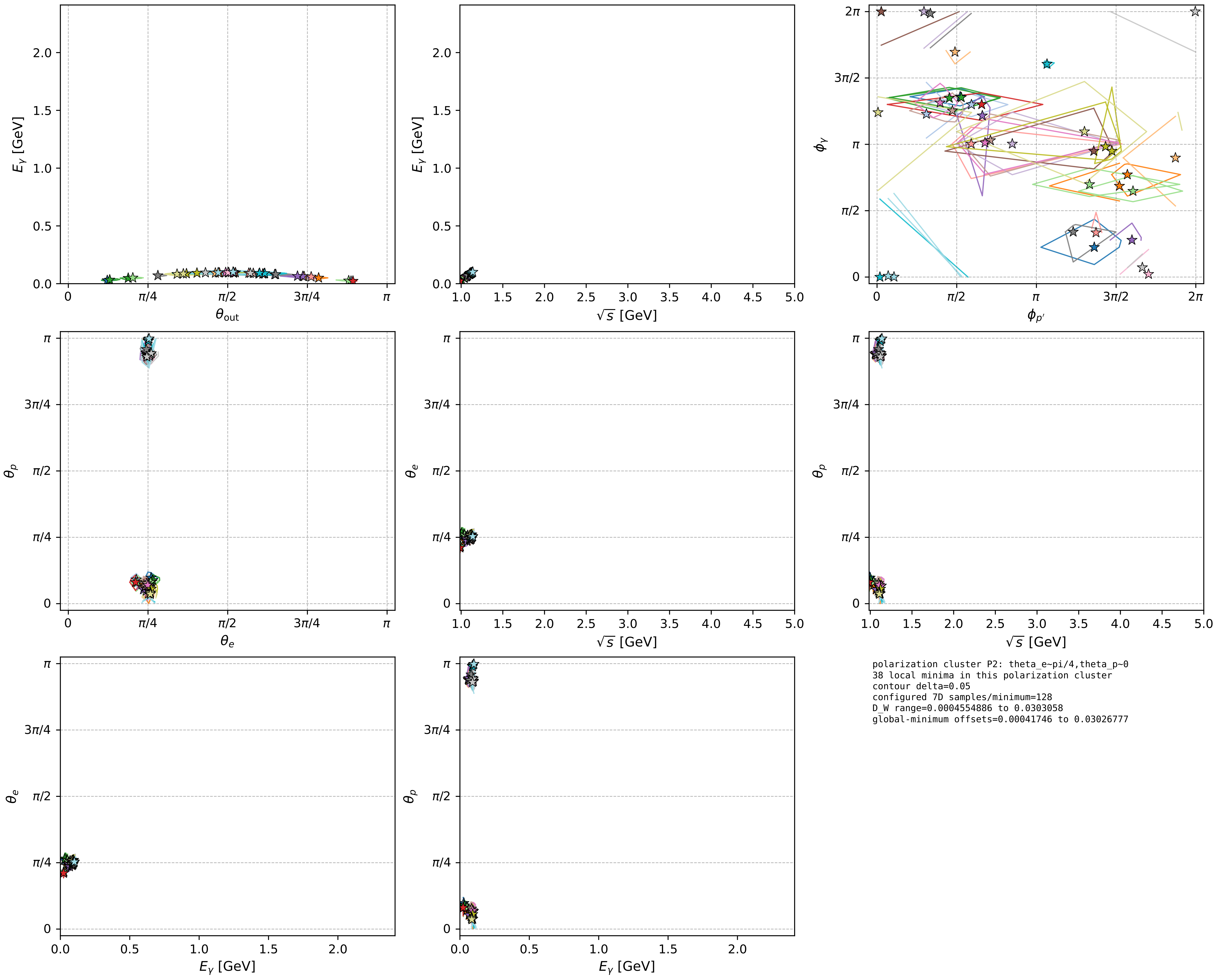
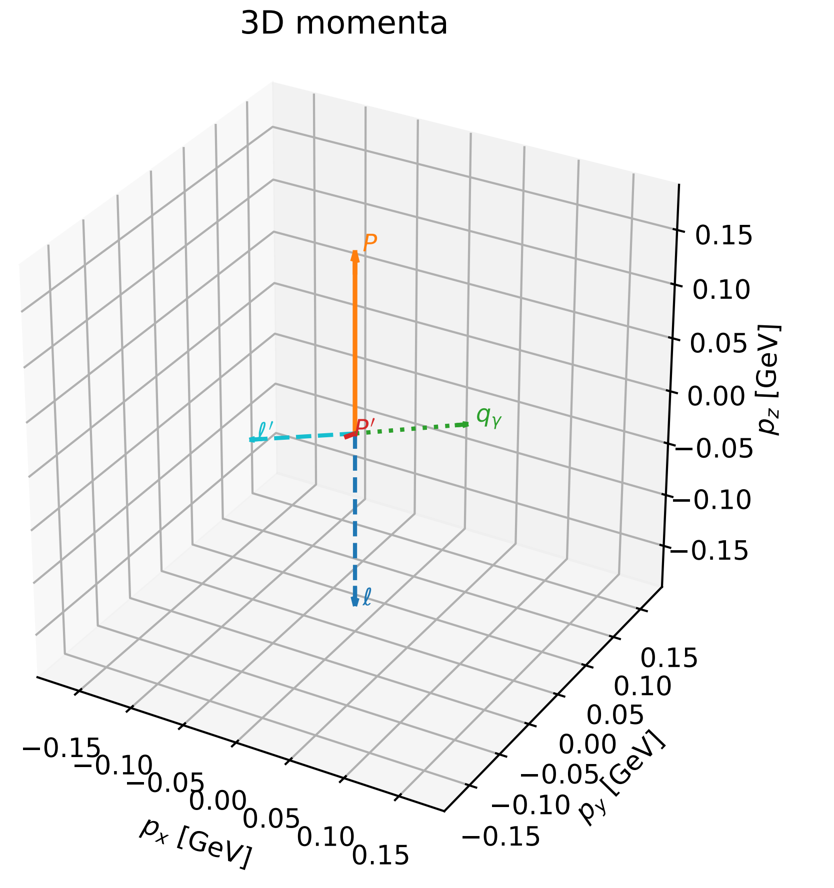
electron: polarization cluster P2; all local-minimum configurations and pairwise projections of their 7D contours



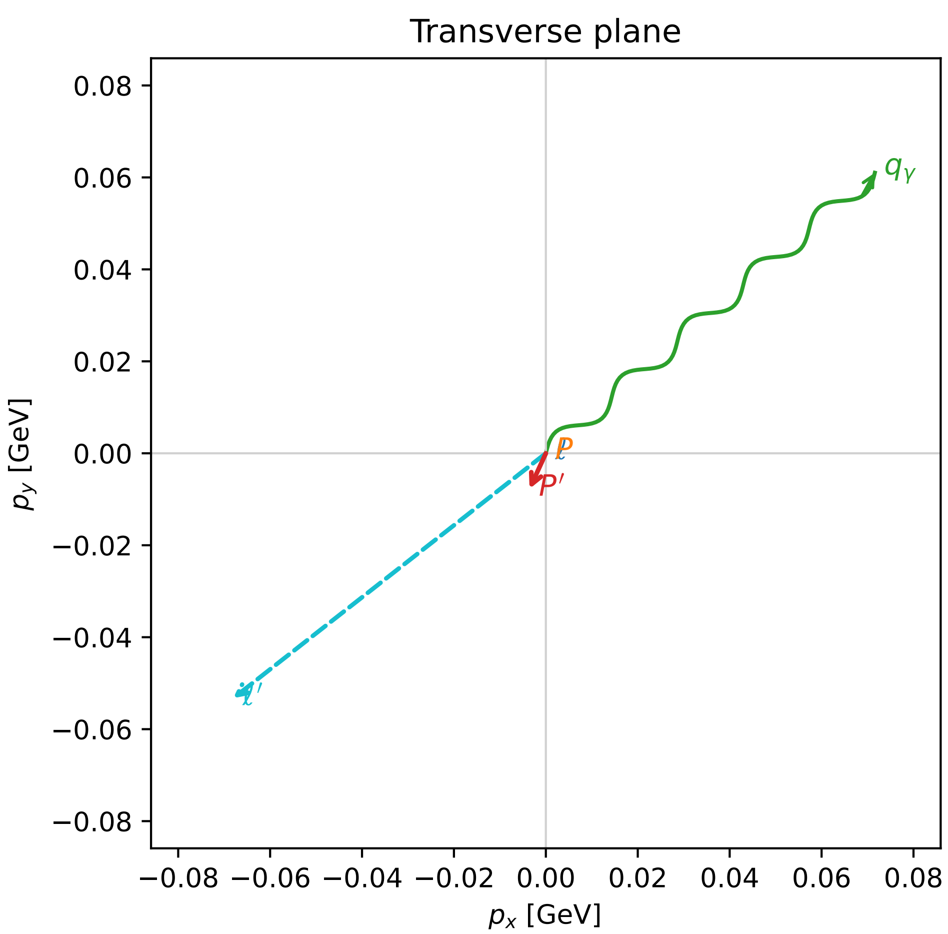
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

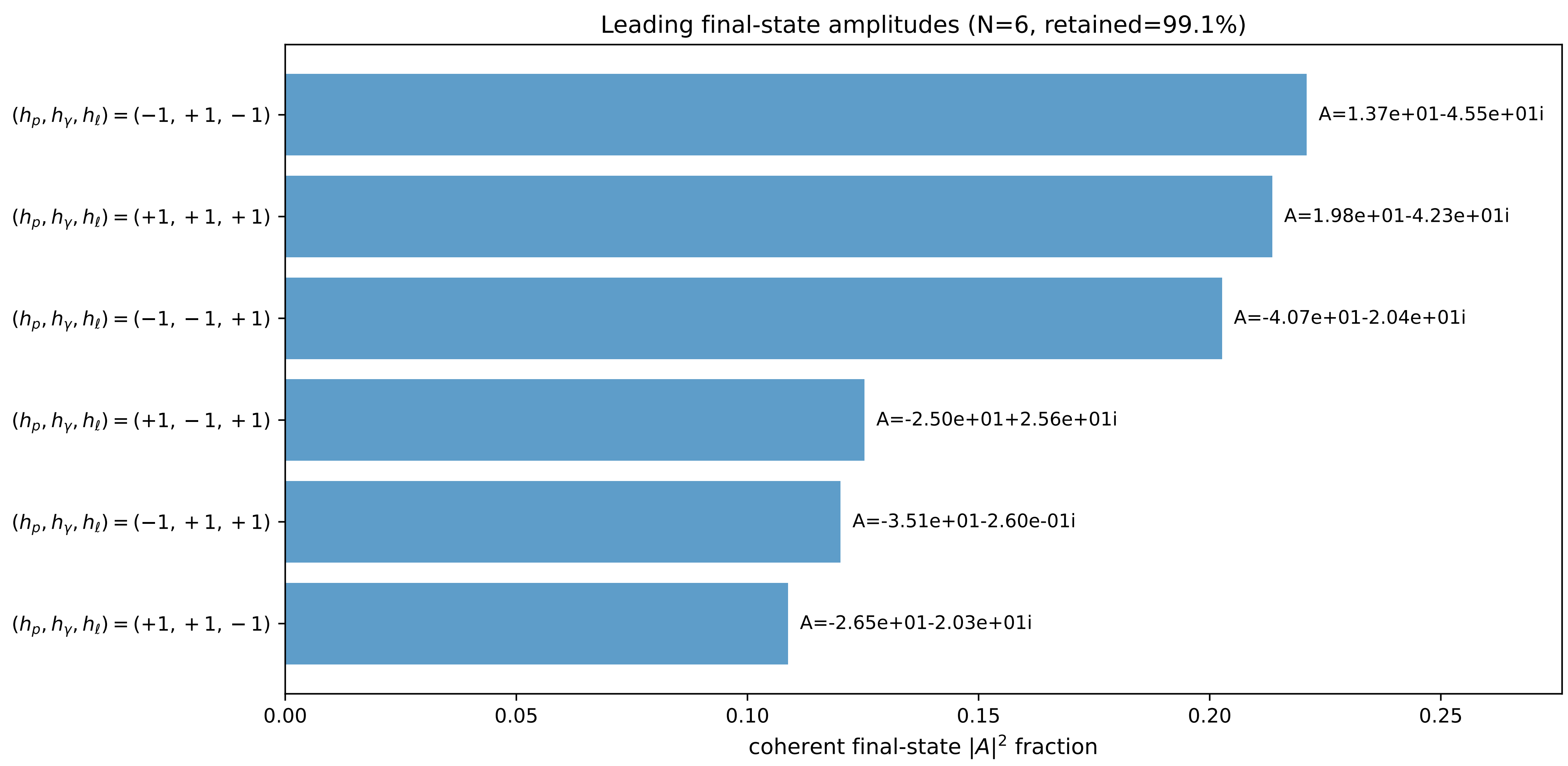


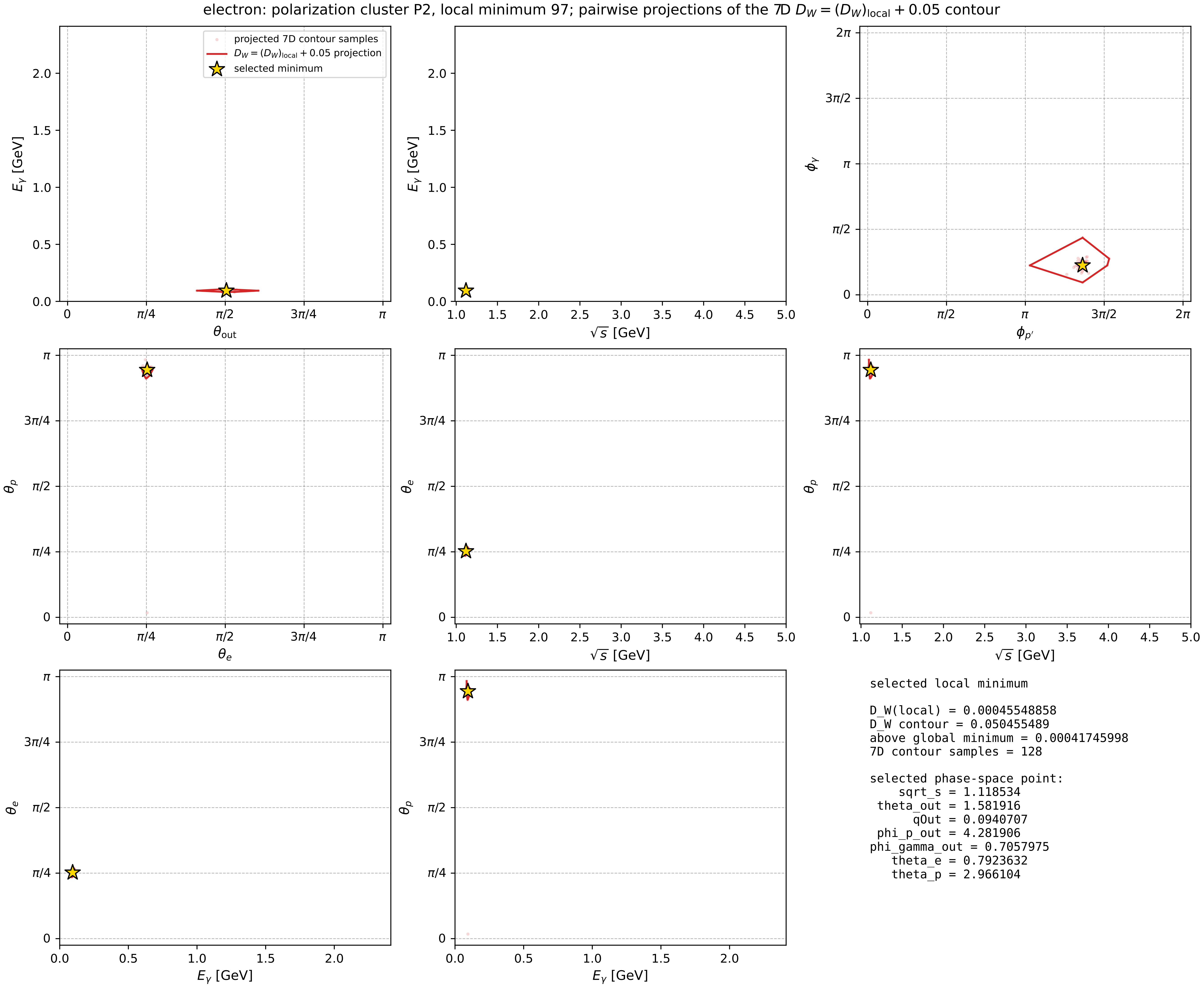

```
dw_mixing_angles_polarization_02_local_minimum_97  D_W=0.000455489
region: polarization_cluster_02_local_minimum_97
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.79236318 rad
incoming lepton: 0.702165|+> +0.712015|->
proton mixing angle: theta_p=2.966104 rad
incoming proton: -0.984641|+> +0.174589|->
```

s=1.25112, $\sqrt{s}$=1.11853, $|\vec{P}|$=0.165964, $|\vec{P}'|$=0.00851551
$\theta_{\rm out}$=1.58192, $\phi_{p'}$=4.28191, ϕ_γ=0.705798
E_γ=0.0940707, $\theta_{\ell'}$=1.5576, $\phi_{\ell'}$=3.8059
Q^2=0.029065, x_B=0.140409, t=-0.0274368, W^2=1.05778, y=0.557545

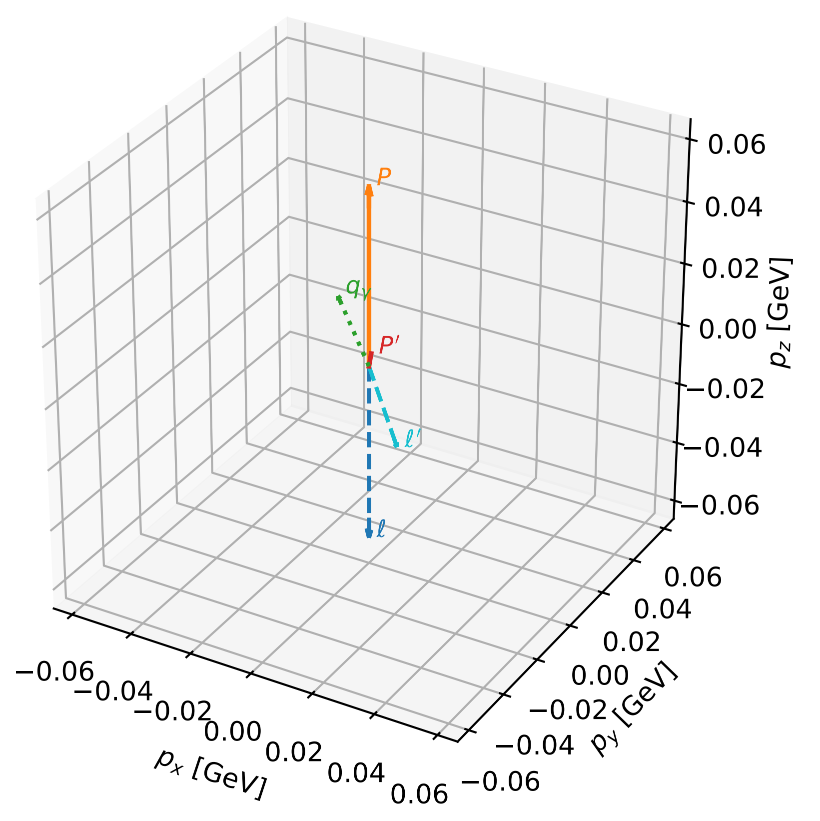
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1660 p_x= 0.0000 p_y= 0.0000 p_z= -0.1660
P E= 0.9526 p_x= 0.0000 p_y= 0.0000 p_z= 0.1660
ℓ' E= 0.0864 p_x= -0.0680 p_y= -0.0533 p_z= 0.0011
P' E= 0.9380 p_x= -0.0036 p_y= -0.0077 p_z= -0.0001
q_γ E= 0.0941 p_x= 0.0716 p_y= 0.0610 p_z= -0.0010






electron: polarization cluster P2, local minimum 97; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.00045548858
D_W contour = 0.050455489
above global minimum = 0.00041745998
7D contour samples = 128
selected phase-space point:
sqrt_s = 1.118534
theta_out = 1.581916
qOut = 0.0940707
phi_p_out = 4.281906
phi_gamma_out = 0.7057975
theta_e = 0.7923632
theta_p = 2.966104

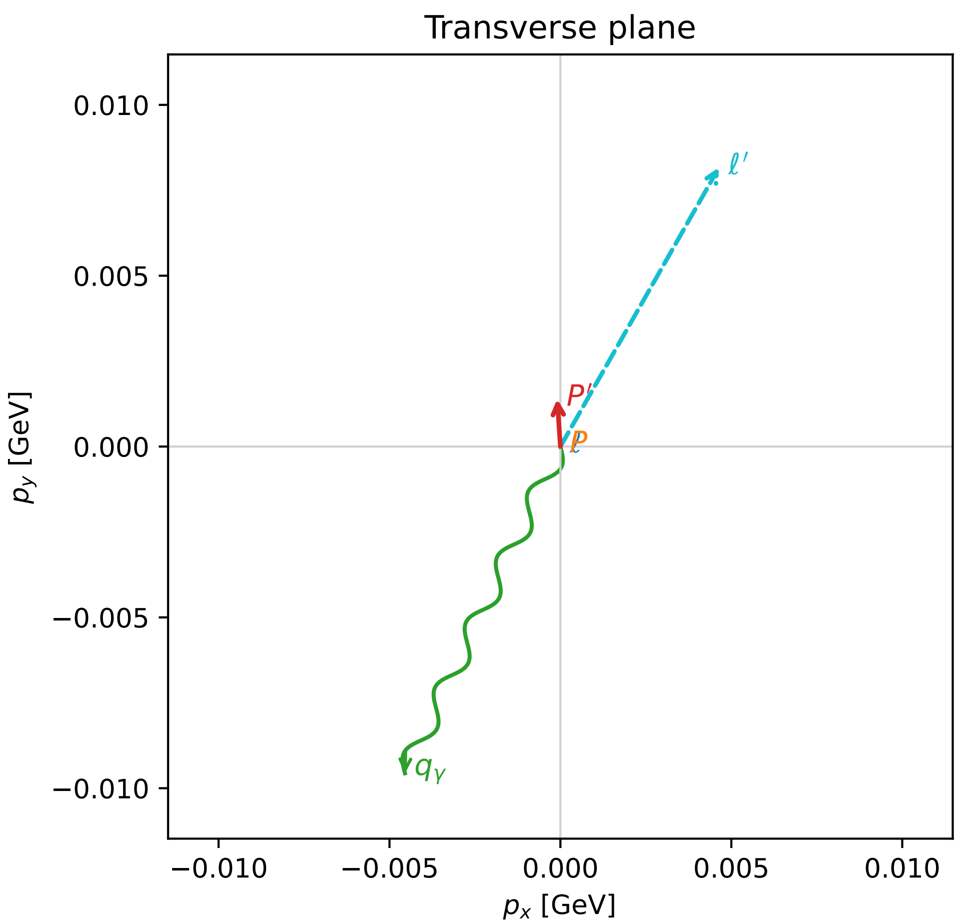
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

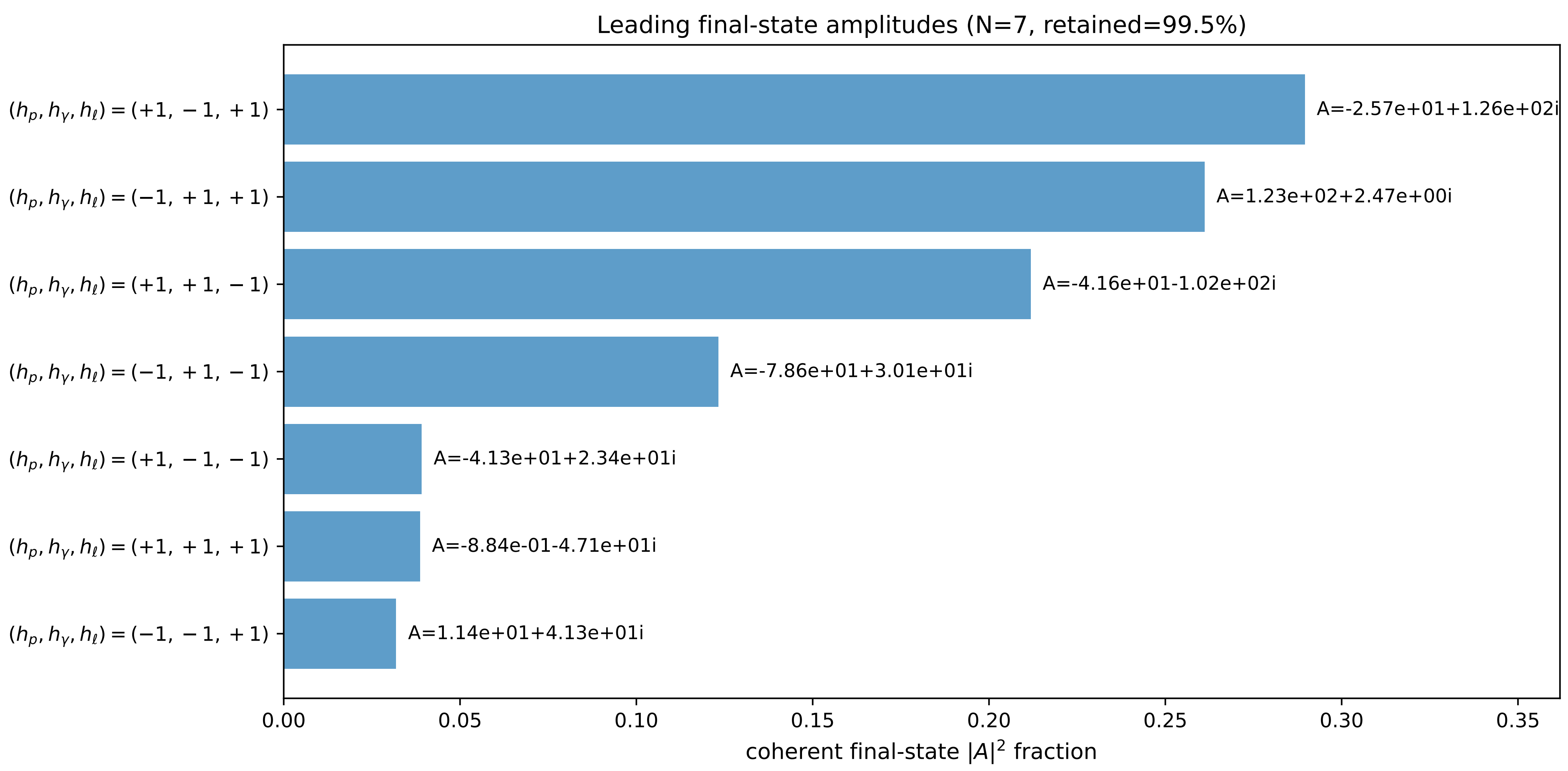


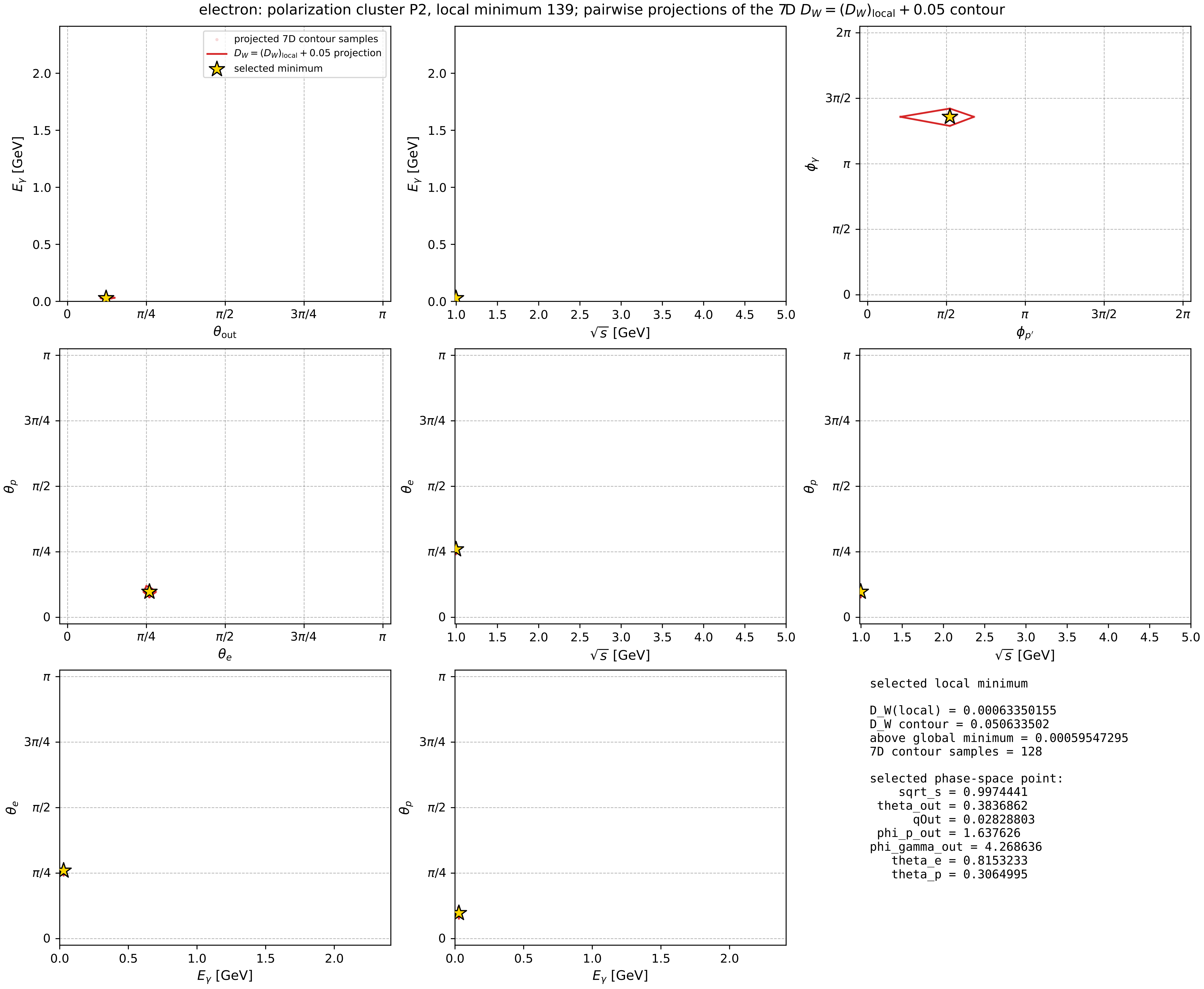

```
dw_mixing_angles_polarization_02_local_minimum_139  D_W=0.000633502
region: polarization_cluster_02_local_minimum_139
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.81532328 rad
incoming lepton: 0.685633|+> +0.727947|->
proton mixing angle: theta_p=0.30649952 rad
incoming proton: 0.953396|+> +0.301723|->
```

s=0.994895, $\sqrt{s}$=0.997444, $|\vec{P}|$=0.0576707, $|\vec{P}'|$=0.00373398
θ_{out}=0.383686, $\phi_{p'}$=1.63763, ϕ_γ=4.26864
E_γ=0.028288, $\theta_{\ell'}$=2.83517, $\phi_{\ell'}$=1.05427
Q^2=0.000167434, x_B=0.00315434, t=-0.00293737, W^2=0.932757, y=0.461366

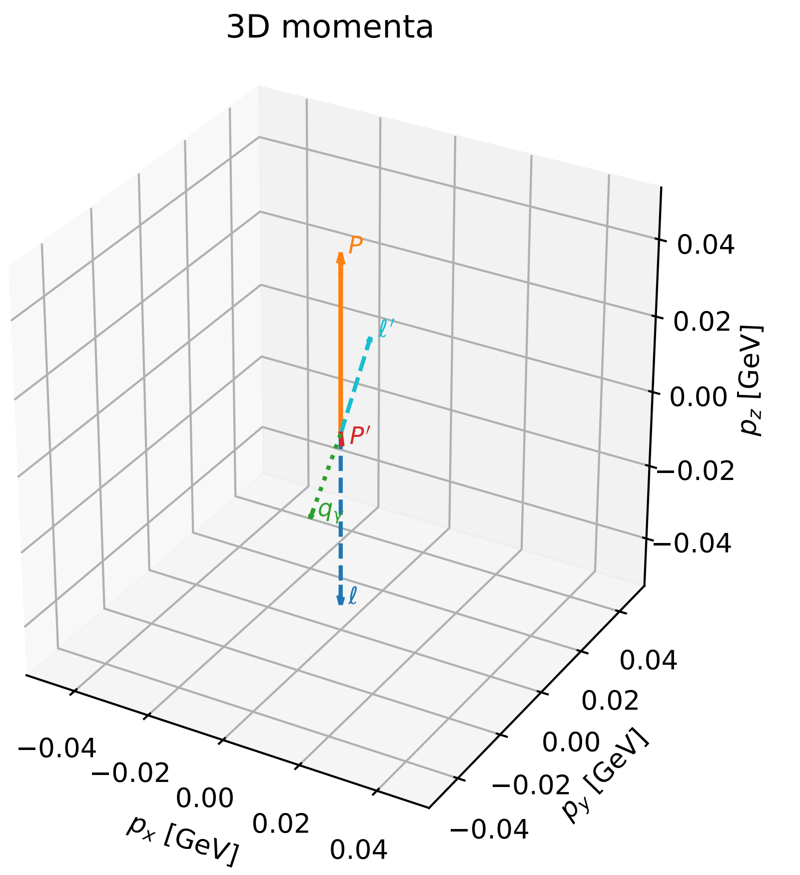
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0577 p_x= 0.0000 p_y= 0.0000 p_z= -0.0577
P E= 0.9398 p_x= 0.0000 p_y= 0.0000 p_z= 0.0577
ℓ' E= 0.0311 p_x= 0.0046 p_y= 0.0082 p_z= -0.0297
P' E= 0.9380 p_x= -0.0001 p_y= 0.0014 p_z= 0.0035
q_γ E= 0.0283 p_x= -0.0045 p_y= -0.0096 p_z= 0.0262






electron: polarization cluster P2, local minimum 139; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.00063350155
D_W contour = 0.050633502
above global minimum = 0.00059547295
7D contour samples = 128
selected phase-space point:
sqrt_s = 0.9974441
theta_out = 0.3836862
qOut = 0.02828803
phi_p_out = 1.637626
phi_gamma_out = 4.268636
theta_e = 0.8153233
theta_p = 0.3064995

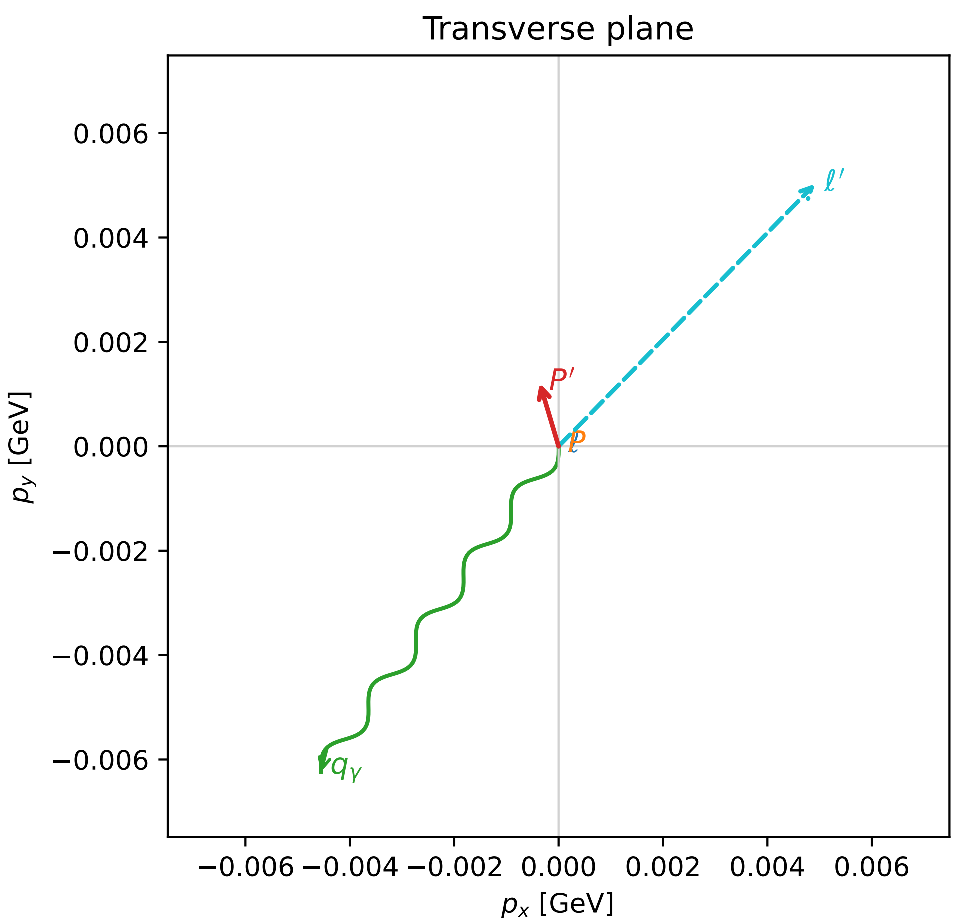
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

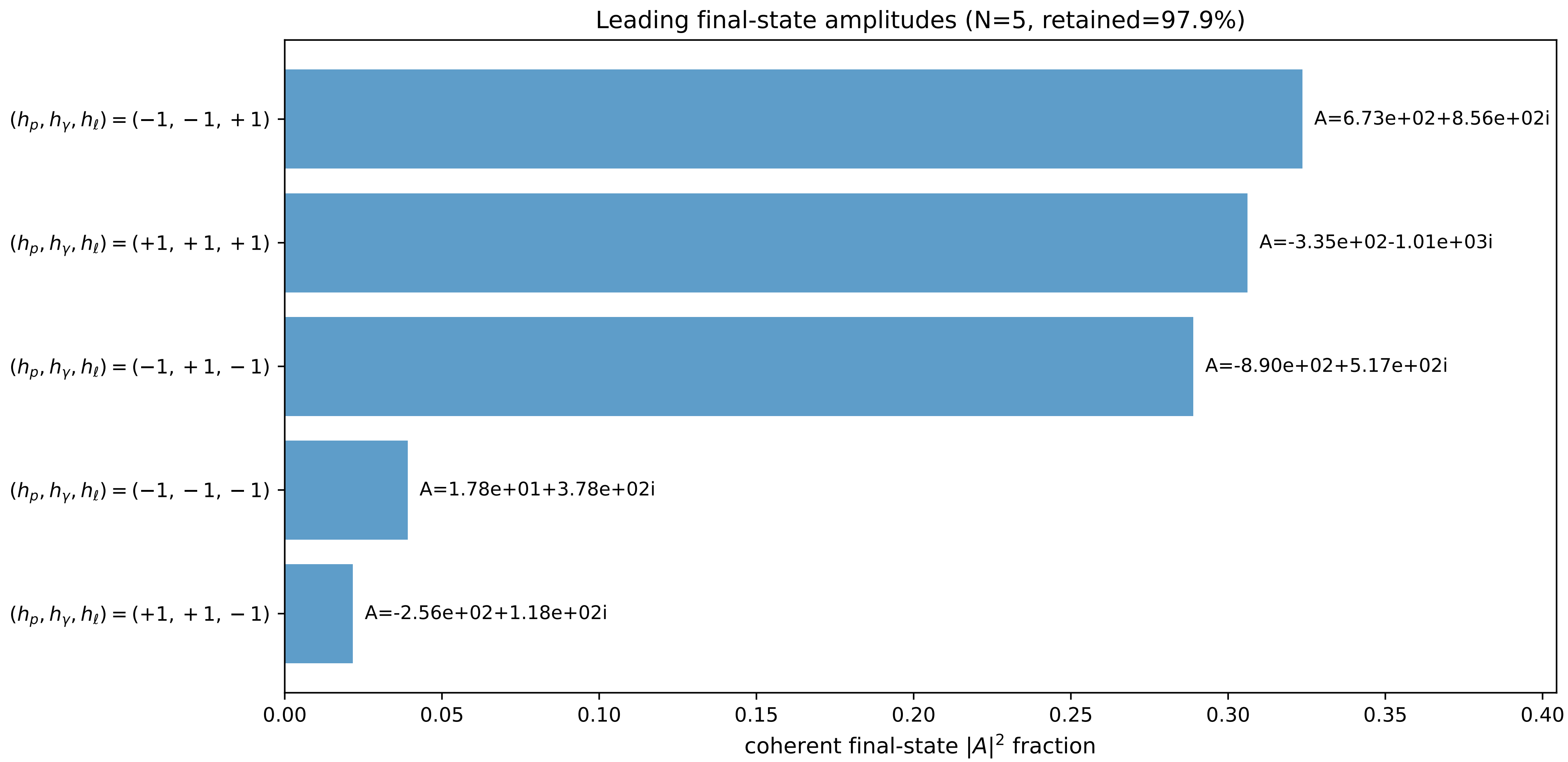


dw_mixing_angles_polarization_02_local_minimum_162 D_W=0.000784921
region: polarization_cluster_02_local_minimum_162
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.67024737 rad
incoming lepton: 0.783668|+> +0.62118|->
proton mixing angle: theta_p=0.28415972 rad
incoming proton: 0.959898|+> +0.280351|->

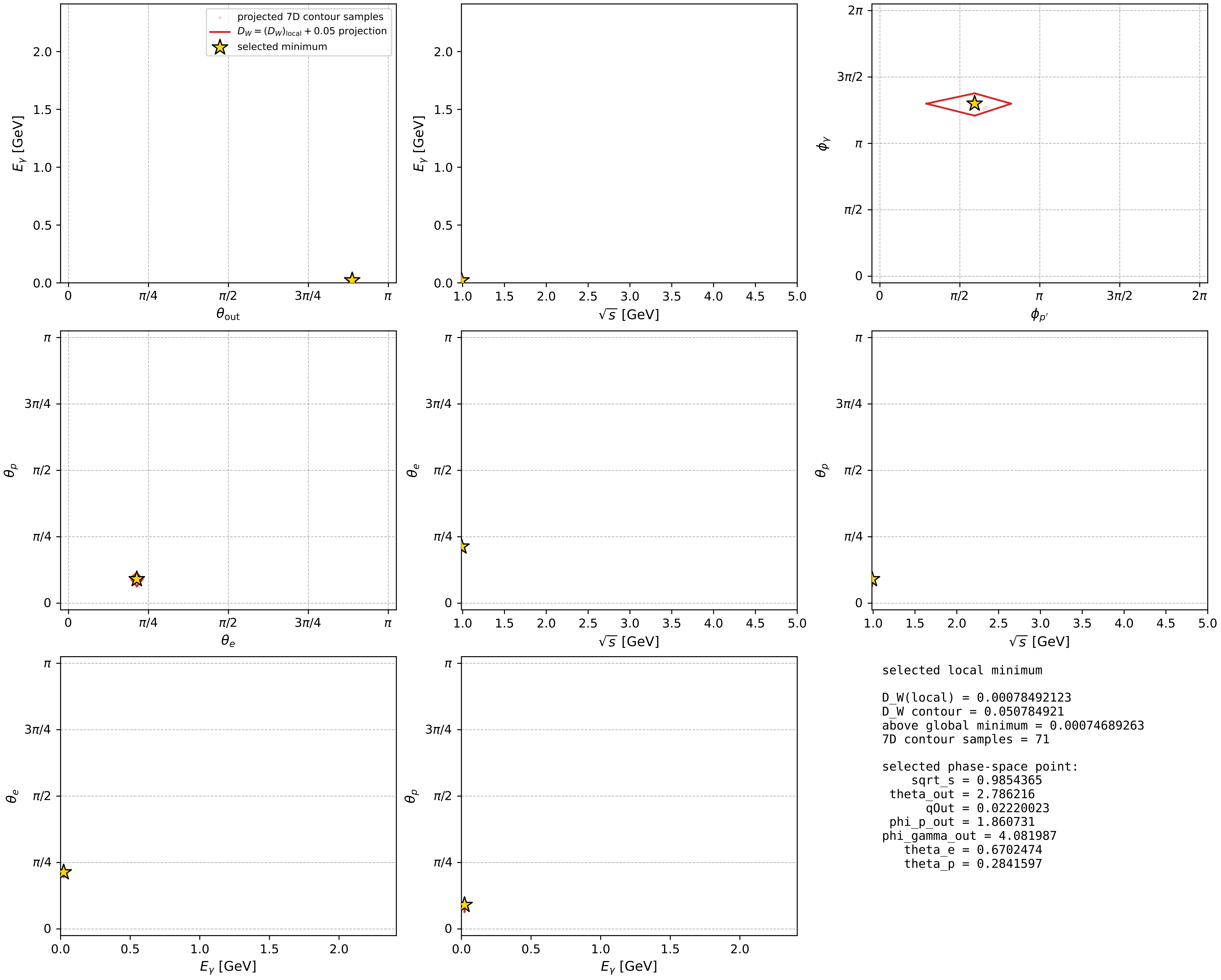
s=0.971085, $\sqrt{s}$=0.985437, $|\vec{P}|$=0.0462921, $|\vec{P}'|$=0.00363888
θ_{out}=2.78622, $\phi_{p'}$=1.86073, ϕ_γ=4.08199
E_γ=0.0222002, $\theta_{\ell'}$=0.282455, $\phi_{\ell'}$=0.796578
Q^2=0.00457827, x_B=0.0993201, t=-0.00247077, W^2=0.921362, y=0.505212

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0463 p_x= 0.0000 p_y= 0.0000 p_z= -0.0463
P E= 0.9391 p_x= 0.0000 p_y= 0.0000 p_z= 0.0463
ℓ' E= 0.0252 p_x= 0.0049 p_y= 0.0050 p_z= 0.0242
P' E= 0.9380 p_x= -0.0004 p_y= 0.0012 p_z= -0.0034
q_γ E= 0.0222 p_x= -0.0046 p_y= -0.0062 p_z= -0.0208

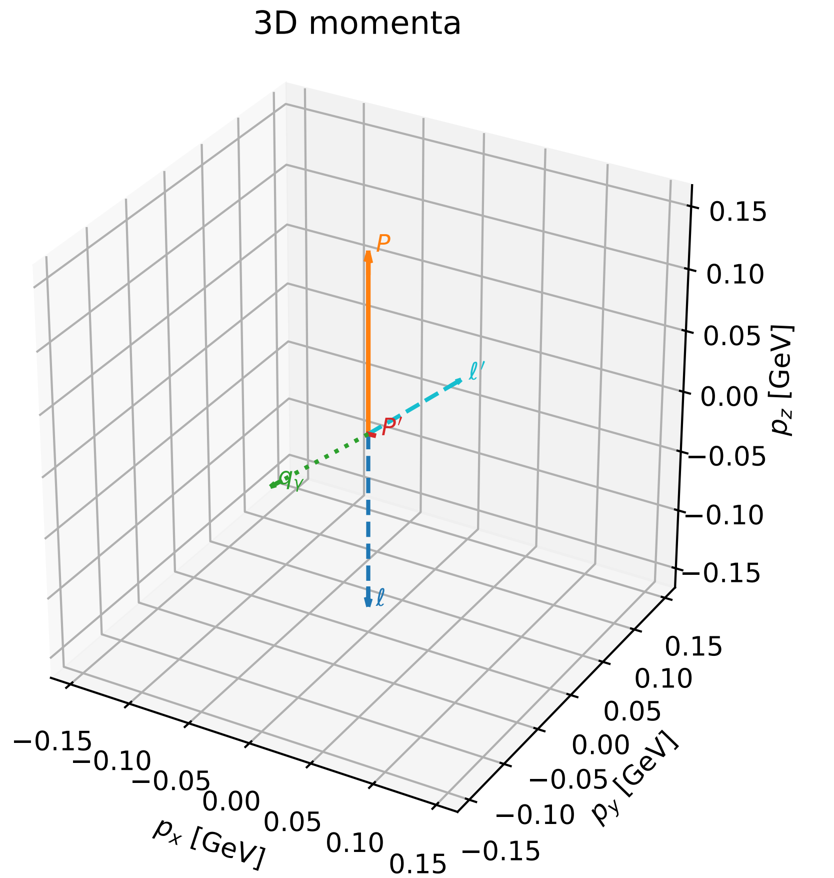




electron: polarization cluster P2, local minimum 162; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



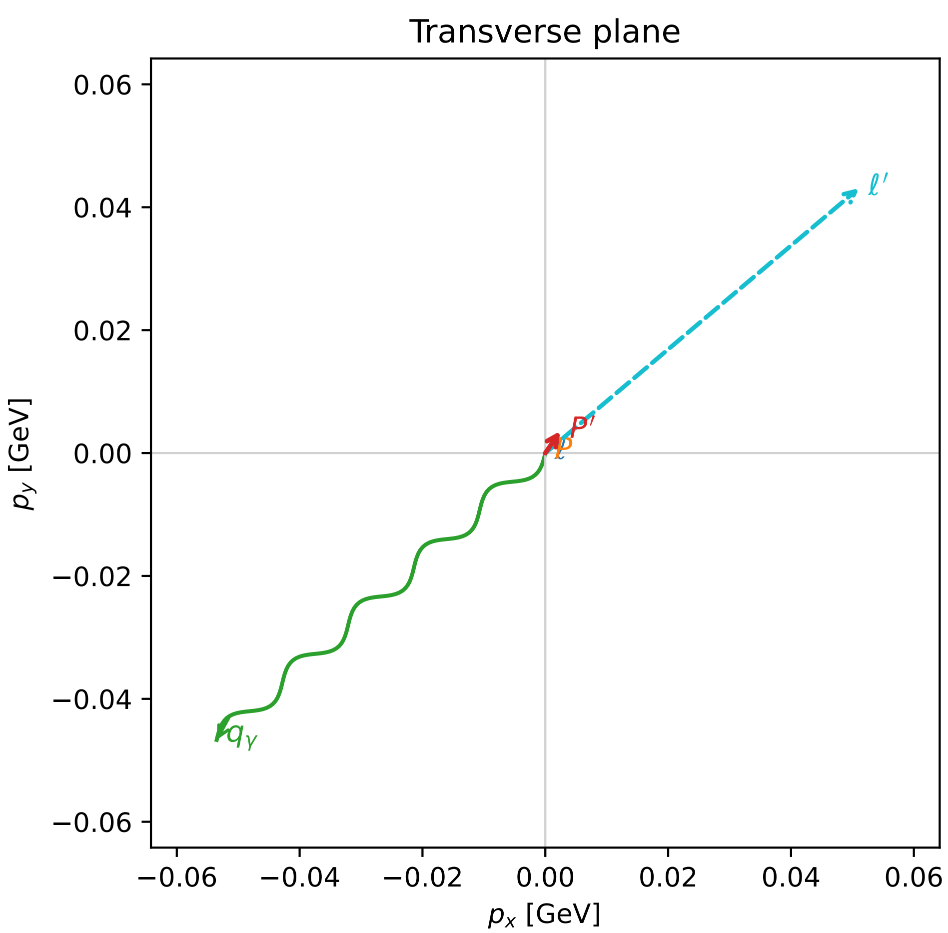
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

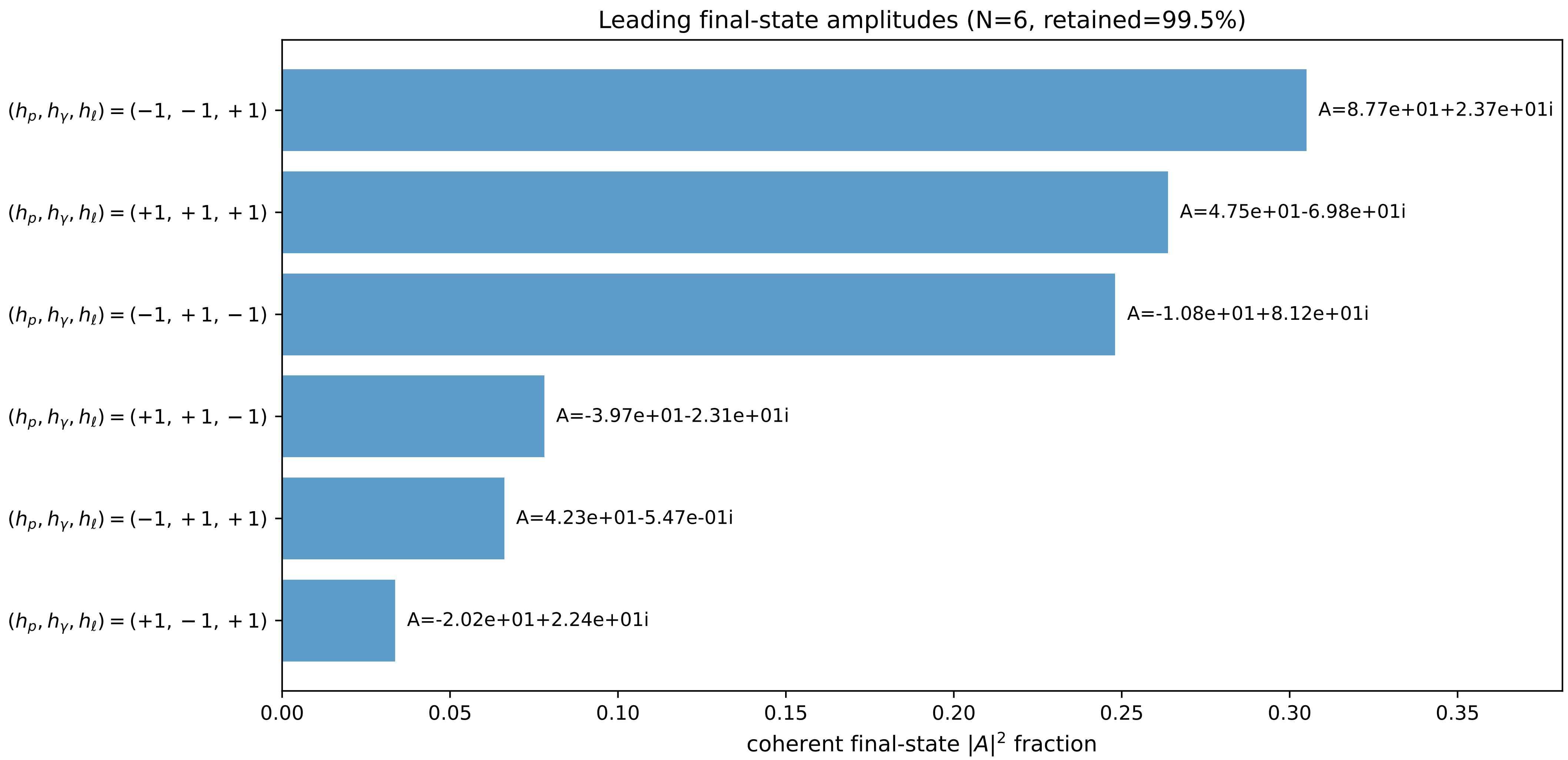


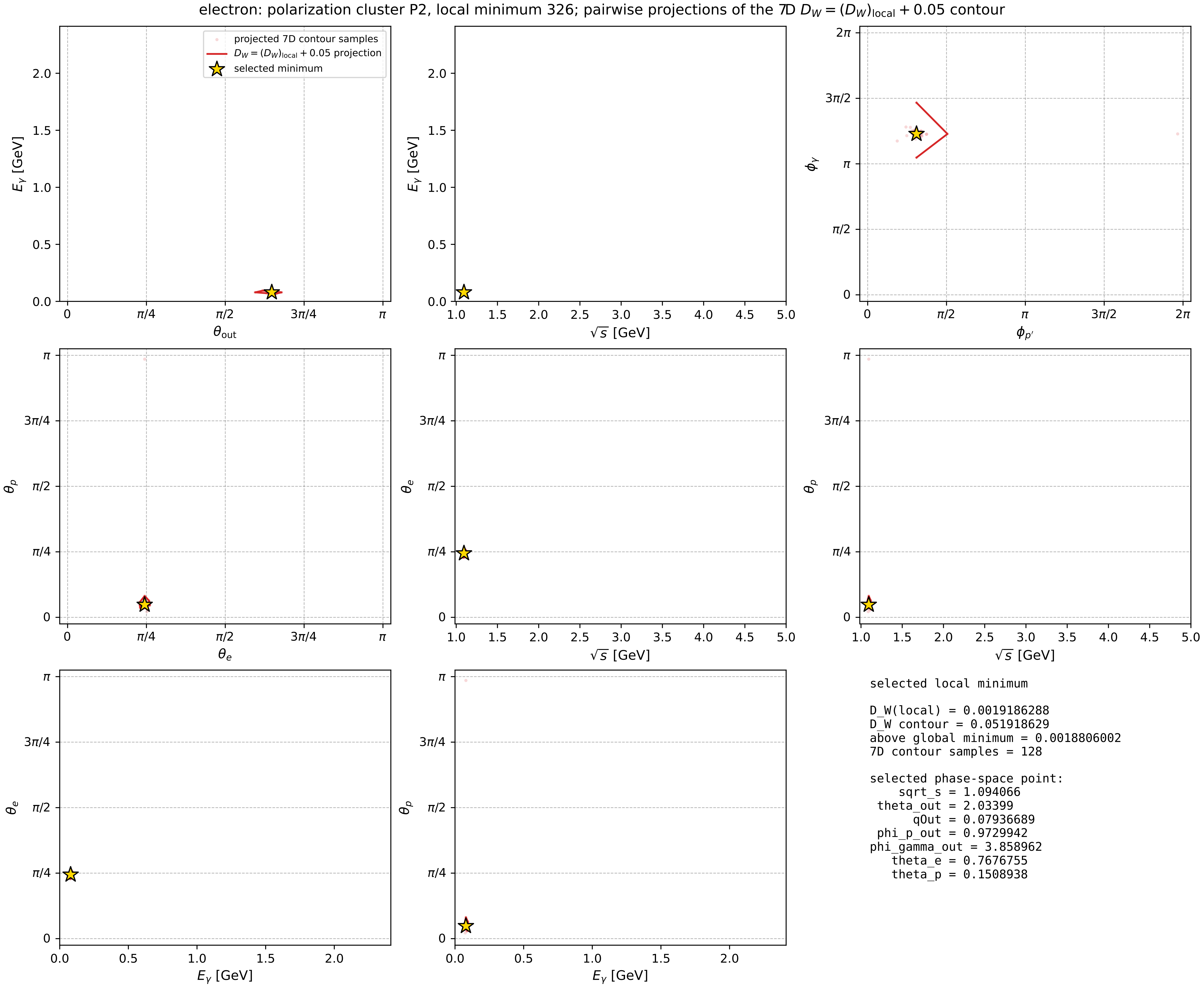
dw_mixing_angles_polarization_02_local_minimum_326 D_W=0.00191863
region: polarization_cluster_02_local_minimum_326
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.76767552 rad
incoming lepton: 0.719527|+> +0.694465|->
proton mixing angle: theta_p=0.1508938 rad
incoming proton: 0.988637|+> +0.150322|->

s=1.19698, $\sqrt{s}$=1.09407, $|\vec{P}|$=0.144934, $|\vec{P}'|$=0.00484418
θ_{out}=2.03399, $\phi_{p'}$=0.972994, ϕ_γ=3.85896
E_γ=0.0793669, $\theta_{\ell'}$=1.05795, $\phi_{\ell'}$=0.70097
Q^2=0.0331354, x_B=0.181592, t=-0.0215332, W^2=1.02918, y=0.575371

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1449 p_x= 0.0000 p_y= 0.0000 p_z= -0.1449
P E= 0.9491 p_x= 0.0000 p_y= 0.0000 p_z= 0.1449
ℓ' E= 0.0767 p_x= 0.0511 p_y= 0.0431 p_z= 0.0376
P' E= 0.9380 p_x= 0.0024 p_y= 0.0036 p_z= -0.0022
q_γ E= 0.0794 p_x= -0.0535 p_y= -0.0467 p_z= -0.0355

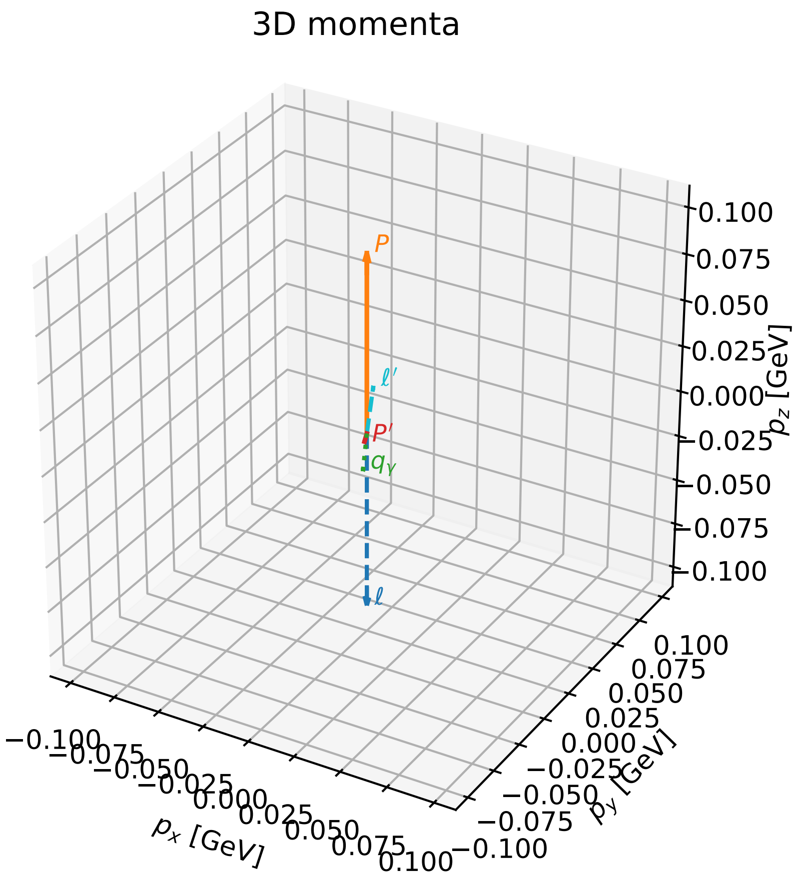






electron: polarization cluster P2, local minimum 326; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.0019186288
D_W contour = 0.051918629
above global minimum = 0.0018806002
7D contour samples = 128
selected phase-space point:
sqrt_s = 1.094066
theta_out = 2.03399
qOut = 0.07936689
phi_p_out = 0.9729942
phi_gamma_out = 3.858962
theta_e = 0.7676755
theta_p = 0.1508938

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)




```
dw_mixing_angles_polarization_02_local_minimum_355  D_W=0.00216699
region: polarization_cluster_02_local_minimum_355
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.74589264 rad
incoming lepton: 0.734482|+> +0.678628|->
proton mixing angle: theta_p=0.22192321 rad
incoming proton: 0.975476|+> +0.220106|->
```

s=1.08139, $\sqrt{s}$=1.0399, $|\vec{P}|$=0.0969061, $|\vec{P}'|$=0.00401962

θ_{out}=2.46683, $\phi_{p'}$=4.77794, ϕ_γ=2.15572

E_γ=0.0503272, $\theta_{\ell'}$=0.604067, $\phi_{\ell'}$=5.33988

Q^2=0.0182182, x_B=0.161906, t=-0.00999044, W^2=0.974149, y=0.558296

four-momenta [E, p_x, p_y, p_z] GeV:

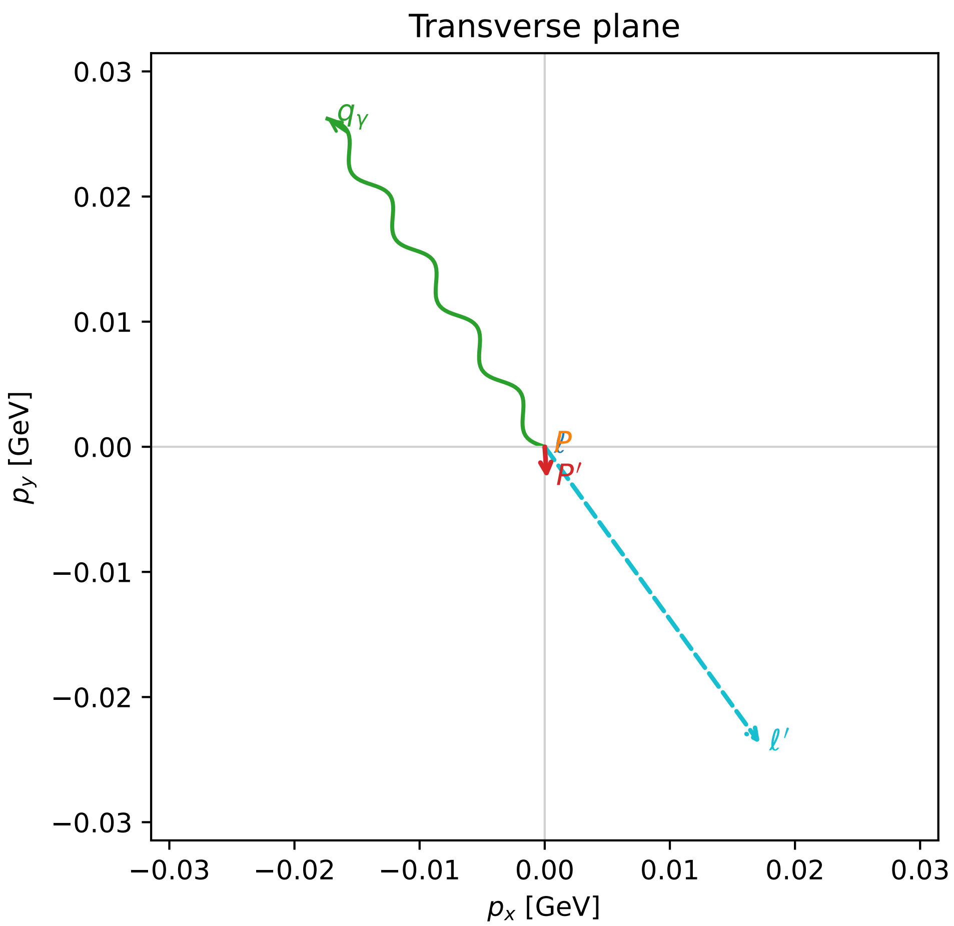
ℓ E= 0.0969 p_x= 0.0000 p_y= 0.0000 p_z= -0.0969

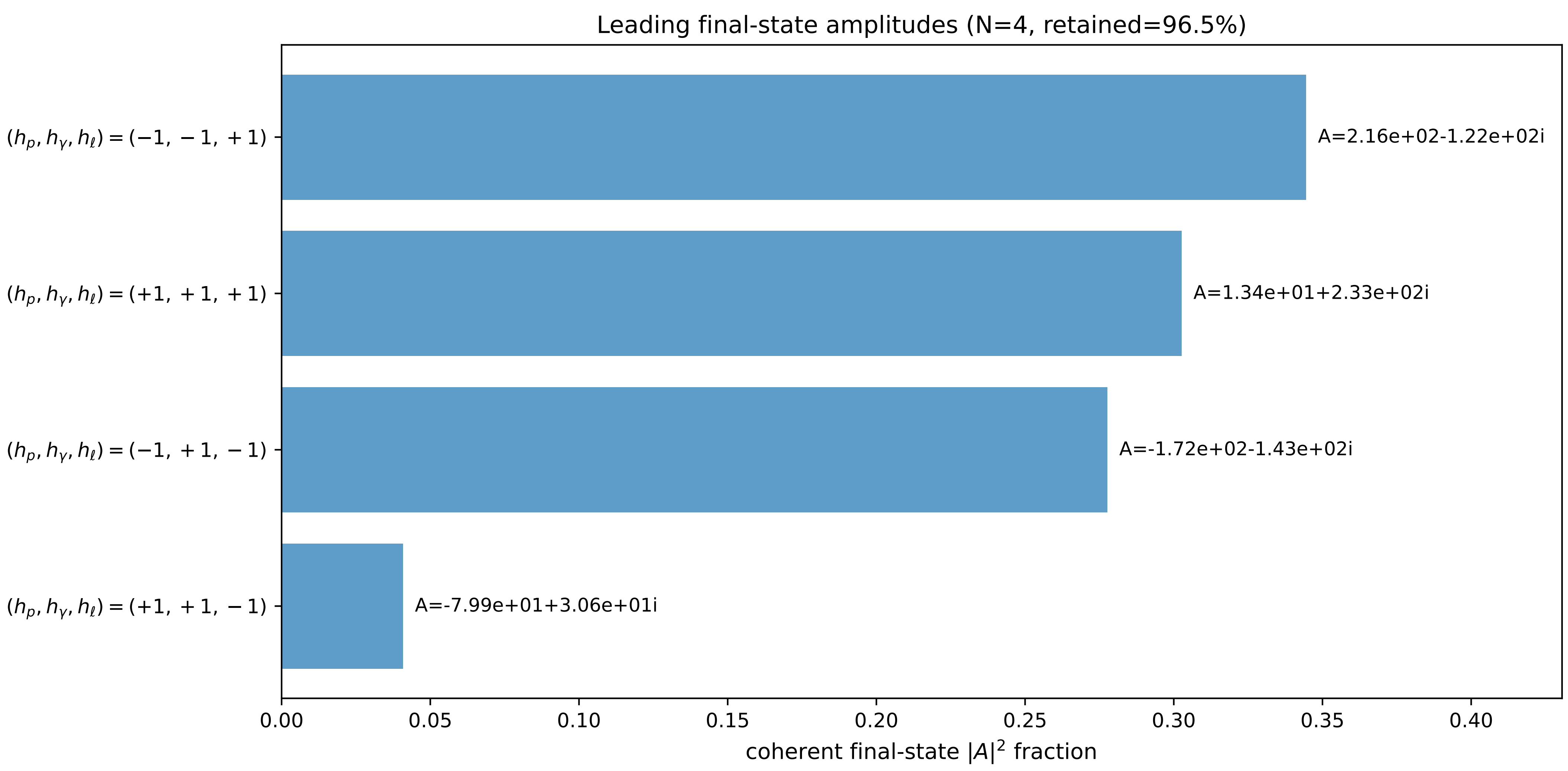
P E= 0.9430 p_x= 0.0000 p_y= 0.0000 p_z= 0.0969

ℓ' E= 0.0516 p_x= 0.0172 p_y= -0.0237 p_z= 0.0424

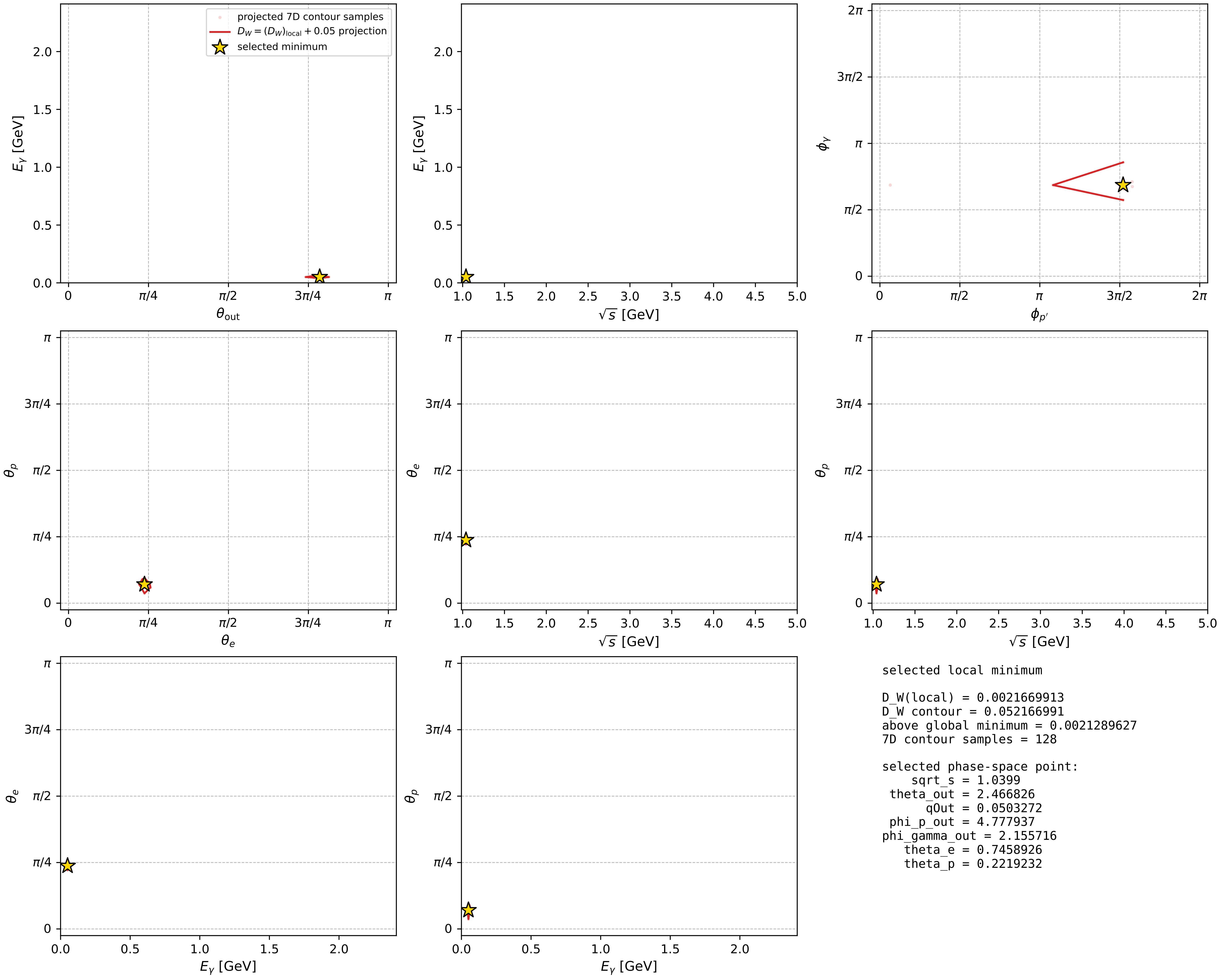
P' E= 0.9380 p_x= 0.0002 p_y= -0.0025 p_z= -0.0031

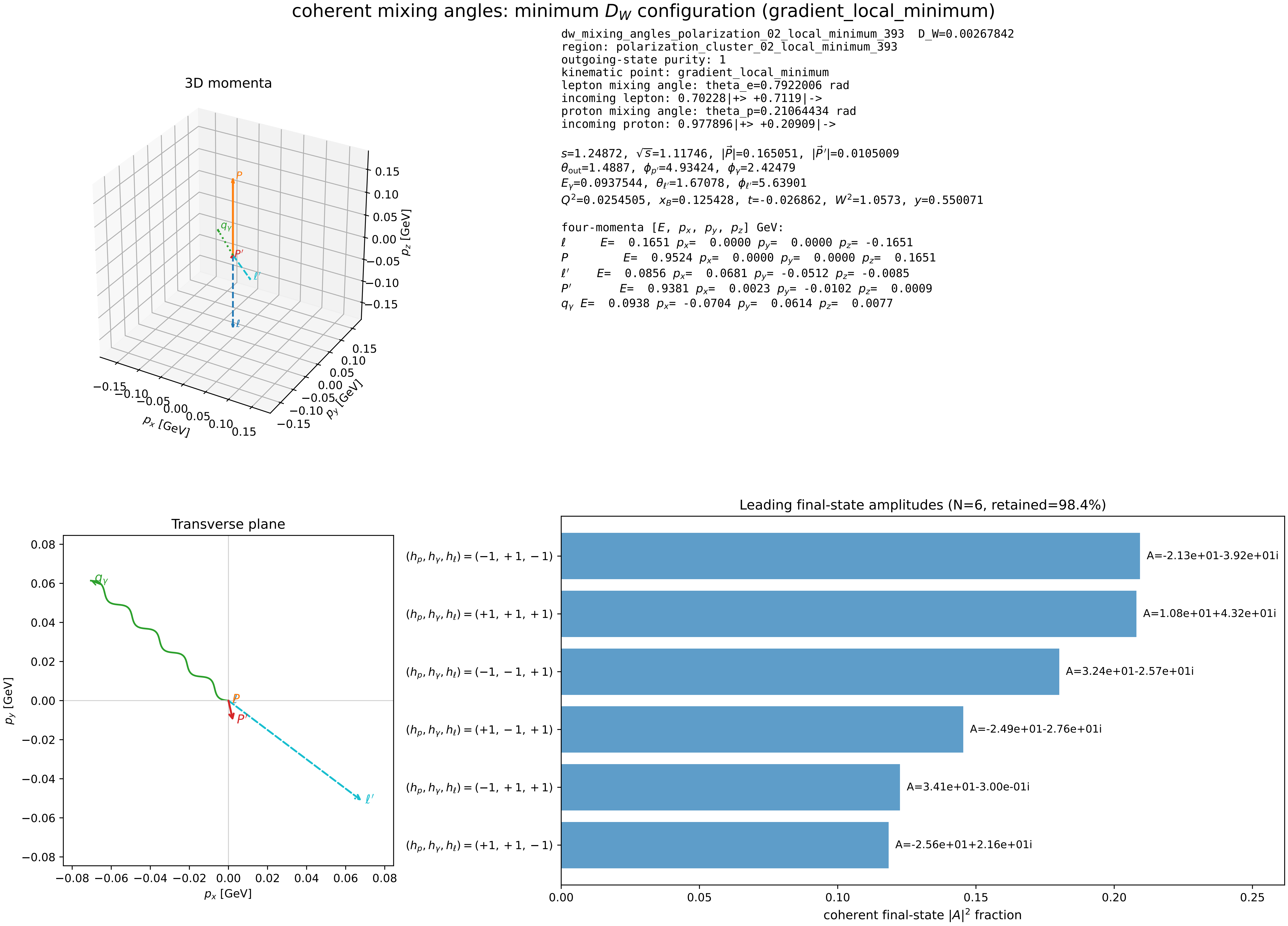
q_γ E= 0.0503 p_x= -0.0174 p_y= 0.0262 p_z= -0.0393

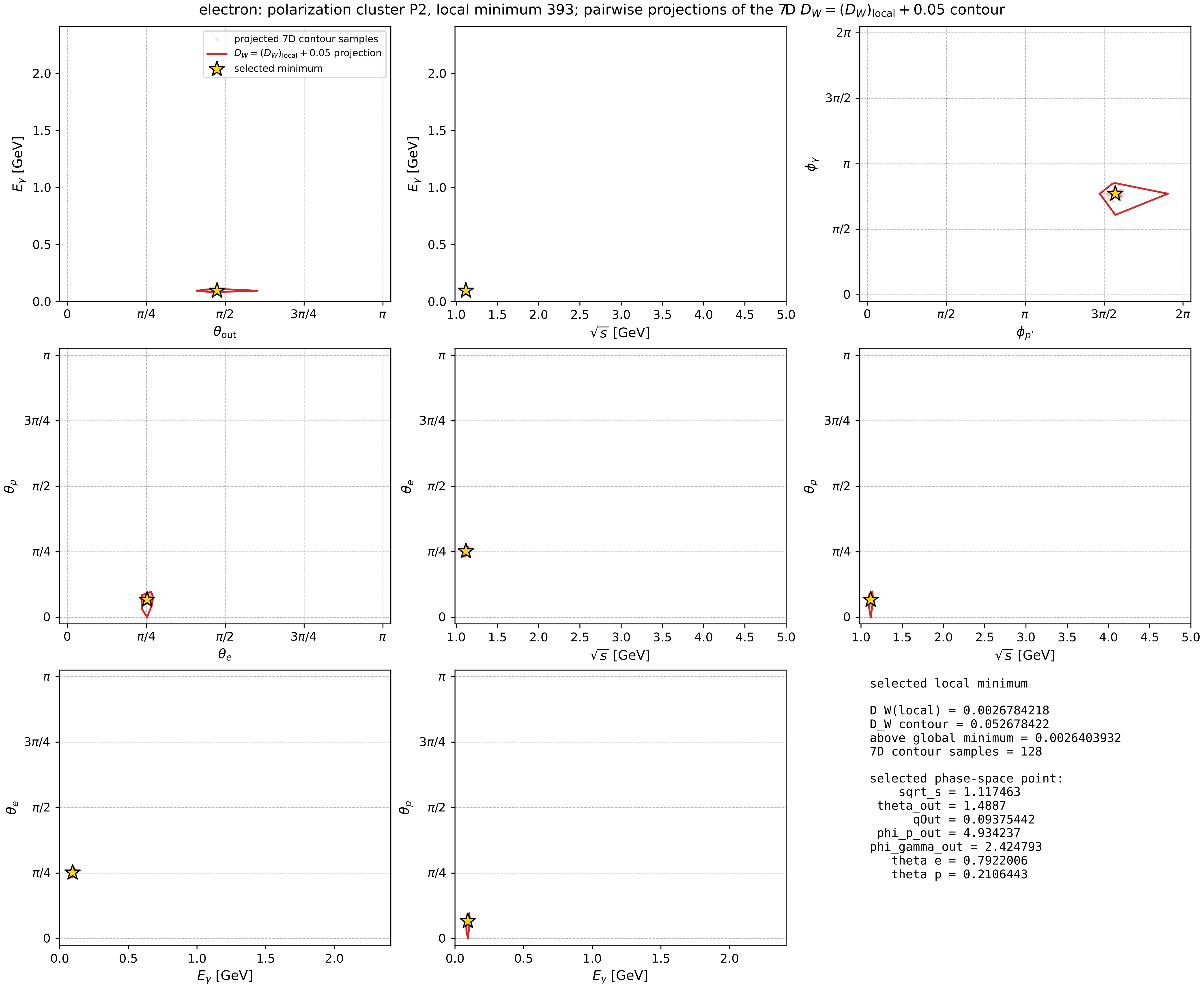




electron: polarization cluster P2, local minimum 355; pairwise projections of the 7D $D_W = (D_W)_{\rm local} + 0.05$ contour

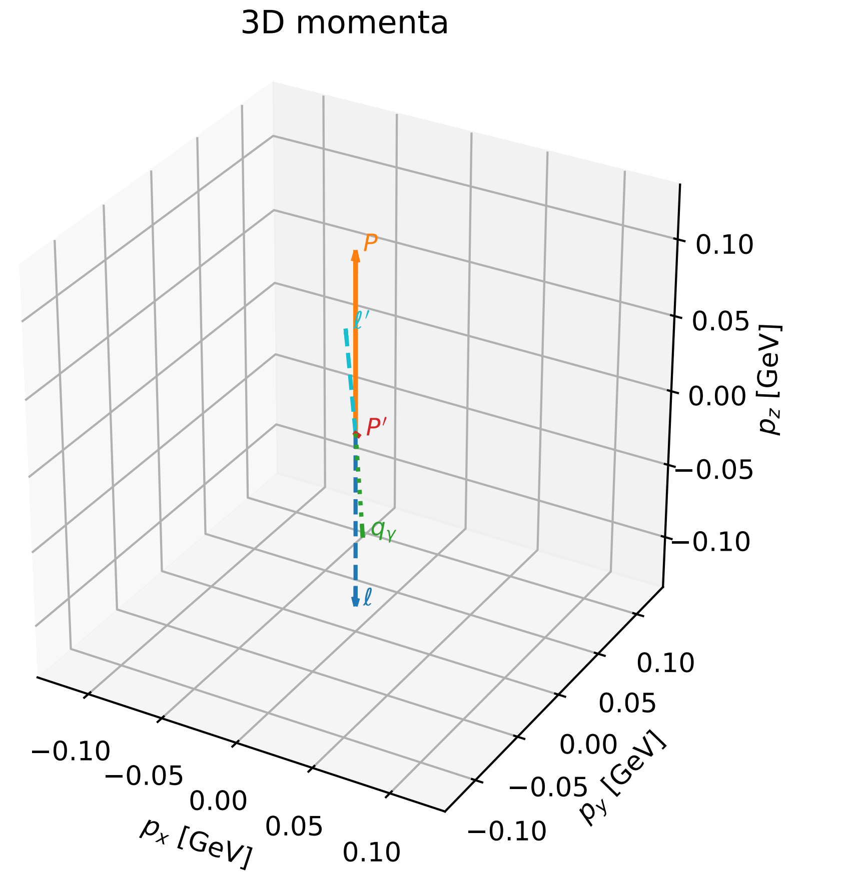



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
dw_mixing_angles_polarization_02_local_minimum_393 D_W=0.00267842
region: polarization_cluster_02_local_minimum_393
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.7922006 rad
incoming lepton: 0.70228|+> +0.7119|->
proton mixing angle: theta_p=0.21064434 rad
incoming proton: 0.977896|+> +0.20909|->
s=1.24872, $\sqrt{s}$=1.11746, $|\vec{P}|$=0.165051, $|\vec{P}'|$=0.0105009
$\theta_{\rm out}$=1.4887, $\phi_{p'}$=4.93424, ϕ_γ=2.42479
E_γ=0.0937544, $\theta_{\ell'}$=1.67078, $\phi_{\ell'}$=5.63901
Q^2=0.0254505, x_B=0.125428, t=-0.026862, W^2=1.0573, y=0.550071
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1651 p_x= 0.0000 p_y= 0.0000 p_z= -0.1651
P E= 0.9524 p_x= 0.0000 p_y= 0.0000 p_z= 0.1651
ℓ' E= 0.0856 p_x= 0.0681 p_y= -0.0512 p_z= -0.0085
P' E= 0.9381 p_x= 0.0023 p_y= -0.0102 p_z= 0.0009
q_γ E= 0.0938 p_x= -0.0704 p_y= 0.0614 p_z= 0.0077
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
q_γ
P
P'
ℓ'
ℓ
Leading final-state amplitudes (N=6, retained=98.4%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-2.13e+01-3.92e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=1.08e+01+4.32e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=3.24e+01-2.57e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=-2.49e+01-2.76e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=3.41e+01-3.00e-01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-2.56e+01+2.16e+01i
coherent final-state $|A|^2$ fraction



electron: polarization cluster P2, local minimum 393; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.0026784218
D_W contour = 0.052678422
above global minimum = 0.0026403932
7D contour samples = 128
selected phase-space point:
sqrt_s = 1.117463
theta_out = 1.4887
qOut = 0.09375442
phi_p_out = 4.934237
phi_gamma_out = 2.424793
theta_e = 0.7922006
theta_p = 0.2106443

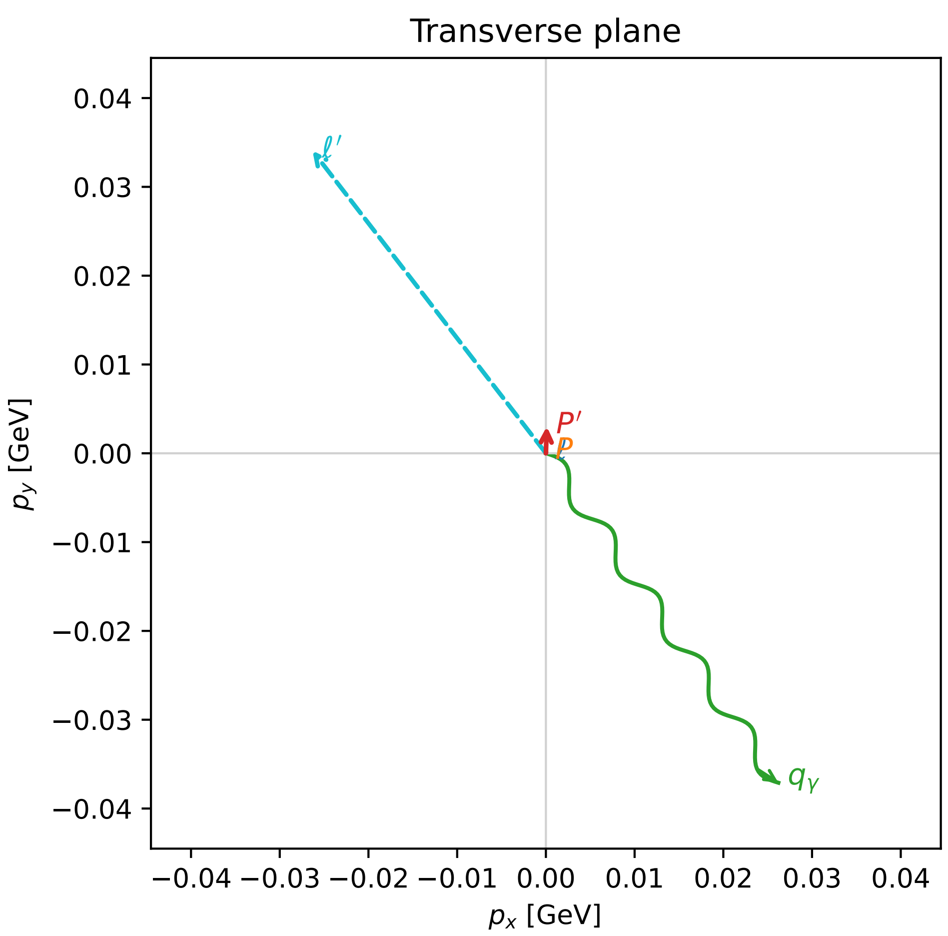
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

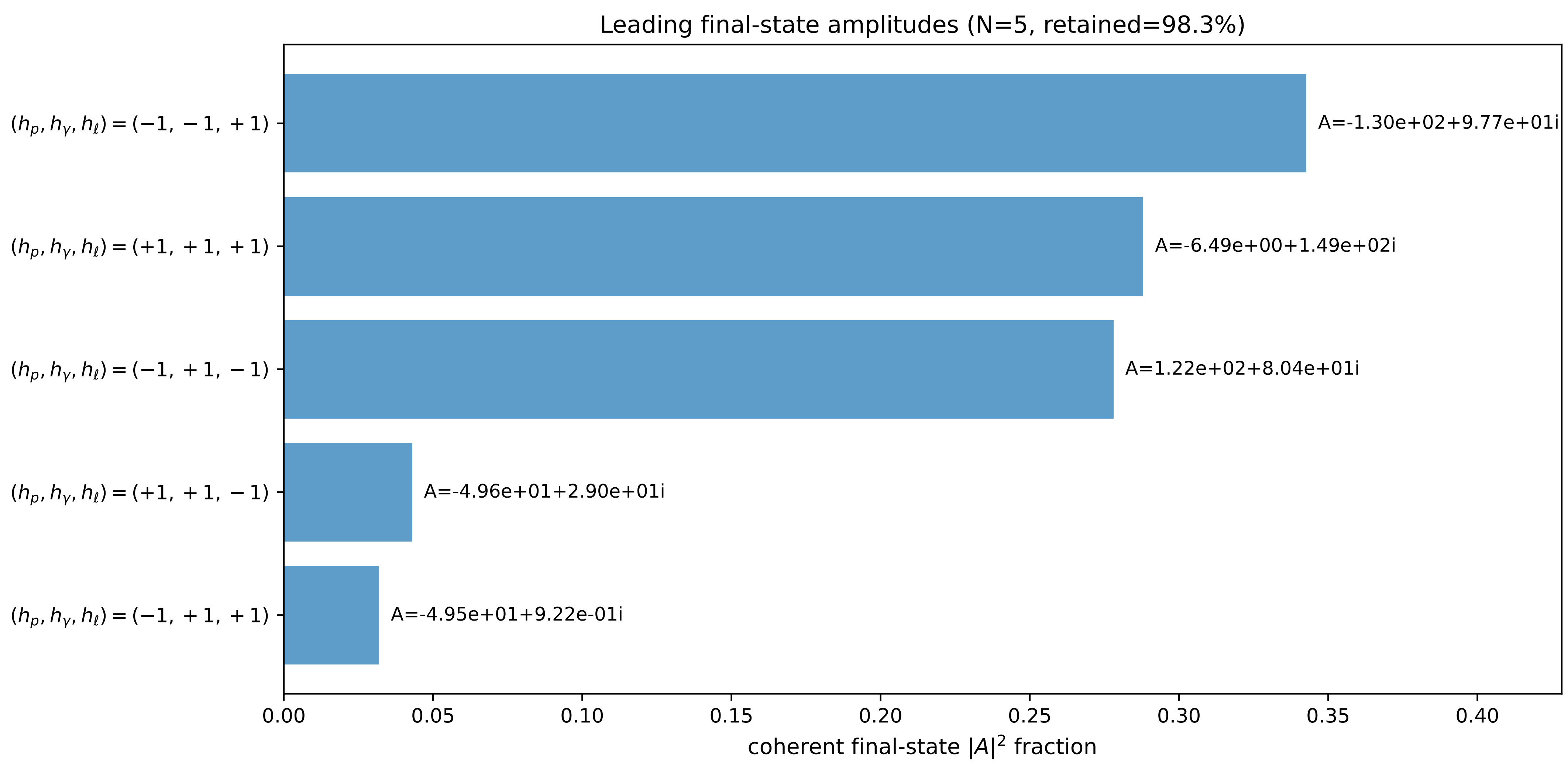


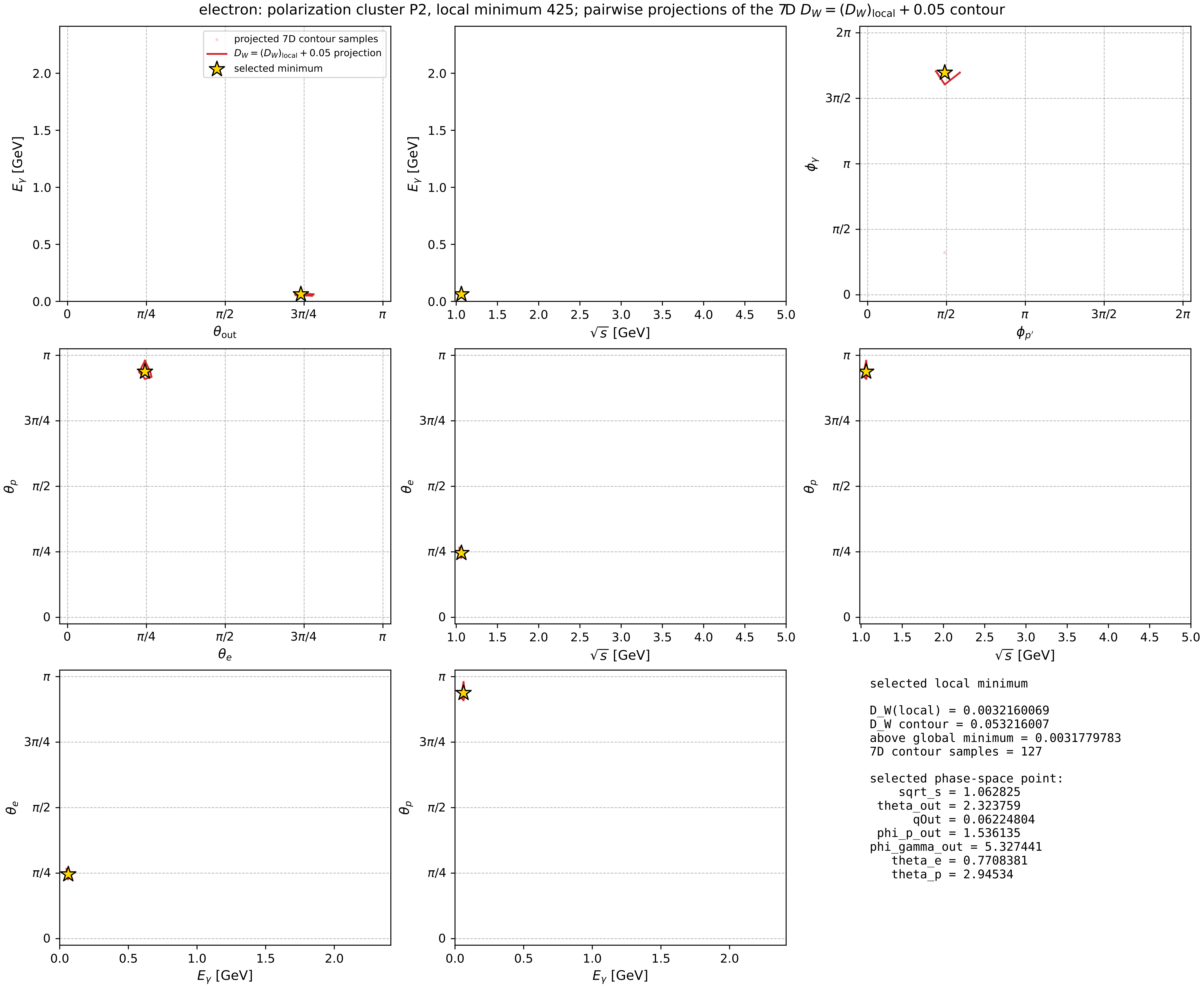
dw_mixing_angles_polarization_02_local_minimum_425 D_W=0.00321601
region: polarization_cluster_02_local_minimum_425
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.77083808 rad
incoming lepton: 0.717327|+> +0.696737|->
proton mixing angle: theta_p=2.9453404 rad
incoming proton: -0.980804|+> +0.194995|->

s=1.1296, $\sqrt{s}$=1.06283, $|\vec{P}|$=0.117494, $|\vec{P}'|$=0.00414224
θ_{out}=2.32376, $\phi_{p'}$=1.53613, ϕ_γ=5.32744
E_γ=0.062248, $\theta_{\ell'}$=0.758888, $\phi_{\ell'}$=2.22833
Q^2=0.0253703, x_B=0.178505, t=-0.014434, W^2=0.9966, y=0.569067

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1175 p_x= 0.0000 p_y= 0.0000 p_z= -0.1175
P E= 0.9453 p_x= 0.0000 p_y= 0.0000 p_z= 0.1175
ℓ' E= 0.0626 p_x= -0.0263 p_y= 0.0341 p_z= 0.0454
P' E= 0.9380 p_x= 0.0001 p_y= 0.0030 p_z= -0.0028
q_γ E= 0.0622 p_x= 0.0262 p_y= -0.0371 p_z= -0.0426

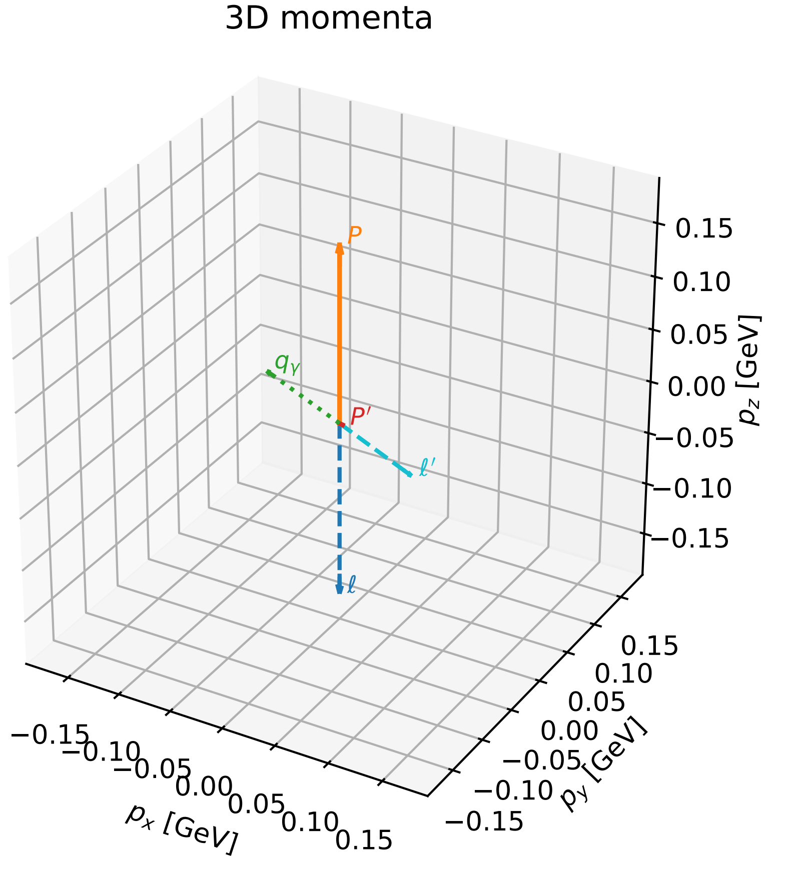






electron: polarization cluster P2, local minimum 425; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.0032160069
D_W contour = 0.053216007
above global minimum = 0.0031779783
7D contour samples = 127
selected phase-space point:
sqrt_s = 1.062825
theta_out = 2.323759
qOut = 0.06224804
phi_p_out = 1.536135
phi_gamma_out = 5.327441
theta_e = 0.7708381
theta_p = 2.94534

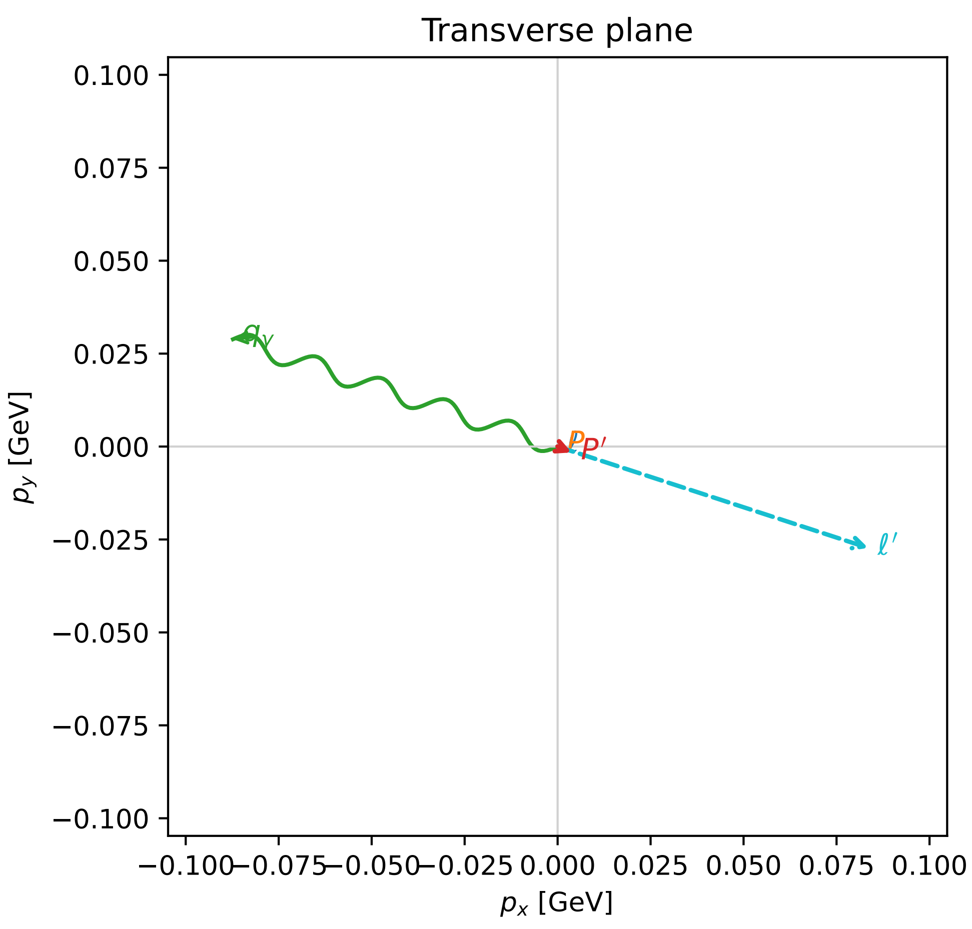
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

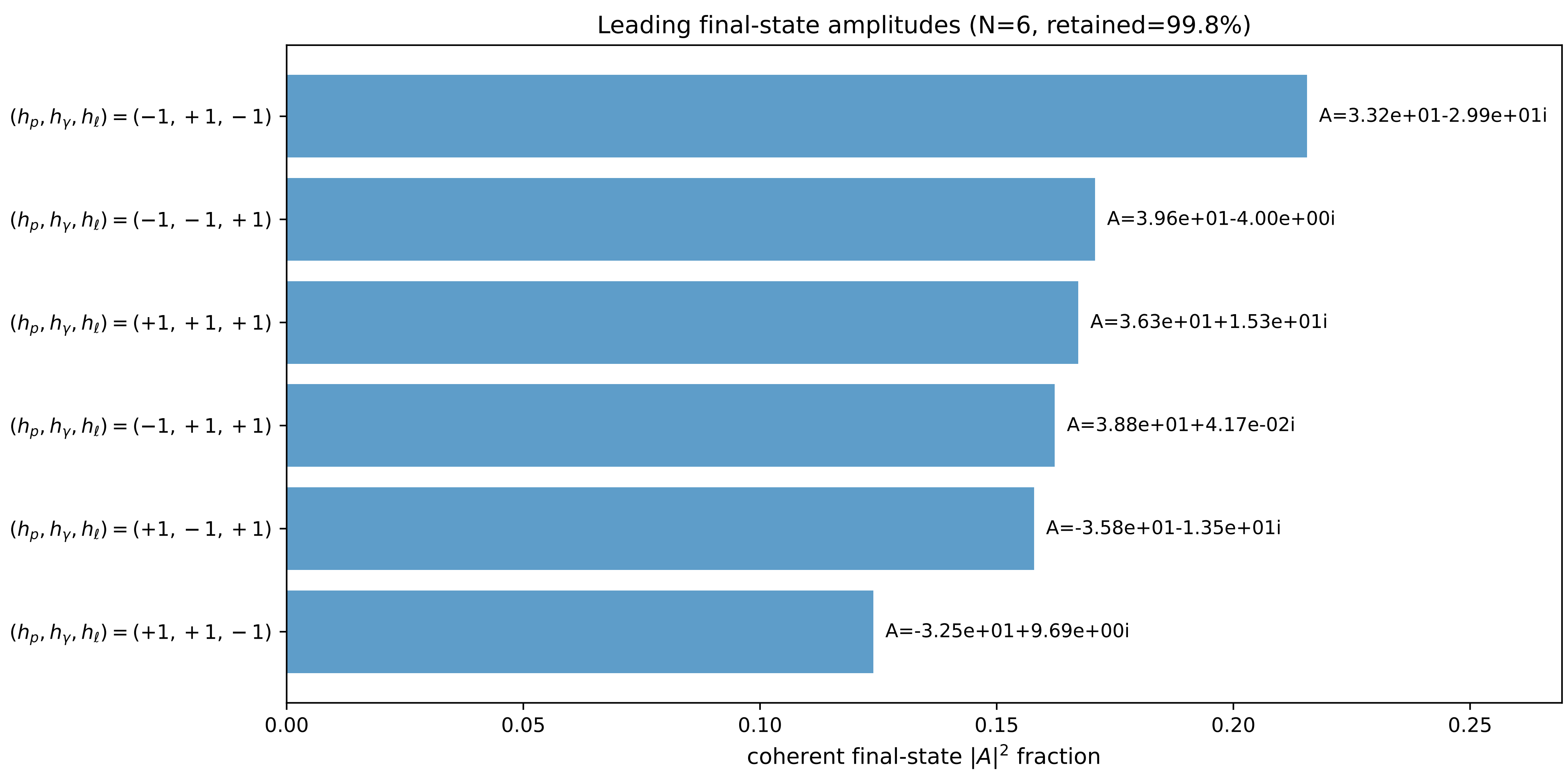


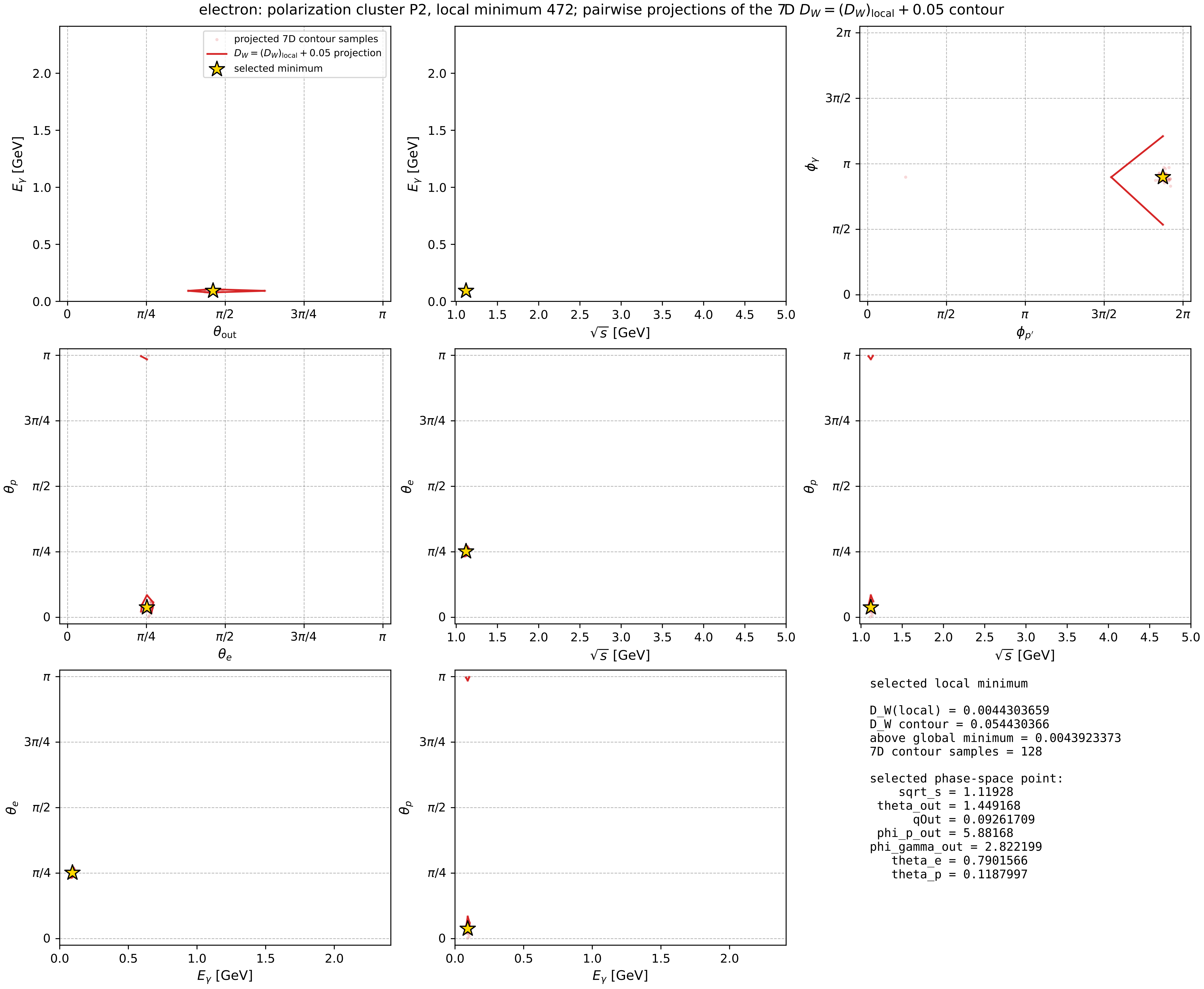
dw_mixing_angles_polarization_02_local_minimum_472 D_W=0.00443037
region: polarization_cluster_02_local_minimum_472
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.79015657 rad
incoming lepton: 0.703734|+> +0.710463|->
proton mixing angle: theta_p=0.11879968 rad
incoming proton: 0.992952|+> +0.11852|->

s=1.25279, $\sqrt{s}$=1.11928, $|\vec{P}|$=0.166599, $|\vec{P}'|$=0.00410572
θ_{out}=1.44917, $\phi_{p'}$=5.88168, ϕ_γ=2.8222
E_γ=0.0926171, $\theta_{\ell'}$=1.70356, $\phi_{\ell'}$=5.9676
Q^2=0.0256286, x_B=0.128069, t=-0.0273908, W^2=1.05433, y=0.536586

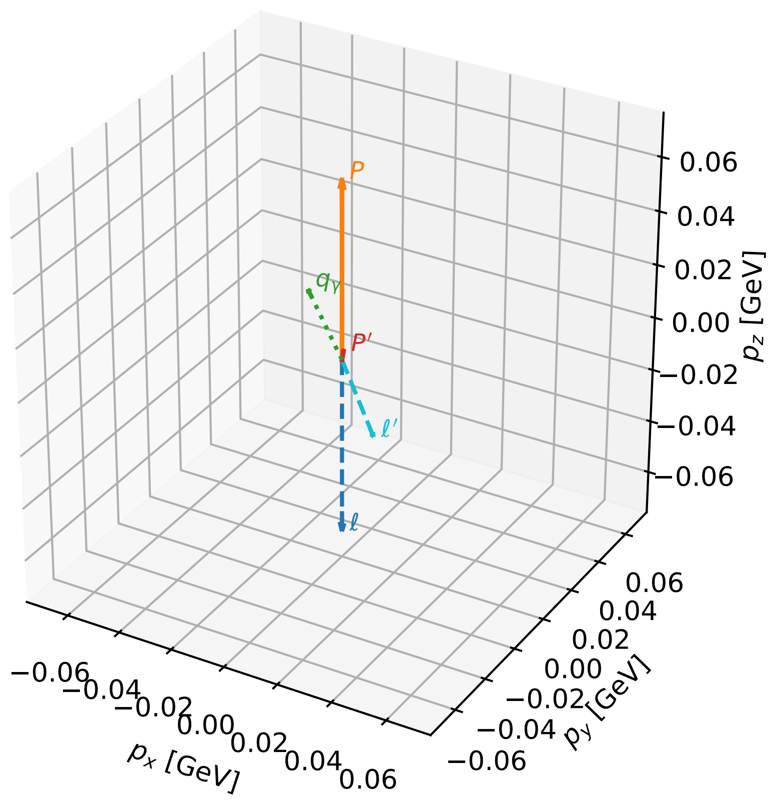
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1666 p_x= 0.0000 p_y= 0.0000 p_z= -0.1666
P E= 0.9527 p_x= 0.0000 p_y= 0.0000 p_z= 0.1666
ℓ' E= 0.0887 p_x= 0.0835 p_y= -0.0273 p_z= -0.0117
P' E= 0.9380 p_x= 0.0038 p_y= -0.0016 p_z= 0.0005
q_γ E= 0.0926 p_x= -0.0873 p_y= 0.0289 p_z= 0.0112






electron: polarization cluster P2, local minimum 472; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.0044303659
D_W contour = 0.054430366
above global minimum = 0.0043923373
7D contour samples = 128
selected phase-space point:
sqrt_s = 1.11928
theta_out = 1.449168
qOut = 0.09261709
phi_p_out = 5.88168
phi_gamma_out = 2.822199
theta_e = 0.7901566
theta_p = 0.1187997

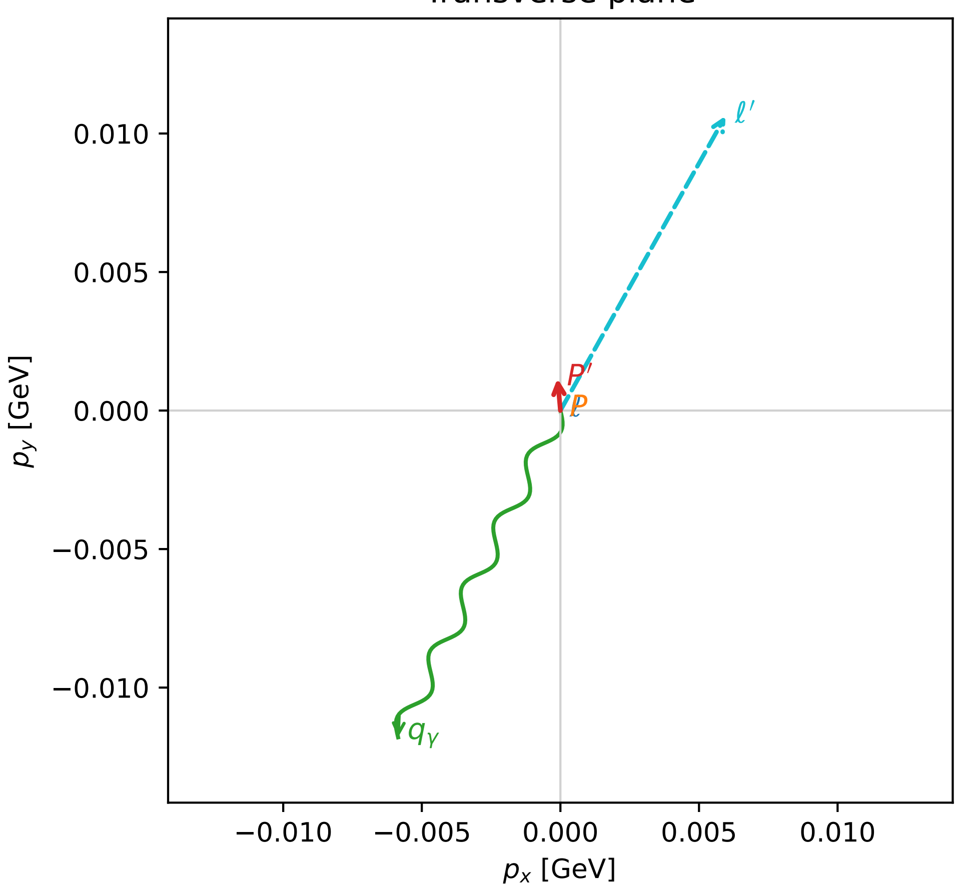
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

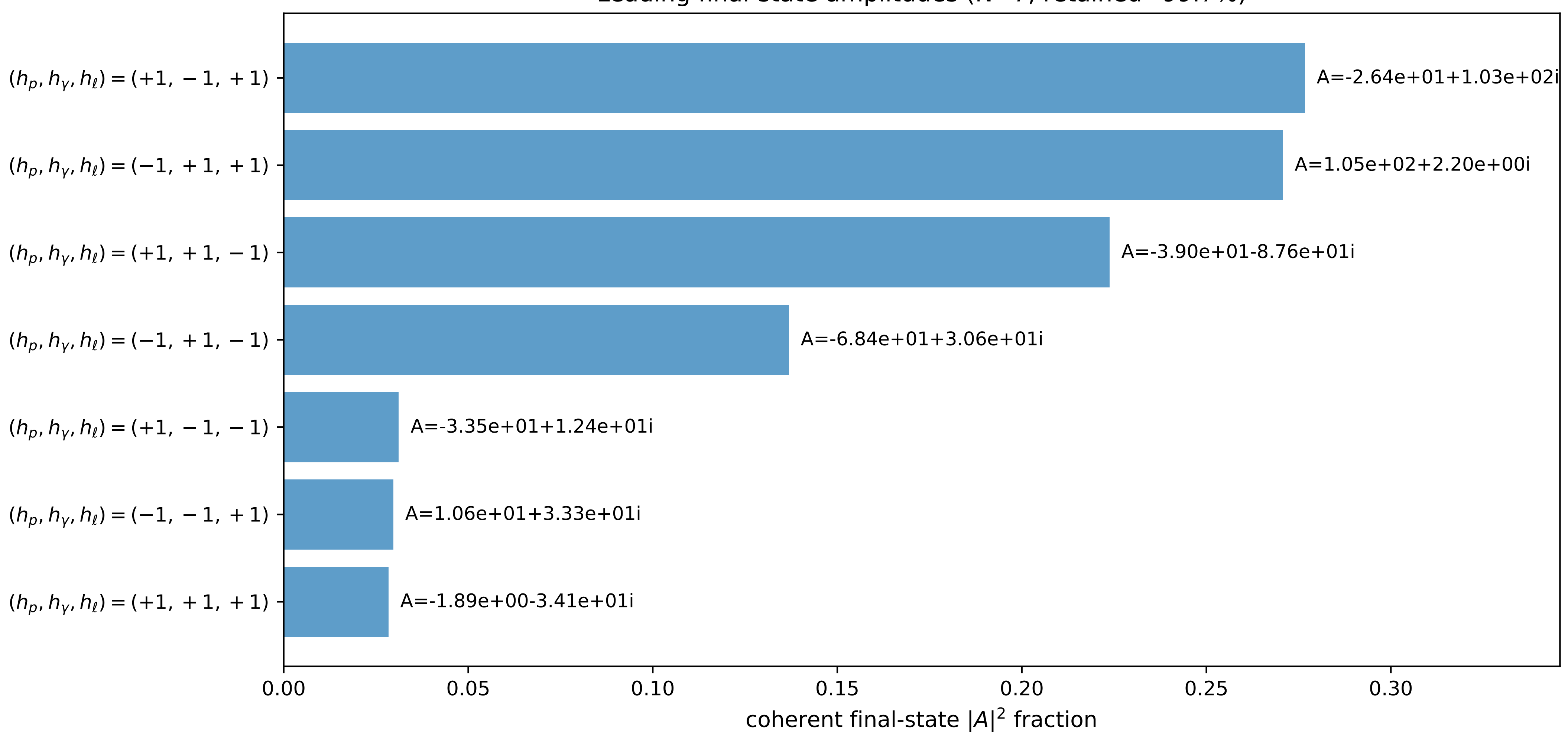


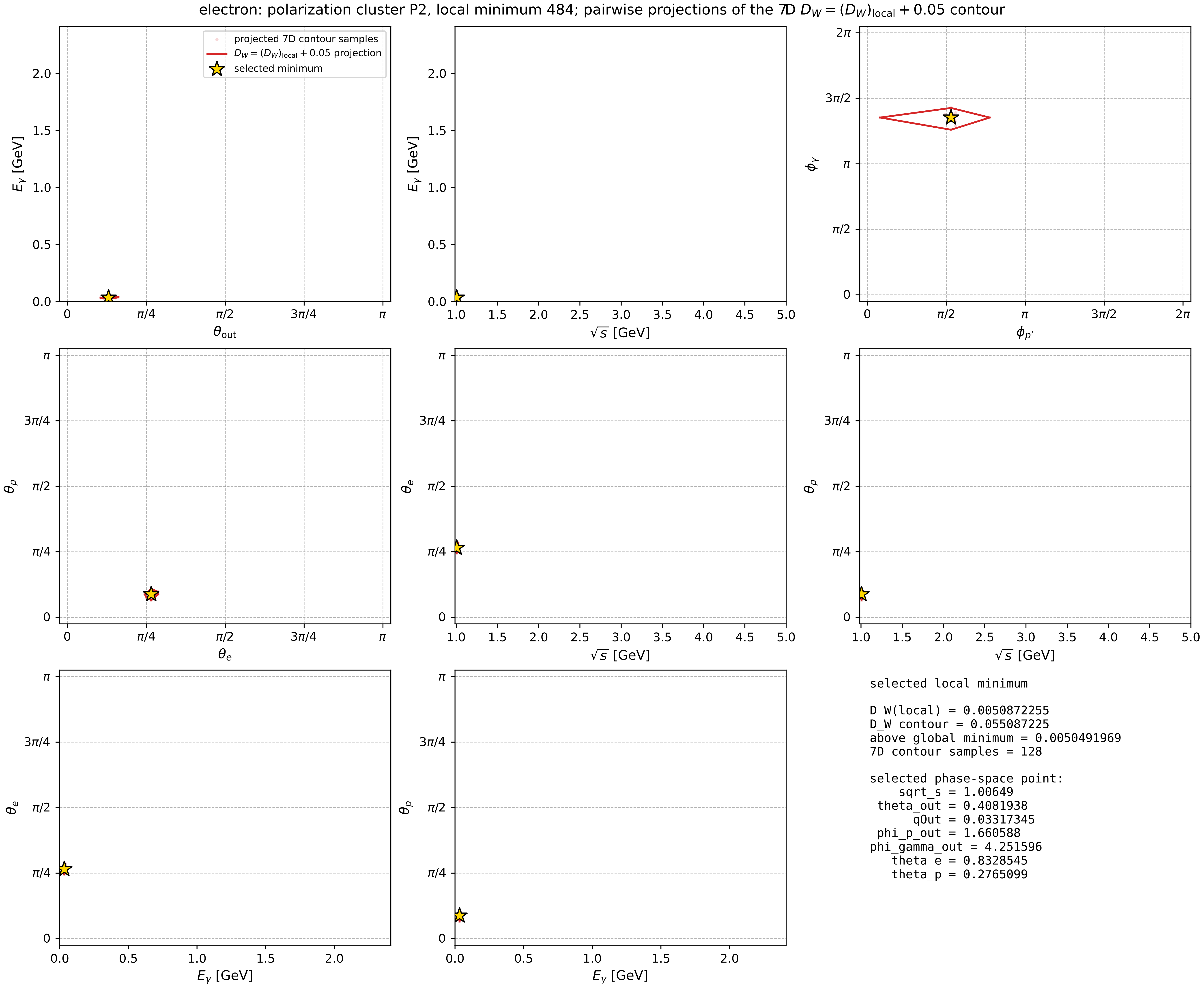
dw_mixing_angles_polarization_02_local_minimum_484 D_W=0.00508723
region: polarization_cluster_02_local_minimum_484
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.83285454 rad
incoming lepton: 0.672767|+> +0.739855|->
proton mixing angle: theta_p=0.27650992 rad
incoming proton: 0.962014|+> +0.273|->

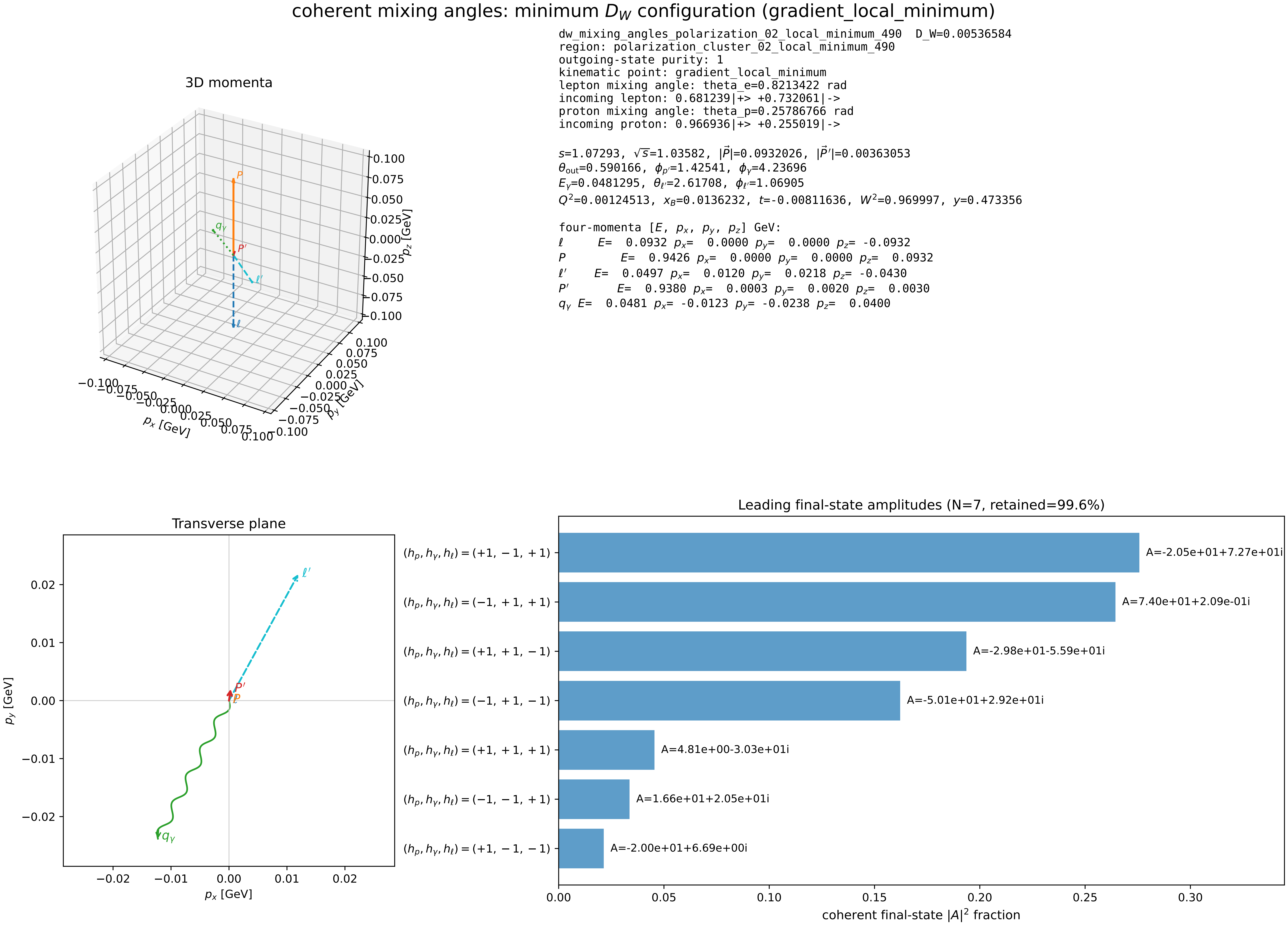
s=1.01302, $\sqrt{s}$=1.00649, $|\vec{P}|$=0.0661576, $|\vec{P}'|$=0.00292872
$\theta_{\rm out}$=0.408194, $\phi_{p'}$=1.66059, ϕ_γ=4.2516
E_γ=0.0331735, $\theta_{\ell'}$=2.78901, $\phi_{\ell'}$=1.0601
Q^2=0.000287498, x_B=0.00460855, t=-0.00402432, W^2=0.94194, y=0.468425

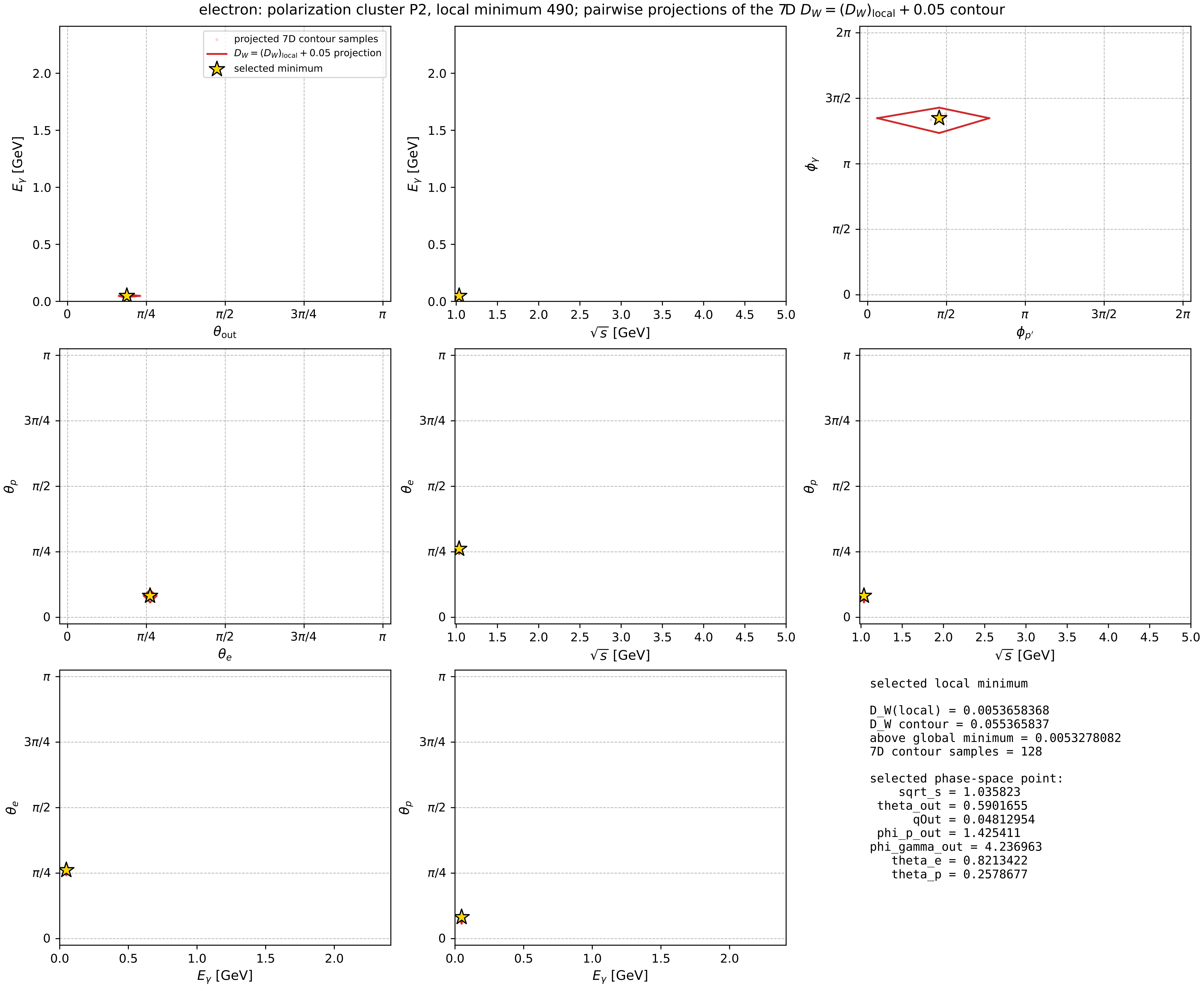
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0662 p_x= 0.0000 p_y= 0.0000 p_z= -0.0662
P E= 0.9403 p_x= 0.0000 p_y= 0.0000 p_z= 0.0662
ℓ' E= 0.0353 p_x= 0.0060 p_y= 0.0106 p_z= -0.0331
P' E= 0.9380 p_x= -0.0001 p_y= 0.0012 p_z= 0.0027
q_γ E= 0.0332 p_x= -0.0059 p_y= -0.0118 p_z= 0.0304



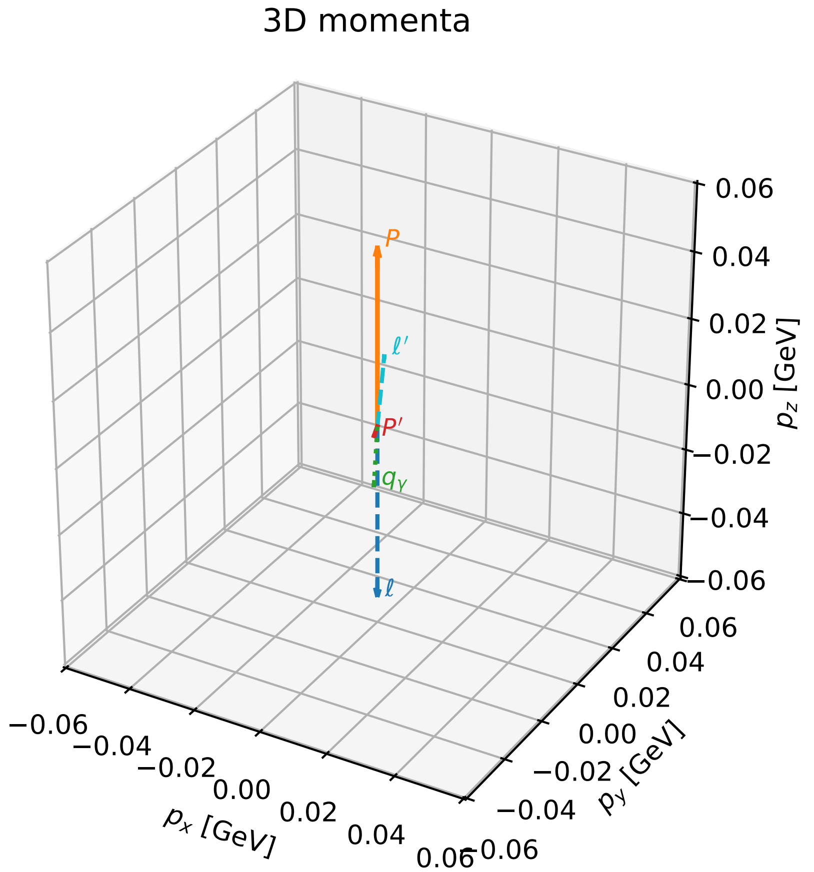



electron: polarization cluster P2, local minimum 484; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.0050872255
D_W contour = 0.055087225
above global minimum = 0.0050491969
7D contour samples = 128
selected phase-space point:
sqrt_s = 1.00649
theta_out = 0.4081938
qOut = 0.03317345
phi_p_out = 1.660588
phi_gamma_out = 4.251596
theta_e = 0.8328545
theta_p = 0.2765099


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_490 D_W=0.00536584
region: polarization_cluster_02_local_minimum_490
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.8213422 rad
incoming lepton: 0.681239|+> +0.732061|->
proton mixing angle: theta_p=0.25786766 rad
incoming proton: 0.966936|+> +0.255019|->
s=1.07293, $\sqrt{s}$=1.03582, $|\vec{P}|$=0.0932026, $|\vec{P}'|$=0.00363053
θ_{out}=0.590166, $\phi_{p'}$=1.42541, ϕ_γ=4.23696
E_γ=0.0481295, $\theta_{\ell'}$=2.61708, $\phi_{\ell'}$=1.06905
Q^2=0.00124513, x_B=0.0136232, t=-0.00811636, W^2=0.969997, y=0.473356
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0932 p_x= 0.0000 p_y= 0.0000 p_z= -0.0932
P E= 0.9426 p_x= 0.0000 p_y= 0.0000 p_z= 0.0932
ℓ' E= 0.0497 p_x= 0.0120 p_y= 0.0218 p_z= -0.0430
P' E= 0.9380 p_x= 0.0003 p_y= 0.0020 p_z= 0.0030
q_γ E= 0.0481 p_x= -0.0123 p_y= -0.0238 p_z= 0.0400
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P
P'
q_γ
ℓ'
ℓ
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=7, retained=99.6%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=-2.05e+01+7.27e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=7.40e+01+2.09e-01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-2.98e+01-5.59e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-5.01e+01+2.92e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=4.81e+00-3.03e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=1.66e+01+2.05e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-2.00e+01+6.69e+00i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P2, local minimum 490; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
0
π/4
π/2
3π/4
π
3π/2
2π
selected local minimum
D_W(local) = 0.0053658368
D_W contour = 0.055365837
above global minimum = 0.0053278082
7D contour samples = 128
selected phase-space point:
sqrt_s = 1.035823
theta_out = 0.5901655
qOut = 0.04812954
phi_p_out = 1.425411
phi_gamma_out = 4.236963
theta_e = 0.8213422
theta_p = 0.2578677

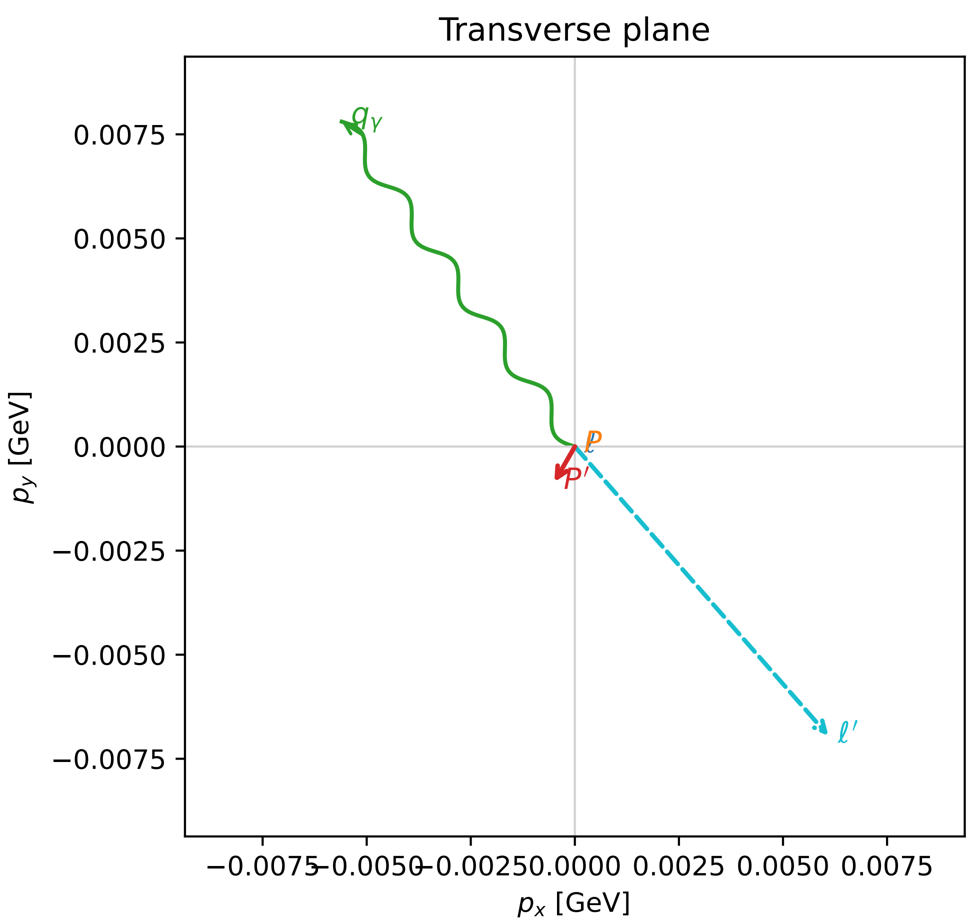
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

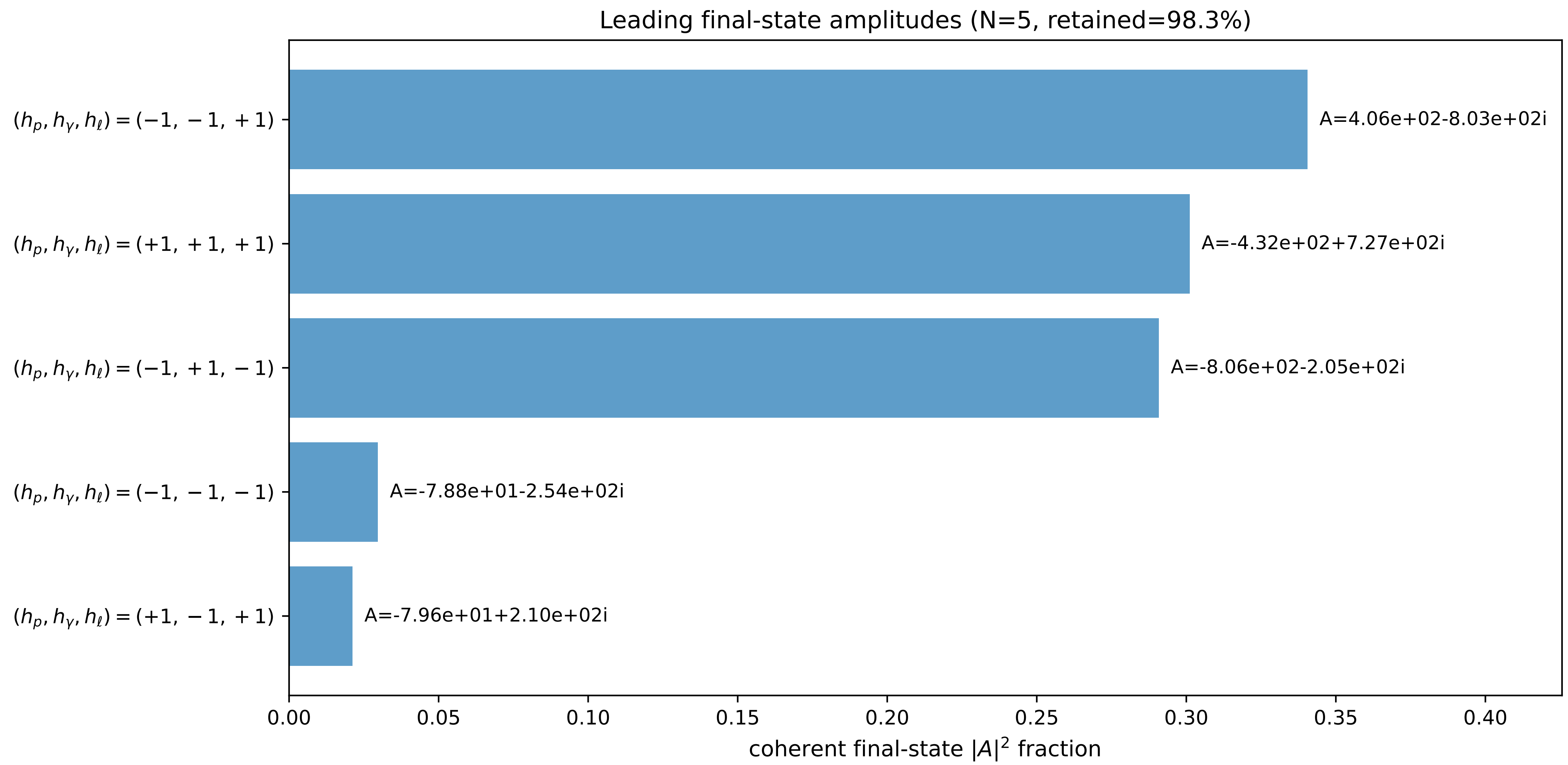


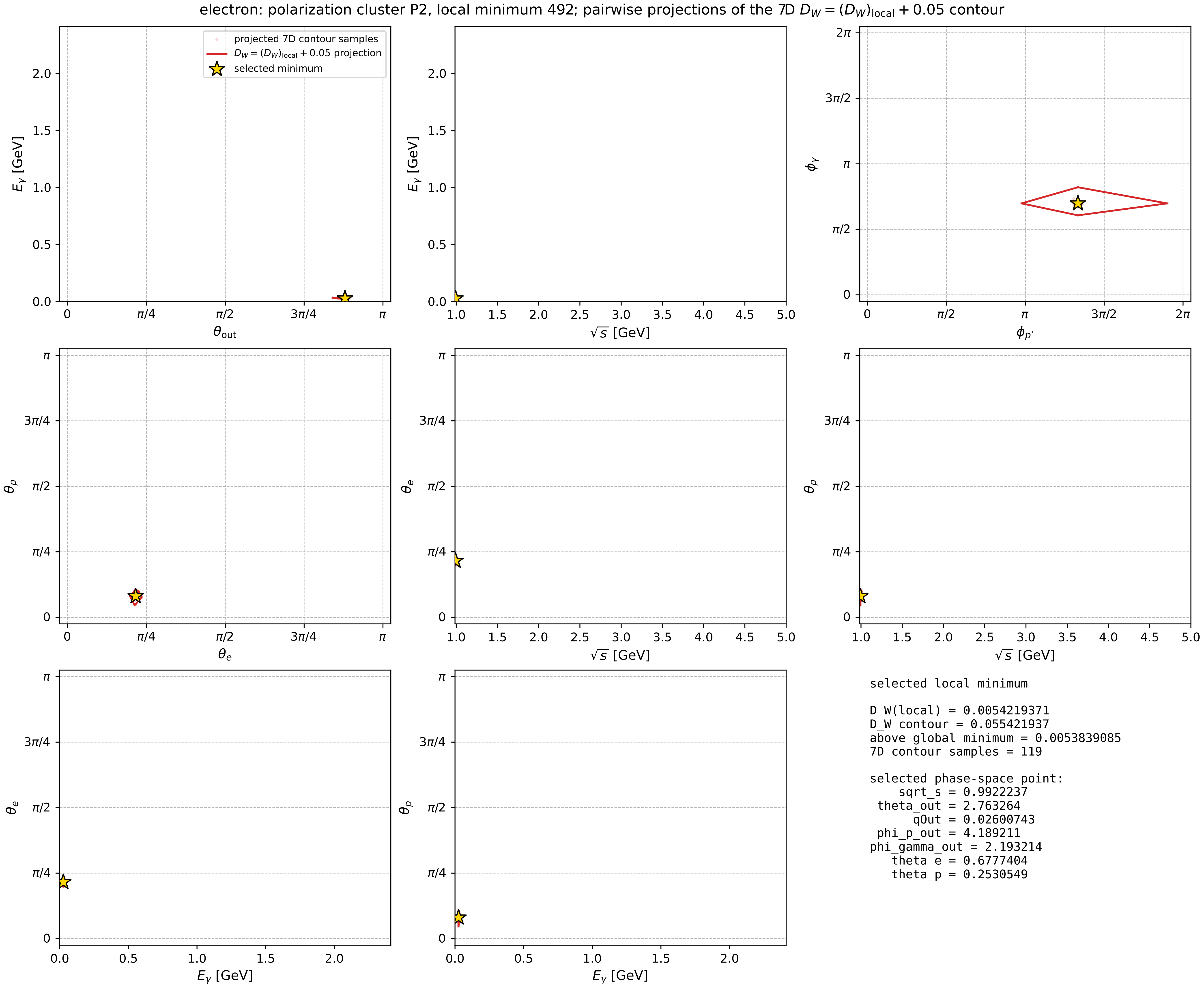

```
dw_mixing_angles_polarization_02_local_minimum_492  D_W=0.00542194
region: polarization_cluster_02_local_minimum_492
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.67774041 rad
incoming lepton: 0.778992|+> +0.627034|->
proton mixing angle: theta_p=0.25305494 rad
incoming proton: 0.968152|+> +0.250363|->
```

s=0.984508, $\sqrt{s}$=0.992224, $|\vec{P}|$=0.0527397, $|\vec{P}'|$=0.0026711
θ_{out}=2.76326, $\phi_{p'}$=4.18921, ϕ_γ=2.19321
E_γ=0.0260074, $\theta_{\ell'}$=0.333846, $\phi_{\ell'}$=5.4322
Q^2=0.00578653, x_B=0.106244, t=-0.00304825, W^2=0.928522, y=0.520377

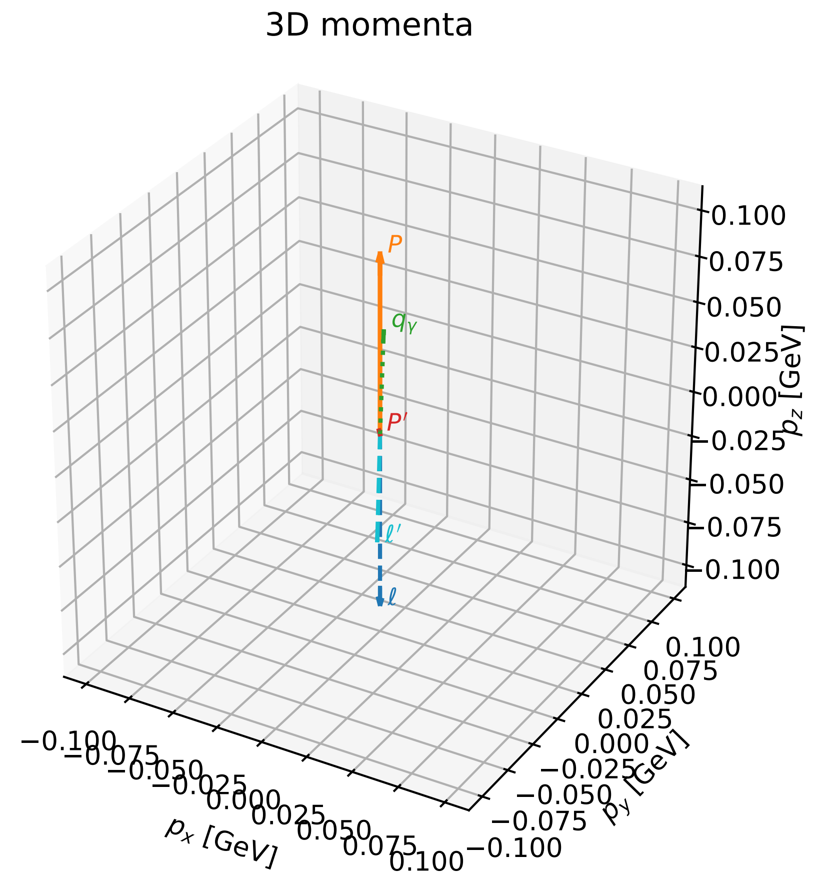
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0527 p_x= 0.0000 p_y= 0.0000 p_z= -0.0527
P E= 0.9395 p_x= 0.0000 p_y= 0.0000 p_z= 0.0527
ℓ' E= 0.0282 p_x= 0.0061 p_y= -0.0070 p_z= 0.0267
P' E= 0.9380 p_x= -0.0005 p_y= -0.0009 p_z= -0.0025
q_γ E= 0.0260 p_x= -0.0056 p_y= 0.0078 p_z= -0.0242






electron: polarization cluster P2, local minimum 492; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.0054219371
D_W contour = 0.055421937
above global minimum = 0.0053839085
7D contour samples = 119
selected phase-space point:
sqrt_s = 0.9922237
theta_out = 2.763264
qOut = 0.02600743
phi_p_out = 4.189211
phi_gamma_out = 2.193214
theta_e = 0.6777404
theta_p = 0.2530549

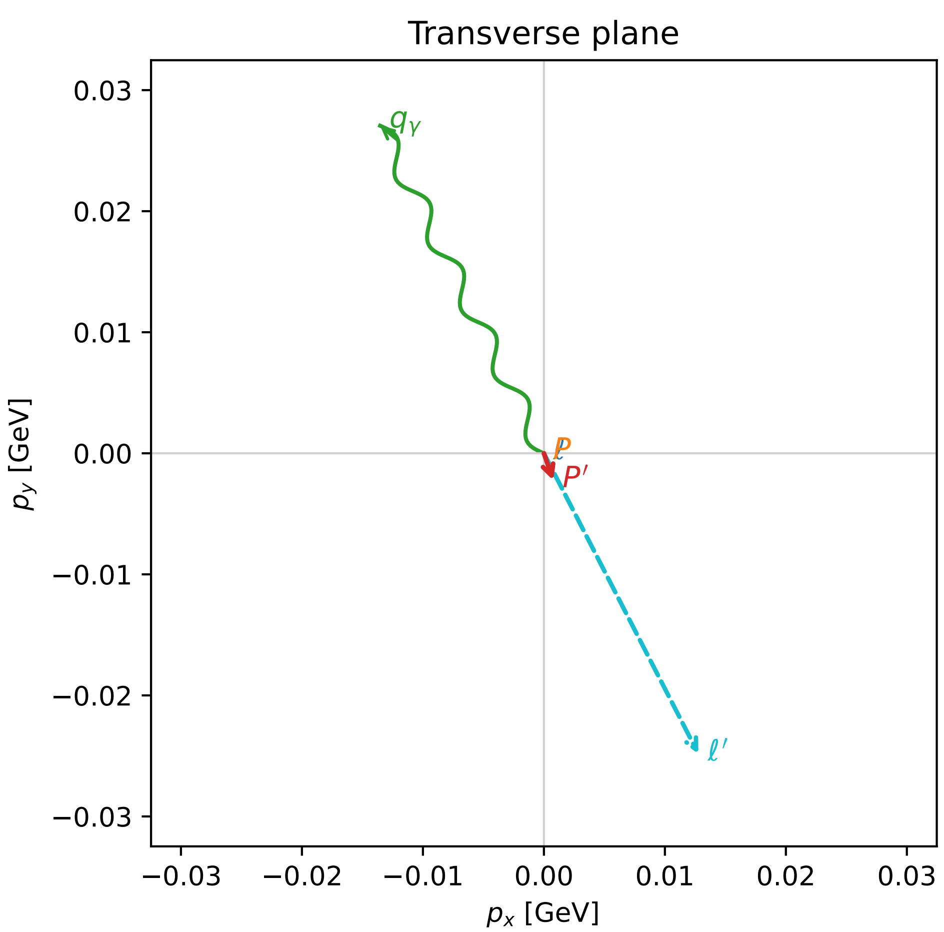
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

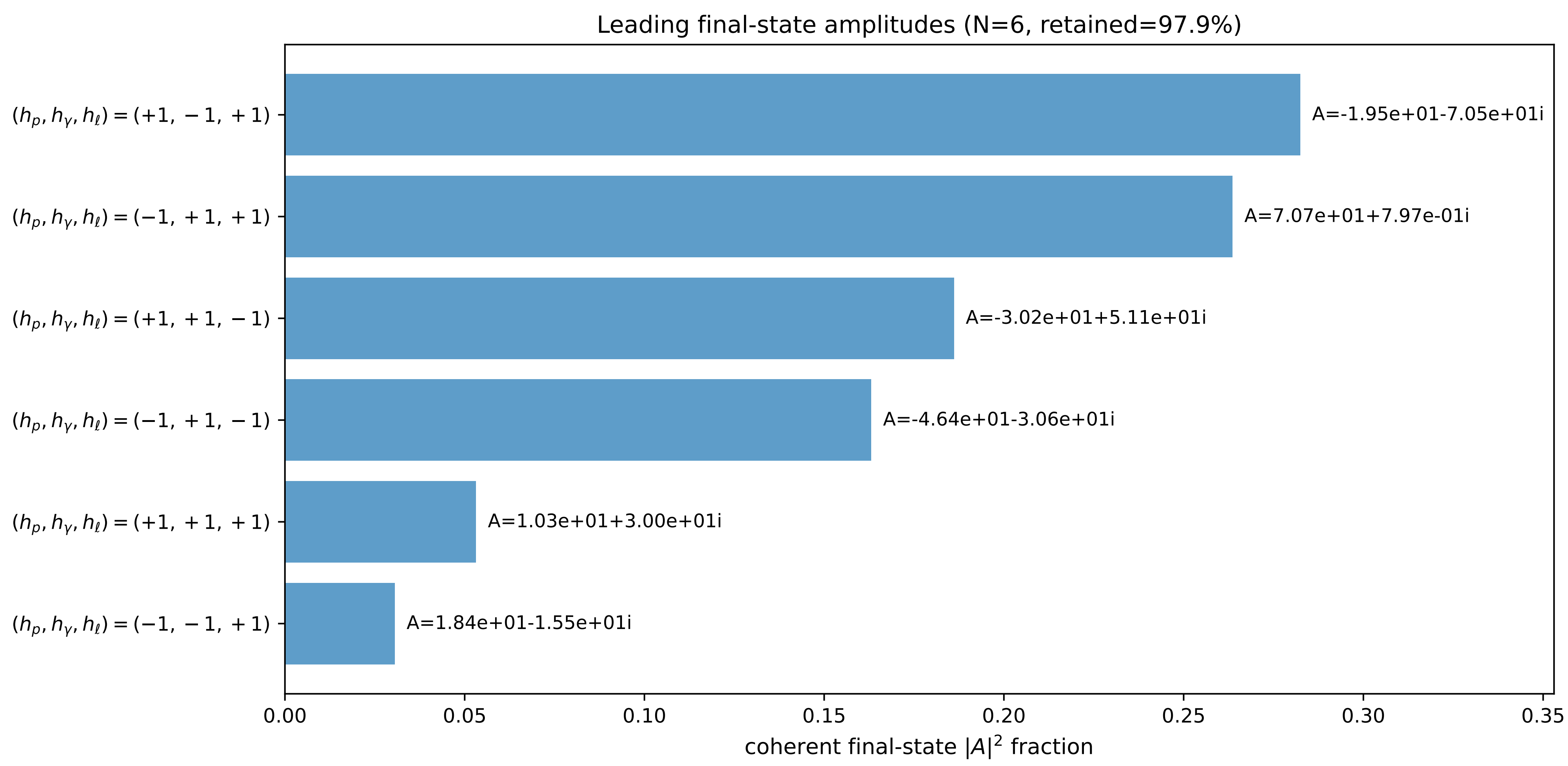



```
dw_mixing_angles_polarization_02_local_minimum_494  D_W=0.00548201
region: polarization_cluster_02_local_minimum_494
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.81078849 rad
incoming lepton: 0.688927|+> +0.724831|->
proton mixing angle: theta_p=0.25470858 rad
incoming proton: 0.967737|+> +0.251963|->
```

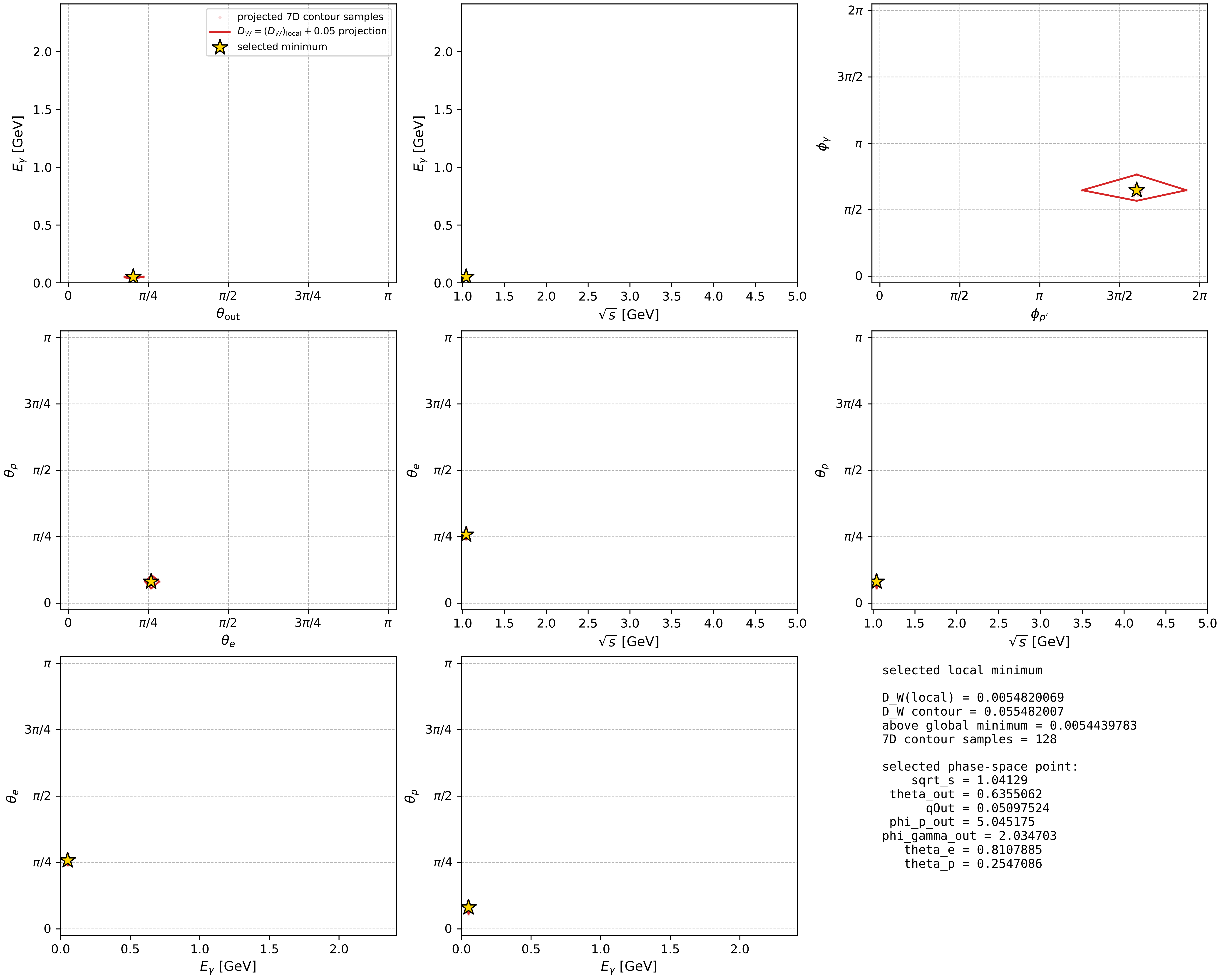
s=1.08428, $\sqrt{s}$=1.04129, $|\vec{P}|$=0.0981656, $|\vec{P}'|$=0.00398744
θ_{out}=0.635506, $\phi_{p'}$=5.04518, ϕ_γ=2.0347
E_γ=0.0509752, $\theta_{\ell'}$=2.57865, $\phi_{\ell'}$=5.18738
Q^2=0.00158472, x_B=0.0163215, t=-0.00899622, W^2=0.975353, y=0.474926

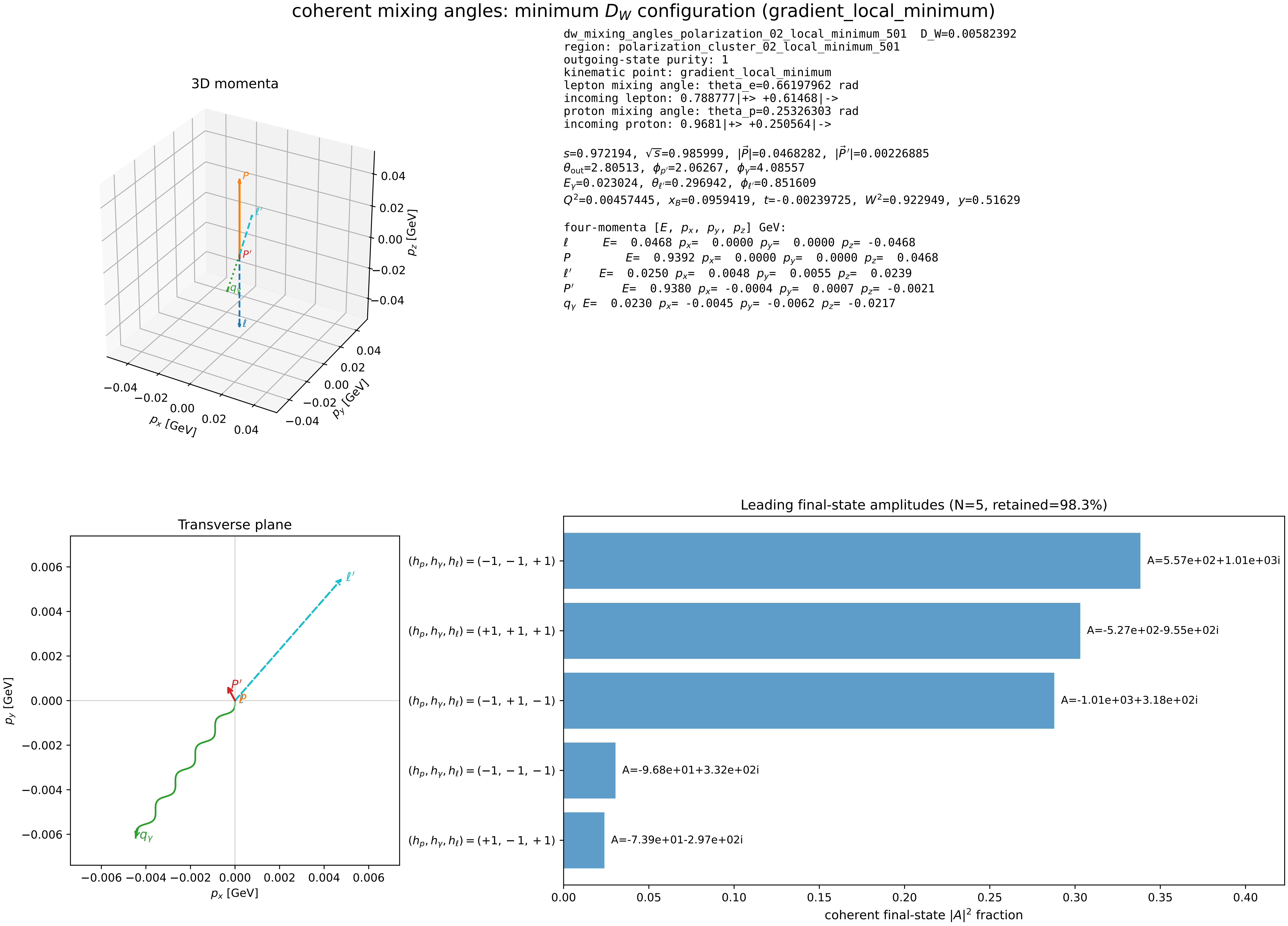
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0982 p_x= 0.0000 p_y= 0.0000 p_z= -0.0982
P E= 0.9431 p_x= 0.0000 p_y= 0.0000 p_z= 0.0982
ℓ' E= 0.0523 p_x= 0.0128 p_y= -0.0248 p_z= -0.0442
P' E= 0.9380 p_x= 0.0008 p_y= -0.0022 p_z= 0.0032
q_γ E= 0.0510 p_x= -0.0135 p_y= 0.0271 p_z= 0.0410

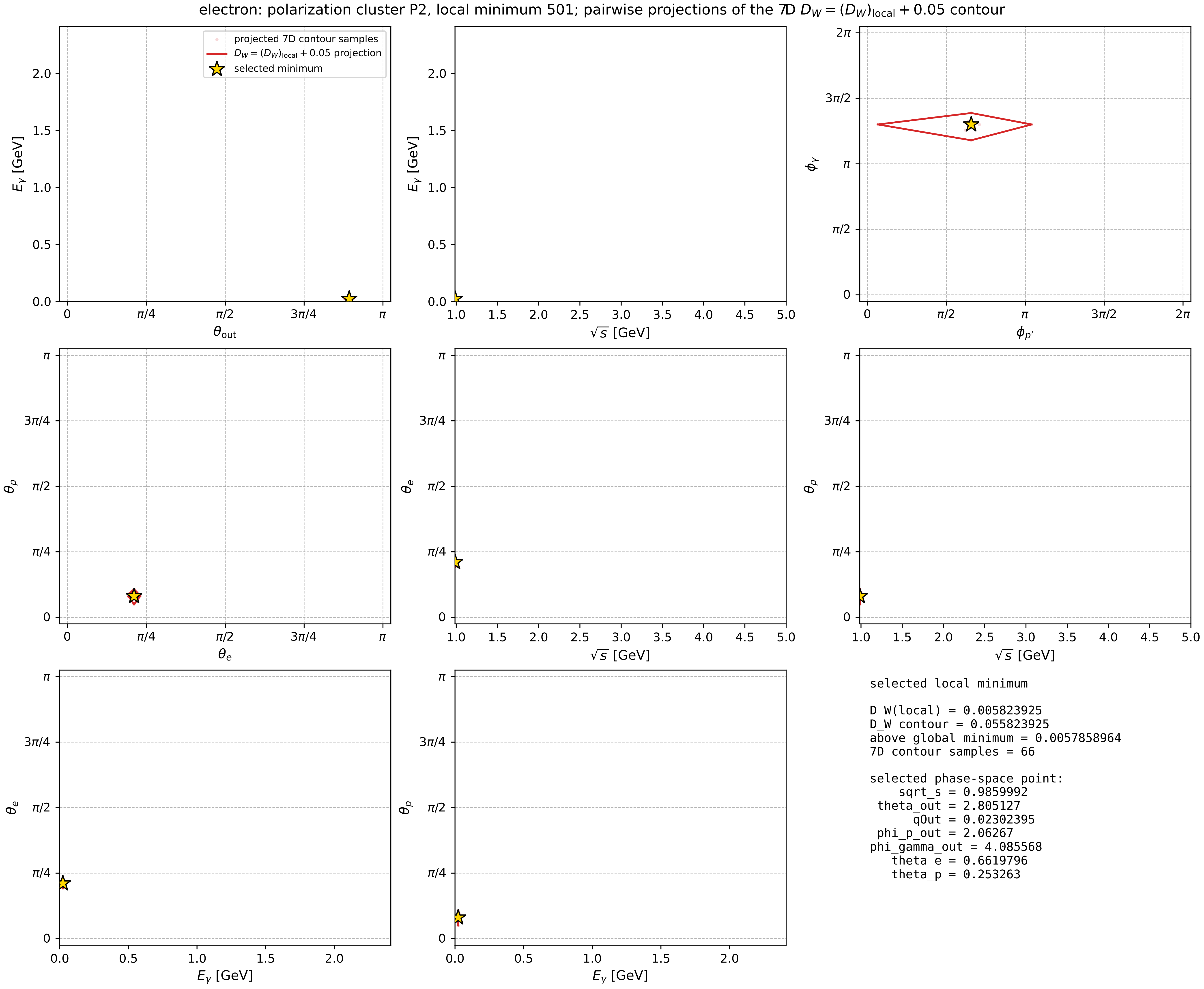




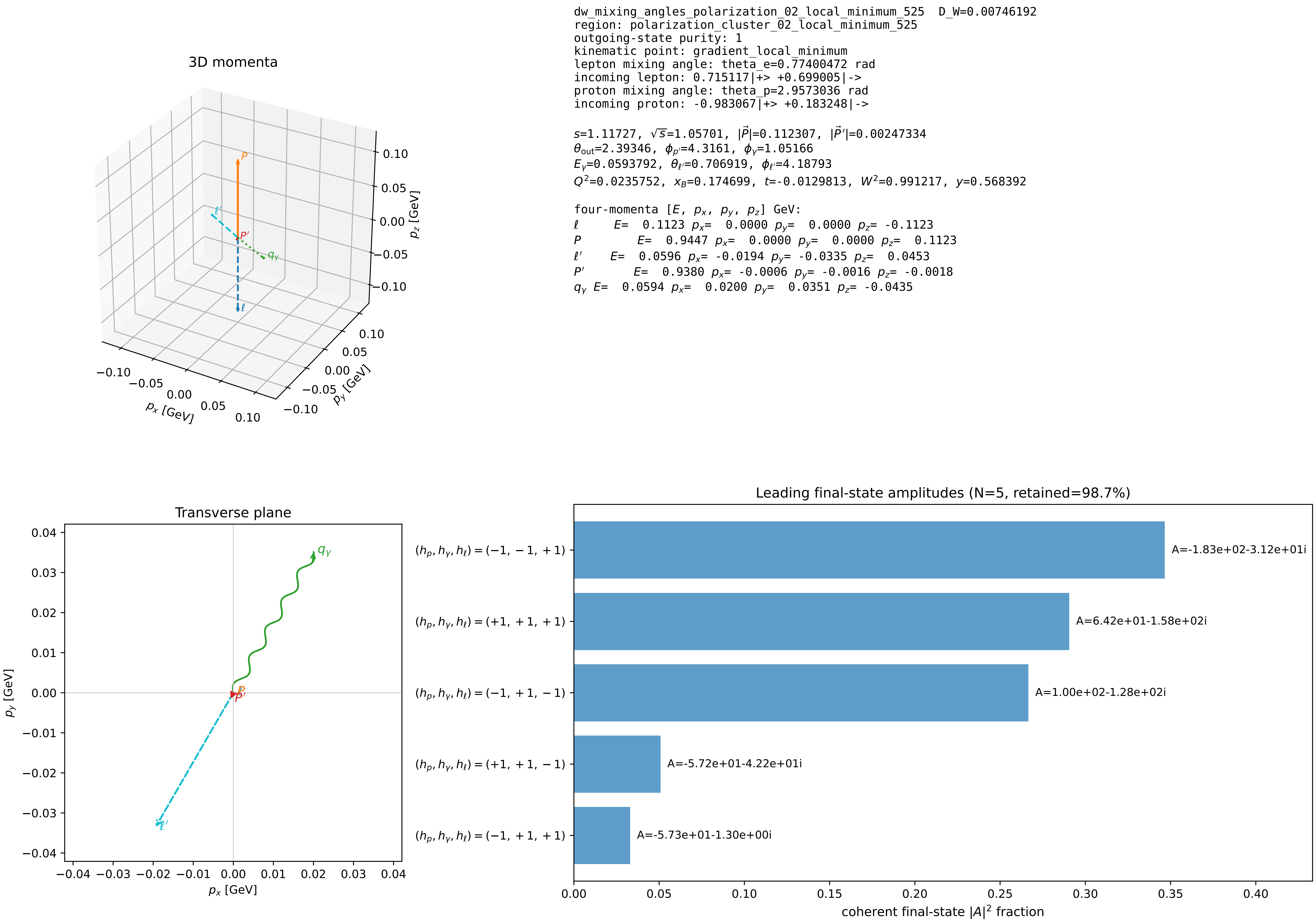
electron: polarization cluster P2, local minimum 494; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour

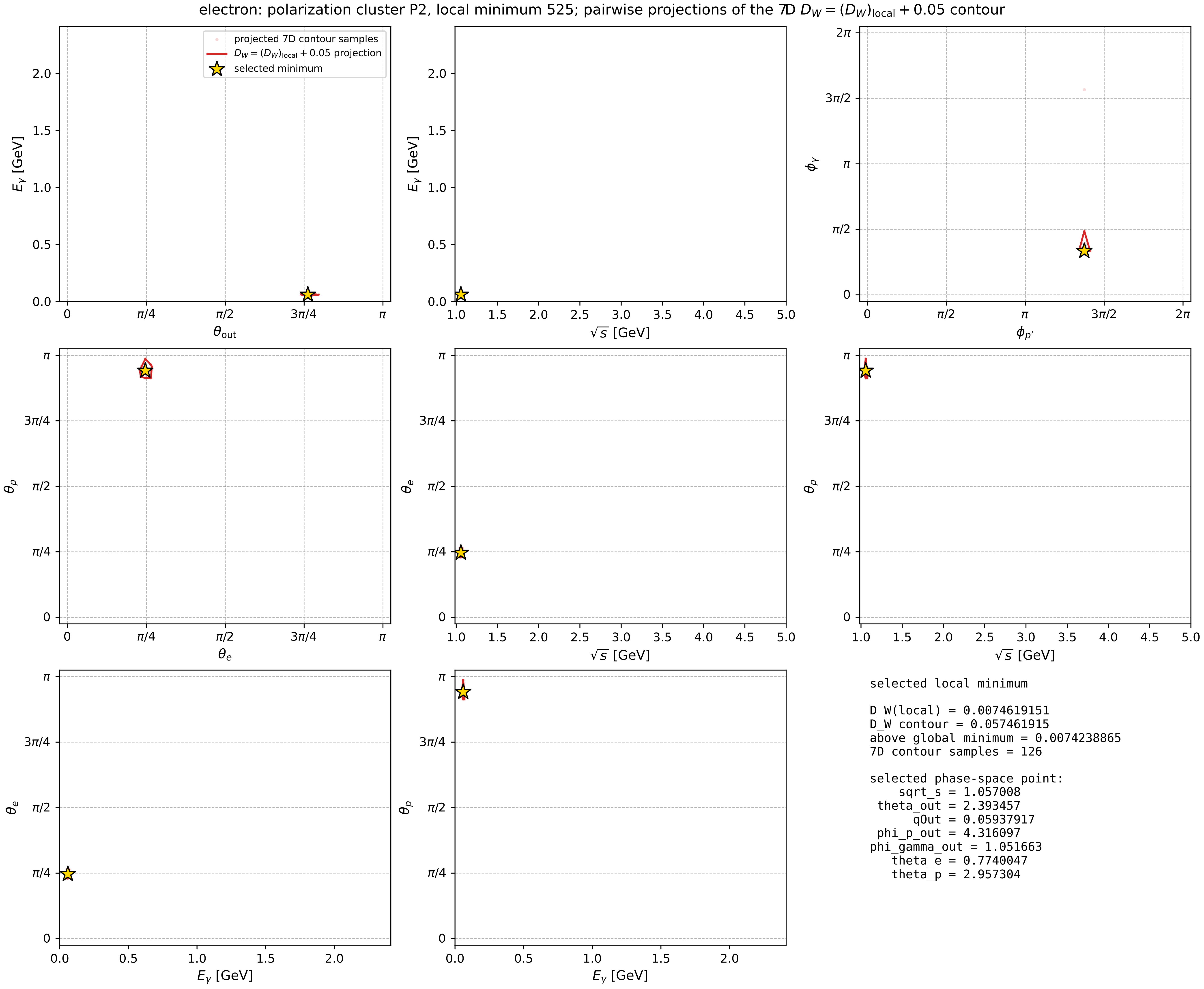


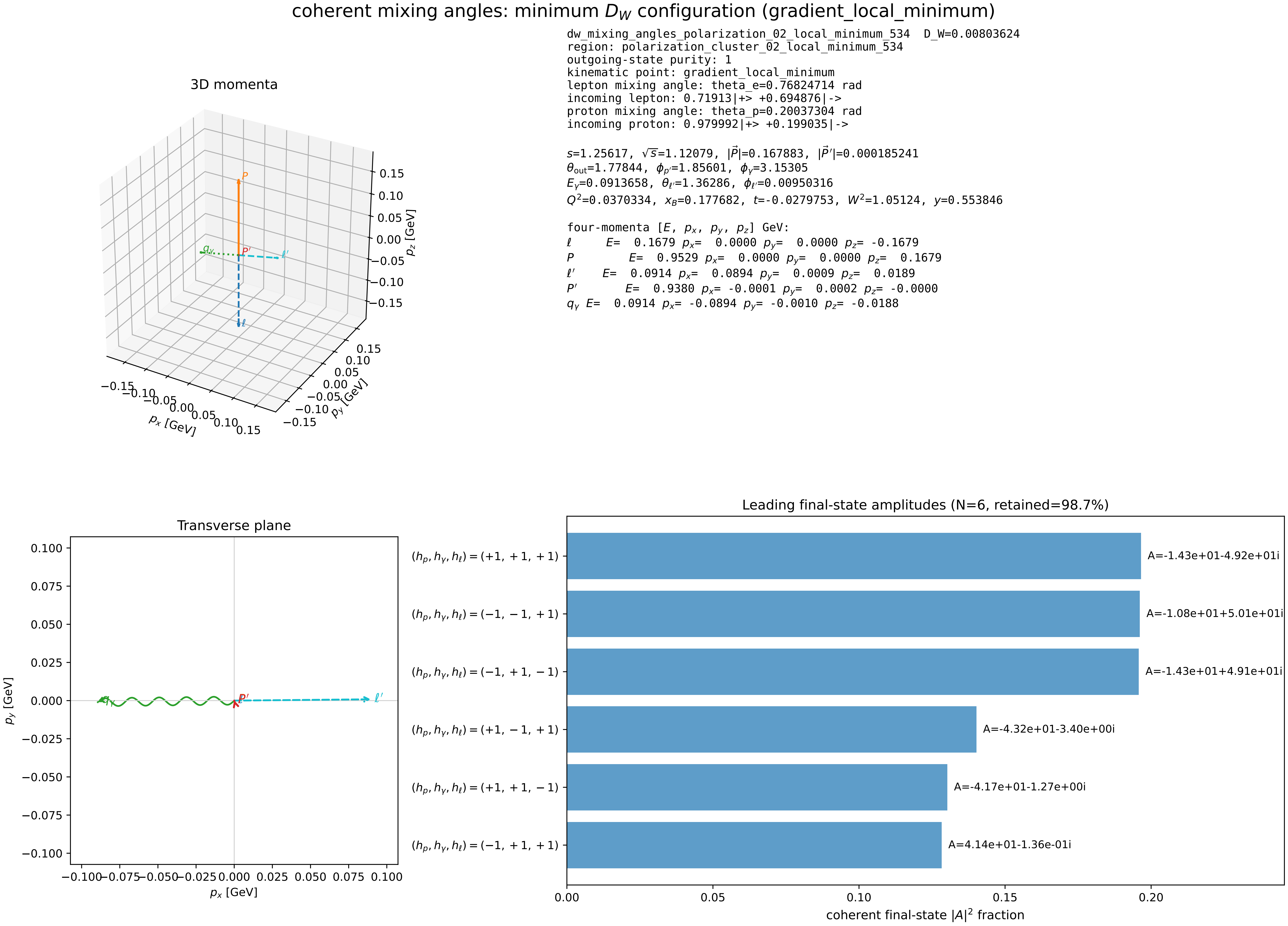

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_501 D_W=0.00582392
region: polarization_cluster_02_local_minimum_501
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.66197962 rad
incoming lepton: 0.788777|+> +0.61468|->
proton mixing angle: theta_p=0.25326303 rad
incoming proton: 0.9681|+> +0.250564|->
s=0.972194, $\sqrt{s}$=0.985999, $|\vec{P}|$=0.0468282, $|\vec{P}'|$=0.00226885
$\theta_{\rm out}$=2.80513, $\phi_{p'}$=2.06267, ϕ_γ=4.08557
E_γ=0.023024, $\theta_{\ell'}$=0.296942, $\phi_{\ell'}$=0.851609
Q^2=0.00457445, x_B=0.0959419, t=-0.00239725, W^2=0.922949, y=0.51629
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0468 p_x= 0.0000 p_y= 0.0000 p_z= -0.0468
P E= 0.9392 p_x= 0.0000 p_y= 0.0000 p_z= 0.0468
ℓ' E= 0.0250 p_x= 0.0048 p_y= 0.0055 p_z= 0.0239
P' E= 0.9380 p_x= -0.0004 p_y= 0.0007 p_z= -0.0021
q_γ E= 0.0230 p_x= -0.0045 p_y= -0.0062 p_z= -0.0217
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=5, retained=98.3%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=5.57e+02+1.01e+03i
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=-5.27e+02-9.55e+02i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-1.01e+03+3.18e+02i
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$ A=-9.68e+01+3.32e+02i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=-7.39e+01-2.97e+02i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P2, local minimum 501; pairwise projections of the 7D $D_W = (D_W)_{\rm local} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\rm local} + 0.05$ projection
selected minimum
E_γ [GeV]
$\theta_{\rm out}$
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.005823925
D_W contour = 0.055823925
above global minimum = 0.0057858964
7D contour samples = 66
selected phase-space point:
sqrt_s = 0.9859992
theta_out = 2.805127
qOut = 0.02302395
phi_p_out = 2.06267
phi_gamma_out = 4.085568
theta_e = 0.6619796
theta_p = 0.253263

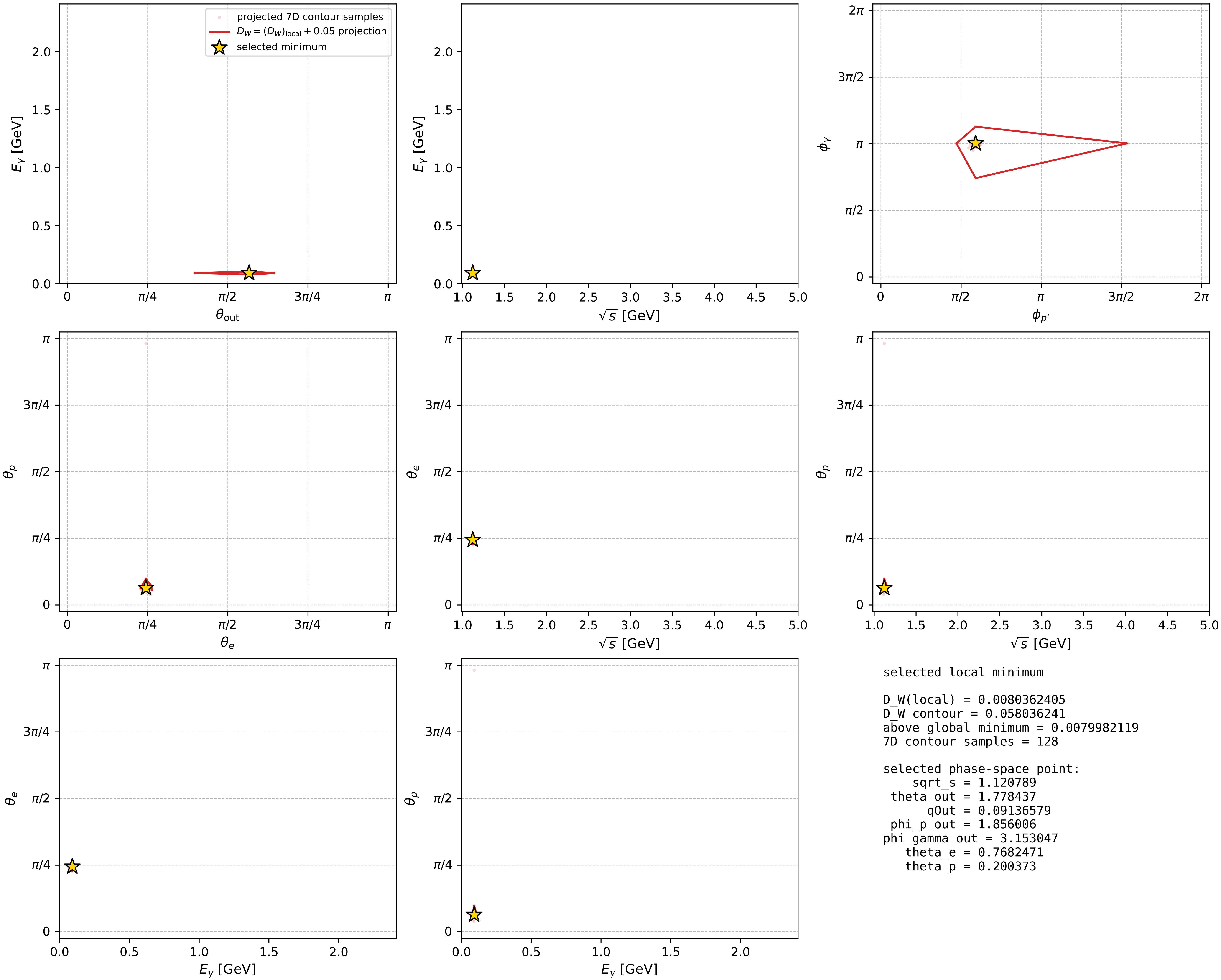
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)



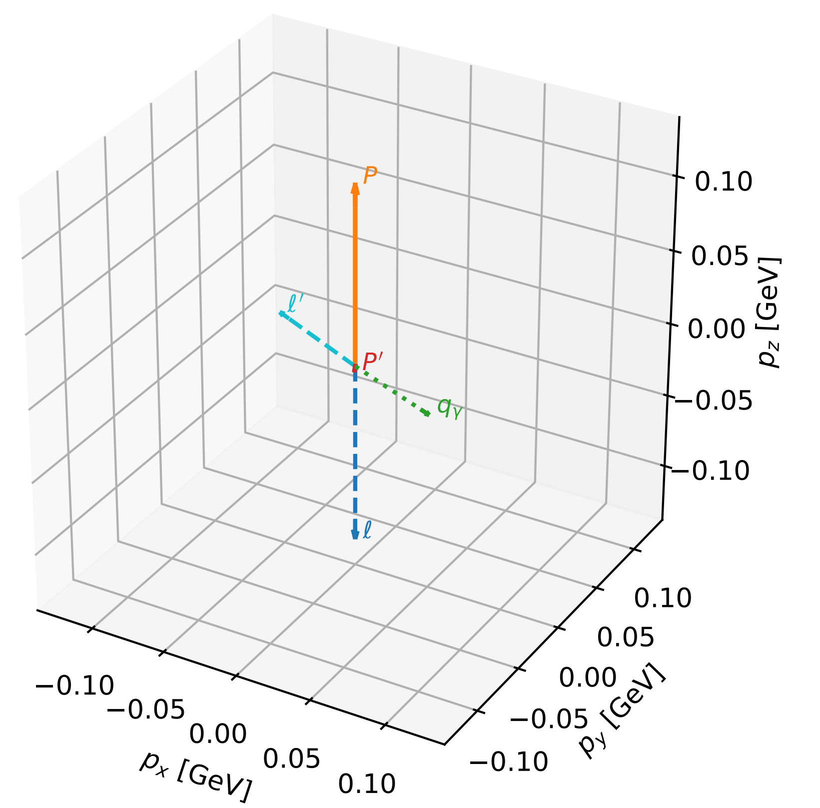

electron: polarization cluster P2, local minimum 525; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.0074619151
D_W contour = 0.057461915
above global minimum = 0.0074238865
7D contour samples = 126
selected phase-space point:
sqrt_s = 1.057008
theta_out = 2.393457
qOut = 0.05937917
phi_p_out = 4.316097
phi_gamma_out = 1.051663
theta_e = 0.7740047
theta_p = 2.957304


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_534 D_W=0.00803624
region: polarization_cluster_02_local_minimum_534
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.76824714 rad
incoming lepton: 0.71913|+> +0.694876|->
proton mixing angle: theta_p=0.20037304 rad
incoming proton: 0.979992|+> +0.199035|->
s=1.25617, $\sqrt{s}$=1.12079, $|\vec{P}|$=0.167883, $|\vec{P}'|$=0.000185241
θ_{out}=1.77844, $\phi_{p'}$=1.85601, ϕ_γ=3.15305
E_γ=0.0913658, $\theta_{\ell'}$=1.36286, $\phi_{\ell'}$=0.00950316
Q^2=0.0370334, x_B=0.177682, t=-0.0279753, W^2=1.05124, y=0.553846
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1679 p_x= 0.0000 p_y= 0.0000 p_z= -0.1679
P E= 0.9529 p_x= 0.0000 p_y= 0.0000 p_z= 0.1679
ℓ' E= 0.0914 p_x= 0.0894 p_y= 0.0009 p_z= 0.0189
P' E= 0.9380 p_x= -0.0001 p_y= 0.0002 p_z= -0.0000
q_γ E= 0.0914 p_x= -0.0894 p_y= -0.0010 p_z= -0.0188
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=6, retained=98.7%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=-1.43e+01-4.92e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-1.08e+01+5.01e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-1.43e+01+4.91e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=-4.32e+01-3.40e+00i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-4.17e+01-1.27e+00i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=4.14e+01-1.36e-01i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P2, local minimum 534; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



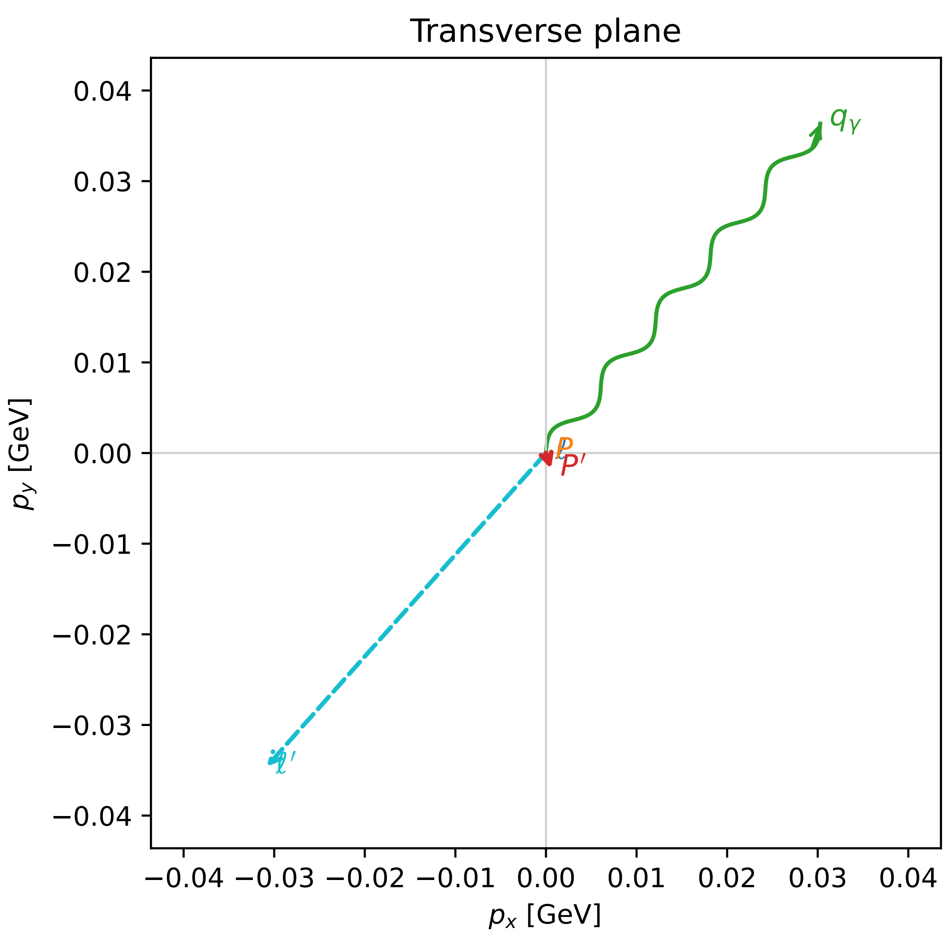
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

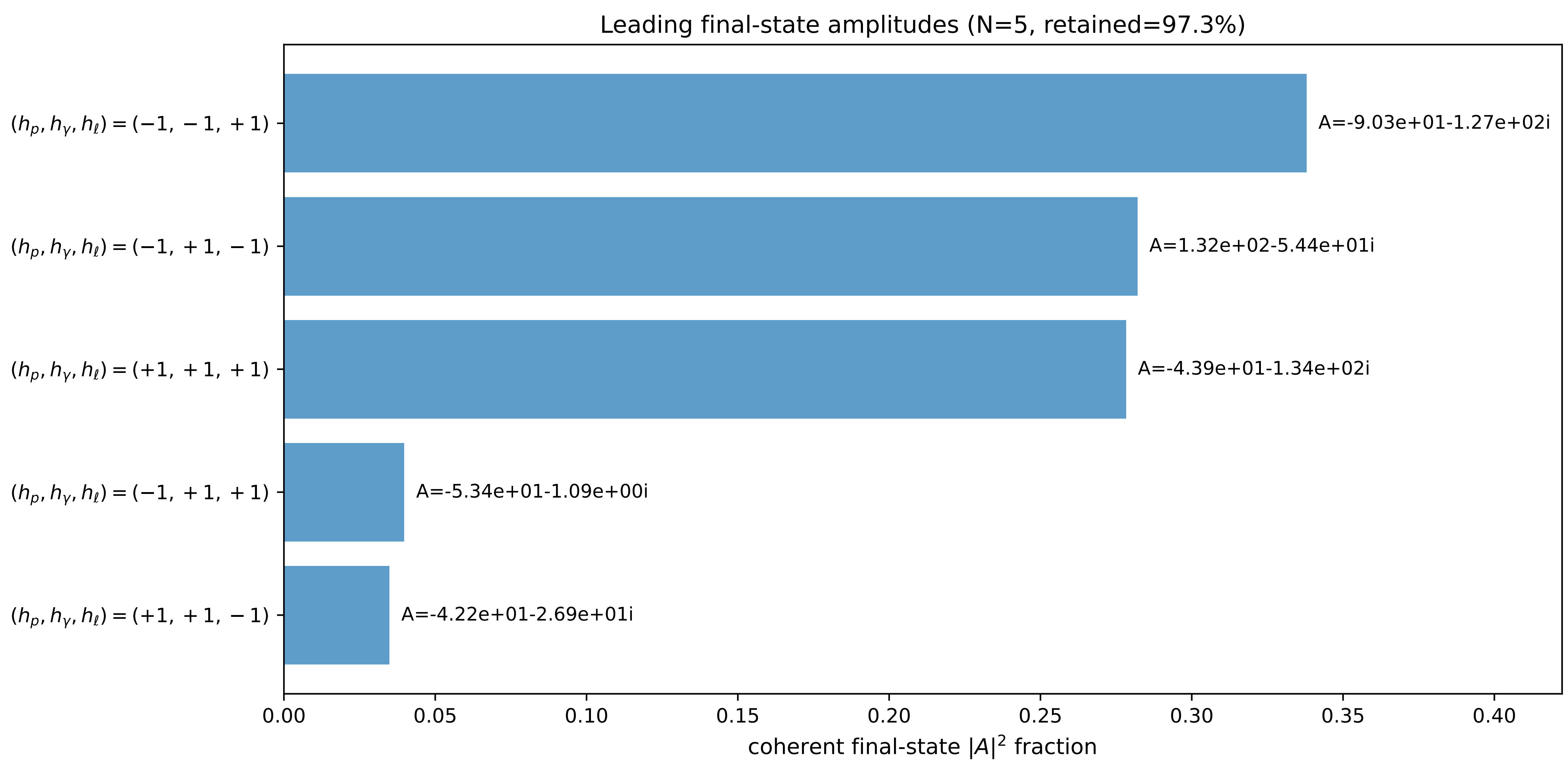


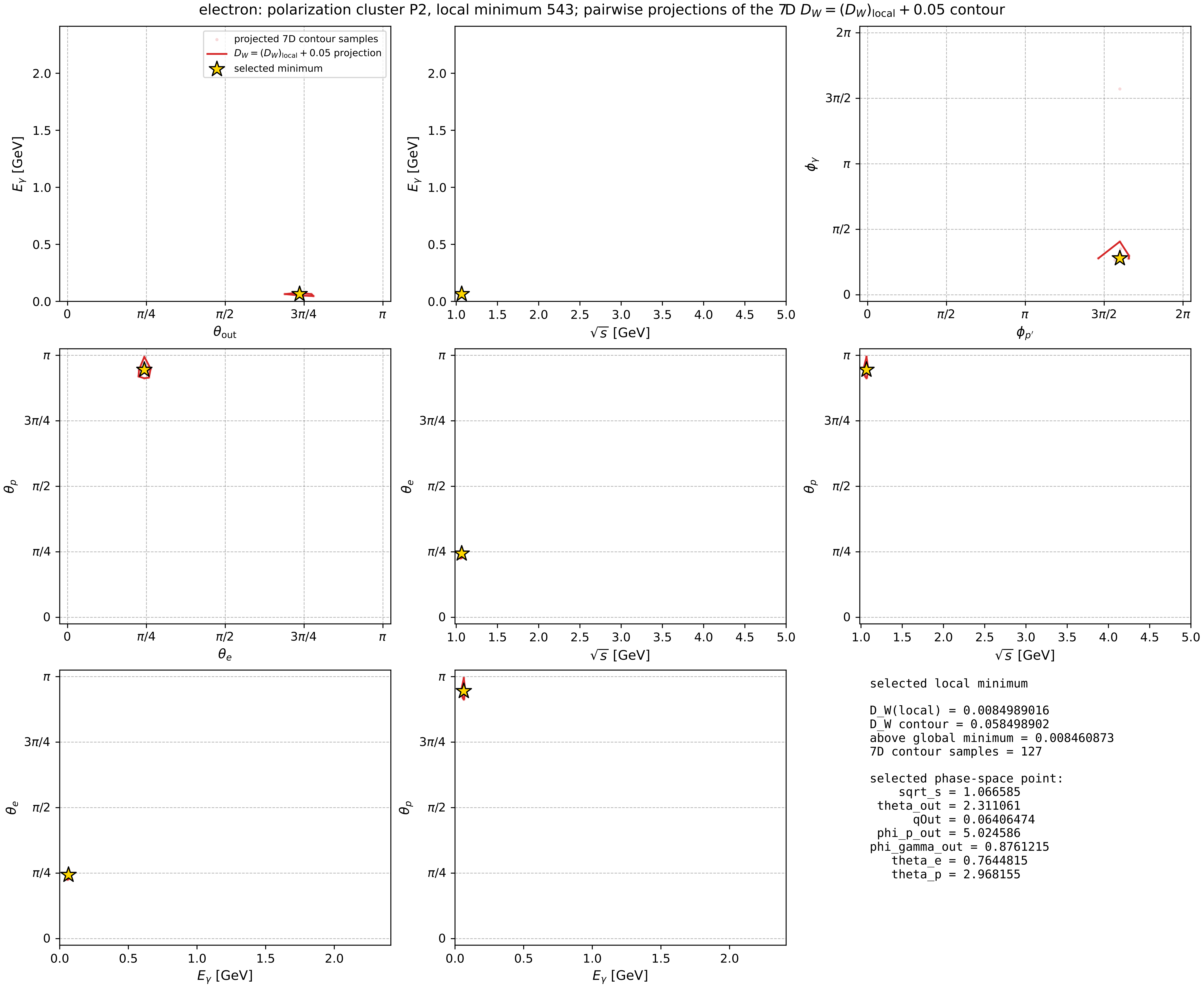
dw_mixing_angles_polarization_02_local_minimum_543 D_W=0.0084989
region: polarization_cluster_02_local_minimum_543
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.76448149 rad
incoming lepton: 0.721741|+> +0.692163|->
proton mixing angle: theta_p=2.9681547 rad
incoming proton: -0.984997|+> +0.17257|->

s=1.1376, $\sqrt{s}$=1.06658, $|\vec{P}|$=0.120833, $|\vec{P}'|$=0.00246895
θ_{out}=2.31106, $\phi_{p'}$=5.02459, ϕ_γ=0.876122
E_γ=0.0640647, $\theta_{\ell'}$=0.801552, $\phi_{\ell'}$=3.98447
Q^2=0.026436, x_B=0.180364, t=-0.0149491, W^2=0.999978, y=0.568634

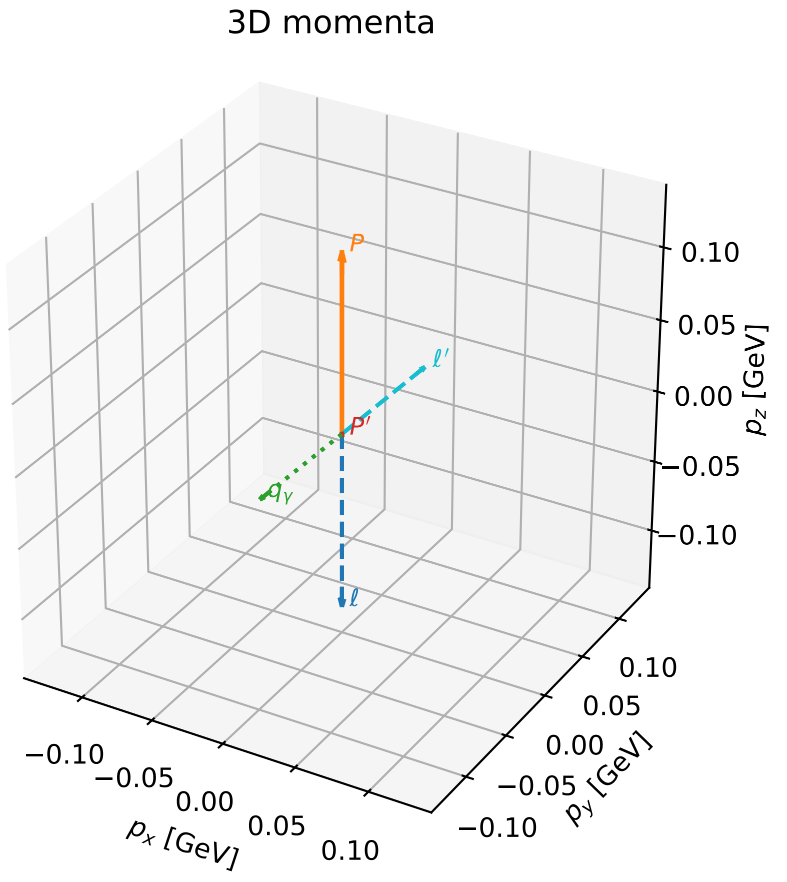
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1208 p_x= 0.0000 p_y= 0.0000 p_z= -0.1208
P E= 0.9458 p_x= 0.0000 p_y= 0.0000 p_z= 0.1208
ℓ' E= 0.0645 p_x= -0.0308 p_y= -0.0346 p_z= 0.0449
P' E= 0.9380 p_x= 0.0006 p_y= -0.0017 p_z= -0.0017
q_γ E= 0.0641 p_x= 0.0303 p_y= 0.0363 p_z= -0.0432






electron: polarization cluster P2, local minimum 543; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.0084989016
D_W contour = 0.058498902
above global minimum = 0.008460873
7D contour samples = 127
selected phase-space point:
sqrt_s = 1.066585
theta_out = 2.311061
qOut = 0.06406474
phi_p_out = 5.024586
phi_gamma_out = 0.8761215
theta_e = 0.7644815
theta_p = 2.968155

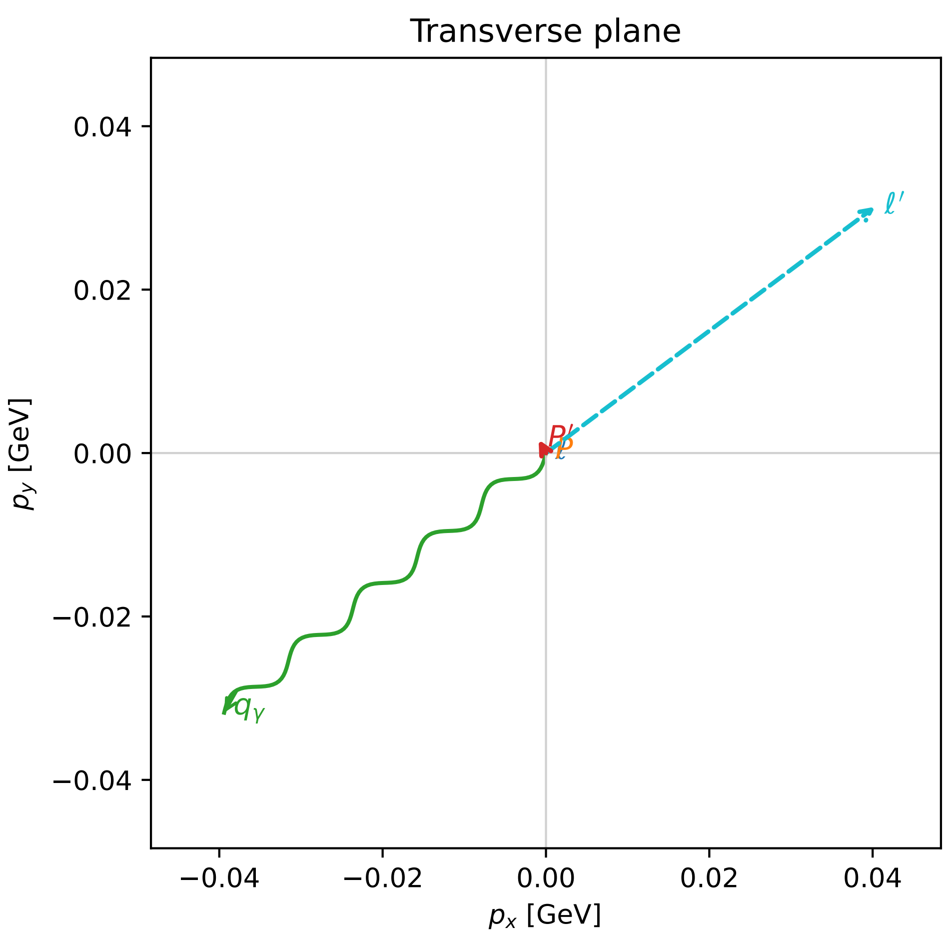
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

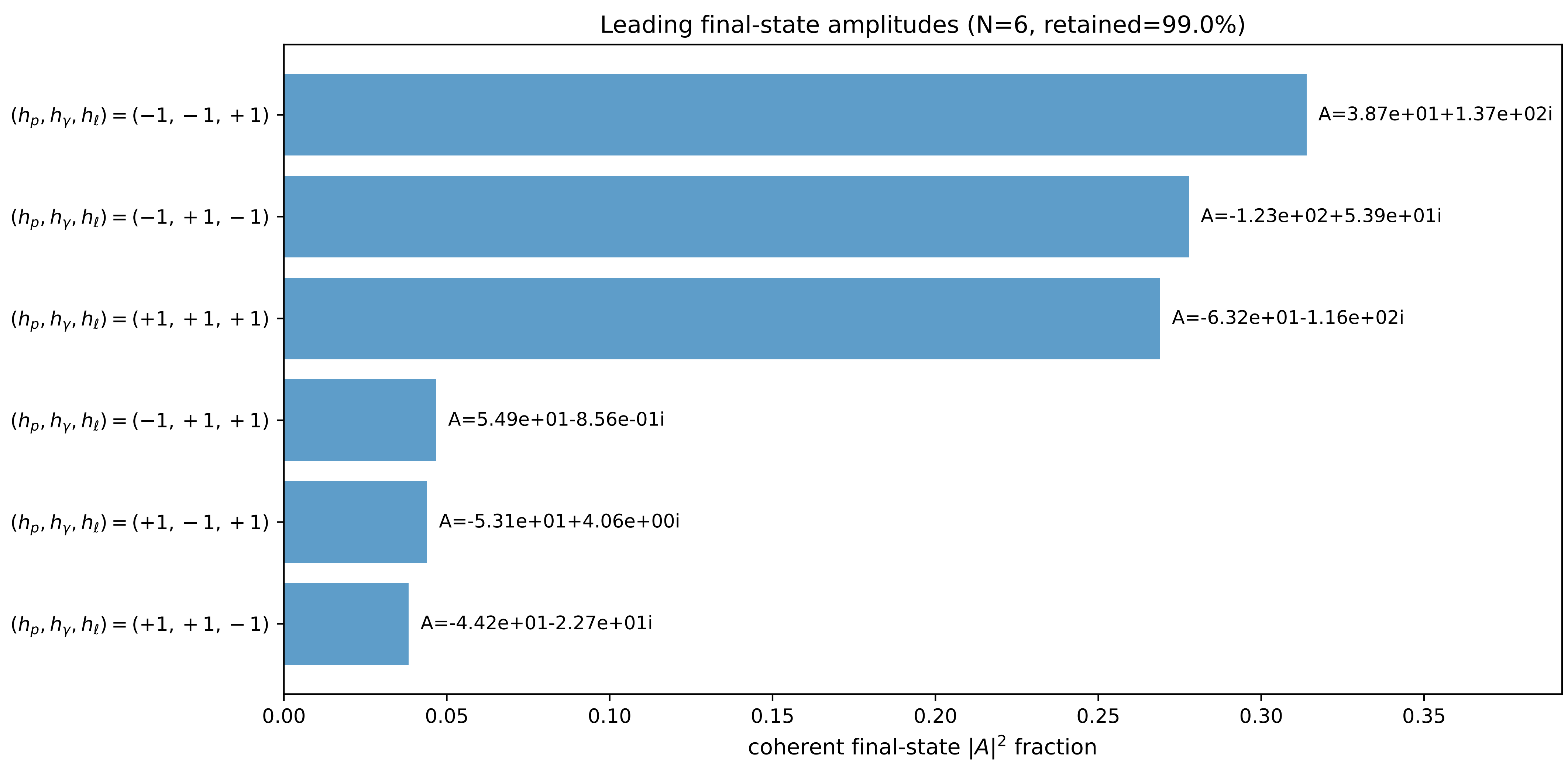


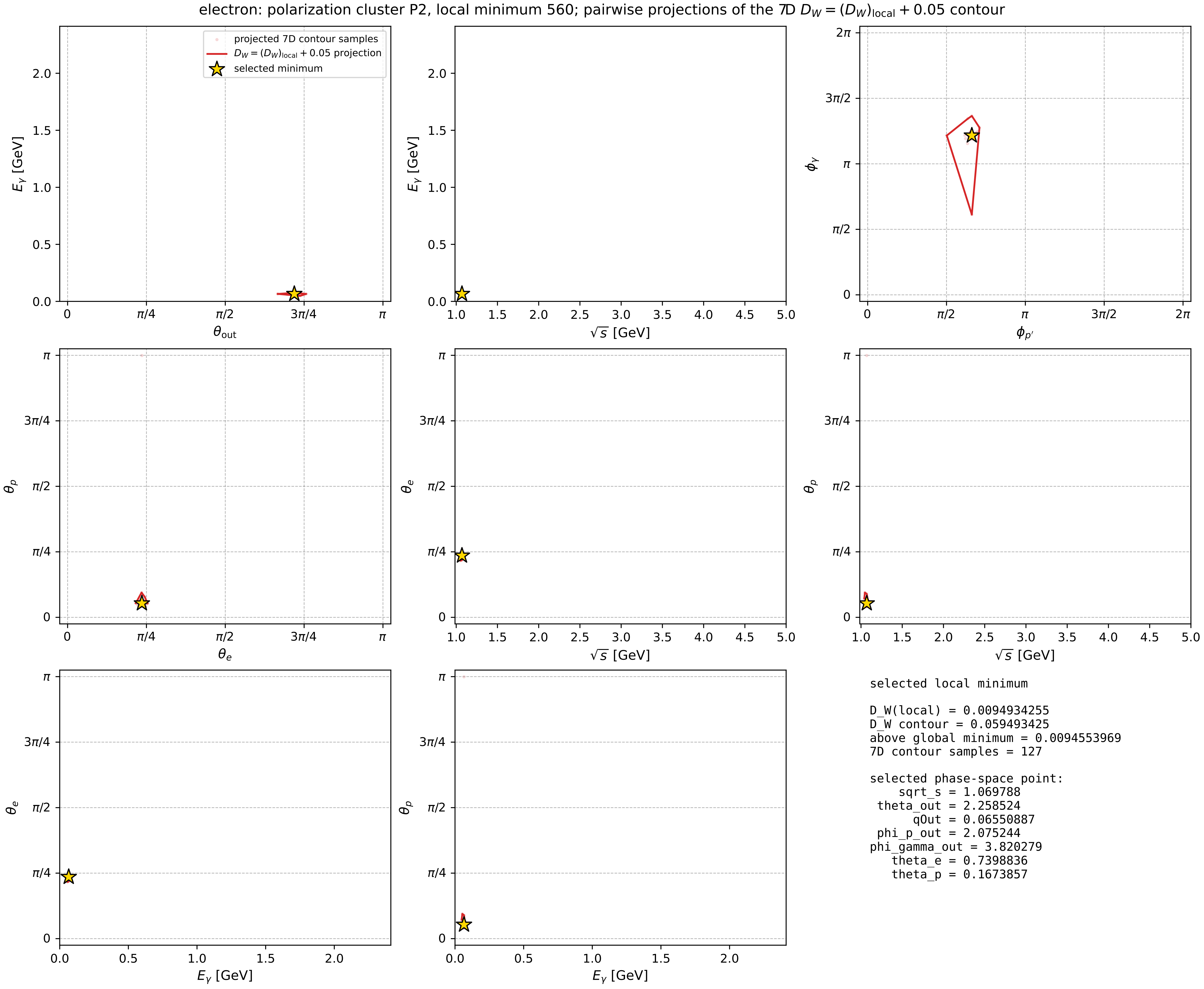

```
dw_mixing_angles_polarization_02_local_minimum_560  D_W=0.00949343
region: polarization_cluster_02_local_minimum_560
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.7398836 rad
incoming lepton: 0.738547|+> +0.674202|->
proton mixing angle: theta_p=0.16738572 rad
incoming proton: 0.986024|+> +0.166605|->
```

s=1.14445, $\sqrt{s}$=1.06979, $|\vec{P}|$=0.123669, $|\vec{P}'|$=0.0024206
θ_{out}=2.25852, $\phi_{p'}$=2.07524, ϕ_γ=3.82028
E_γ=0.0655089, $\theta_{\ell'}$=0.86237, $\phi_{\ell'}$=0.642076
Q^2=0.0270576, x_B=0.180555, t=-0.0156142, W^2=1.00264, y=0.566351

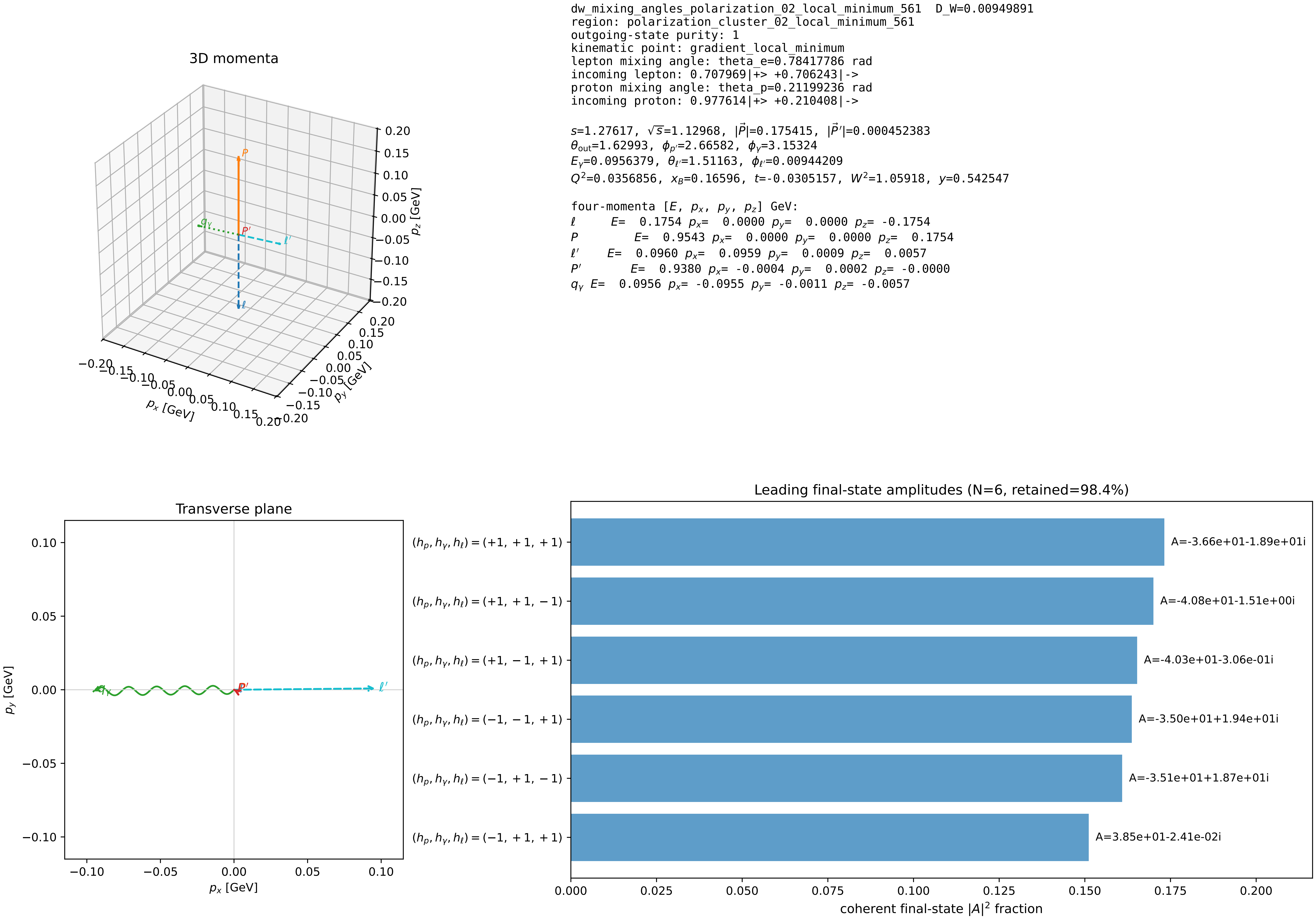
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1237 p_x= 0.0000 p_y= 0.0000 p_z= -0.1237
P E= 0.9461 p_x= 0.0000 p_y= 0.0000 p_z= 0.1237
ℓ' E= 0.0663 p_x= 0.0403 p_y= 0.0301 p_z= 0.0431
P' E= 0.9380 p_x= -0.0009 p_y= 0.0016 p_z= -0.0015
q_γ E= 0.0655 p_x= -0.0394 p_y= -0.0318 p_z= -0.0416



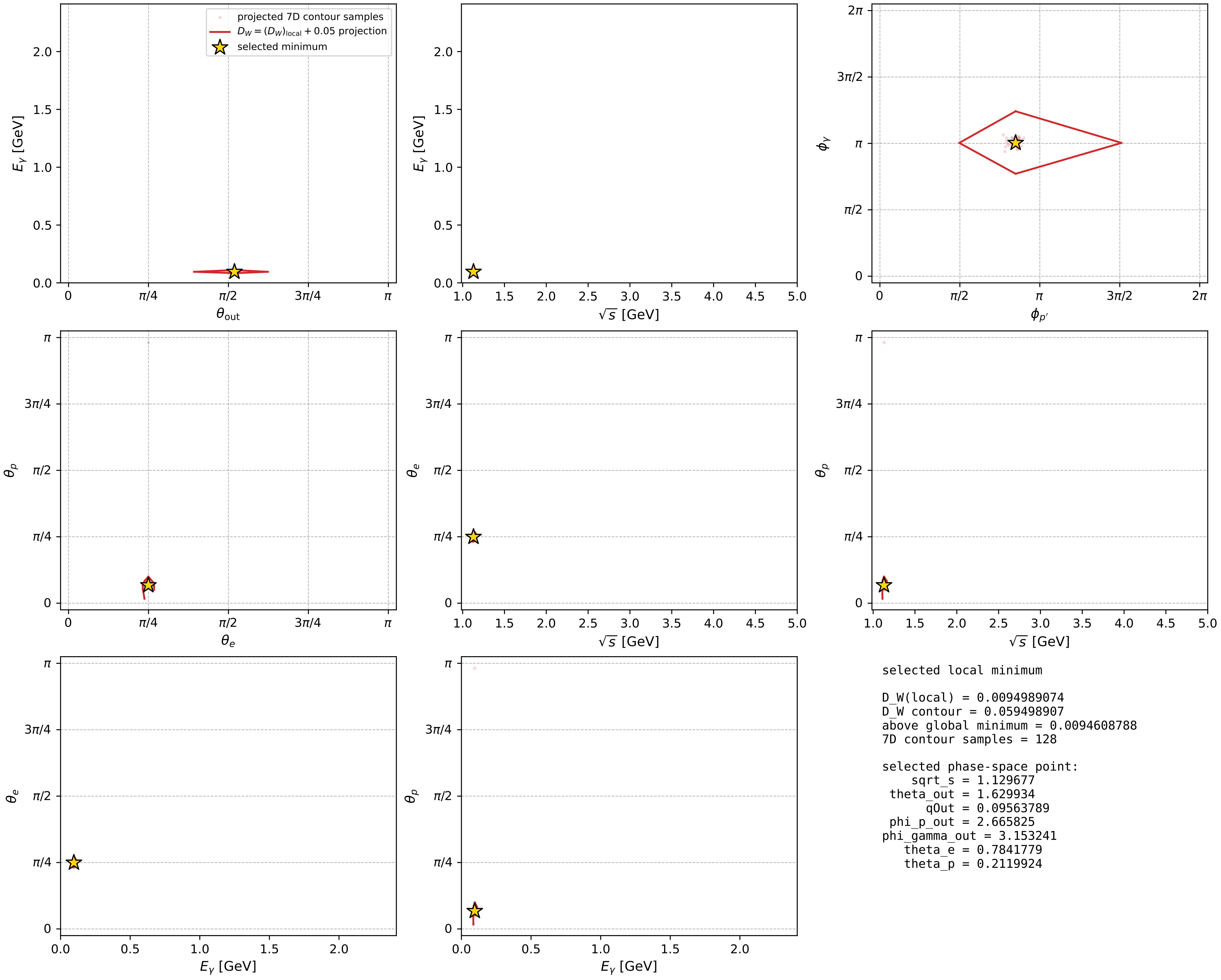



electron: polarization cluster P2, local minimum 560; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
0
$\pi/4$
$\pi/2$
$3\pi/4$
π
$3\pi/2$
2π
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
selected local minimum
D_W(local) = 0.0094934255
D_W contour = 0.059493425
above global minimum = 0.0094553969
7D contour samples = 127
selected phase-space point:
sqrt_s = 1.069788
theta_out = 2.258524
qOut = 0.06550887
phi_p_out = 2.075244
phi_gamma_out = 3.820279
theta_e = 0.7398836
theta_p = 0.1673857

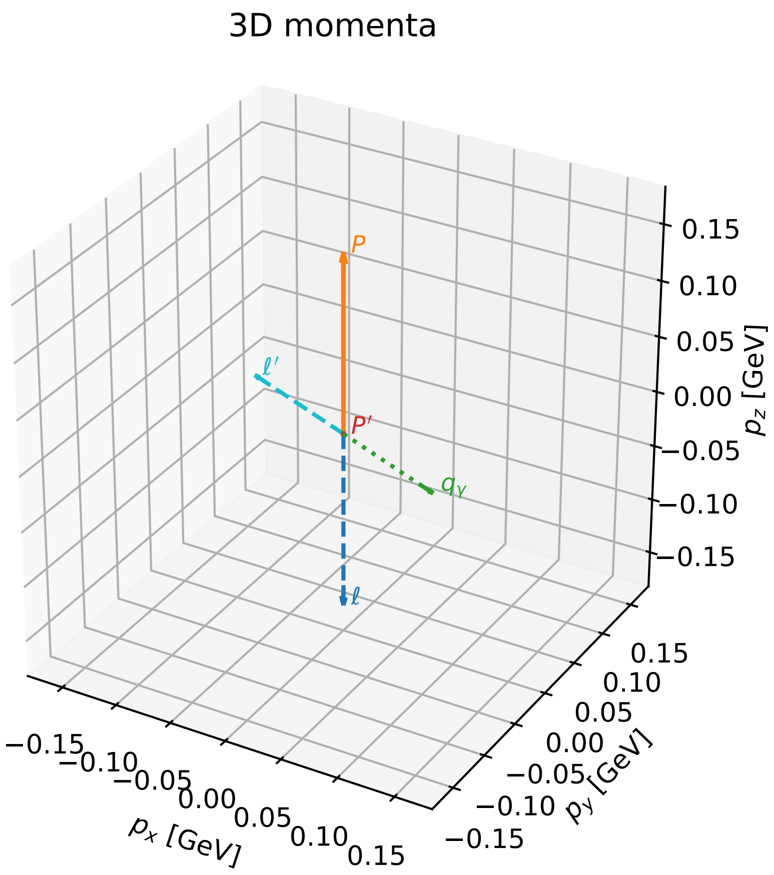
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)



electron: polarization cluster P2, local minimum 561; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



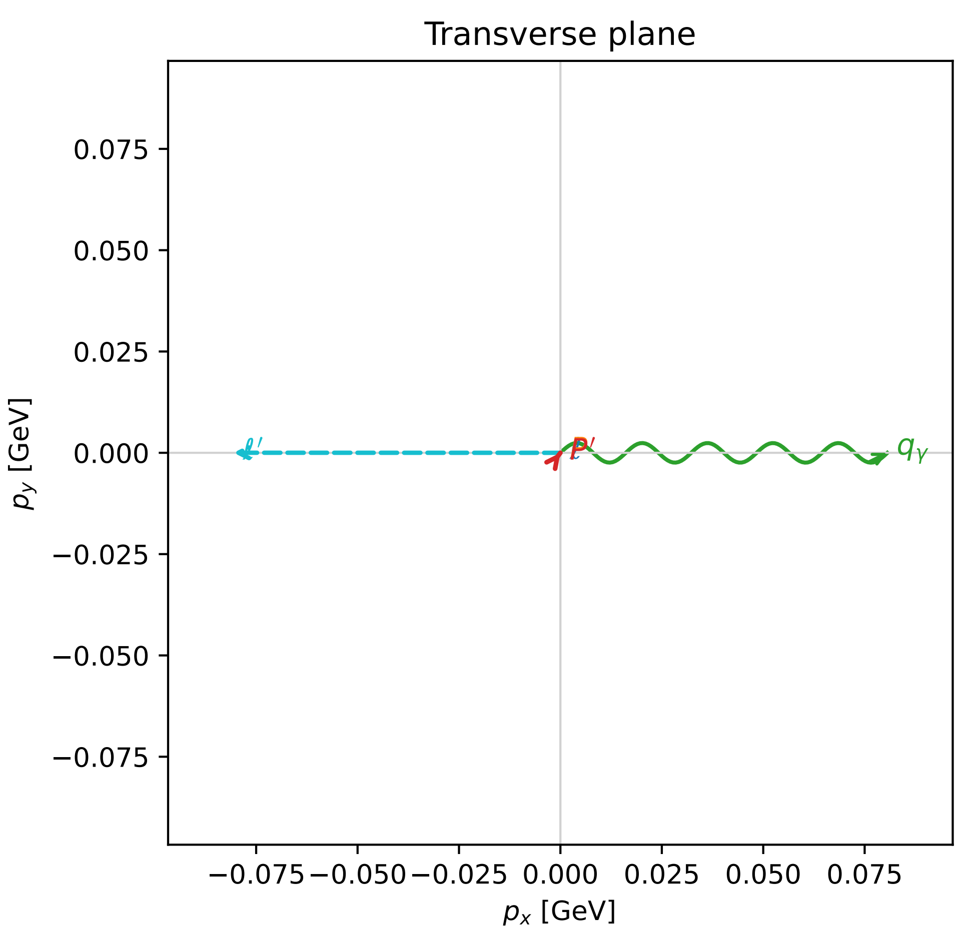
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

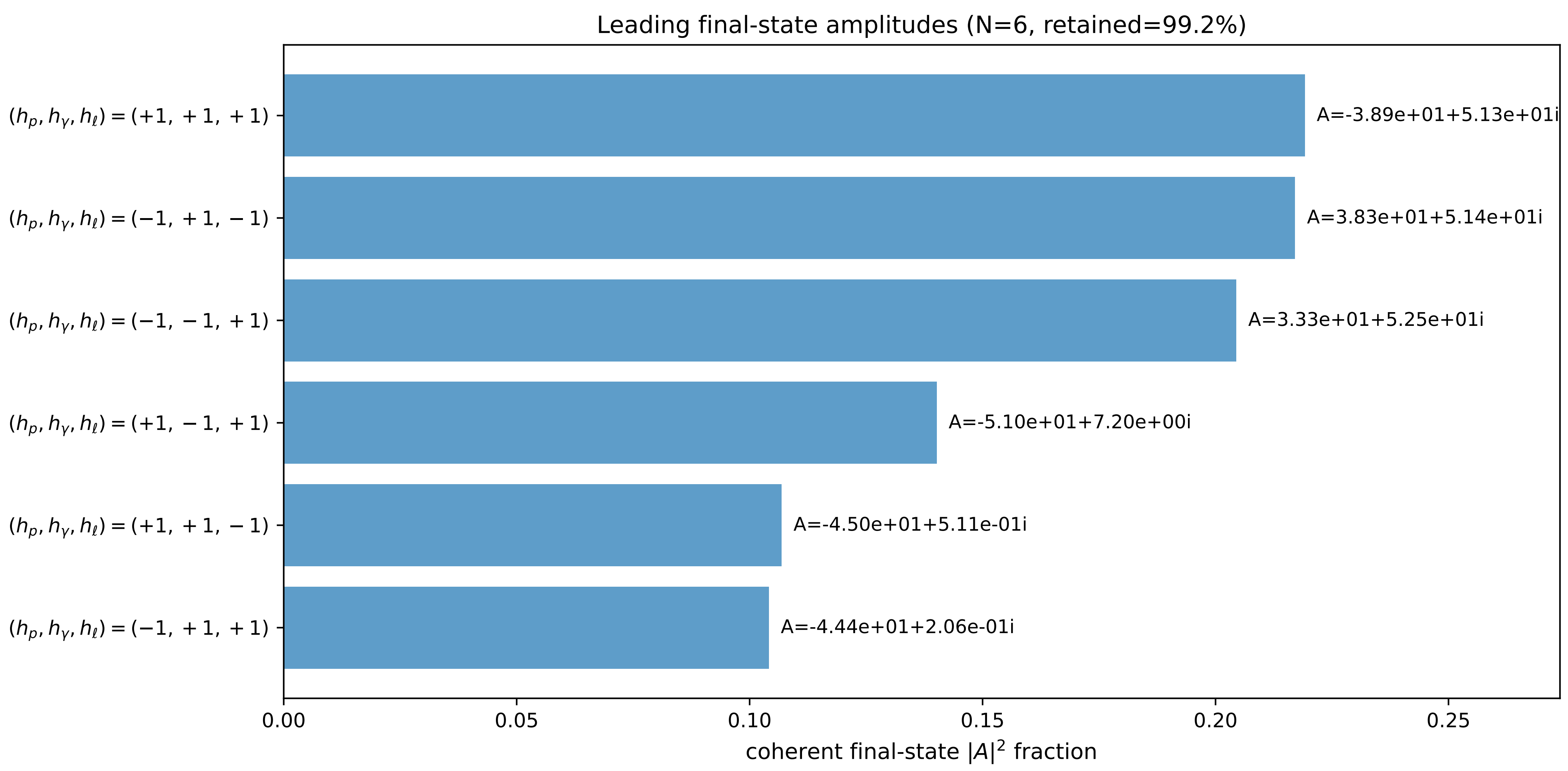


dw_mixing_angles_polarization_02_local_minimum_566 D_W=0.0096585
region: polarization_cluster_02_local_minimum_566
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.76981104 rad
incoming lepton: 0.718042|+> +0.696|->
proton mixing angle: theta_p=2.9737989 rad
incoming proton: -0.985956|+> +0.167007|->

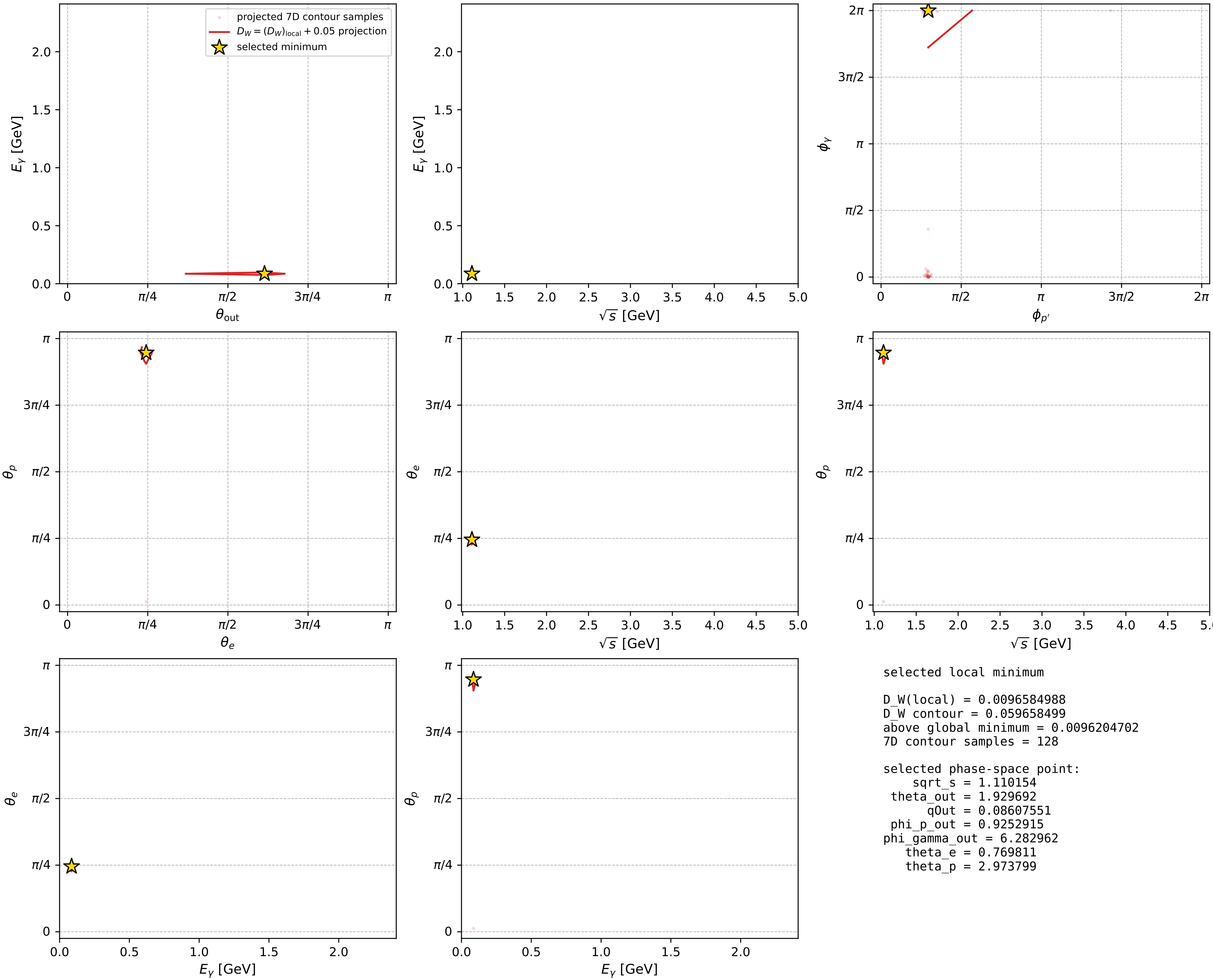
s=1.23244, $\sqrt{s}$=1.11015, $|\vec{P}|$=0.158805, $|\vec{P}'|$=2.82985e-06
θ_{out}=1.92969, $\phi_{p'}$=0.925292, ϕ_γ=6.28296
E_γ=0.0860755, $\theta_{\ell'}$=1.2119, $\phi_{\ell'}$=3.1414
Q^2=0.0369419, x_B=0.186181, t=-0.0250413, W^2=1.04132, y=0.562734

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1588 p_x= 0.0000 p_y= 0.0000 p_z= -0.1588
P E= 0.9513 p_x= 0.0000 p_y= 0.0000 p_z= 0.1588
ℓ' E= 0.0861 p_x= -0.0806 p_y= 0.0000 p_z= 0.0302
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= -0.0000
q_γ E= 0.0861 p_x= 0.0806 p_y= -0.0000 p_z= -0.0302

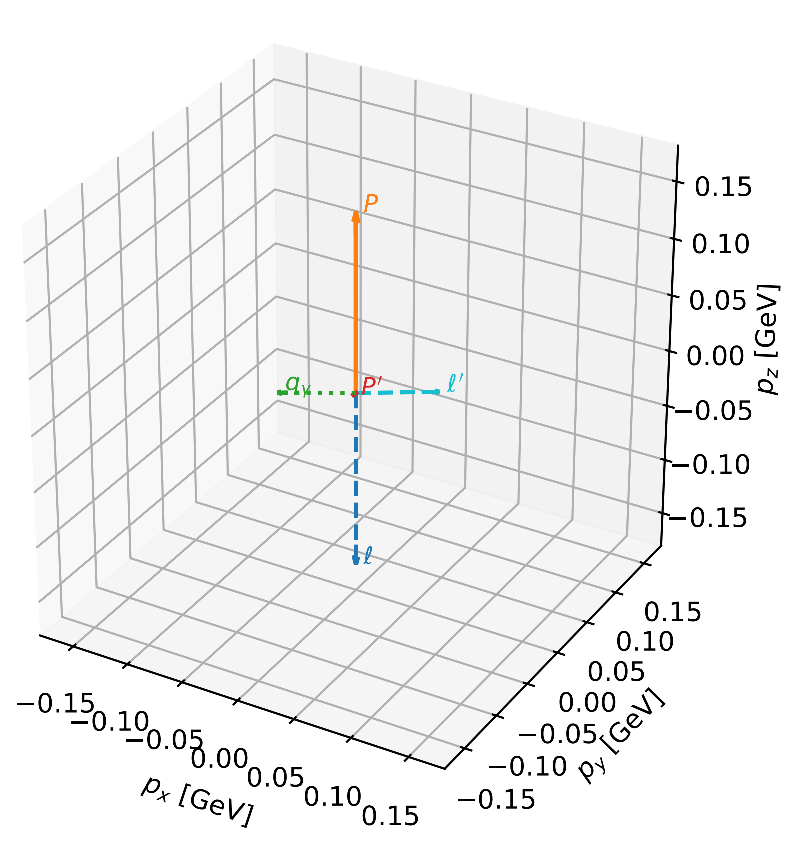




electron: polarization cluster P2, local minimum 566; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



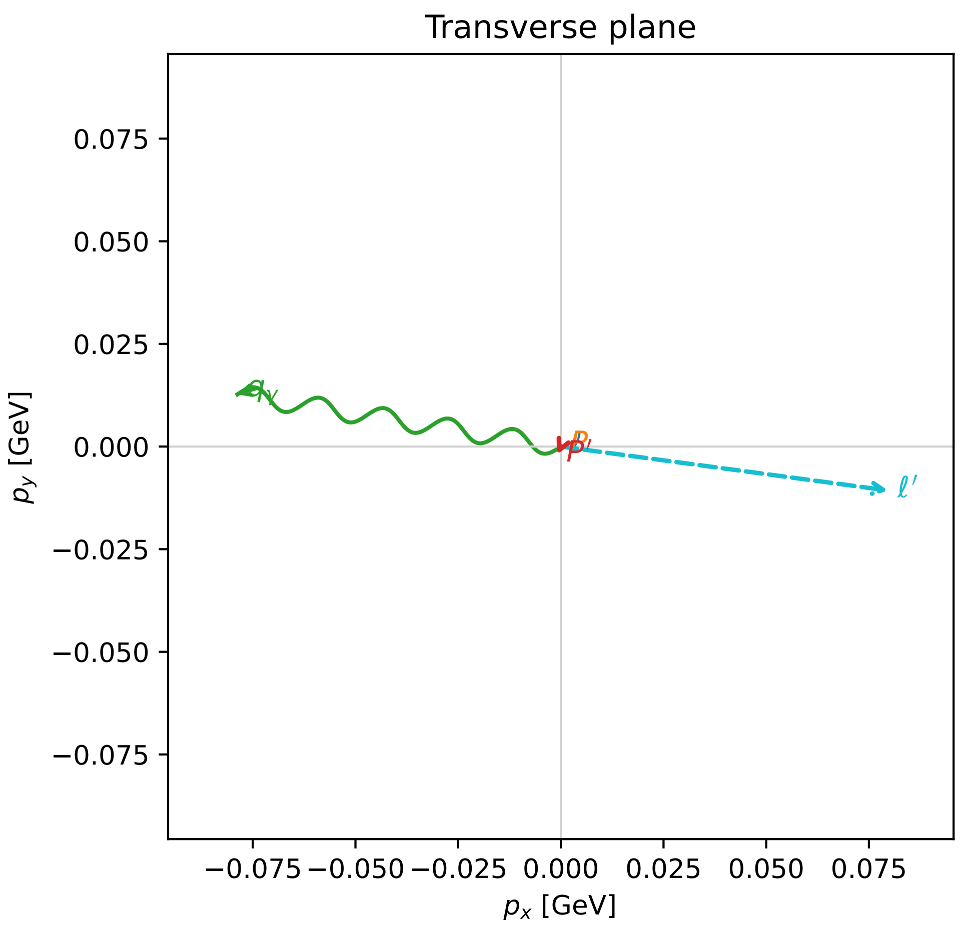
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

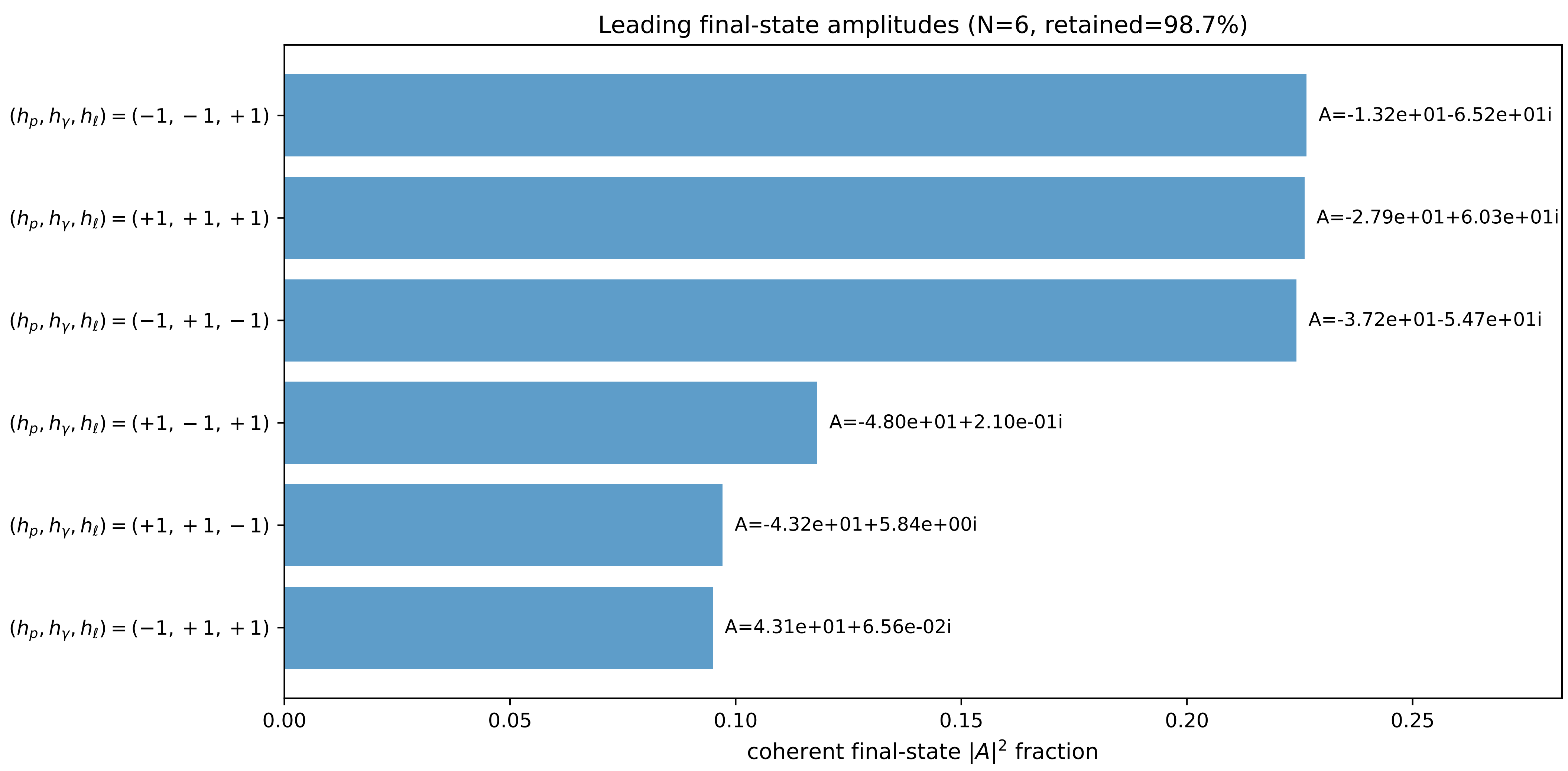


dw_mixing_angles_polarization_02_local_minimum_567 D_W=0.00969125
region: polarization_cluster_02_local_minimum_567
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.7527473 rad
incoming lepton: 0.729813|+> +0.683646|->
proton mixing angle: theta_p=0.19585389 rad
incoming proton: 0.980882|+> +0.194604|->

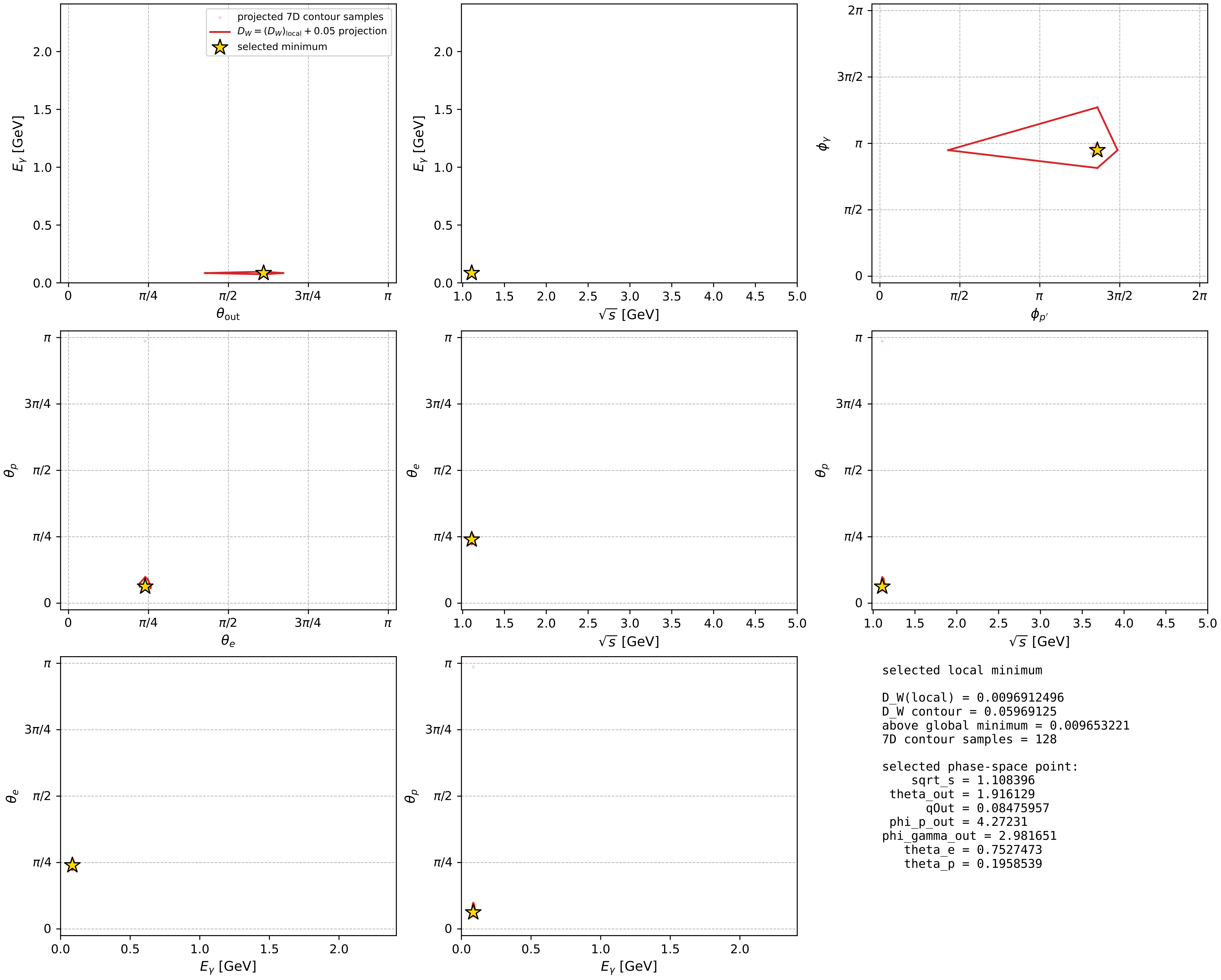
s=1.22854, $\sqrt{s}$=1.1084, $|\vec{P}|$=0.157298, $|\vec{P}'|$=0.00234931
θ_{out}=1.91613, $\phi_{p'}$=4.27231, ϕ_γ=2.98165
E_γ=0.0847596, $\theta_{\ell'}$=1.21925, $\phi_{\ell'}$=6.14967
Q^2=0.036216, x_B=0.185645, t=-0.0248268, W^2=1.03871, y=0.55946

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1573 p_x= 0.0000 p_y= 0.0000 p_z= -0.1573
P E= 0.9511 p_x= 0.0000 p_y= 0.0000 p_z= 0.1573
ℓ' E= 0.0856 p_x= 0.0797 p_y= -0.0107 p_z= 0.0295
P' E= 0.9380 p_x= -0.0009 p_y= -0.0020 p_z= -0.0008
q_γ E= 0.0848 p_x= -0.0787 p_y= 0.0127 p_z= -0.0287

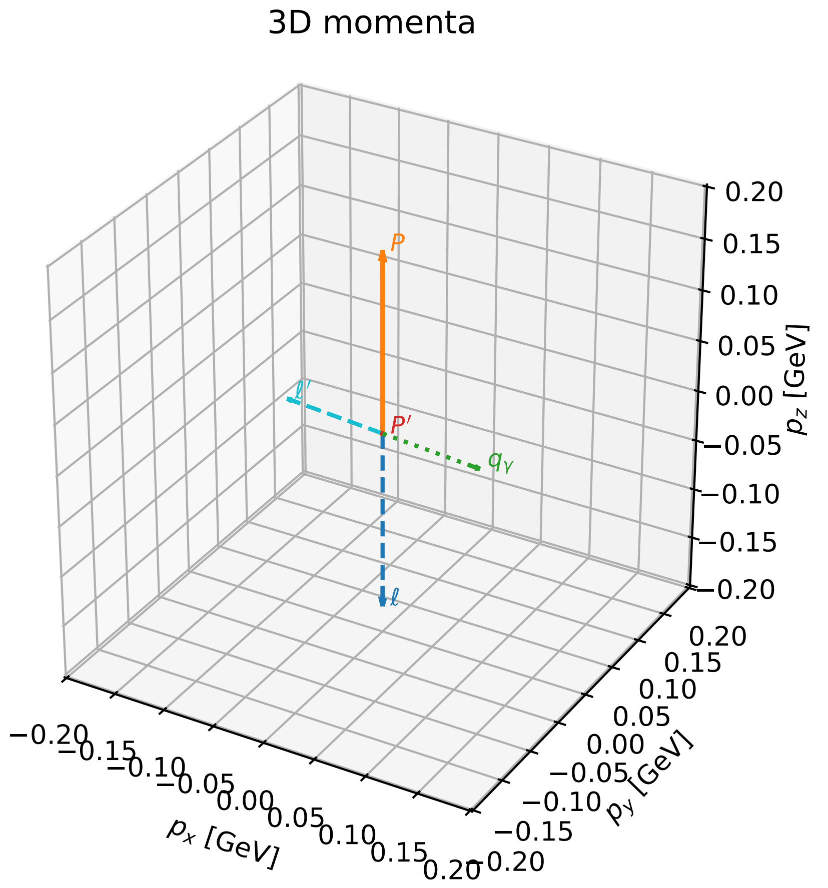




electron: polarization cluster P2, local minimum 567; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



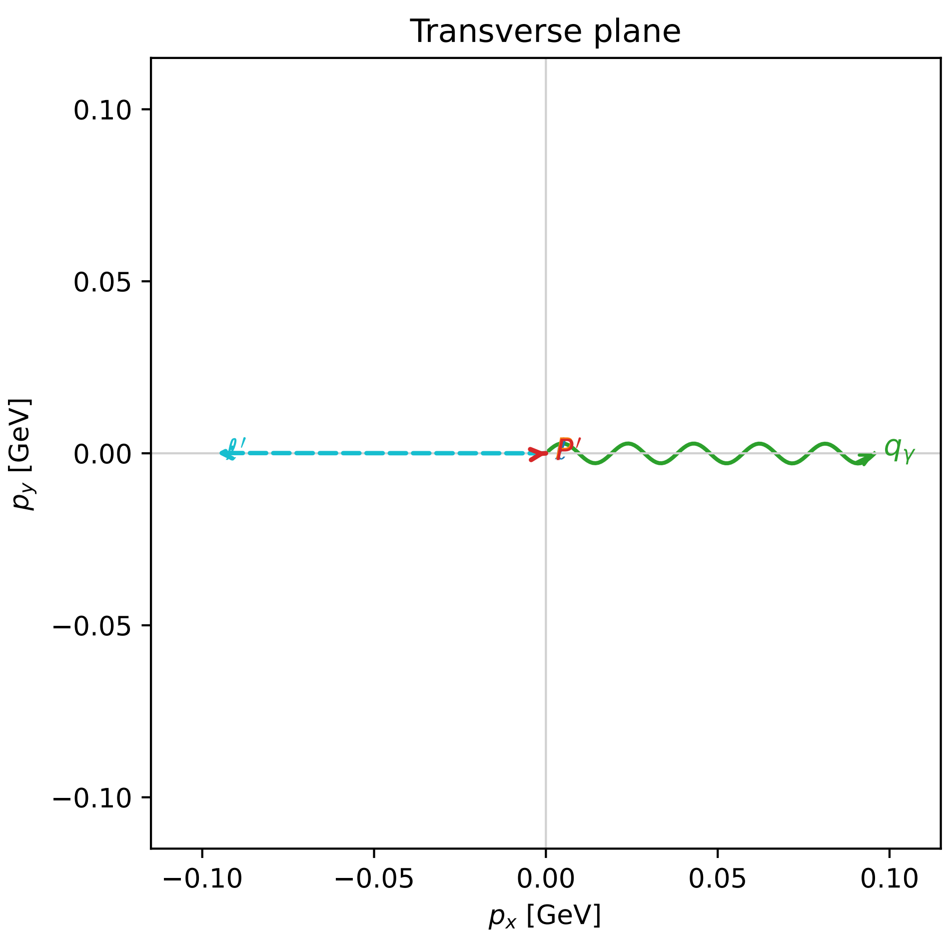
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

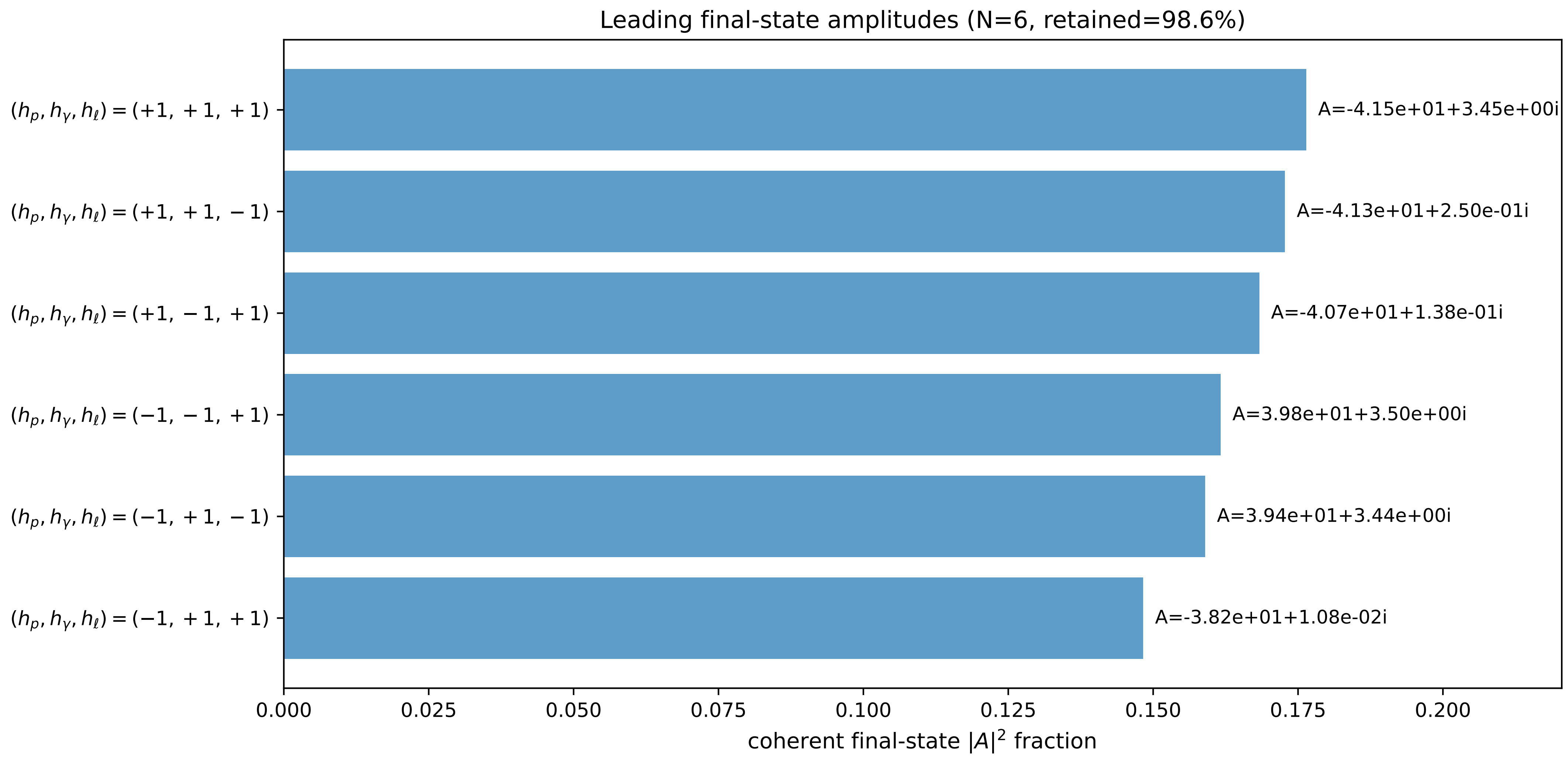


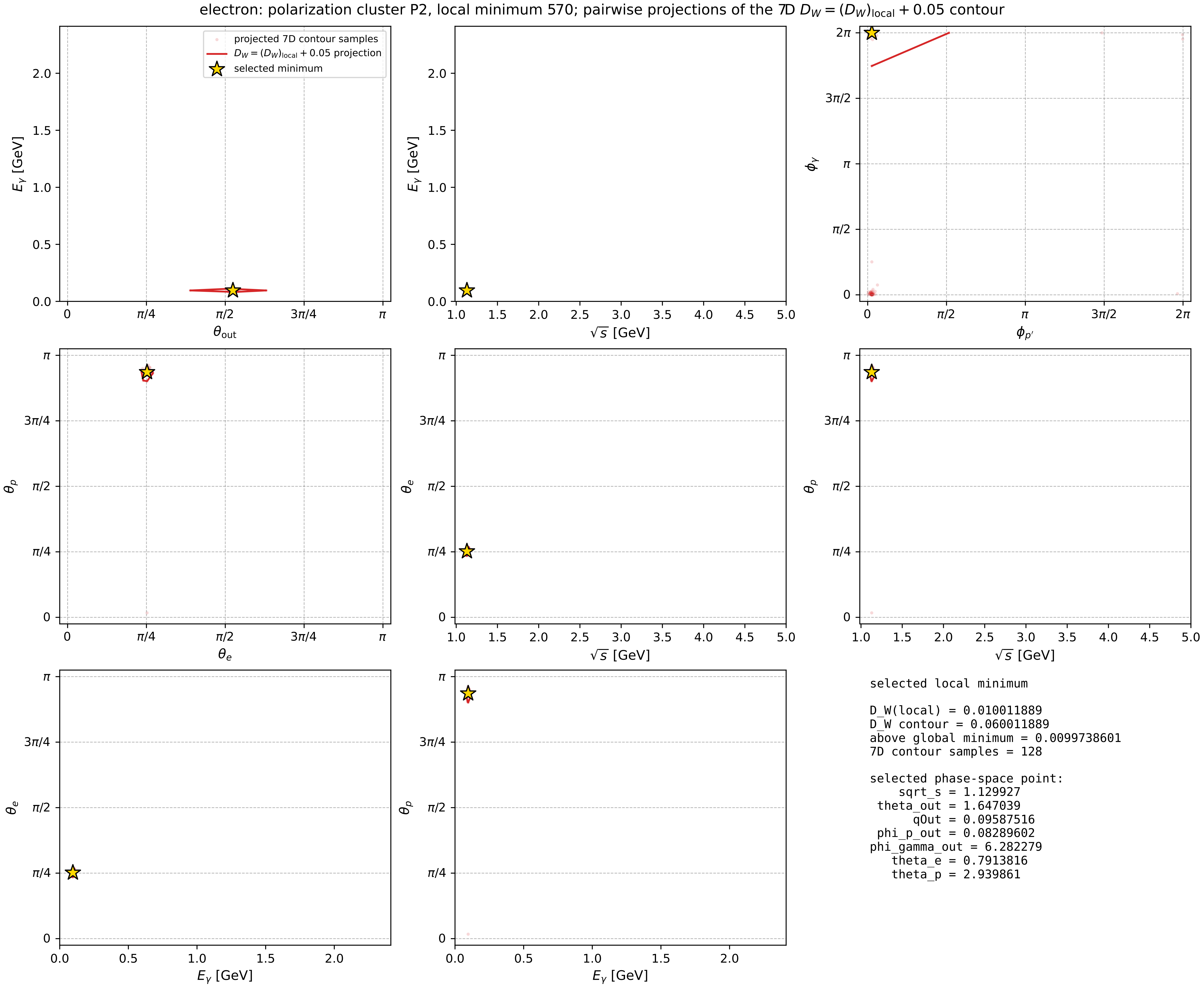
dw_mixing_angles_polarization_02_local_minimum_570 D_W=0.0100119
region: polarization_cluster_02_local_minimum_570
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.79138164 rad
incoming lepton: 0.702863|+> +0.711325|->
proton mixing angle: theta_p=2.9398606 rad
incoming proton: -0.979721|+> +0.200367|->

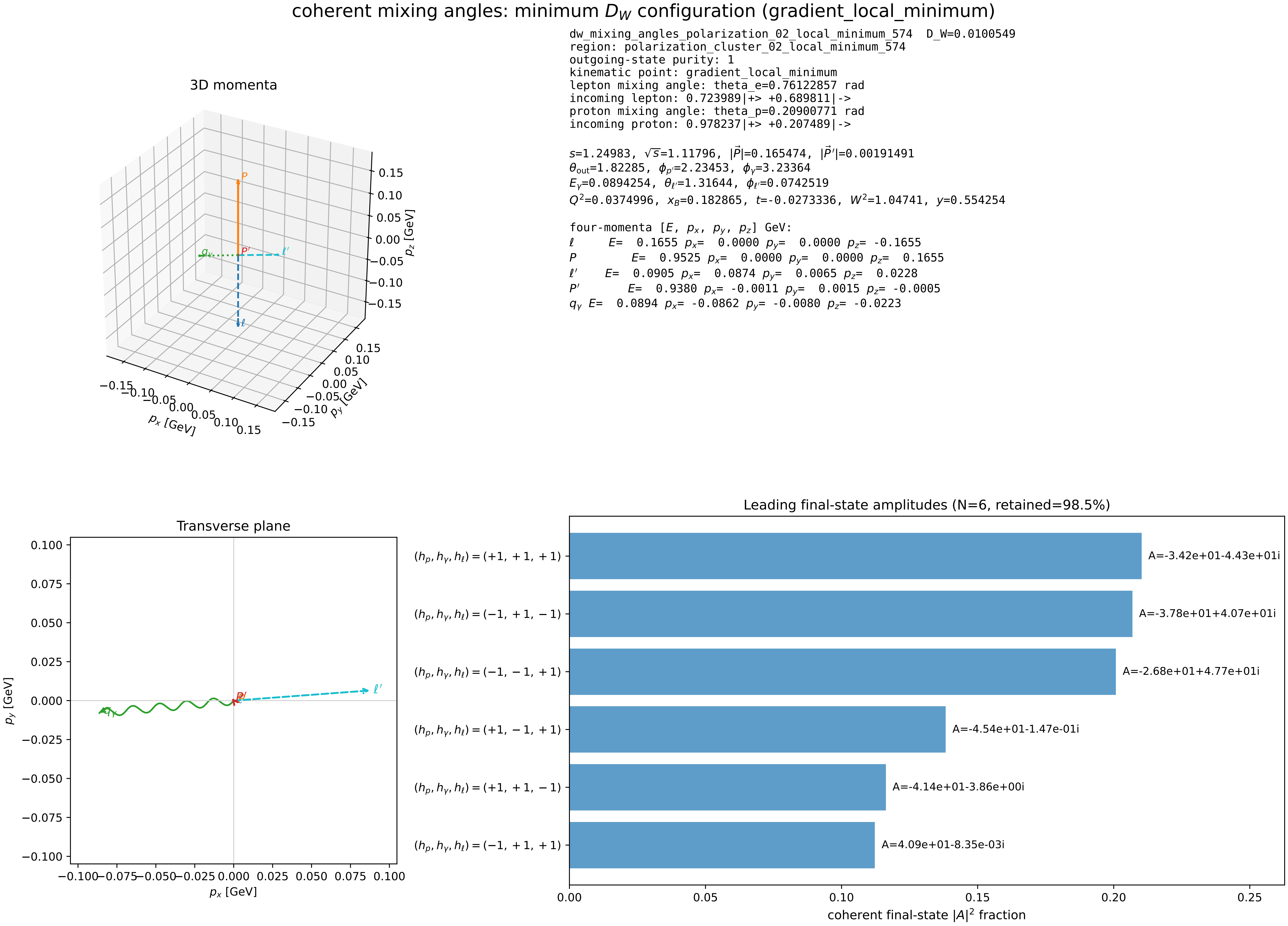
s=1.27674, $\sqrt{s}$=1.12993, $|\vec{P}|$=0.175626, $|\vec{P}'|$=0.000175908
θ_{out}=1.64704, $\phi_{p'}$=0.082896, ϕ_γ=6.28228
E_γ=0.0958752, $\theta_{\ell'}$=1.49455, $\phi_{\ell'}$=3.14084
Q^2=0.0363078, x_B=0.167986, t=-0.0305836, W^2=1.05967, y=0.544573

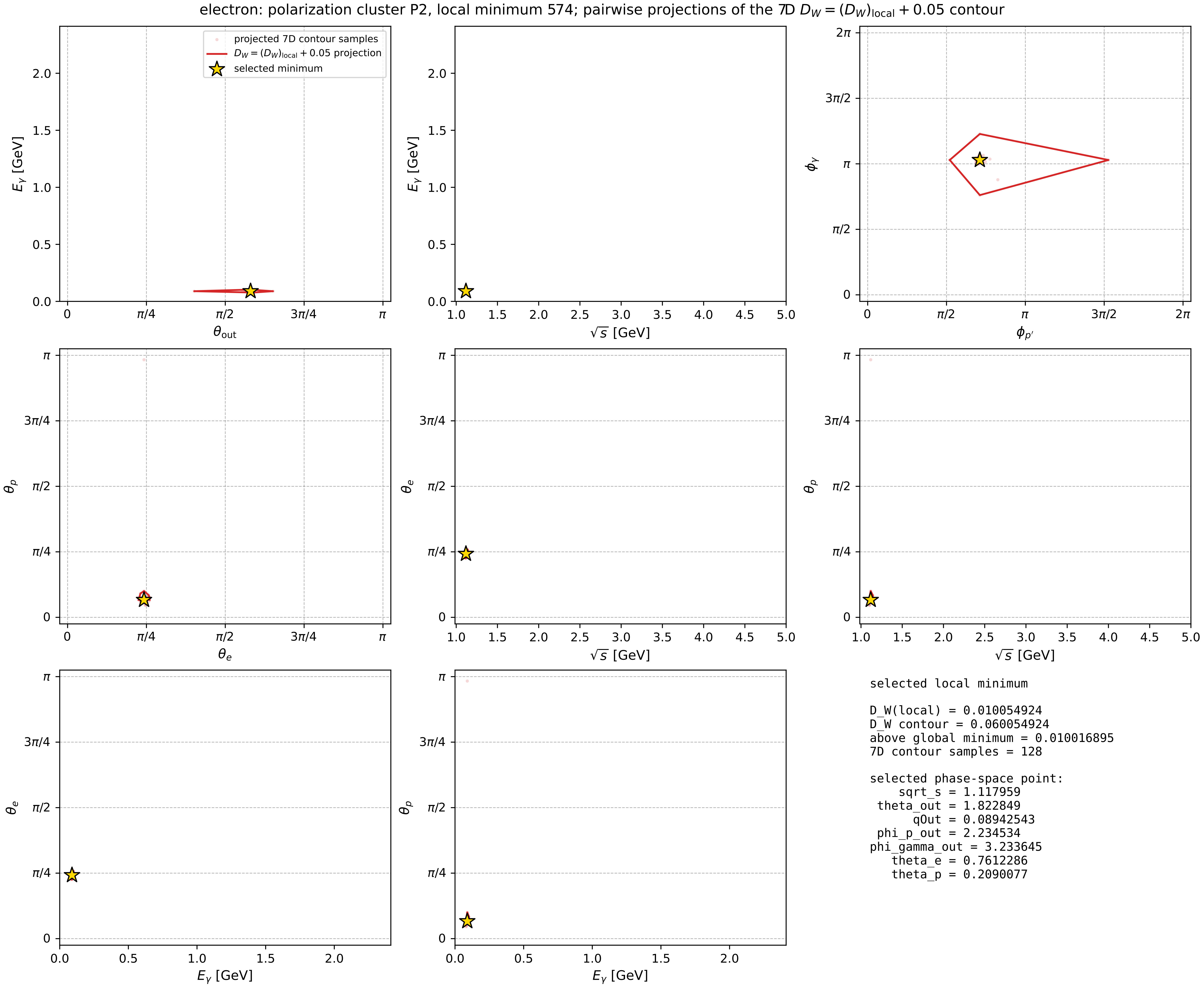
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1756 p_x= 0.0000 p_y= 0.0000 p_z= -0.1756
P E= 0.9543 p_x= 0.0000 p_y= 0.0000 p_z= 0.1756
ℓ' E= 0.0961 p_x= -0.0958 p_y= 0.0001 p_z= 0.0073
P' E= 0.9380 p_x= 0.0002 p_y= 0.0000 p_z= -0.0000
q_γ E= 0.0959 p_x= 0.0956 p_y= -0.0001 p_z= -0.0073



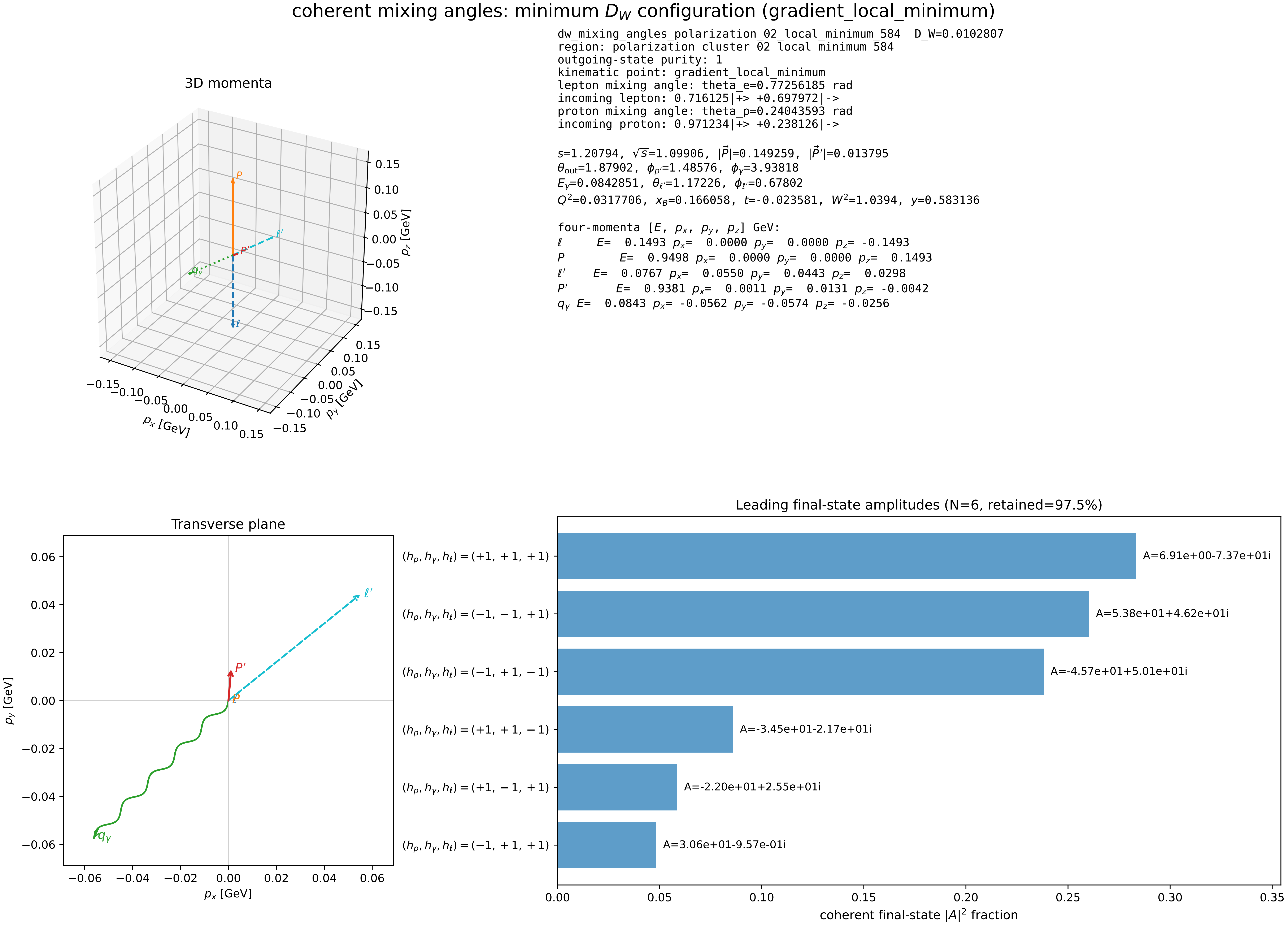



electron: polarization cluster P2, local minimum 570; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.010011889
D_W contour = 0.060011889
above global minimum = 0.0099738601
7D contour samples = 128
selected phase-space point:
sqrt_s = 1.129927
theta_out = 1.647039
qOut = 0.09587516
phi_p_out = 0.08289602
phi_gamma_out = 6.282279
theta_e = 0.7913816
theta_p = 2.939861

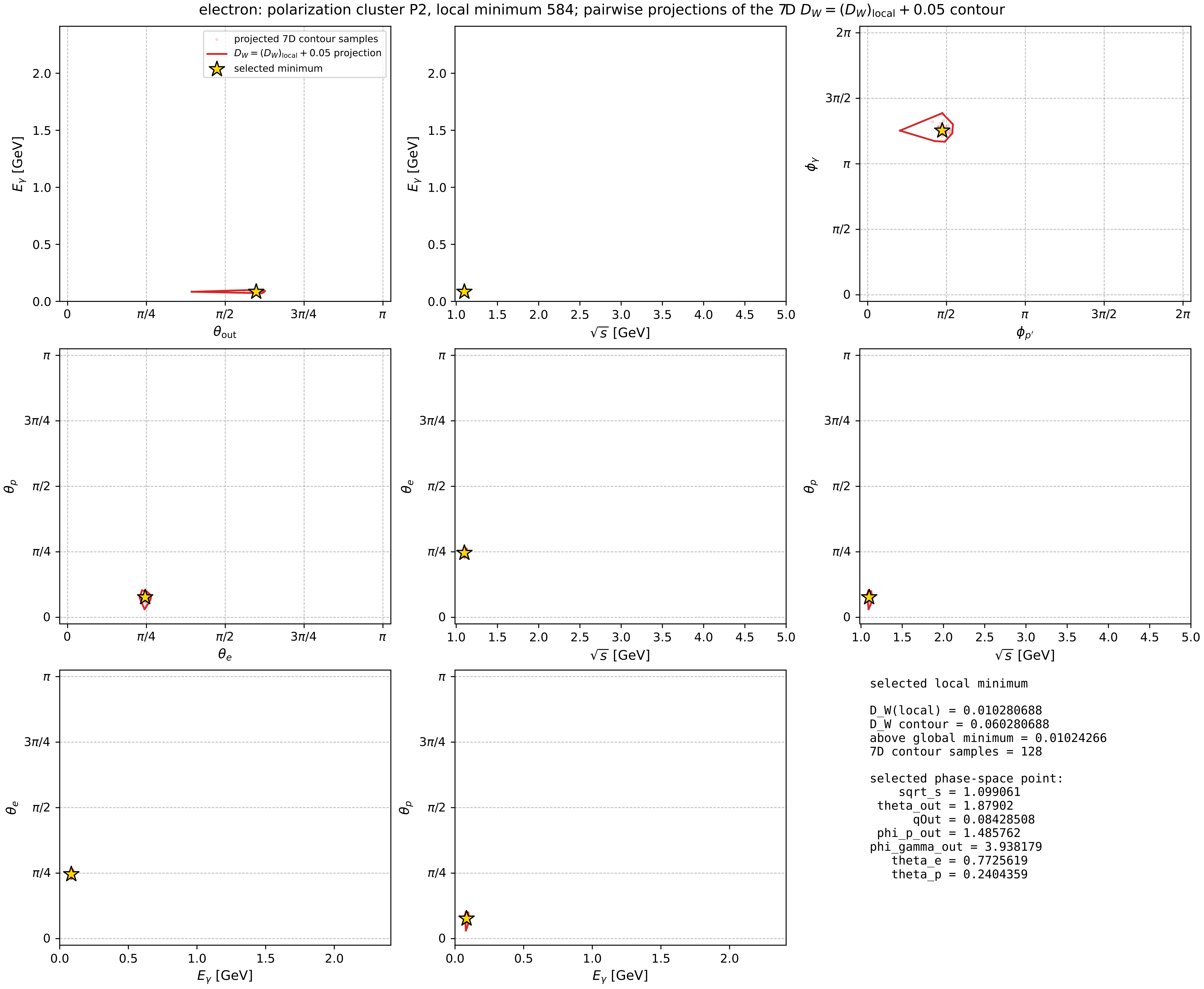

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
dw_mixing_angles_polarization_02_local_minimum_574 D_W=0.0100549
region: polarization_cluster_02_local_minimum_574
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.76122857 rad
incoming lepton: 0.723989|+> +0.689811|->
proton mixing angle: theta_p=0.20900771 rad
incoming proton: 0.978237|+> +0.207489|->
s=1.24983, $\sqrt{s}$=1.11796, $|\vec{P}|$=0.165474, $|\vec{P}'|$=0.00191491
$\theta_{\rm out}$=1.82285, $\phi_{p'}$=2.23453, ϕ_γ=3.23364
E_γ=0.0894254, $\theta_{\ell'}$=1.31644, $\phi_{\ell'}$=0.0742519
Q^2=0.0374996, x_B=0.182865, t=-0.0273336, W^2=1.04741, y=0.554254
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1655 p_x= 0.0000 p_y= 0.0000 p_z= -0.1655
P E= 0.9525 p_x= 0.0000 p_y= 0.0000 p_z= 0.1655
ℓ' E= 0.0905 p_x= 0.0874 p_y= 0.0065 p_z= 0.0228
P' E= 0.9380 p_x= -0.0011 p_y= 0.0015 p_z= -0.0005
q_γ E= 0.0894 p_x= -0.0862 p_y= -0.0080 p_z= -0.0223
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
Leading final-state amplitudes (N=6, retained=98.5%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=-3.42e+01-4.43e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-3.78e+01+4.07e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-2.68e+01+4.77e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=-4.54e+01-1.47e-01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-4.14e+01-3.86e+00i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=4.09e+01-8.35e-03i
coherent final-state $|A|^2$ fraction



electron: polarization cluster P2, local minimum 574; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.010054924
D_W contour = 0.060054924
above global minimum = 0.010016895
7D contour samples = 128
selected phase-space point:
sqrt_s = 1.117959
theta_out = 1.822849
qOut = 0.08942543
phi_p_out = 2.234534
phi_gamma_out = 3.233645
theta_e = 0.7612286
theta_p = 0.2090077

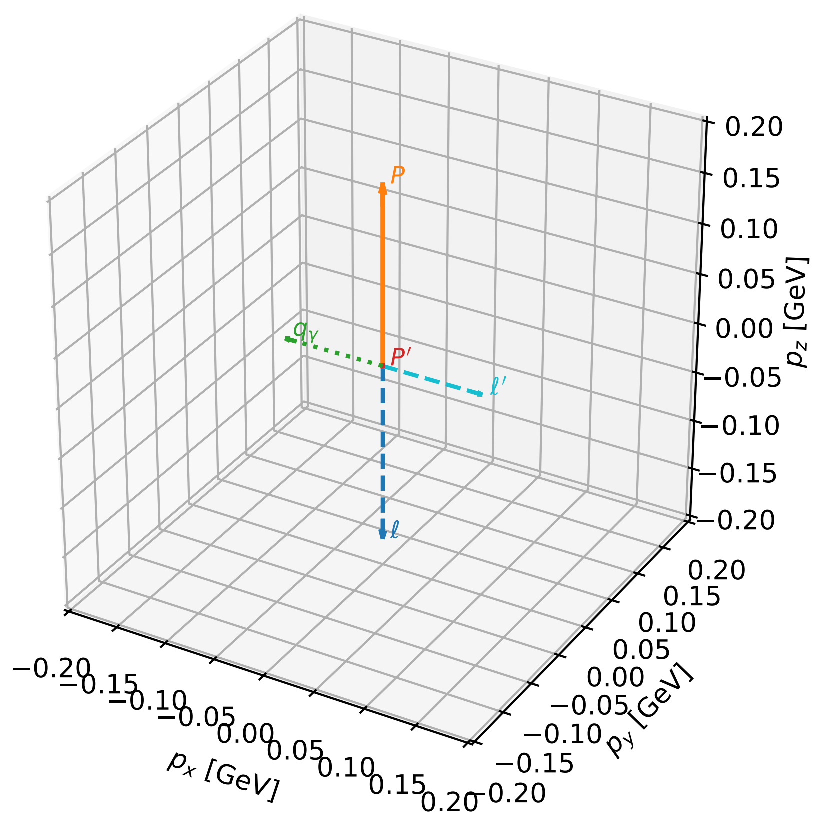


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_584 D_W=0.0102807
region: polarization_cluster_02_local_minimum_584
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.77256185 rad
incoming lepton: 0.716125|+> +0.697972|->
proton mixing angle: theta_p=0.24043593 rad
incoming proton: 0.971234|+> +0.238126|->
s=1.20794, $\sqrt{s}$=1.09906, $|\vec{P}|$=0.149259, $|\vec{P}'|$=0.013795
θ_{out}=1.87902, $\phi_{p'}$=1.48576, ϕ_γ=3.93818
E_γ=0.0842851, $\theta_{\ell'}$=1.17226, $\phi_{\ell'}$=0.67802
Q^2=0.0317706, x_B=0.166058, t=-0.023581, W^2=1.0394, y=0.583136
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1493 p_x= 0.0000 p_y= 0.0000 p_z= -0.1493
P E= 0.9498 p_x= 0.0000 p_y= 0.0000 p_z= 0.1493
ℓ' E= 0.0767 p_x= 0.0550 p_y= 0.0443 p_z= 0.0298
P' E= 0.9381 p_x= 0.0011 p_y= 0.0131 p_z= -0.0042
q_γ E= 0.0843 p_x= -0.0562 p_y= -0.0574 p_z= -0.0256
3D momenta
P
ℓ'
P'
q_γ
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=6, retained=97.5%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=6.91e+00-7.37e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=5.38e+01+4.62e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-4.57e+01+5.01e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-3.45e+01-2.17e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=-2.20e+01+2.55e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=3.06e+01-9.57e-01i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P2, local minimum 584; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.010280688
D_W contour = 0.060280688
above global minimum = 0.01024266
7D contour samples = 128
selected phase-space point:
sqrt_s = 1.099061
theta_out = 1.87902
qOut = 0.08428508
phi_p_out = 1.485762
phi_gamma_out = 3.938179
theta_e = 0.7725619
theta_p = 0.2404359

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

3D momenta

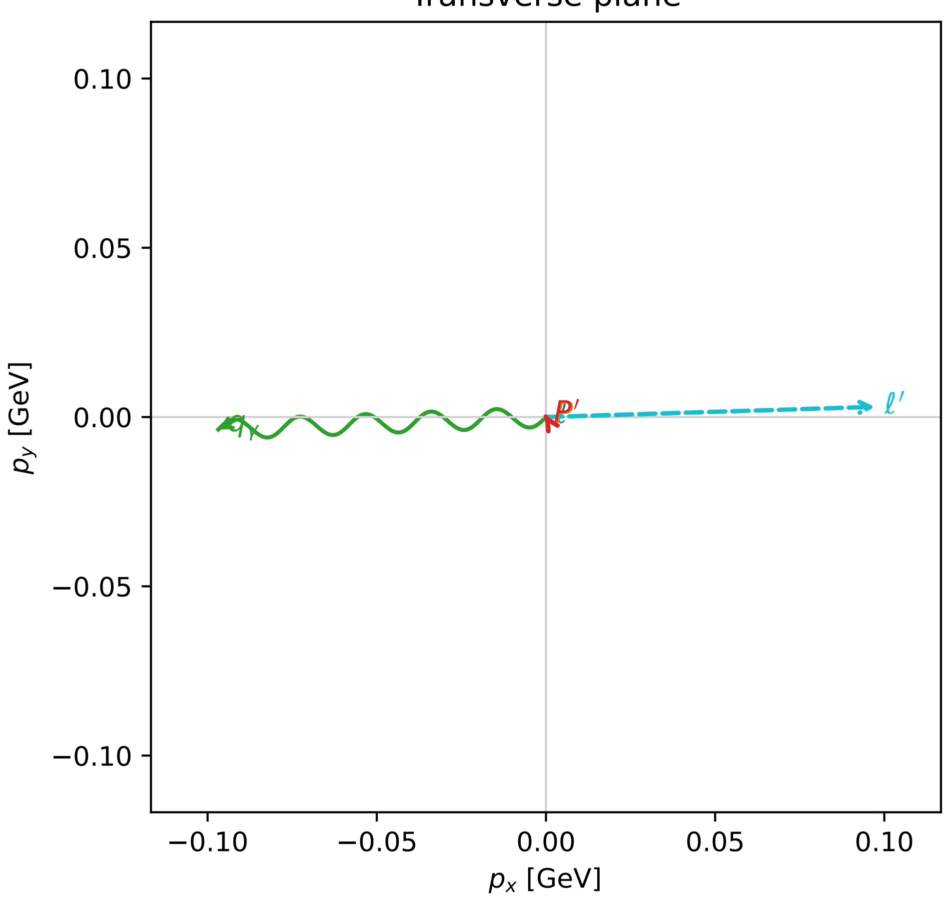


dw_mixing_angles_polarization_02_local_minimum_586 D_W=0.0104003
region: polarization_cluster_02_local_minimum_586
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.79402422 rad
incoming lepton: 0.700981|+> +0.71318|->
proton mixing angle: theta_p=0.21220994 rad
incoming proton: 0.977568|+> +0.210621|->

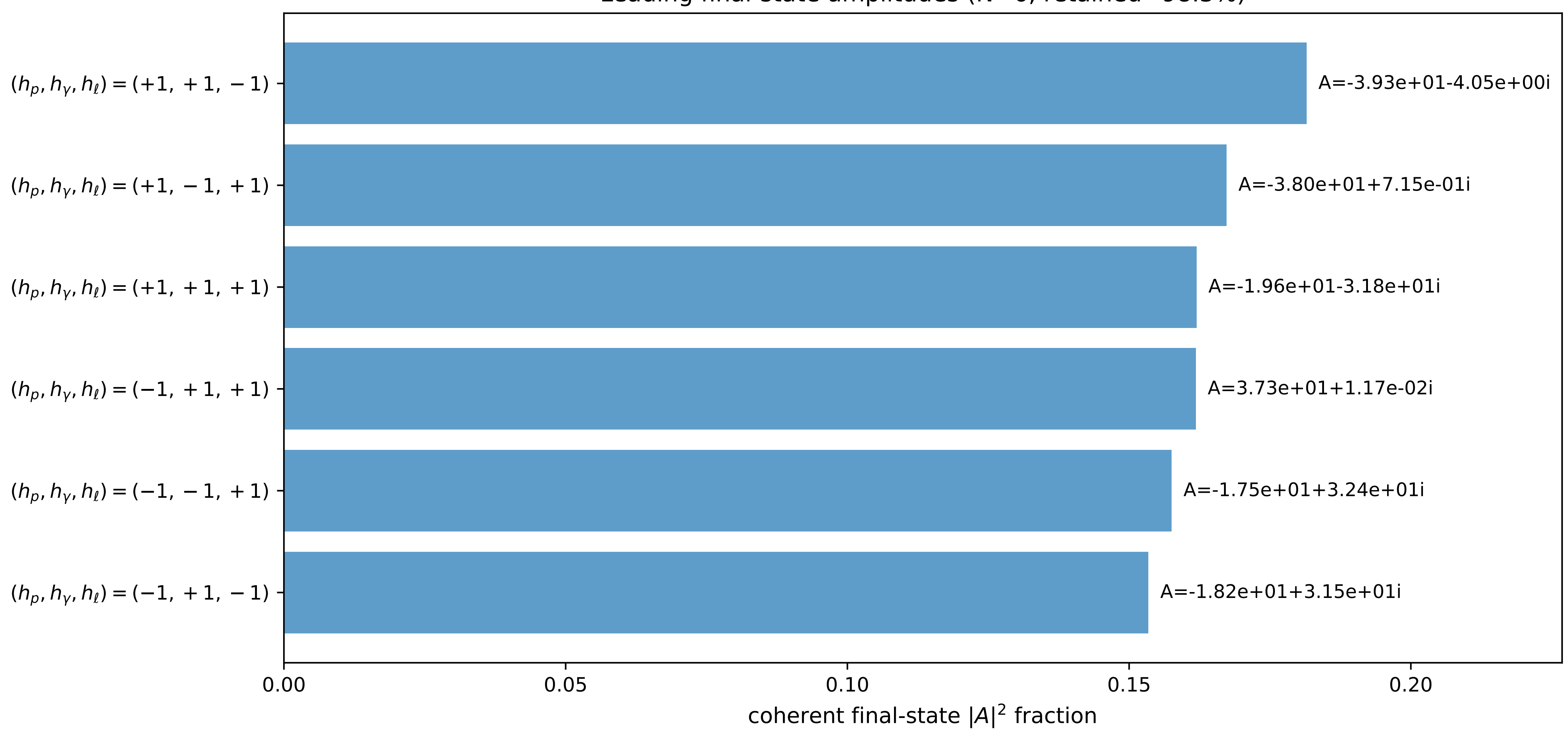
s=1.28213, $\sqrt{s}$=1.13231, $|\vec{P}|$=0.177637, $|\vec{P}'|$=0.000791841
θ_{out}=1.55058, $\phi_{p'}$=2.12765, ϕ_γ=3.17969
E_γ=0.0969564, $\theta_{\ell'}$=1.5911, $\phi_{\ell'}$=0.0310382
Q^2=0.0338845, x_B=0.157092, t=-0.0312718, W^2=1.06166, y=0.536189

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1776 p_x= 0.0000 p_y= 0.0000 p_z= -0.1776
P E= 0.9547 p_x= 0.0000 p_y= 0.0000 p_z= 0.1776
ℓ' E= 0.0974 p_x= 0.0973 p_y= 0.0030 p_z= -0.0020
P' E= 0.9380 p_x= -0.0004 p_y= 0.0007 p_z= 0.0000
q_γ E= 0.0970 p_x= -0.0969 p_y= -0.0037 p_z= 0.0020

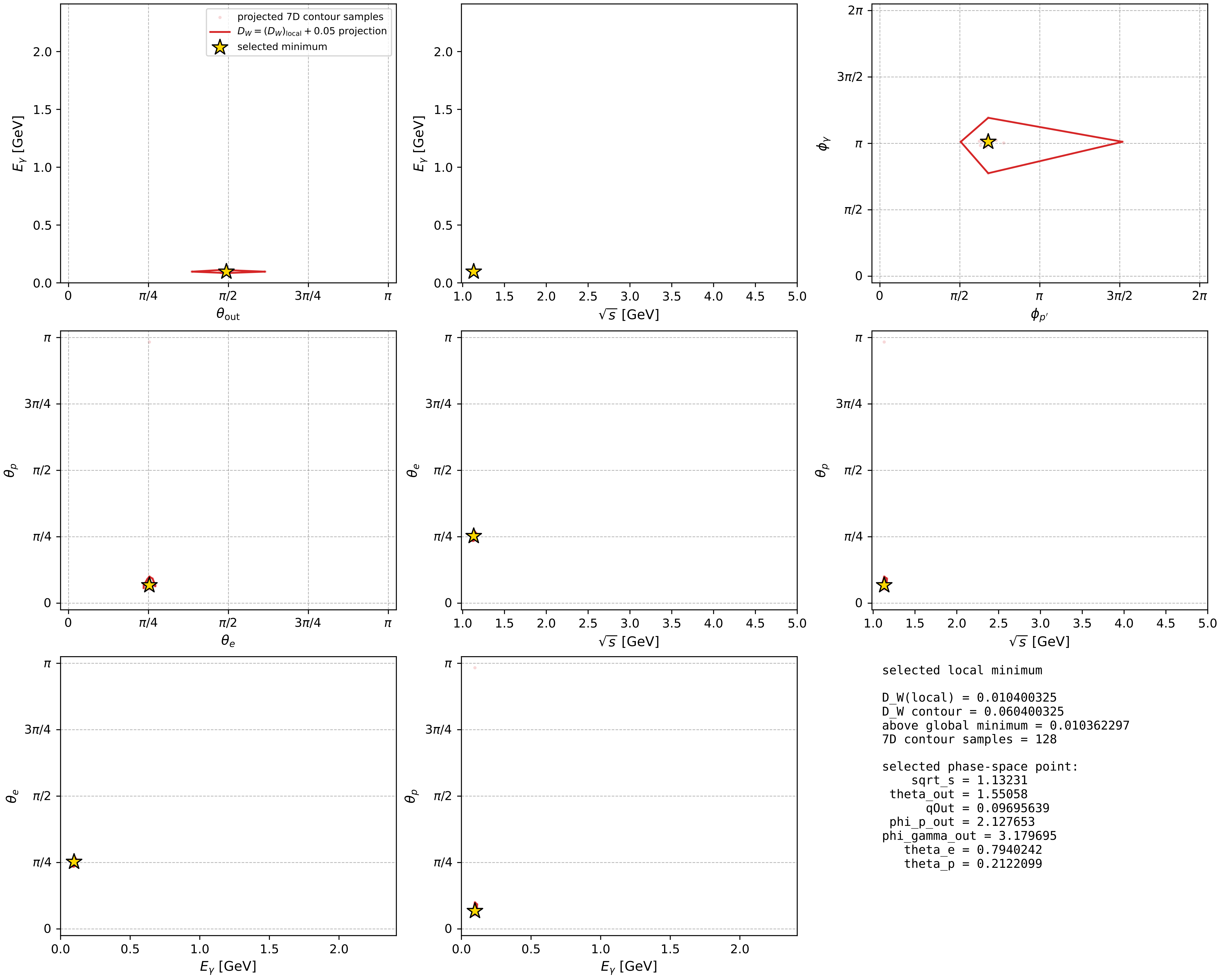
Transverse plane

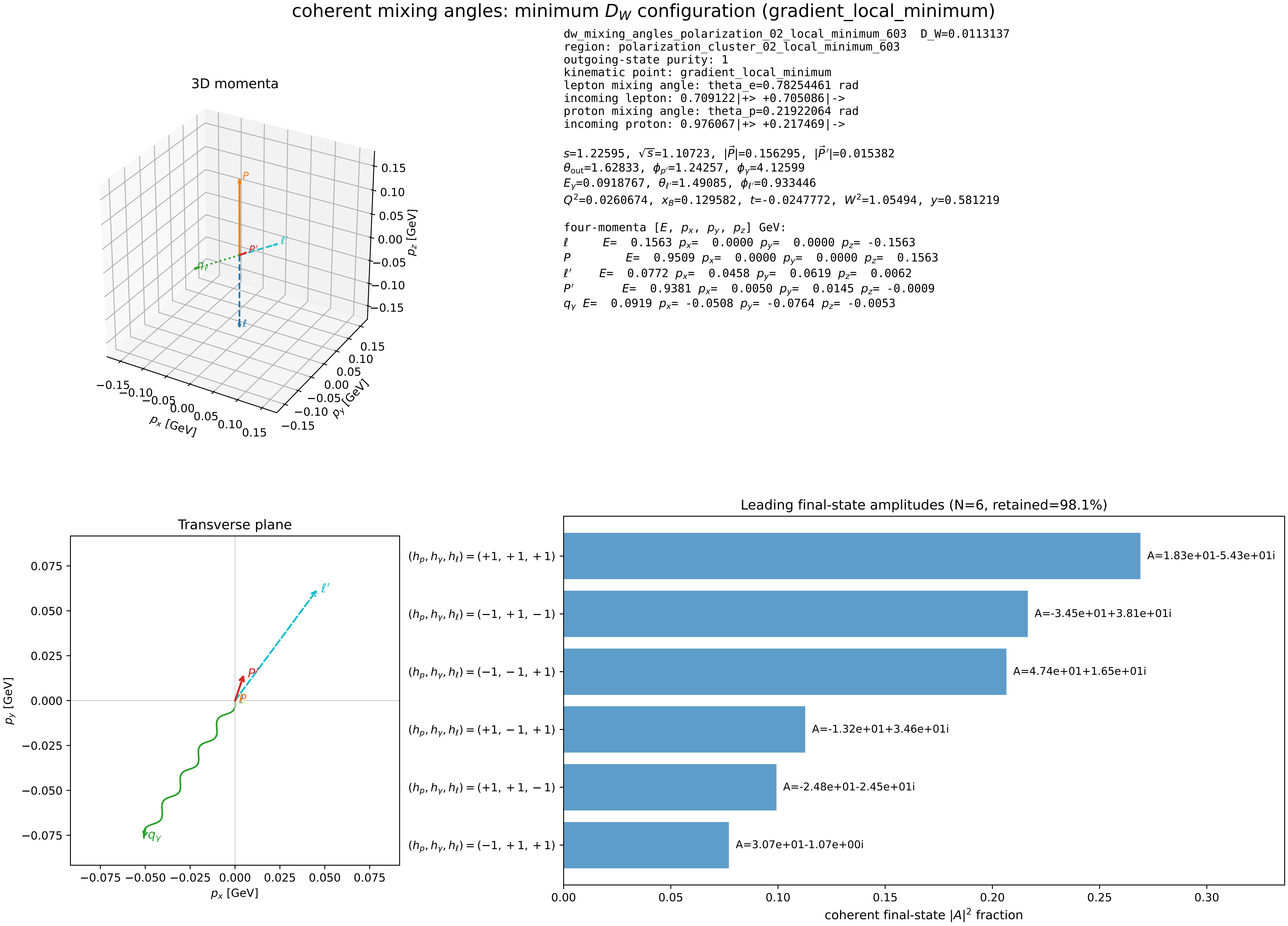


Leading final-state amplitudes (N=6, retained=98.3%)

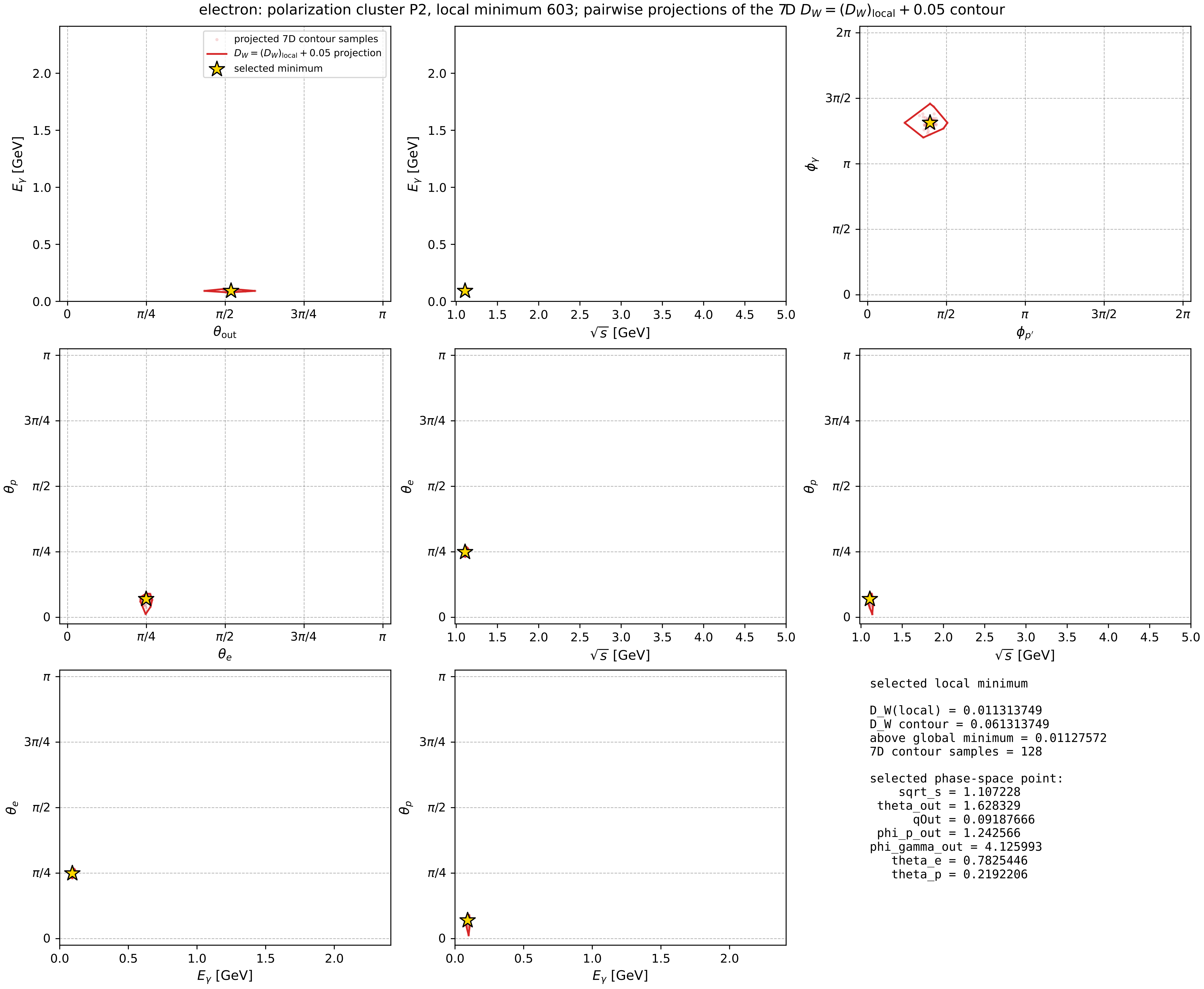


electron: polarization cluster P2, local minimum 586; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour

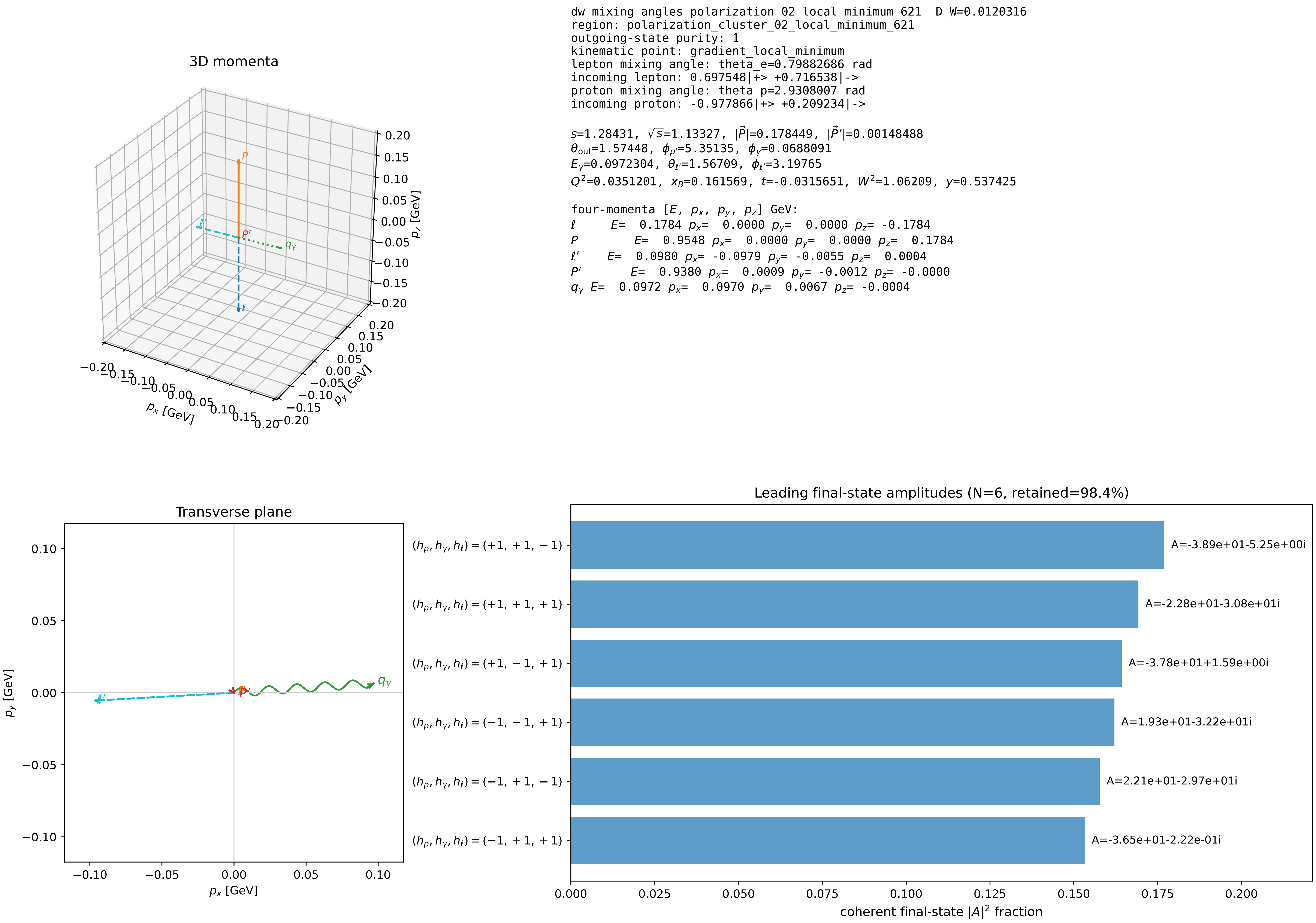


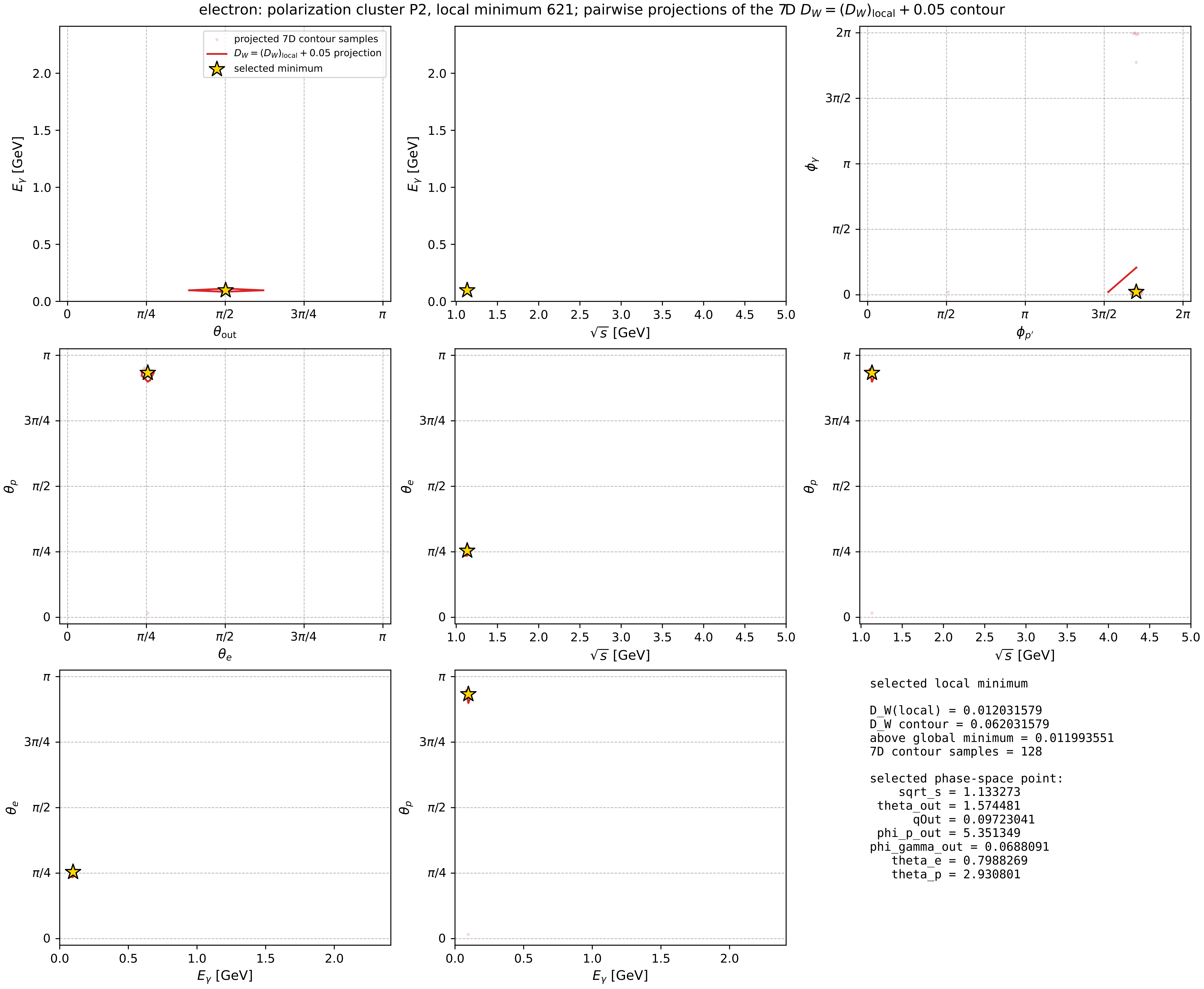


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
dw_mixing_angles_polarization_02_local_minimum_603 D_W=0.0113137
region: polarization_cluster_02_local_minimum_603
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.78254461 rad
incoming lepton: 0.709122|+> +0.705086|->
proton mixing angle: theta_p=0.21922064 rad
incoming proton: 0.976067|+> +0.217469|->
s=1.22595, $\sqrt{s}$=1.10723, $|\vec{P}|$=0.156295, $|\vec{P}'|$=0.015382
θ_{out}=1.62833, $\phi_{p'}$=1.24257, ϕ_γ=4.12599
E_γ=0.0918767, $\theta_{\ell'}$=1.49085, $\phi_{\ell'}$=0.933446
Q^2=0.0260674, x_B=0.129582, t=-0.0247772, W^2=1.05494, y=0.581219
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1563 p_x= 0.0000 p_y= 0.0000 p_z= -0.1563
P E= 0.9509 p_x= 0.0000 p_y= 0.0000 p_z= 0.1563
ℓ' E= 0.0772 p_x= 0.0458 p_y= 0.0619 p_z= 0.0062
P' E= 0.9381 p_x= 0.0050 p_y= 0.0145 p_z= -0.0009
q_γ E= 0.0919 p_x= -0.0508 p_y= -0.0764 p_z= -0.0053
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
Leading final-state amplitudes (N=6, retained=98.1%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=1.83e+01-5.43e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-3.45e+01+3.81e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=4.74e+01+1.65e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=-1.32e+01+3.46e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-2.48e+01-2.45e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=3.07e+01-1.07e+00i
coherent final-state $|A|^2$ fraction

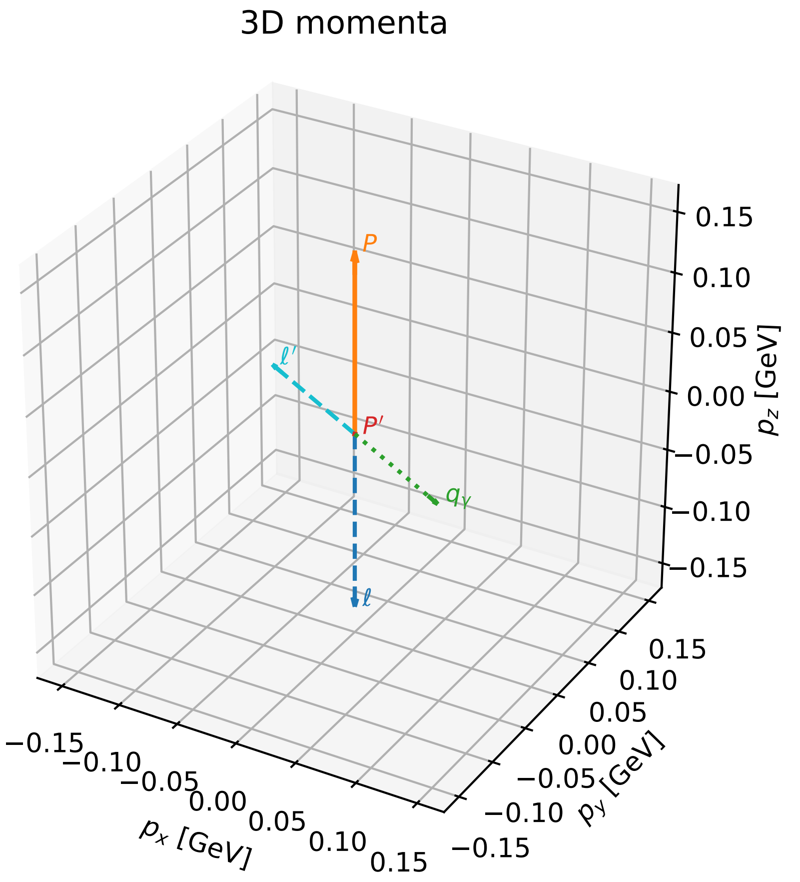

electron: polarization cluster P2, local minimum 603; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.011313749
D_W contour = 0.061313749
above global minimum = 0.01127572
7D contour samples = 128
selected phase-space point:
sqrt_s = 1.107228
theta_out = 1.628329
qOut = 0.09187666
phi_p_out = 1.242566
phi_gamma_out = 4.125993
theta_e = 0.7825446
theta_p = 0.2192206

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)




electron: polarization cluster P2, local minimum 621; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.012031579
D_W contour = 0.062031579
above global minimum = 0.011993551
7D contour samples = 128
selected phase-space point:
sqrt_s = 1.133273
theta_out = 1.574481
qOut = 0.09723041
phi_p_out = 5.351349
phi_gamma_out = 0.0688091
theta_e = 0.7988269
theta_p = 2.930801

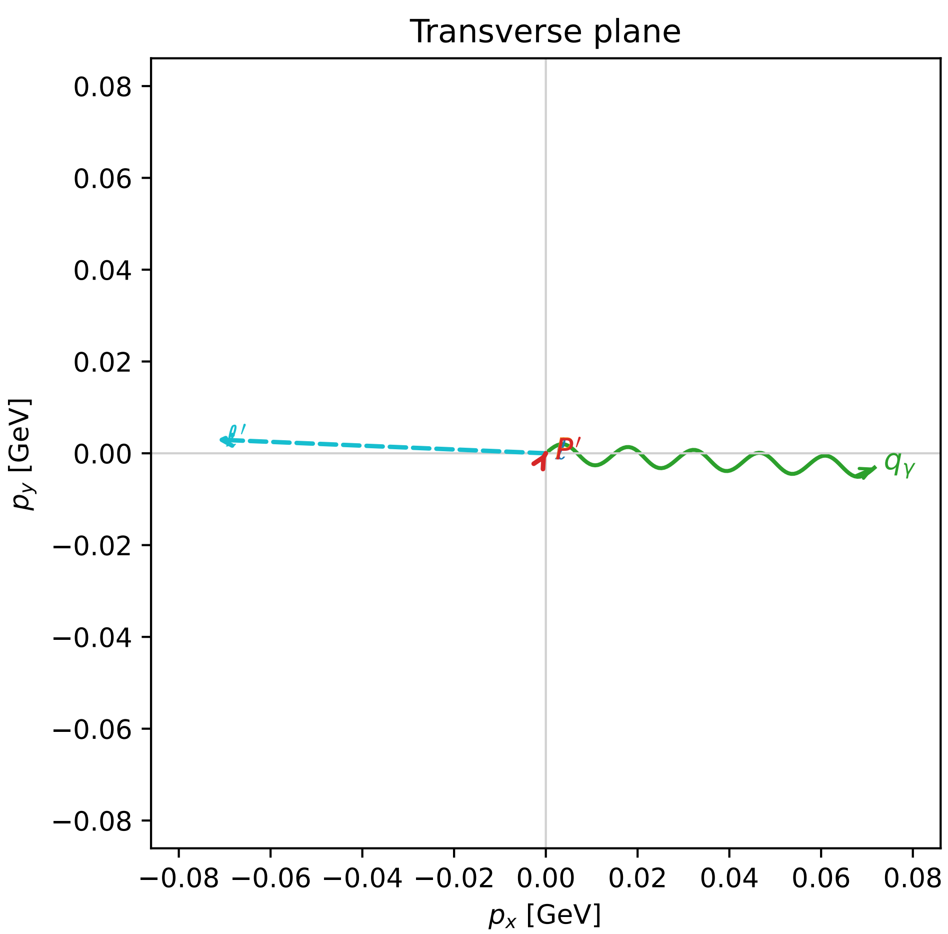
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

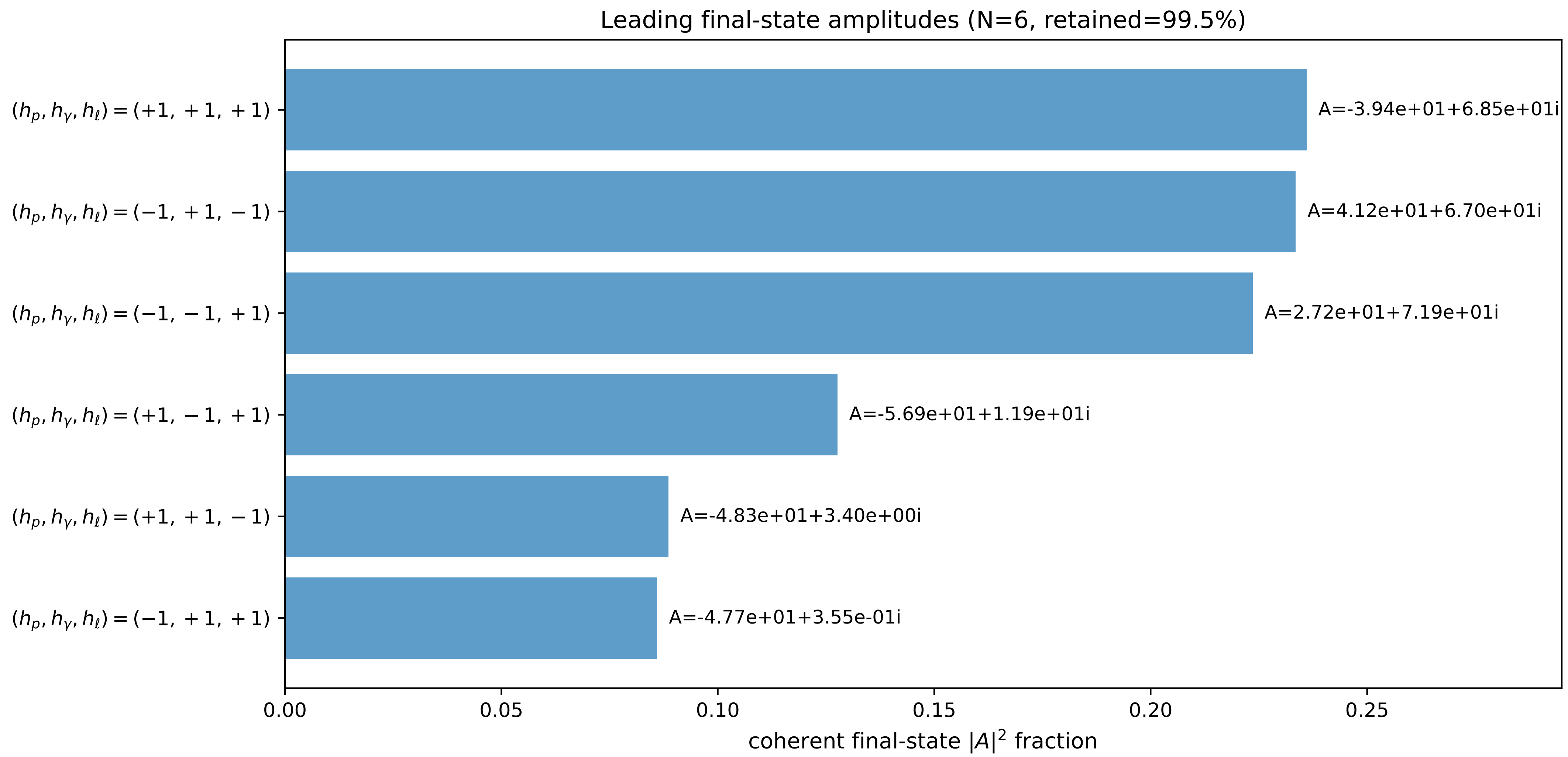



```
dw_mixing_angles_polarization_02_local_minimum_624  D_W=0.0120591
region: polarization_cluster_02_local_minimum_624
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.76902926 rad
incoming lepton: 0.718586|+> +0.695438|->
proton mixing angle: theta_p=3.0112165 rad
incoming proton: -0.991513|+> +0.130007|->
```

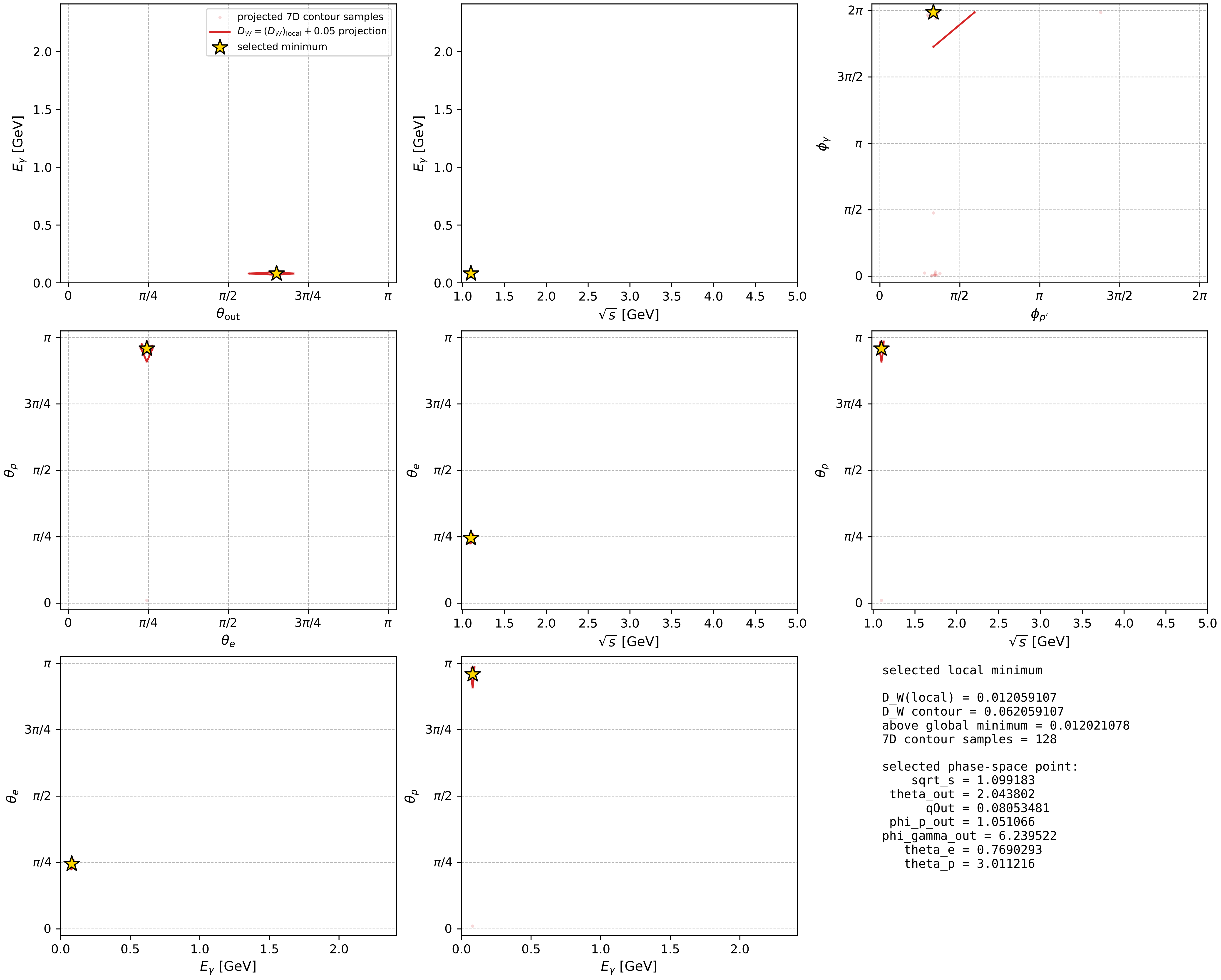
s=1.2082, $\sqrt{s}$=1.09918, $|\vec{P}|$=0.149365, $|\vec{P}'|$=0.000195743
θ_{out}=2.0438, $\phi_{p'}$=1.05107, ϕ_γ=6.23952
E_γ=0.0805348, $\theta_{\ell'}$=1.09726, $\phi_{\ell'}$=3.10009
Q^2=0.0350782, x_B=0.188447, t=-0.0221968, W^2=1.03091, y=0.56689

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1494 p_x= 0.0000 p_y= 0.0000 p_z= -0.1494
P E= 0.9498 p_x= 0.0000 p_y= 0.0000 p_z= 0.1494
ℓ' E= 0.0806 p_x= -0.0717 p_y= 0.0030 p_z= 0.0368
P' E= 0.9380 p_x= 0.0001 p_y= 0.0002 p_z= -0.0001
q_γ E= 0.0805 p_x= 0.0716 p_y= -0.0031 p_z= -0.0367

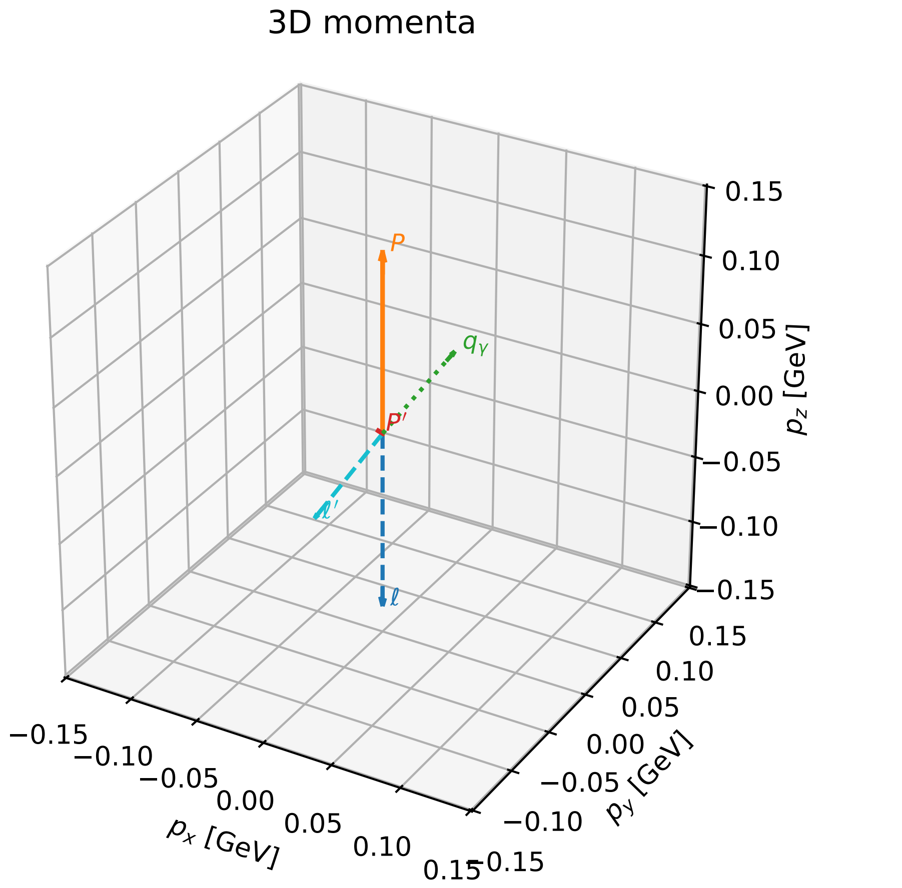




electron: polarization cluster P2, local minimum 624; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



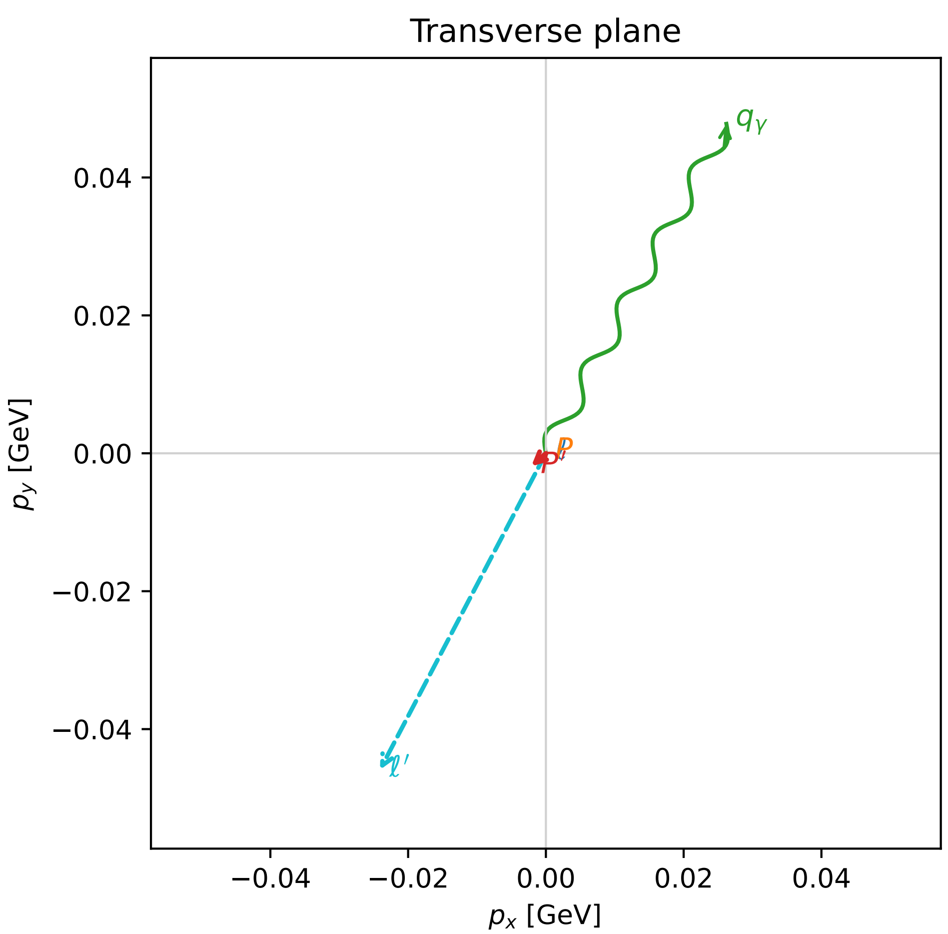
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

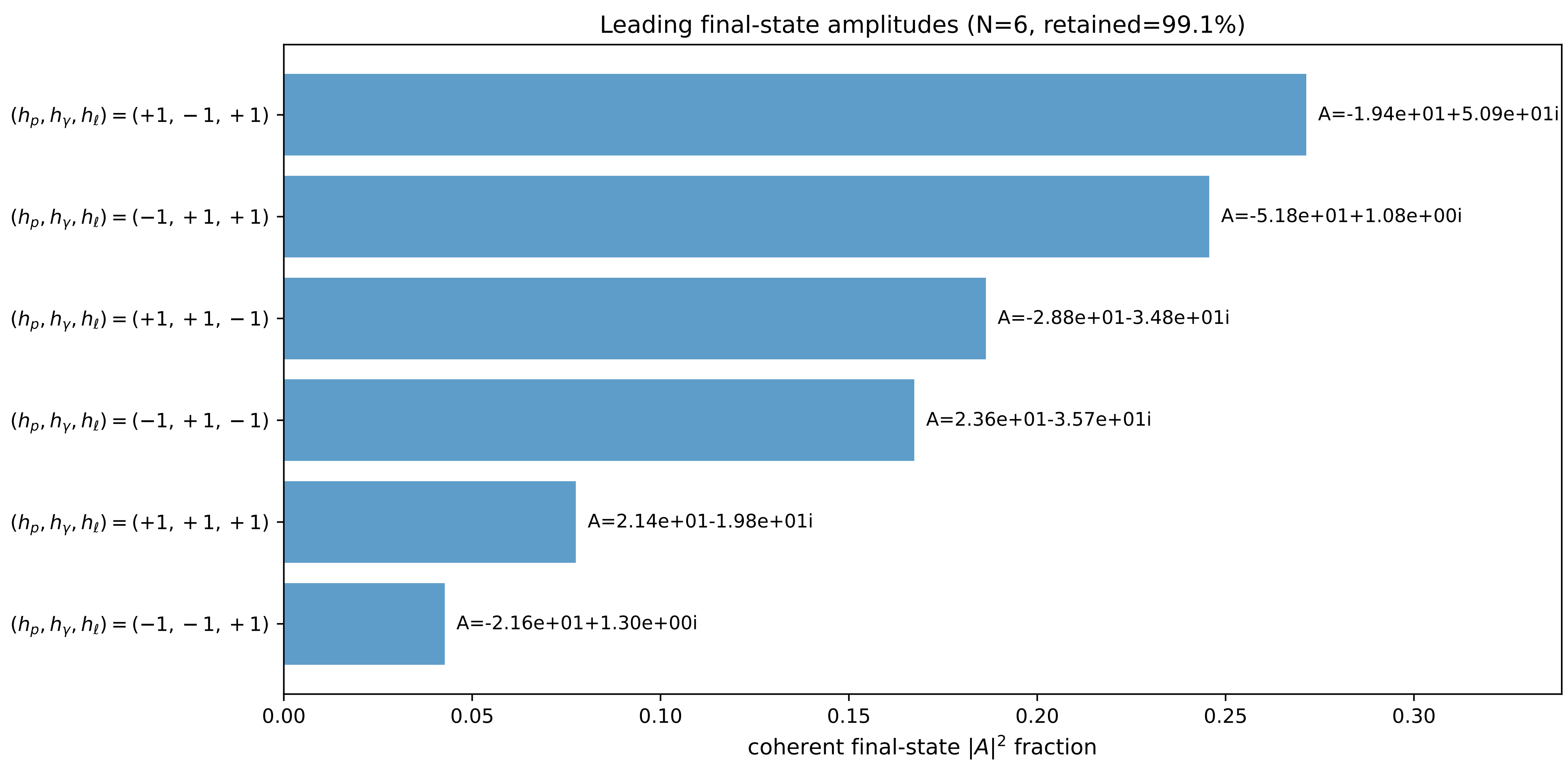



```
dw_mixing_angles_polarization_02_local_minimum_638  D_W=0.0133373
region: polarization_cluster_02_local_minimum_638
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.80475679 rad
incoming lepton: 0.693287|+> +0.720662|->
proton mixing angle: theta_p=2.9458893 rad
incoming proton: -0.980911|+> +0.194457|->
```

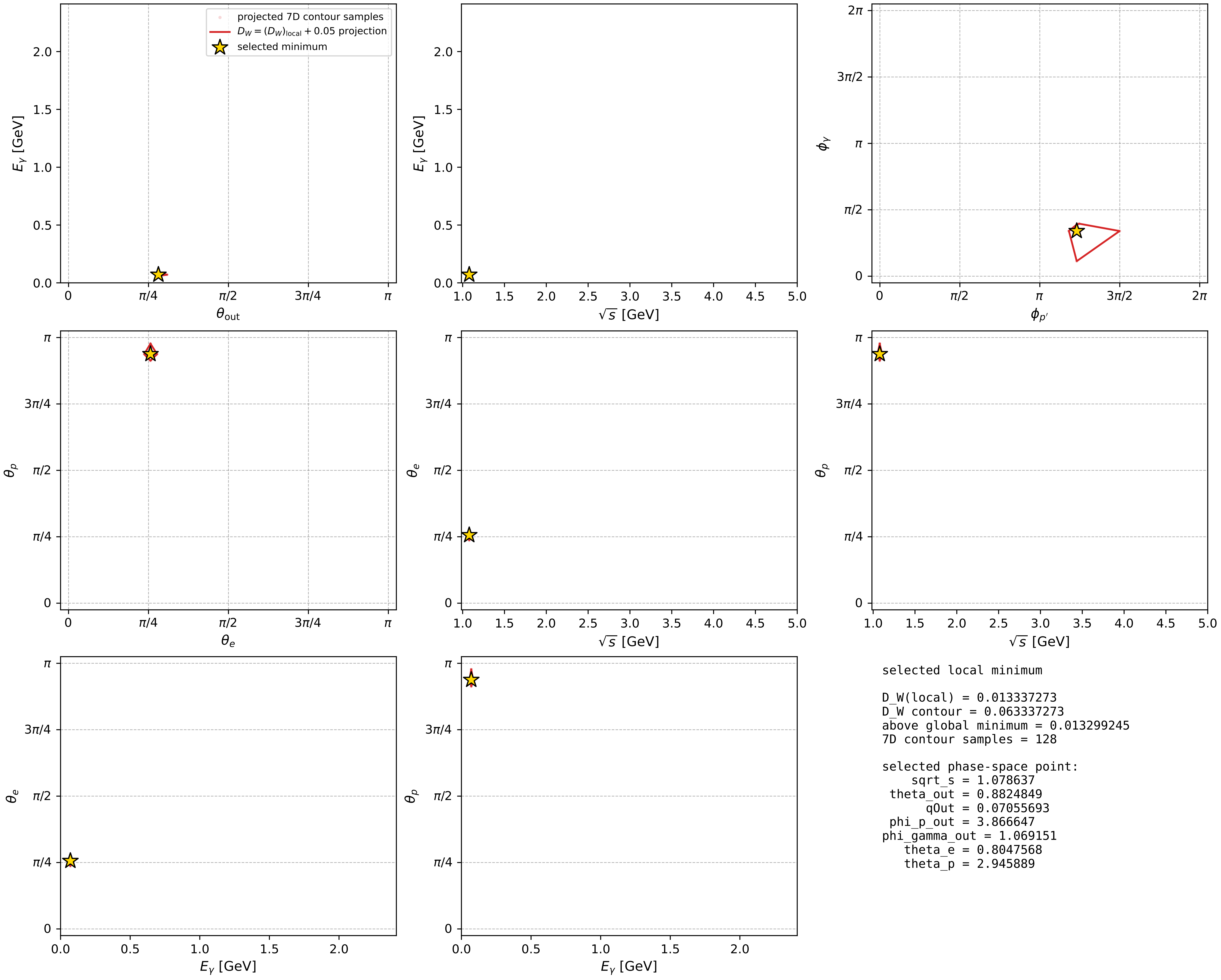
s=1.16346, $\sqrt{s}$=1.07864, $|\vec{P}|$=0.131467, $|\vec{P}'|$=0.00367195
θ_{out}=0.882485, $\phi_{p'}$=3.86665, ϕ_γ=1.06915
E_γ=0.0705569, $\theta_{\ell'}$=2.30896, $\phi_{\ell'}$=4.2292
Q^2=0.00602604, x_B=0.0435176, t=-0.0165999, W^2=1.01229, y=0.488249

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1315 p_x= 0.0000 p_y= 0.0000 p_z= -0.1315
P E= 0.9472 p_x= 0.0000 p_y= 0.0000 p_z= 0.1315
ℓ' E= 0.0701 p_x= -0.0241 p_y= -0.0459 p_z= -0.0472
P' E= 0.9380 p_x= -0.0021 p_y= -0.0019 p_z= 0.0023
q_γ E= 0.0706 p_x= 0.0262 p_y= 0.0478 p_z= 0.0448

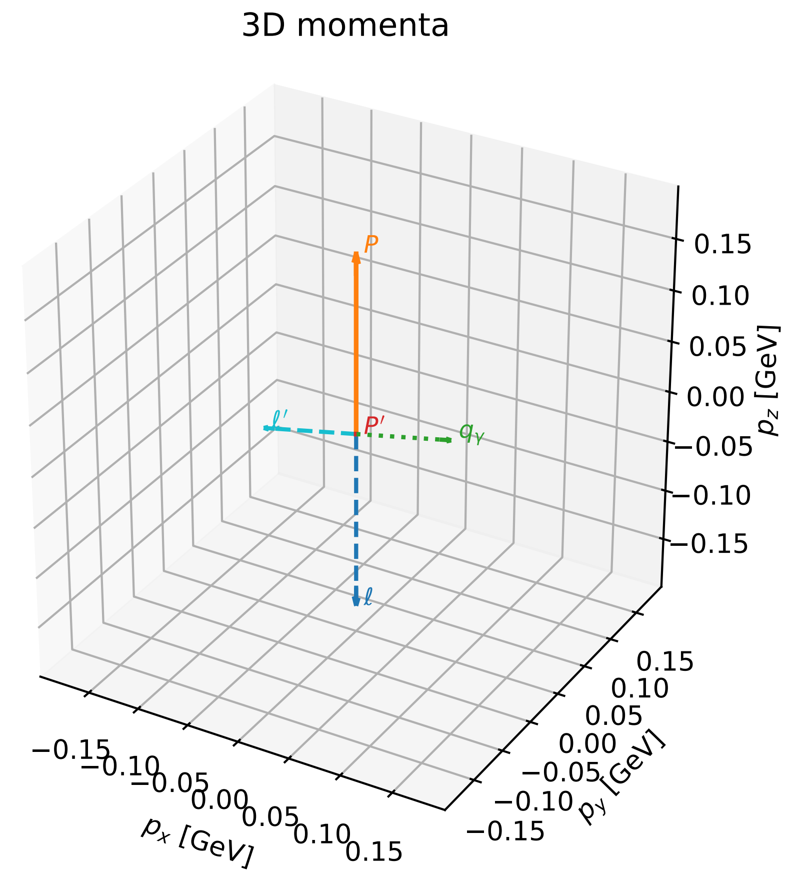




electron: polarization cluster P2, local minimum 638; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



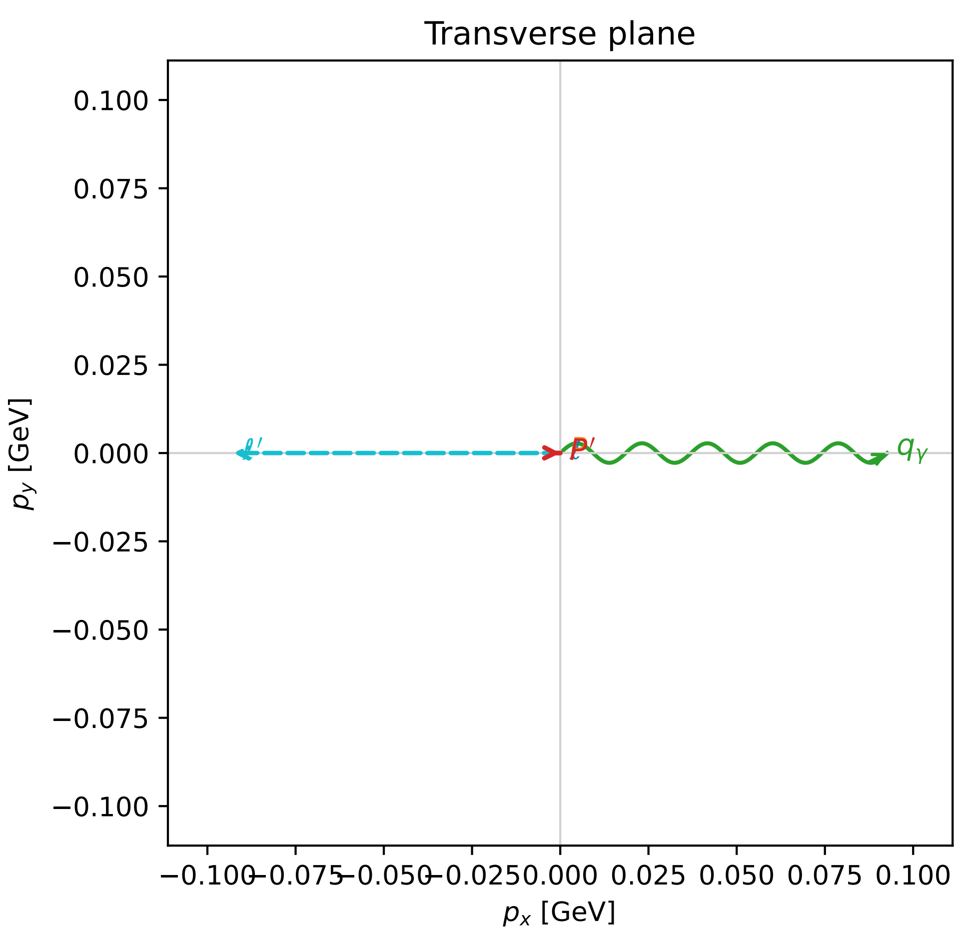
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

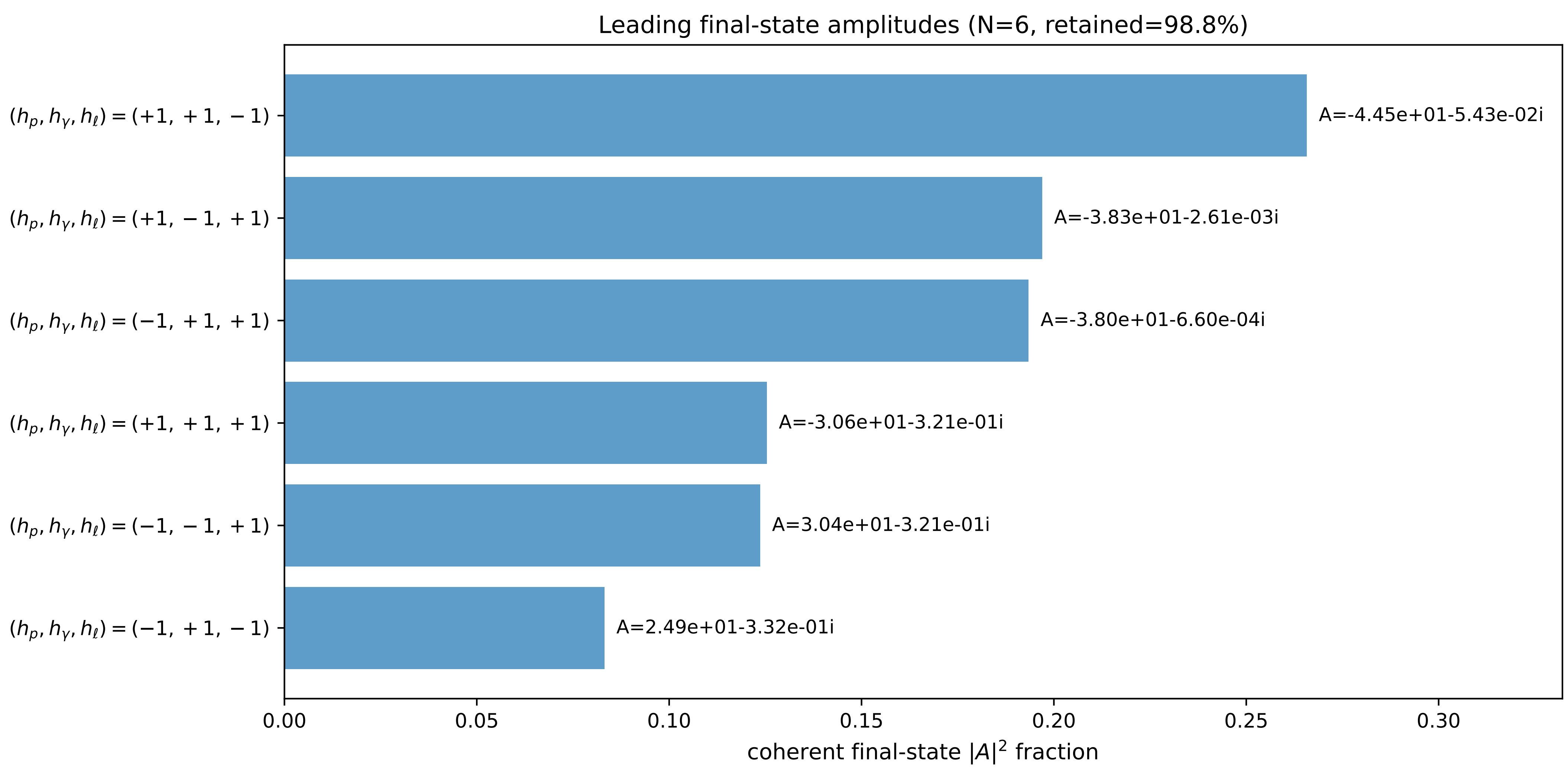


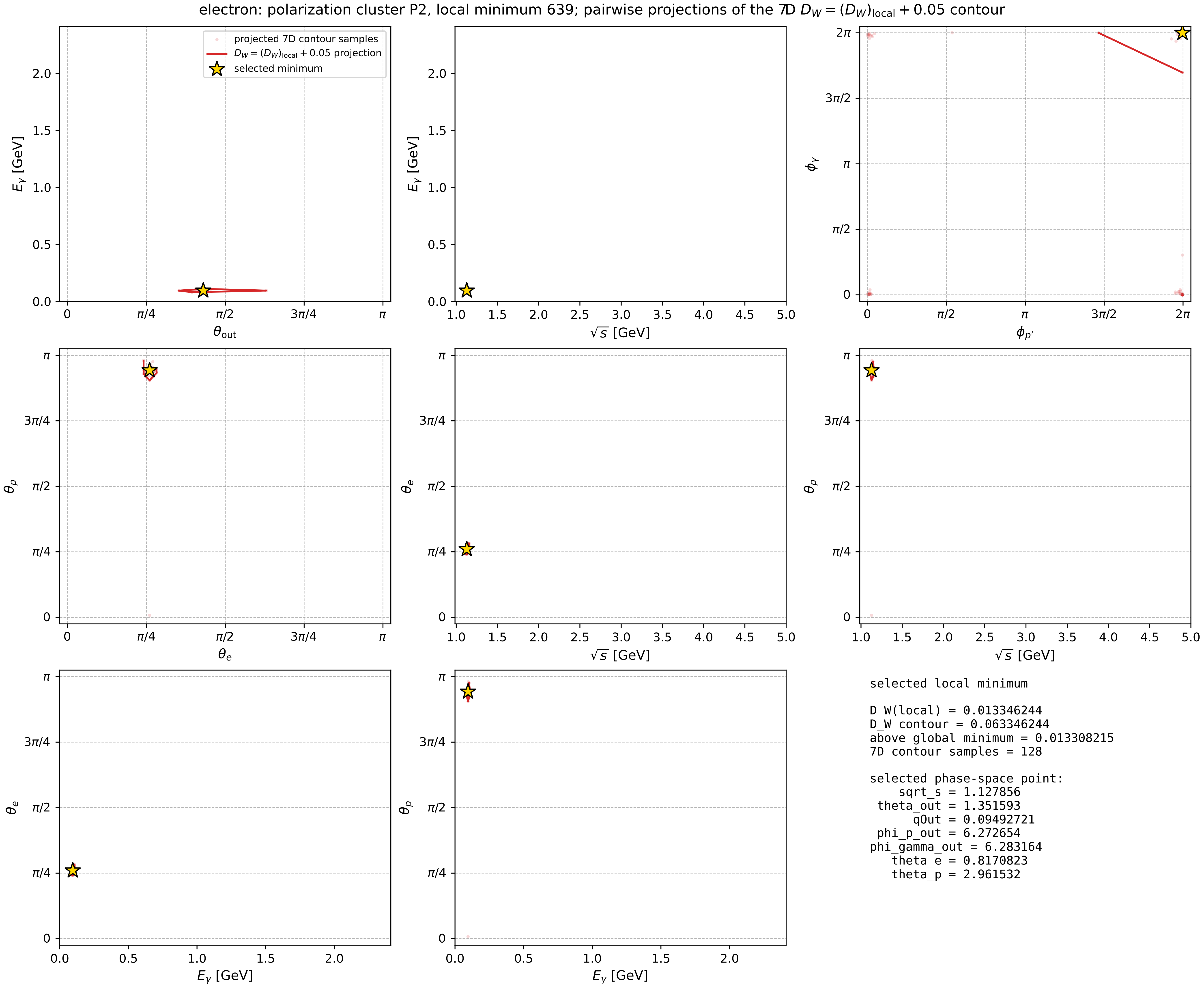
dw_mixing_angles_polarization_02_local_minimum_639 D_W=0.0133462
region: polarization_cluster_02_local_minimum_639
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.81708234 rad
incoming lepton: 0.684352|+> +0.729152|->
proton mixing angle: theta_p=2.9615322 rad
incoming proton: -0.983833|+> +0.179089|->

s=1.27206, $\sqrt{s}$=1.12786, $|\vec{P}|$=0.173876, $|\vec{P}'|$=2.41068e-07
θ_{out}=1.35159, $\phi_{p'}$=6.27265, ϕ_γ=6.28316
E_γ=0.0949272, $\theta_{\ell'}$=1.79, $\phi_{\ell'}$=3.14157
Q^2=0.0258329, x_B=0.126684, t=-0.0299774, W^2=1.05793, y=0.51991

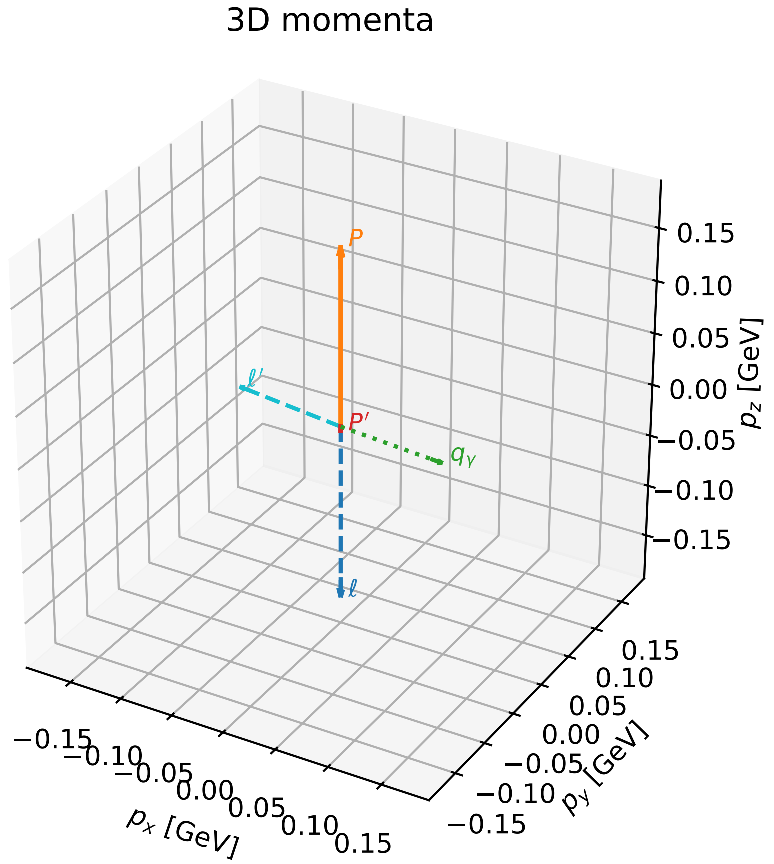
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1739 p_x= 0.0000 p_y= 0.0000 p_z= -0.1739
P E= 0.9540 p_x= 0.0000 p_y= 0.0000 p_z= 0.1739
ℓ' E= 0.0949 p_x= -0.0927 p_y= 0.0000 p_z= -0.0206
P' E= 0.9380 p_x= 0.0000 p_y= -0.0000 p_z= 0.0000
q_γ E= 0.0949 p_x= 0.0927 p_y= -0.0000 p_z= 0.0206






electron: polarization cluster P2, local minimum 639; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.013346244
D_W contour = 0.063346244
above global minimum = 0.013308215
7D contour samples = 128
selected phase-space point:
sqrt_s = 1.127856
theta_out = 1.351593
qOut = 0.09492721
phi_p_out = 6.272654
phi_gamma_out = 6.283164
theta_e = 0.8170823
theta_p = 2.961532

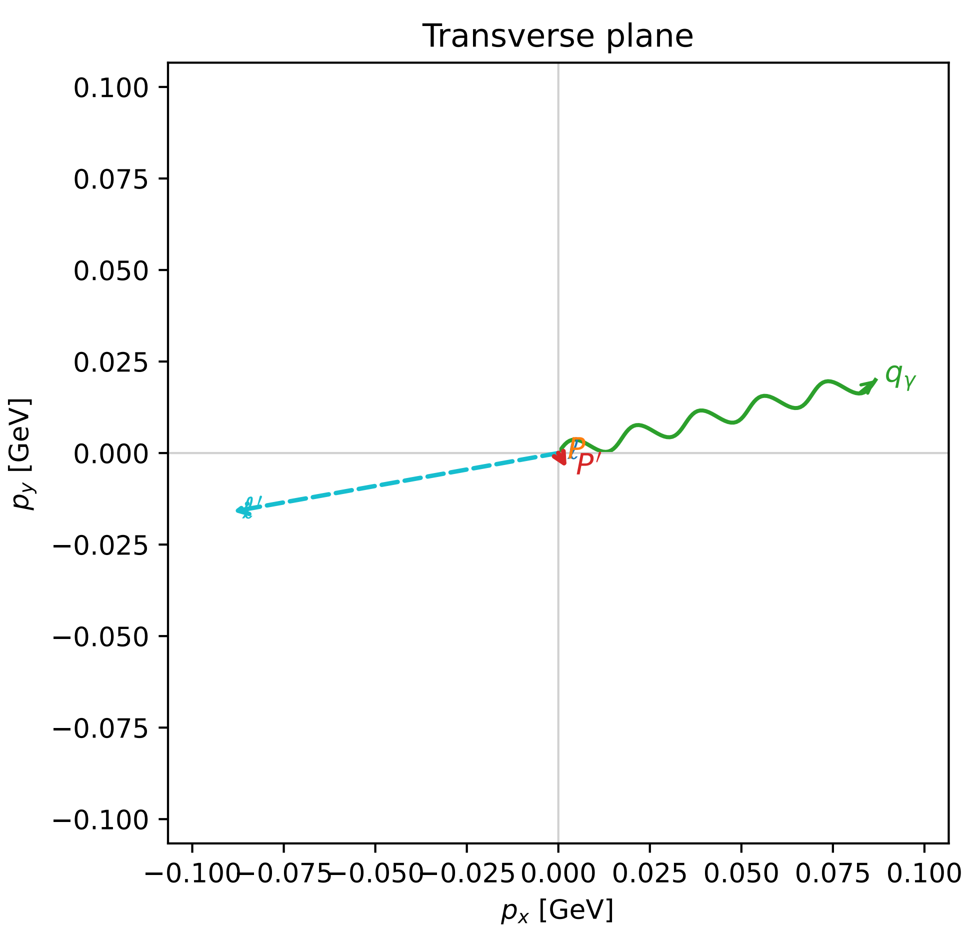
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

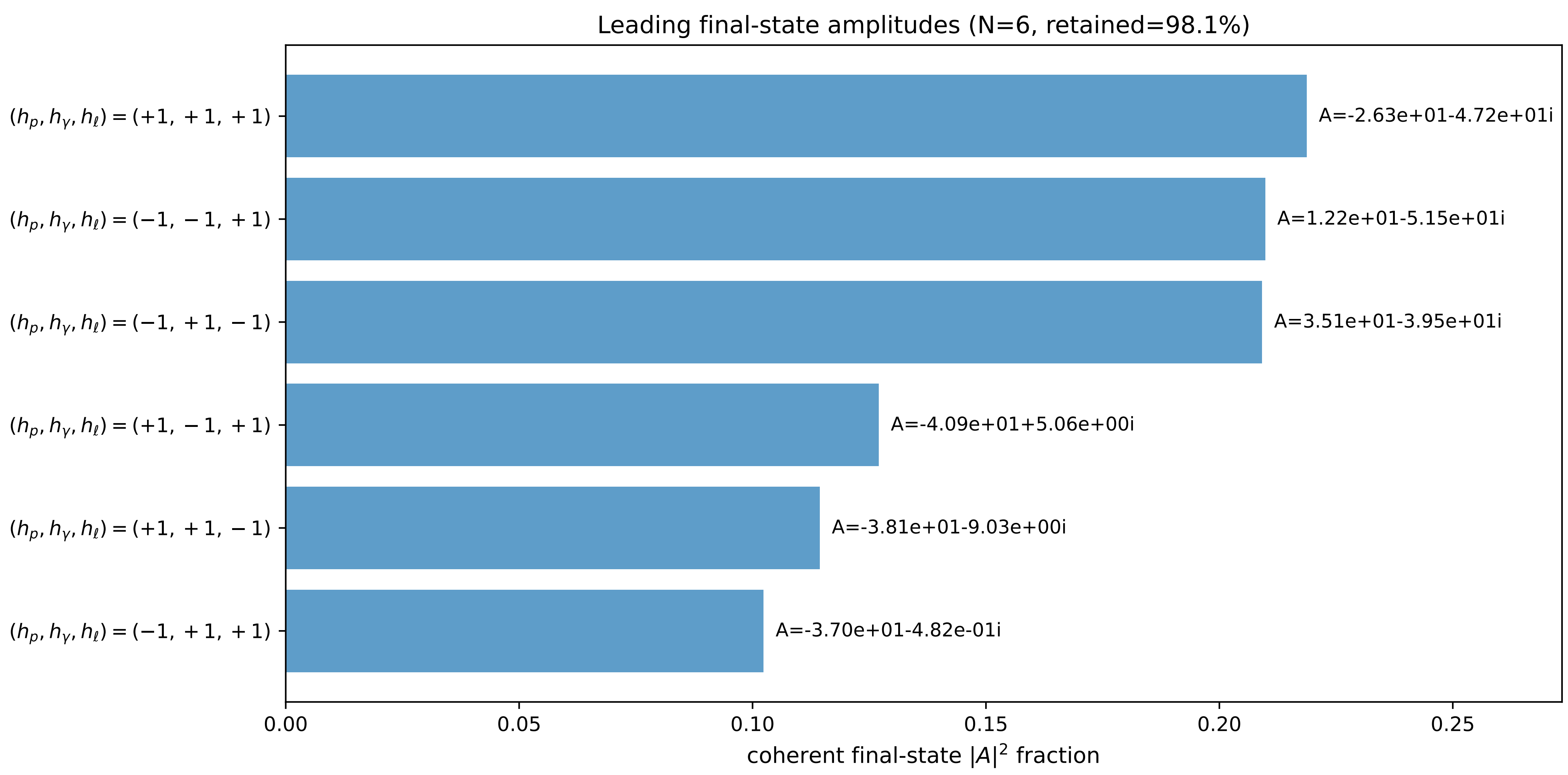


dw_mixing_angles_polarization_02_local_minimum_640 D_W=0.0134134
region: polarization_cluster_02_local_minimum_640
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.77364916 rad
incoming lepton: 0.715366|+> +0.69875|->
proton mixing angle: theta_p=2.9219019 rad
incoming proton: -0.975965|+> +0.217928|->

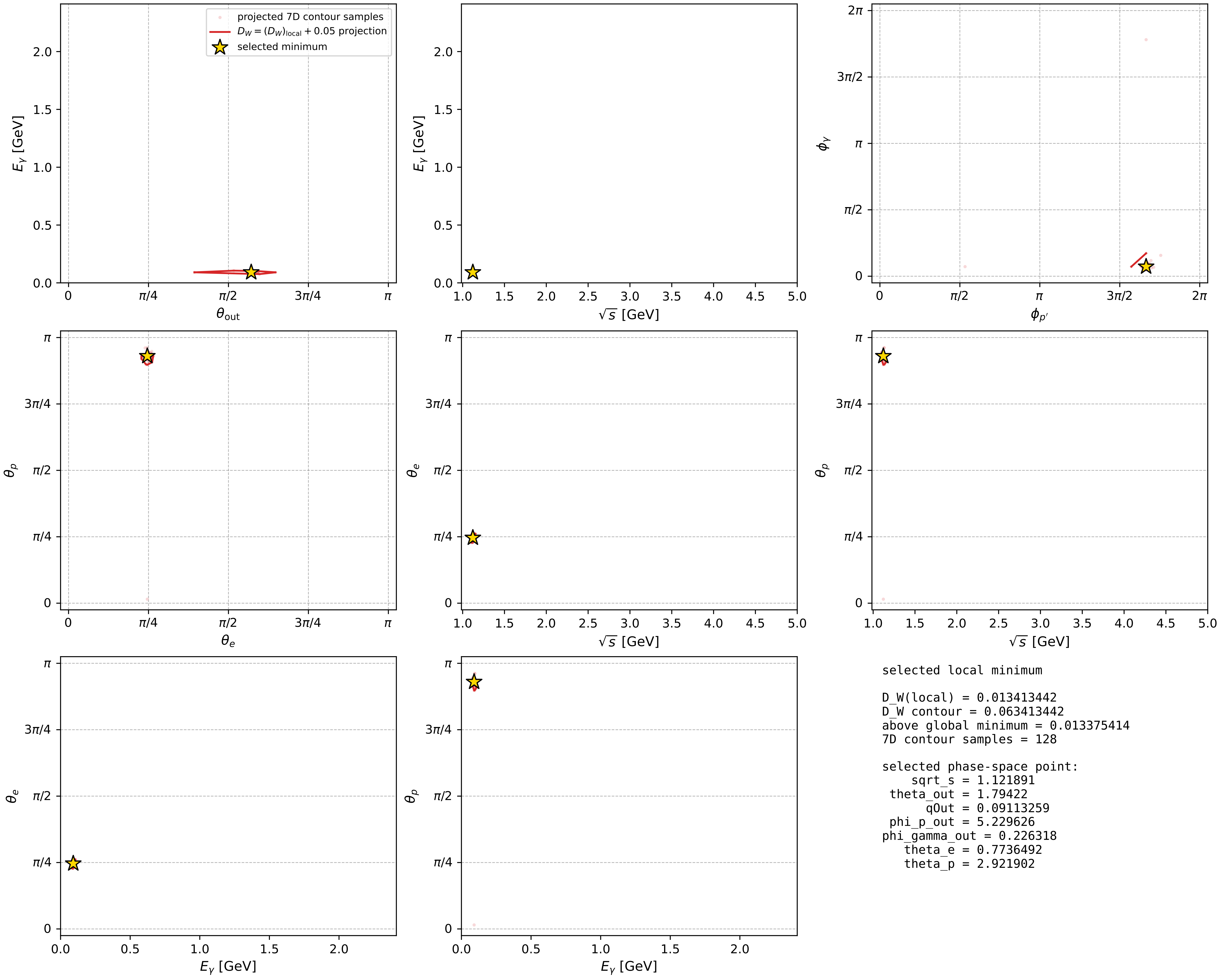
s=1.25864, $\sqrt{s}$=1.12189, $|\vec{P}|$=0.168819, $|\vec{P}'|$=0.00468137
θ_{out}=1.79422, $\phi_{p'}$=5.22963, ϕ_γ=0.226318
E_γ=0.0911326, $\theta_{\ell'}$=1.33985, $\phi_{\ell'}$=3.31945
Q^2=0.0384824, x_B=0.183972, t=-0.0286453, W^2=1.05054, y=0.552211

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1688 p_x= 0.0000 p_y= 0.0000 p_z= -0.1688
P E= 0.9531 p_x= 0.0000 p_y= 0.0000 p_z= 0.1688
ℓ' E= 0.0927 p_x= -0.0889 p_y= -0.0160 p_z= 0.0212
P' E= 0.9380 p_x= 0.0023 p_y= -0.0040 p_z= -0.0010
q_γ E= 0.0911 p_x= 0.0866 p_y= 0.0199 p_z= -0.0202



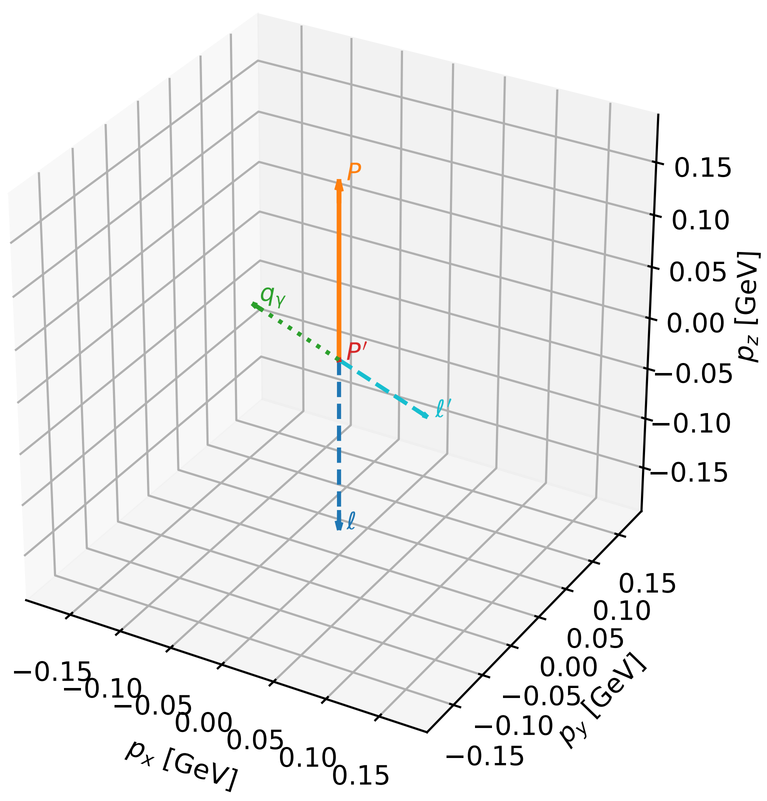


electron: polarization cluster P2, local minimum 640; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

3D momenta




```
dw_mixing_angles_polarization_02_local_minimum_645  D_W=0.0140218
region: polarization_cluster_02_local_minimum_645
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.81651273 rad
incoming lepton: 0.684767|+> +0.728762|->
proton mixing angle: theta_p=0.17488097 rad
incoming proton: 0.984747|+> +0.173991|->
```

s=1.25979, $\sqrt{s}$=1.1224, $|\vec{P}|$=0.169253, $|\vec{P}'|$=0.000521138

$\theta_{\rm out}$=1.26763, $\phi_{p'}$=4.50397, ϕ_γ=3.08667

E_γ=0.0921398, $\theta_{\ell'}$=1.87532, $\phi_{\ell'}$=6.23385

Q^2=0.0218665, x_B=0.112309, t=-0.0283648, W^2=1.05268, y=0.512445

four-momenta [E, p_x, p_y, p_z] GeV:

ℓ E= 0.1693 p_x= 0.0000 p_y= 0.0000 p_z= -0.1693

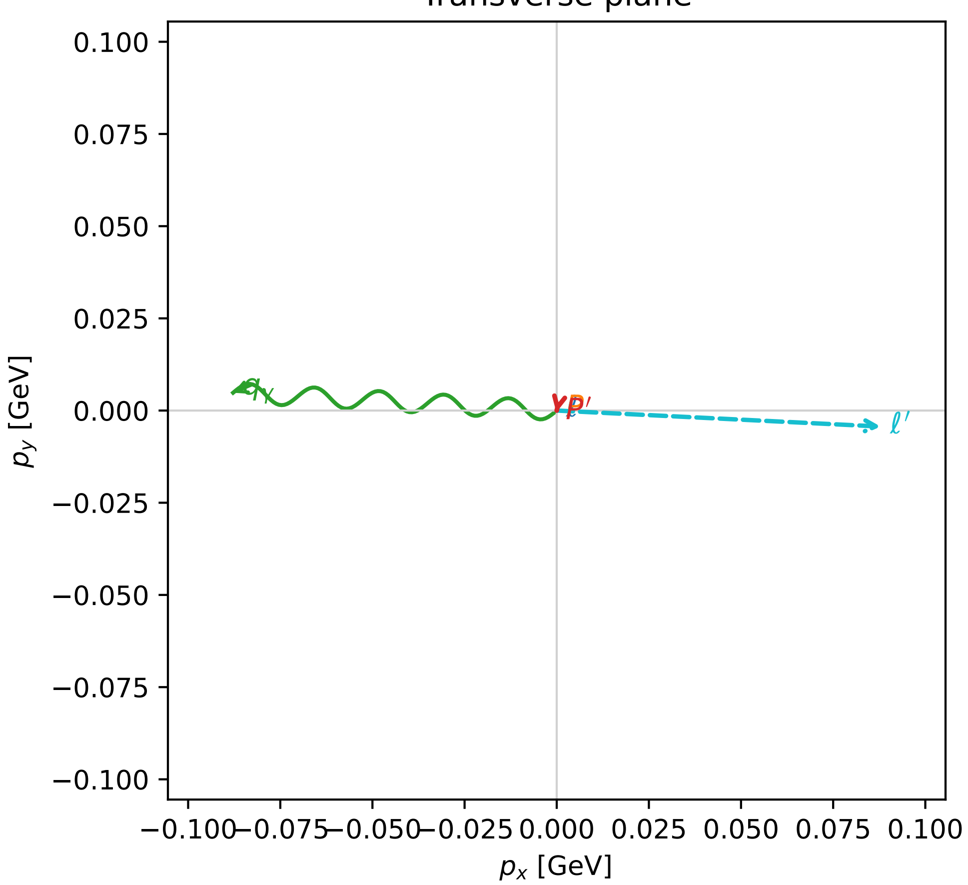
P E= 0.9531 p_x= 0.0000 p_y= 0.0000 p_z= 0.1693

ℓ' E= 0.0923 p_x= 0.0879 p_y= -0.0043 p_z= -0.0277

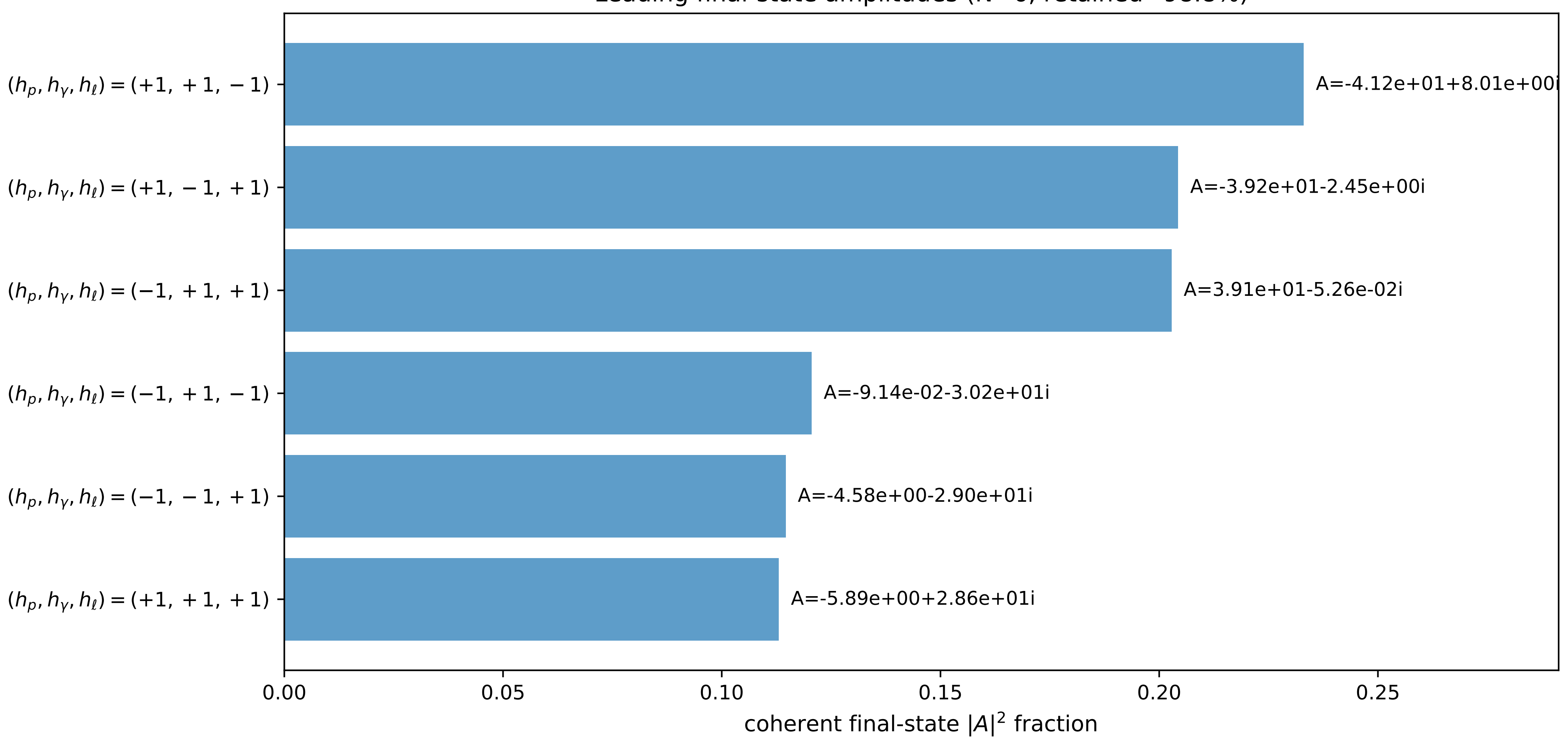
P' E= 0.9380 p_x= -0.0001 p_y= -0.0005 p_z= 0.0002

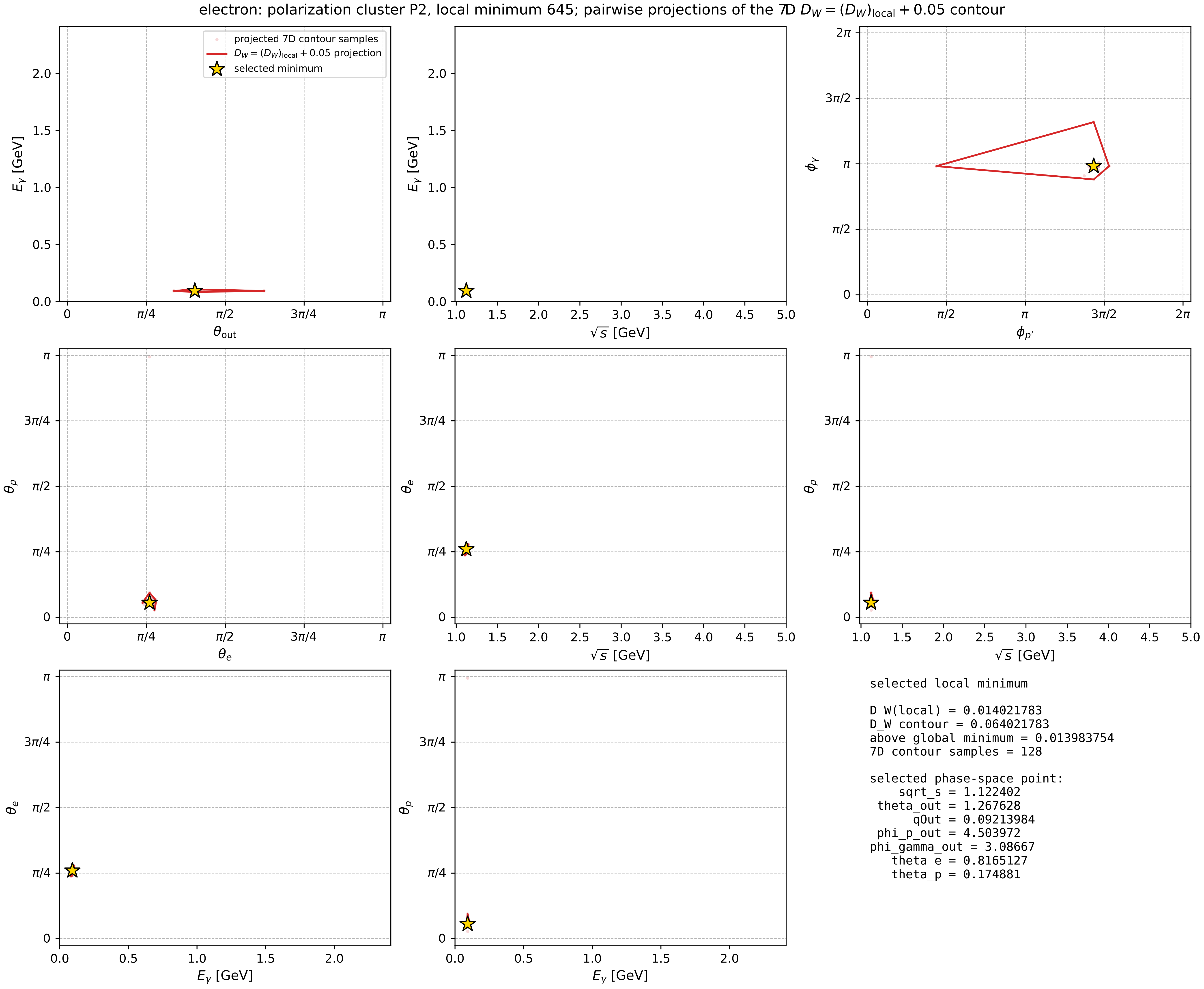
q_γ E= 0.0921 p_x= -0.0878 p_y= 0.0048 p_z= 0.0275

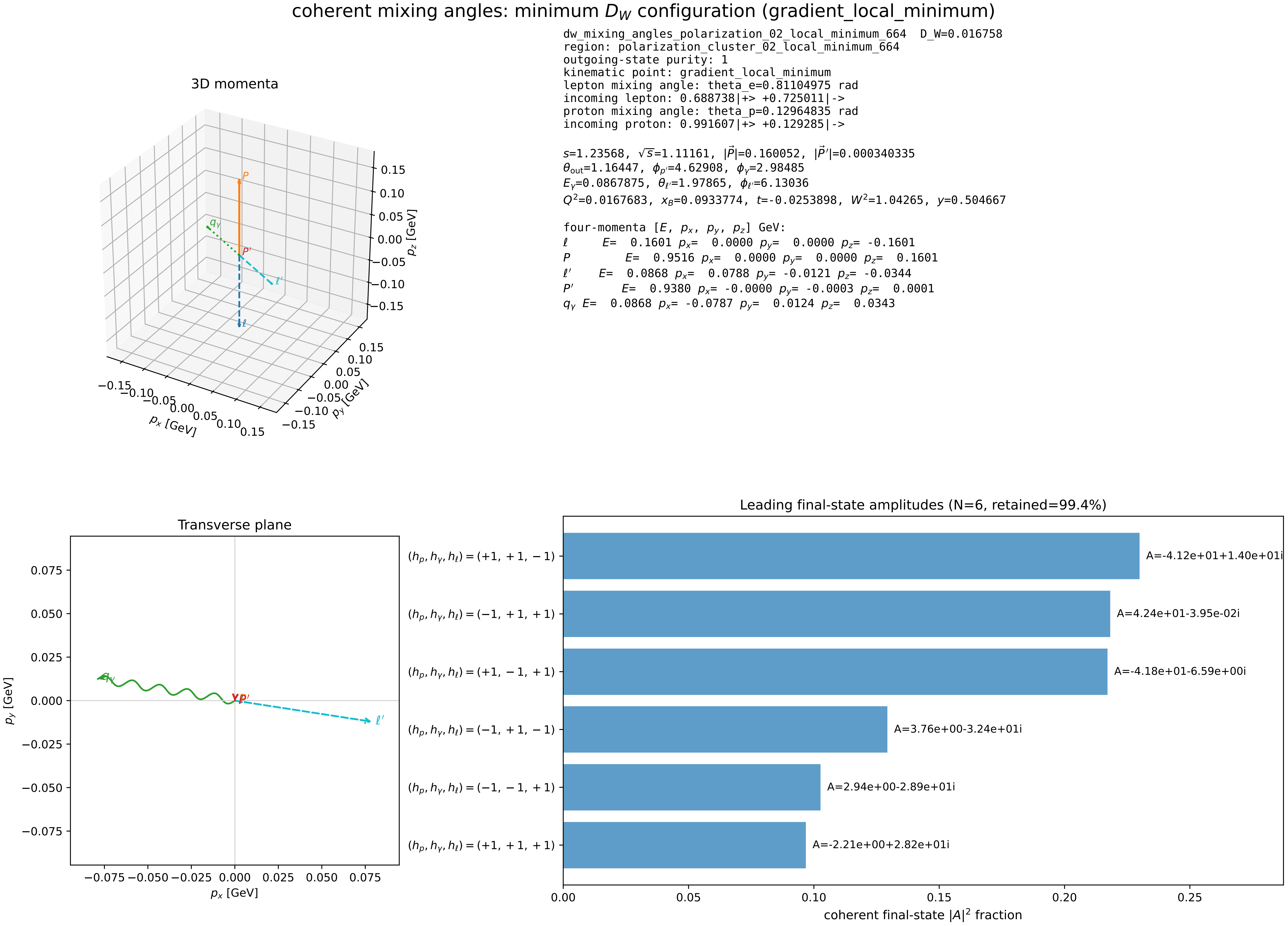
Transverse plane



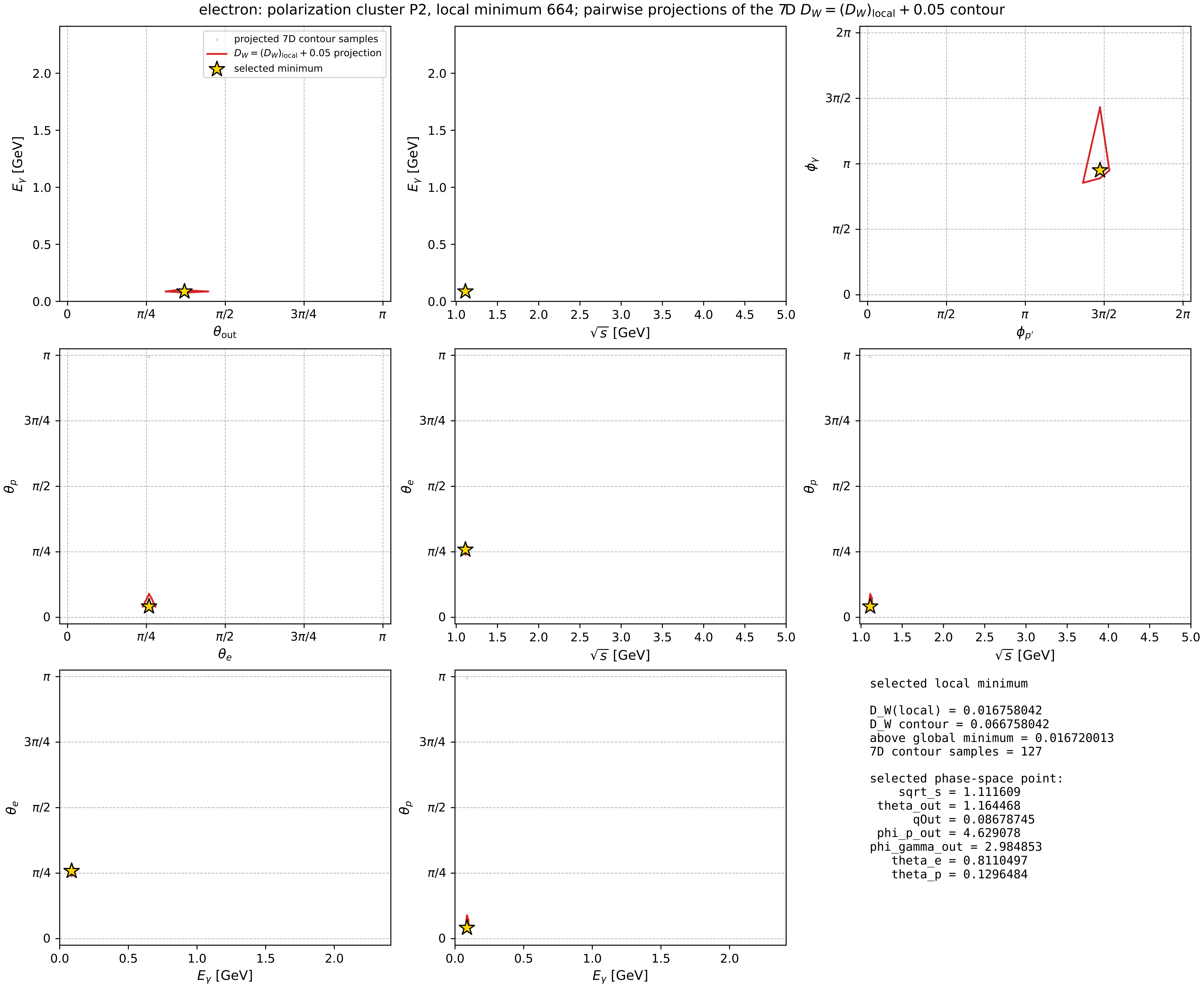
Leading final-state amplitudes (N=6, retained=98.8%)



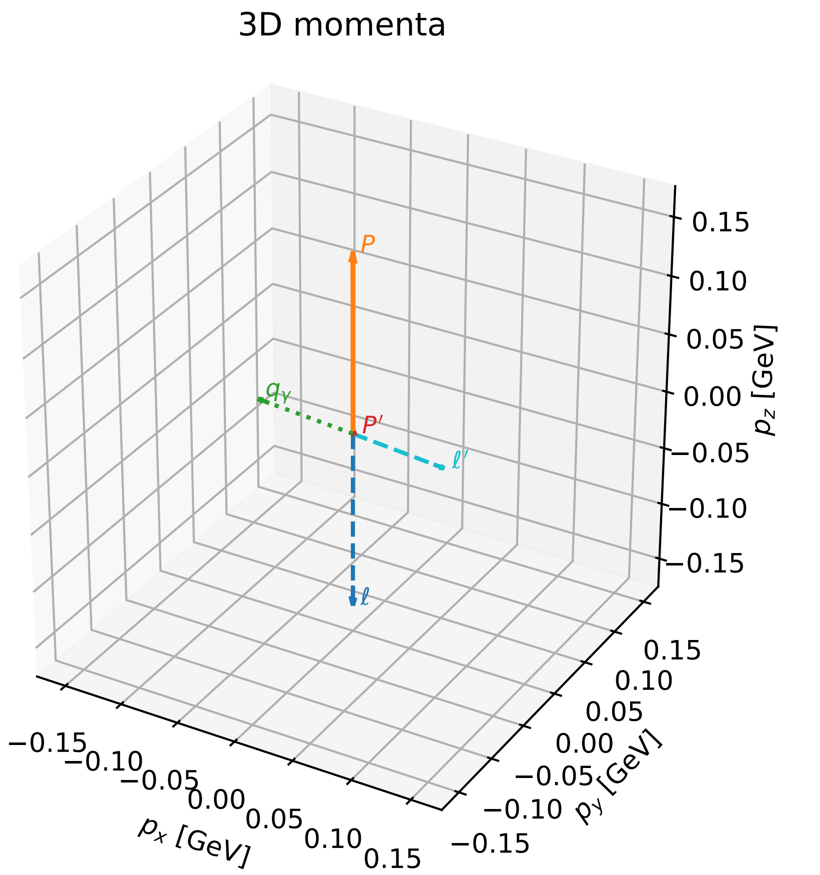

electron: polarization cluster P2, local minimum 645; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.014021783
D_W contour = 0.064021783
above global minimum = 0.013983754
7D contour samples = 128
selected phase-space point:
sqrt_s = 1.122402
theta_out = 1.267628
qOut = 0.09213984
phi_p_out = 4.503972
phi_gamma_out = 3.08667
theta_e = 0.8165127
theta_p = 0.174881



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_664 D_W=0.016758
region: polarization_cluster_02_local_minimum_664
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.81104975 rad
incoming lepton: 0.688738|+> +0.725011|->
proton mixing angle: theta_p=0.12964835 rad
incoming proton: 0.991607|+> +0.129285|->
s=1.23568, $\sqrt{s}$=1.11161, $|\vec{P}|$=0.160052, $|\vec{P}'|$=0.000340335
θ_{out}=1.16447, $\phi_{p'}$=4.62908, ϕ_γ=2.98485
E_γ=0.0867875, $\theta_{\ell'}$=1.97865, $\phi_{\ell'}$=6.13036
Q^2=0.0167683, x_B=0.0933774, t=-0.0253898, W^2=1.04265, y=0.504667
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1601 p_x= 0.0000 p_y= 0.0000 p_z= -0.1601
P E= 0.9516 p_x= 0.0000 p_y= 0.0000 p_z= 0.1601
ℓ' E= 0.0868 p_x= 0.0788 p_y= -0.0121 p_z= -0.0344
P' E= 0.9380 p_x= -0.0000 p_y= -0.0003 p_z= 0.0001
q_γ E= 0.0868 p_x= -0.0787 p_y= 0.0124 p_z= 0.0343
3D momenta
P
q_γ
P'
ℓ'
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
q_γ
P'
ℓ'
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=6, retained=99.4%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-4.12e+01+1.40e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=4.24e+01-3.95e-02i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=-4.18e+01-6.59e+00i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=3.76e+00-3.24e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=2.94e+00-2.89e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=-2.21e+00+2.82e+01i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P2, local minimum 664; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.016758042
D_W contour = 0.066758042
above global minimum = 0.016720013
7D contour samples = 127
selected phase-space point:
sqrt_s = 1.111609
theta_out = 1.164468
qOut = 0.08678745
phi_p_out = 4.629078
phi_gamma_out = 2.984853
theta_e = 0.8110497
theta_p = 0.1296484

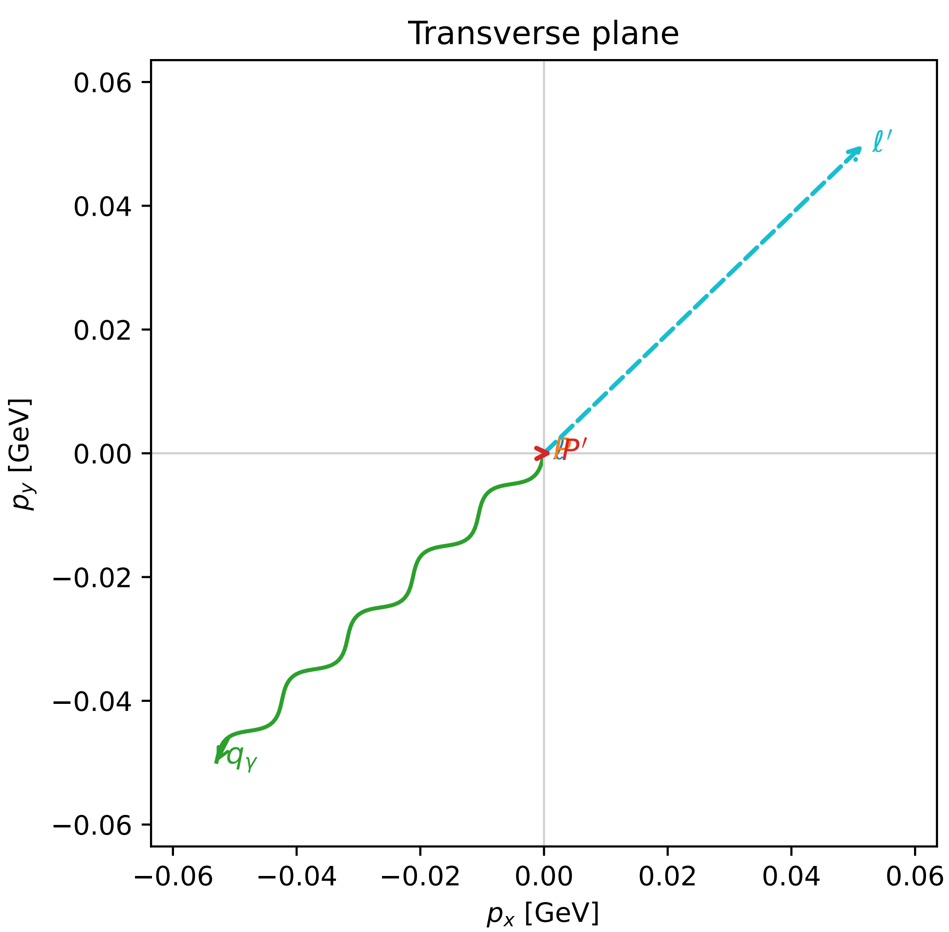
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

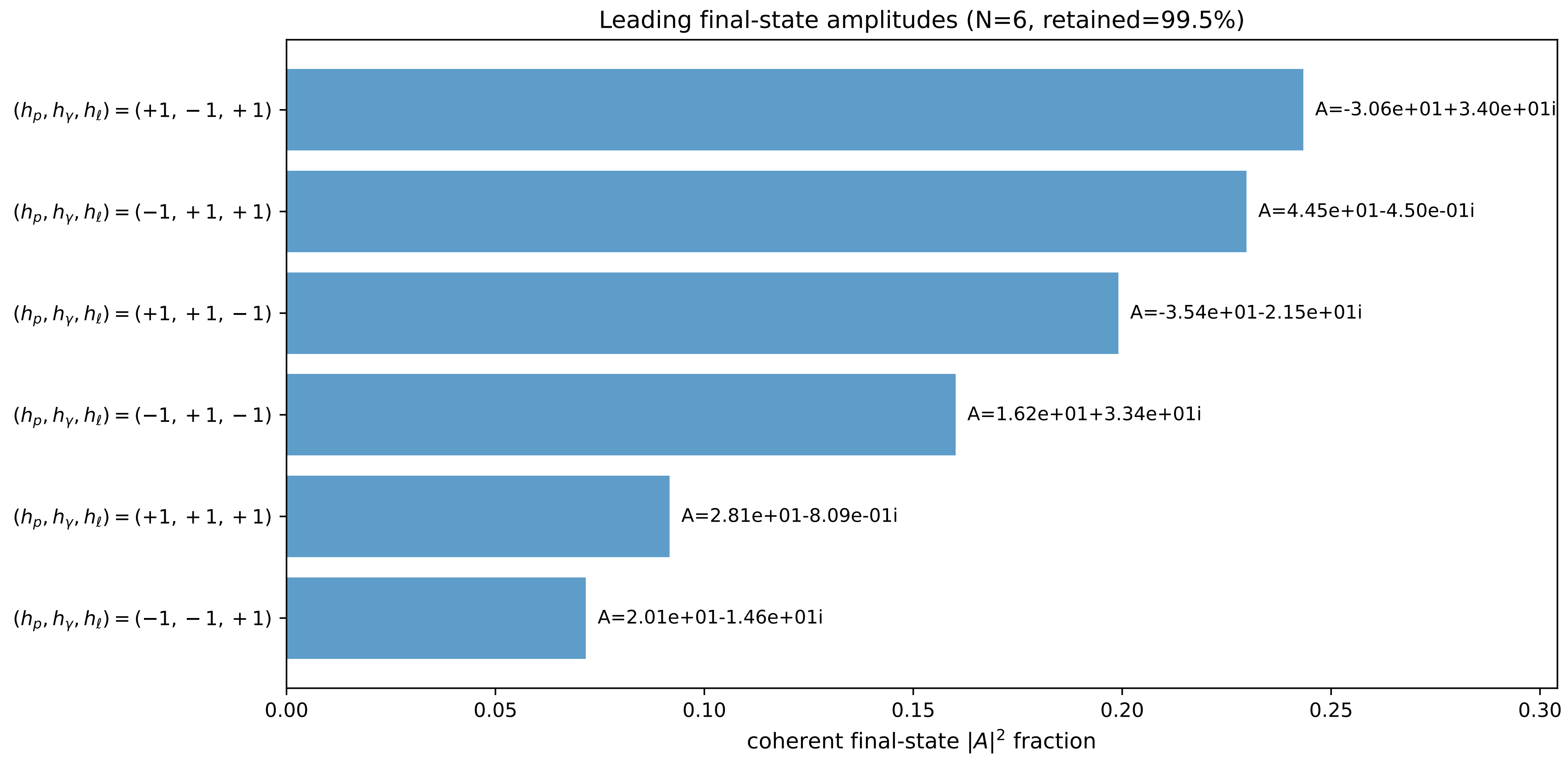


dw_mixing_angles_polarization_02_local_minimum_670 D_W=0.017341
region: polarization_cluster_02_local_minimum_670
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.80616447 rad
incoming lepton: 0.692271|+> +0.721637|->
proton mixing angle: theta_p=0.1287758 rad
incoming proton: 0.99172|+> +0.12842|->

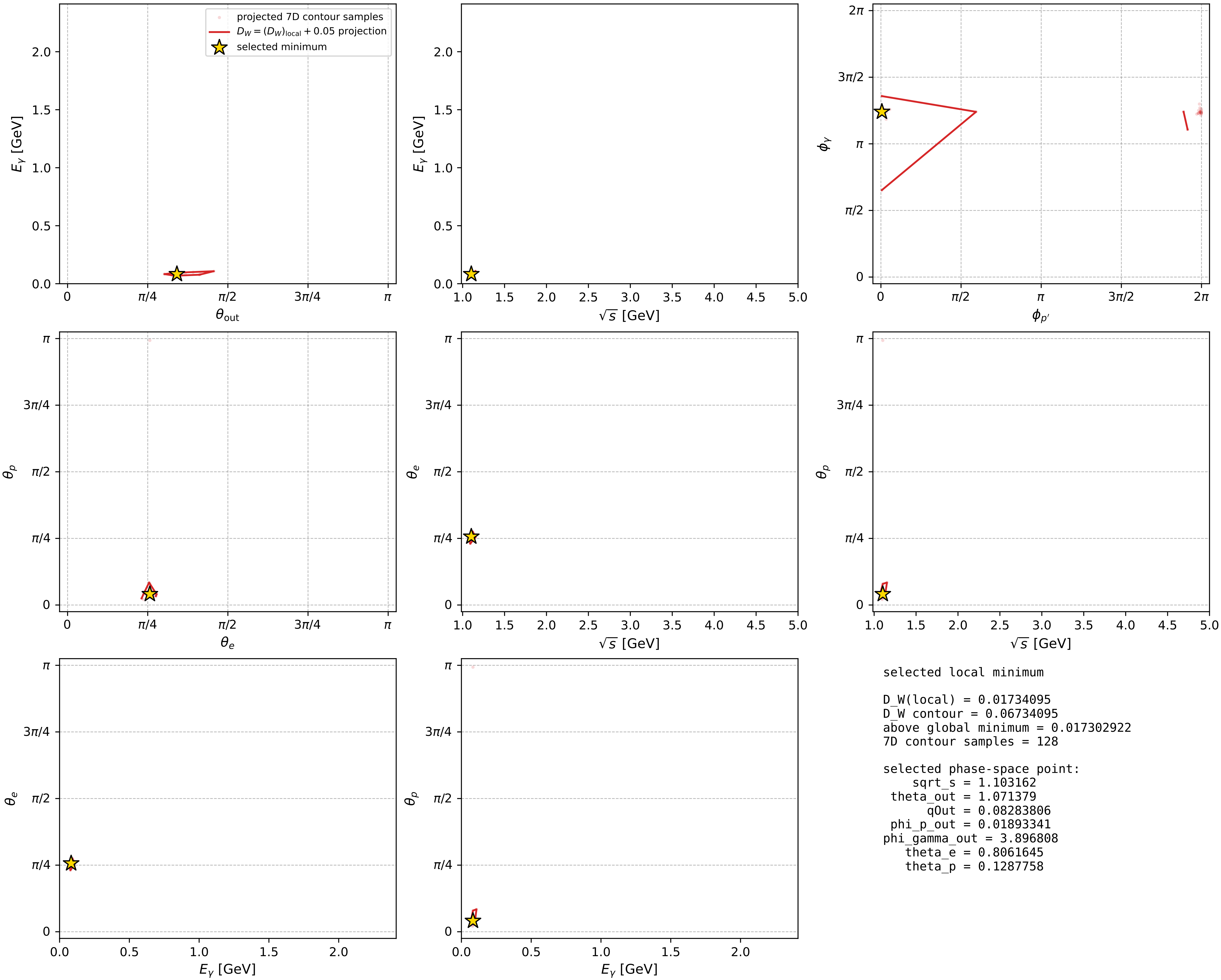
s=1.21697, $\sqrt{s}$=1.10316, $|\vec{P}|$=0.152797, $|\vec{P}'|$=0.00155207
θ_{out}=1.07138, $\phi_{p'}$=0.0189334, ϕ_γ=3.89681
E_γ=0.0828381, $\theta_{\ell'}$=2.08398, $\phi_{\ell'}$=0.767974
Q^2=0.0128061, x_B=0.0760914, t=-0.0229694, W^2=1.03534, y=0.499222

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1528 p_x= 0.0000 p_y= 0.0000 p_z= -0.1528
P E= 0.9504 p_x= 0.0000 p_y= 0.0000 p_z= 0.1528
ℓ' E= 0.0823 p_x= 0.0516 p_y= 0.0498 p_z= -0.0404
P' E= 0.9380 p_x= 0.0014 p_y= 0.0000 p_z= 0.0007
q_γ E= 0.0828 p_x= -0.0529 p_y= -0.0498 p_z= 0.0397

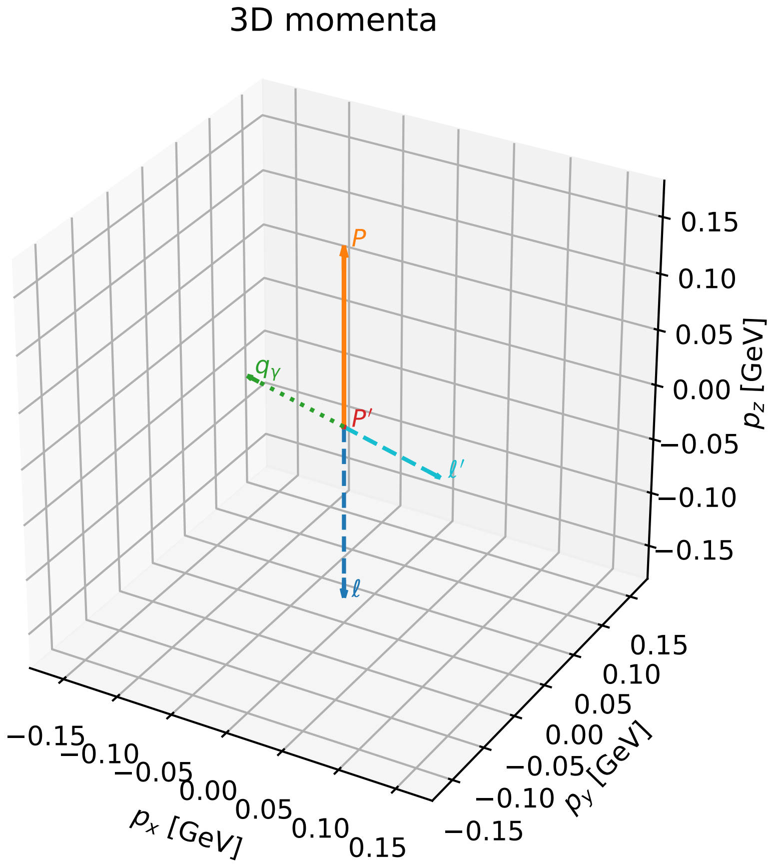




electron: polarization cluster P2, local minimum 670; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)




```
dw_mixing_angles_polarization_02_local_minimum_682  D_W=0.018174
region: polarization_cluster_02_local_minimum_682
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.80835444 rad
incoming lepton: 0.690689|+> +0.723152|->
proton mixing angle: theta_p=0.11341603 rad
incoming proton: 0.993575|+> +0.113173|->
```

s=1.22998, $\sqrt{s}$=1.10905, $|\vec{P}|$=0.157854, $|\vec{P}'|$=2.21459e-06

$\theta_{\rm out}$=1.13361, $\phi_{p'}$=4.08862, ϕ_γ=3.44304

E_γ=0.0855209, $\theta_{\ell'}$=2.00799, $\phi_{\ell'}$=0.301464

Q^2=0.0155686, x_B=0.088455, t=-0.0247438, W^2=1.04028, y=0.502676

four-momenta [E, p_x, p_y, p_z] GeV:

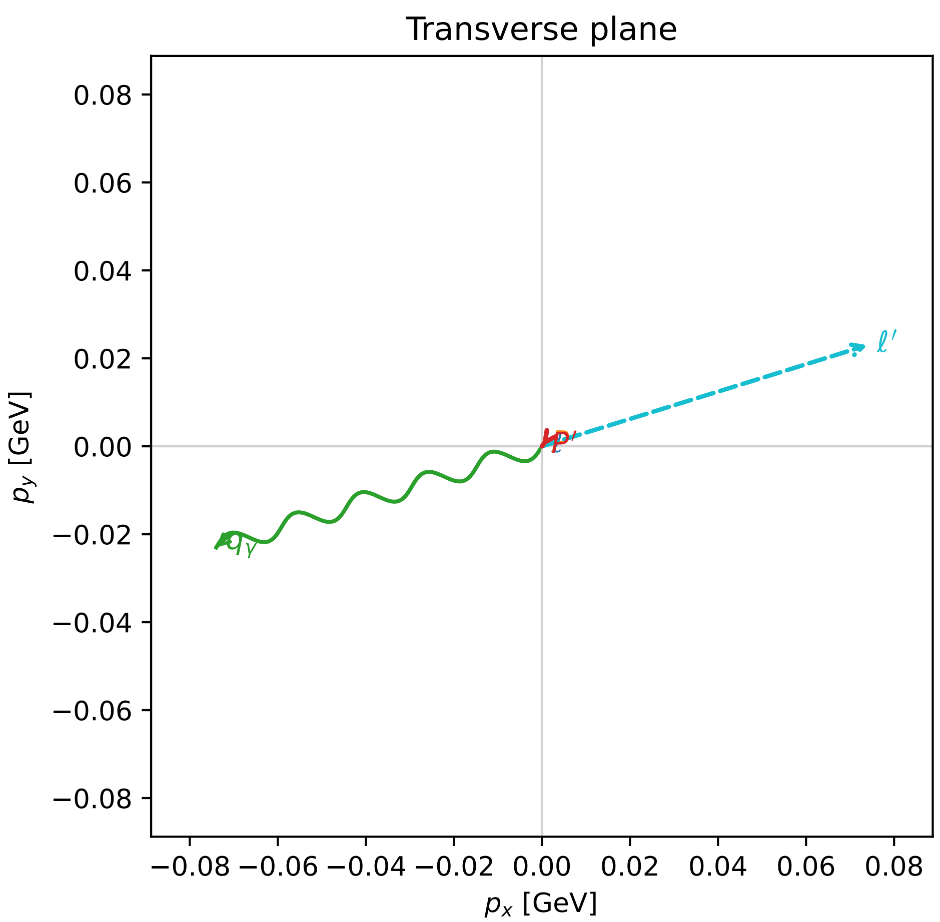
ℓ E= 0.1579 p_x= 0.0000 p_y= 0.0000 p_z= -0.1579

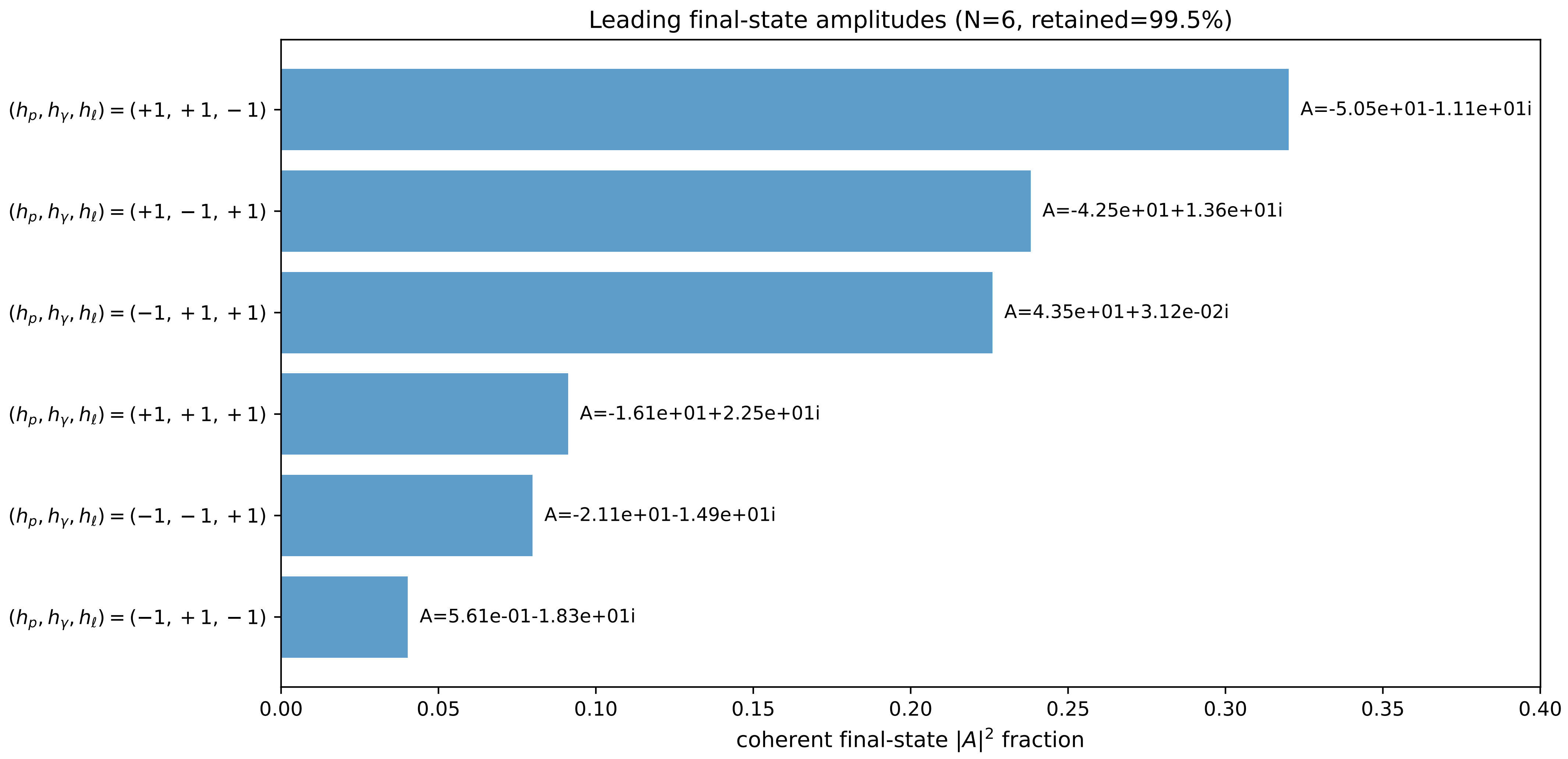
P E= 0.9512 p_x= 0.0000 p_y= 0.0000 p_z= 0.1579

ℓ' E= 0.0855 p_x= 0.0740 p_y= 0.0230 p_z= -0.0362

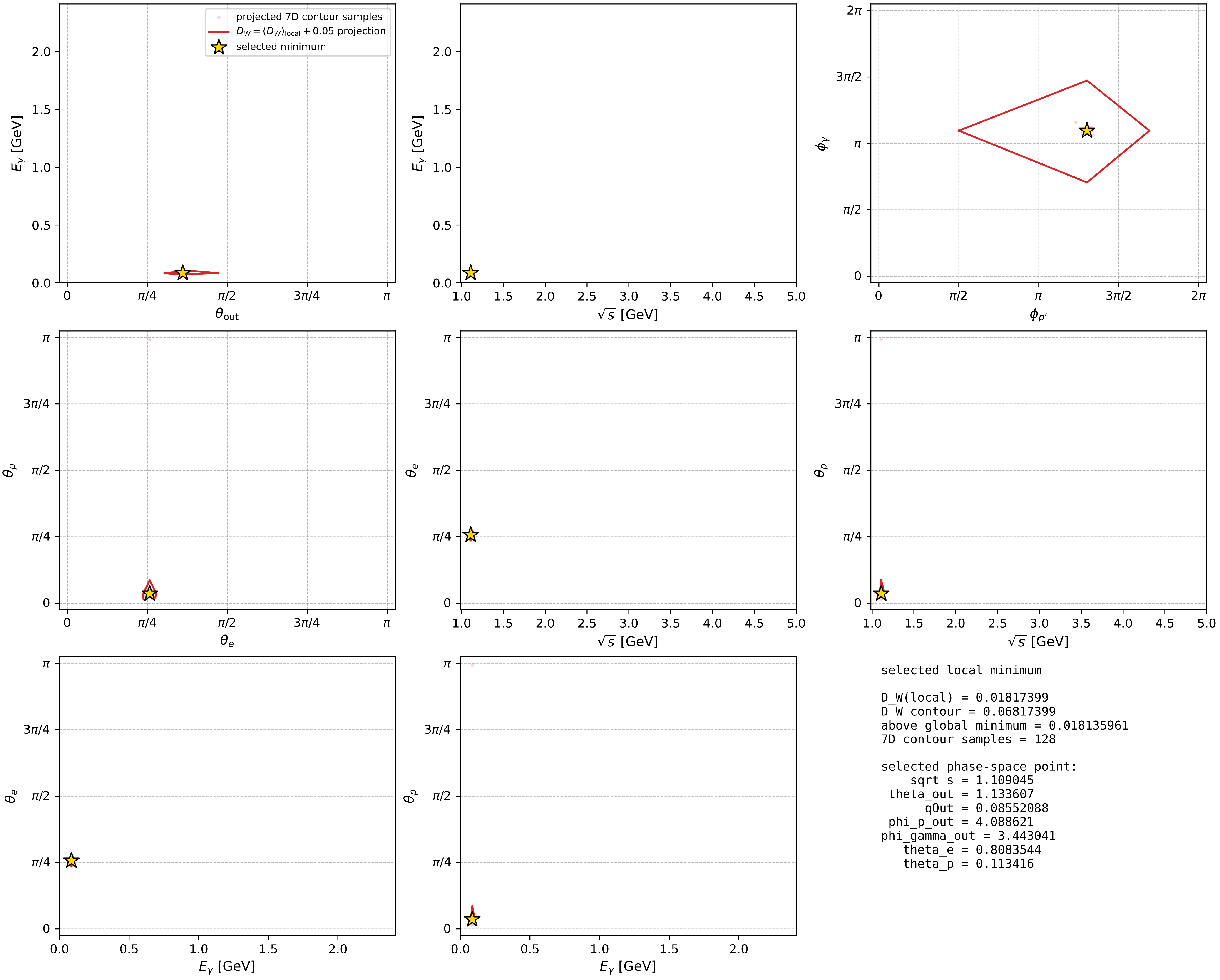
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= 0.0000

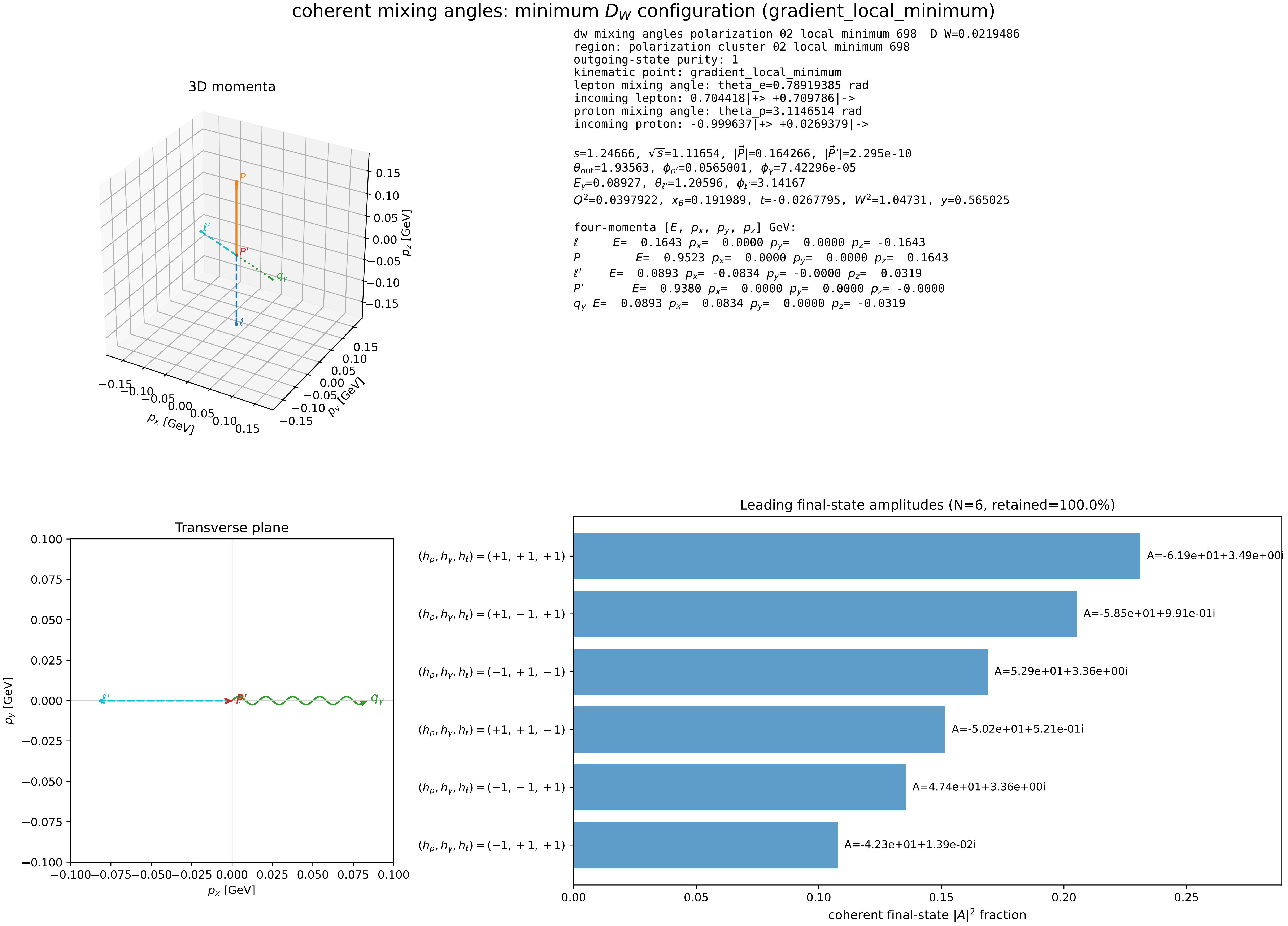
q_γ E= 0.0855 p_x= -0.0740 p_y= -0.0230 p_z= 0.0362

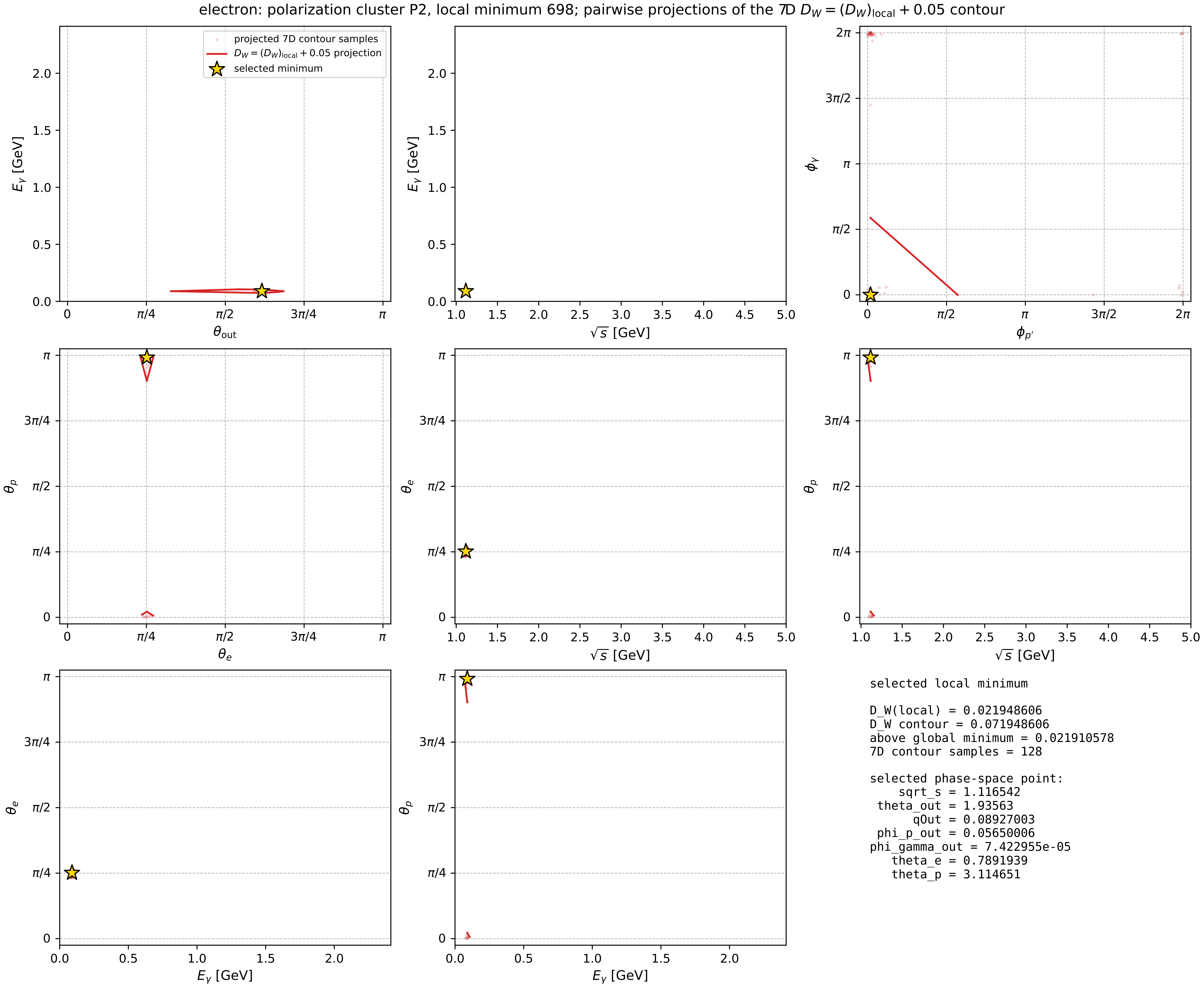




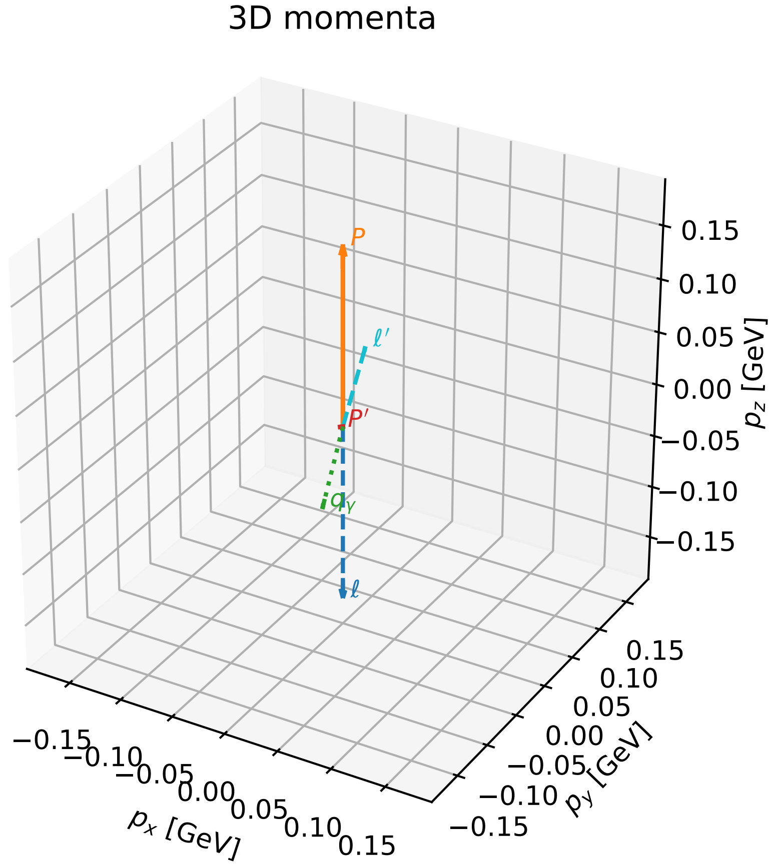
electron: polarization cluster P2, local minimum 682; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour




coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_698 D_W=0.0219486
region: polarization_cluster_02_local_minimum_698
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.78919385 rad
incoming lepton: 0.704418|+> +0.709786|->
proton mixing angle: theta_p=3.1146514 rad
incoming proton: -0.999637|+> +0.0269379|->
s=1.24666, $\sqrt{s}$=1.11654, $|\vec{P}|$=0.164266, $|\vec{P}'|$=2.295e-10
θ_{out}=1.93563, $\phi_{p'}$=0.0565001, ϕ_γ=7.42296e-05
E_γ=0.08927, $\theta_{\ell'}$=1.20596, $\phi_{\ell'}$=3.14167
Q^2=0.0397922, x_B=0.191989, t=-0.0267795, W^2=1.04731, y=0.565025
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1643 p_x= 0.0000 p_y= 0.0000 p_z= -0.1643
P E= 0.9523 p_x= 0.0000 p_y= 0.0000 p_z= 0.1643
ℓ' E= 0.0893 p_x= -0.0834 p_y= -0.0000 p_z= 0.0319
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= -0.0000
q_γ E= 0.0893 p_x= 0.0834 p_y= 0.0000 p_z= -0.0319
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P
ℓ'
P'
q_γ
ℓ
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=6, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=-6.19e+01+3.49e+00i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=-5.85e+01+9.91e-01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=5.29e+01+3.36e+00i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-5.02e+01+5.21e-01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=4.74e+01+3.36e+00i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=-4.23e+01+1.39e-02i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P2, local minimum 698; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.021948606
D_W contour = 0.071948606
above global minimum = 0.021910578
7D contour samples = 128
selected phase-space point:
sqrt_s = 1.116542
theta_out = 1.93563
qOut = 0.08927003
phi_p_out = 0.05650006
phi_gamma_out = 7.422955e-05
theta_e = 0.7891939
theta_p = 3.114651

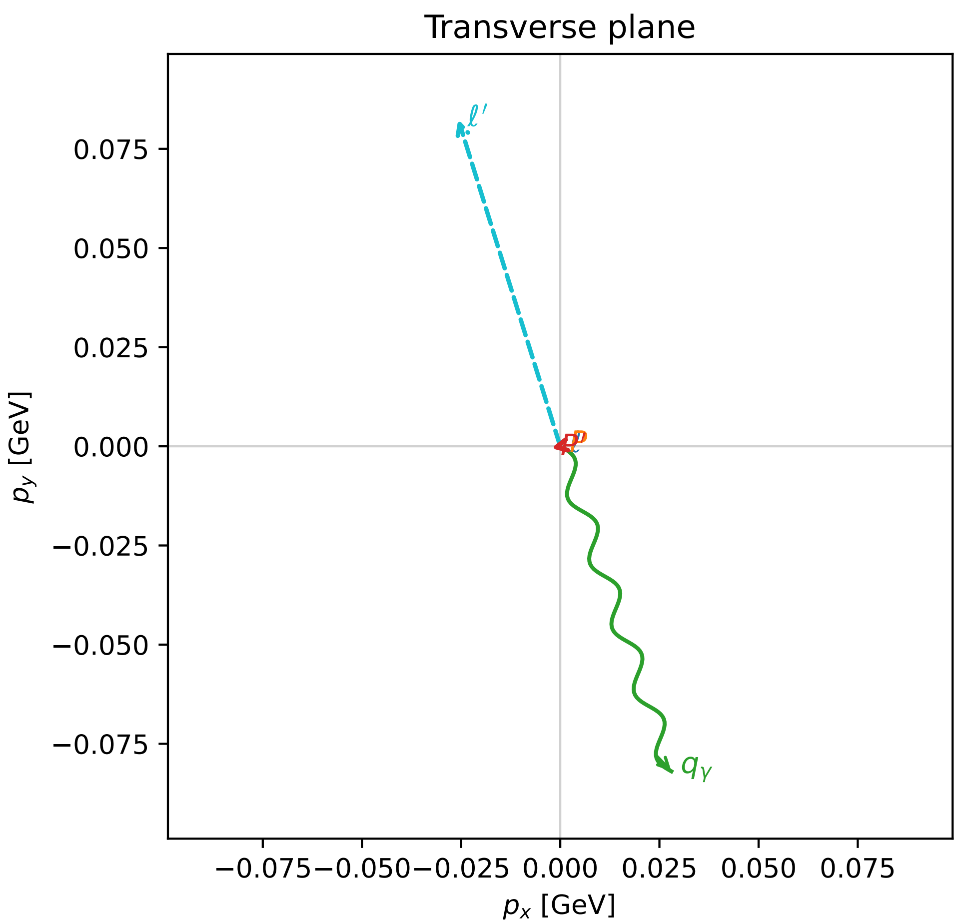
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

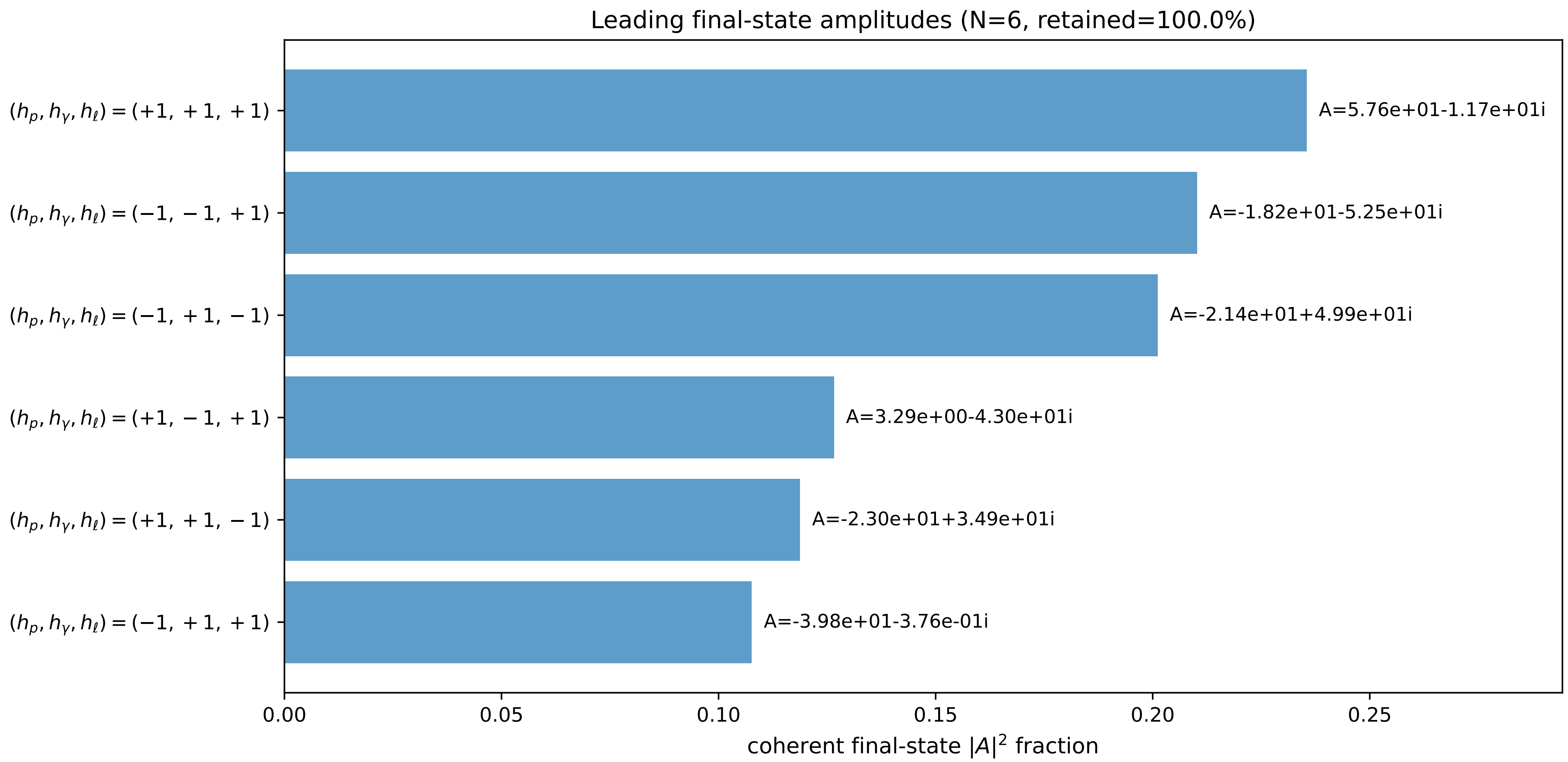


dw_mixing_angles_polarization_02_local_minimum_711 D_W=0.0244391
region: polarization_cluster_02_local_minimum_711
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.7874636 rad
incoming lepton: 0.705645|+> +0.708566|->
proton mixing angle: theta_p=3.1067705 rad
incoming proton: -0.999394|+> +0.0348151|->

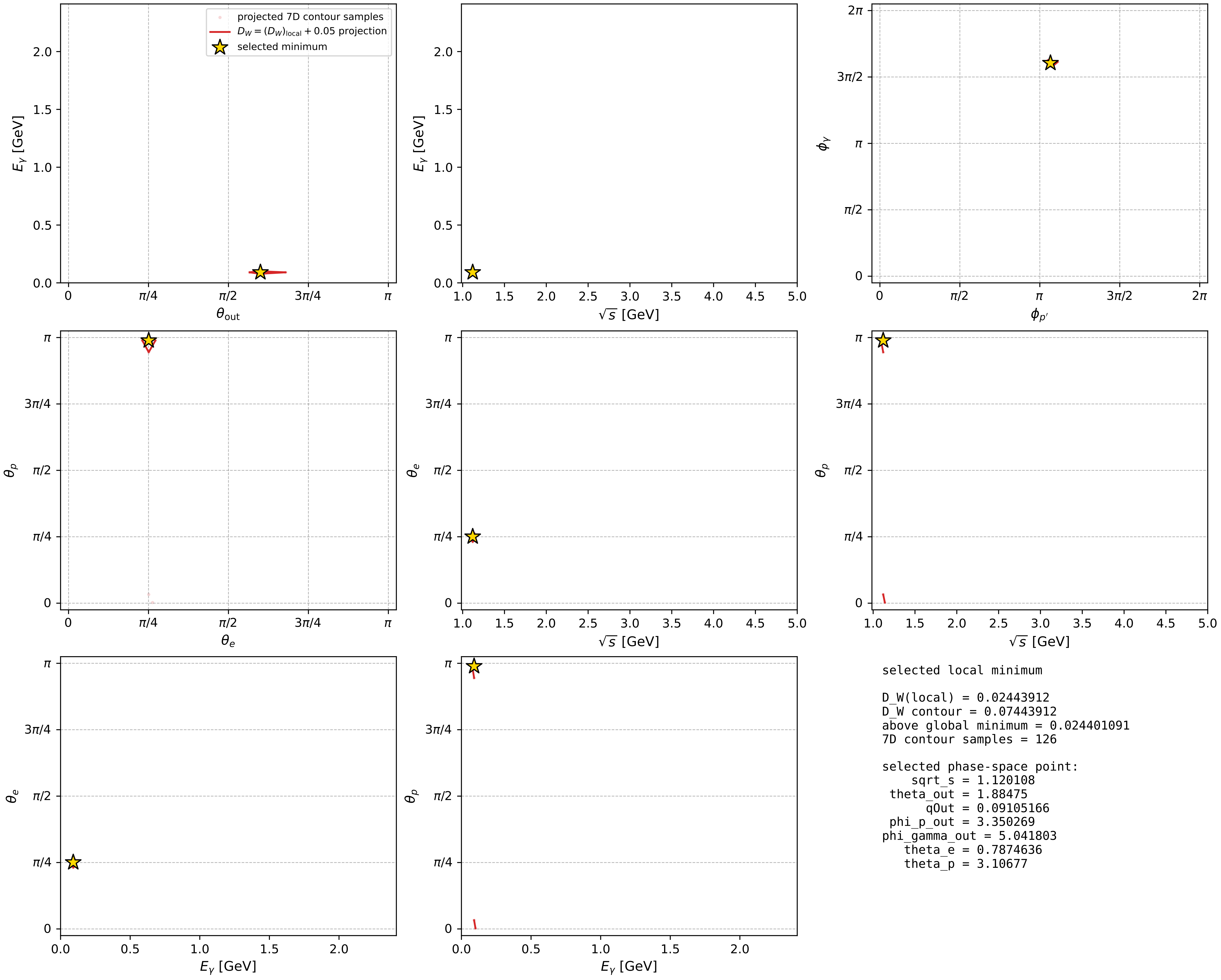
s=1.25464, $\sqrt{s}$=1.12011, $|\vec{P}|$=0.167304, $|\vec{P}'|$=0.00246305
θ_{out}=1.88475, $\phi_{p'}$=3.35027, ϕ_γ=5.0418
E_γ=0.0910517, $\theta_{\ell'}$=1.24805, $\phi_{\ell'}$=1.87328
Q^2=0.0401298, x_B=0.190234, t=-0.028032, W^2=1.05066, y=0.562836

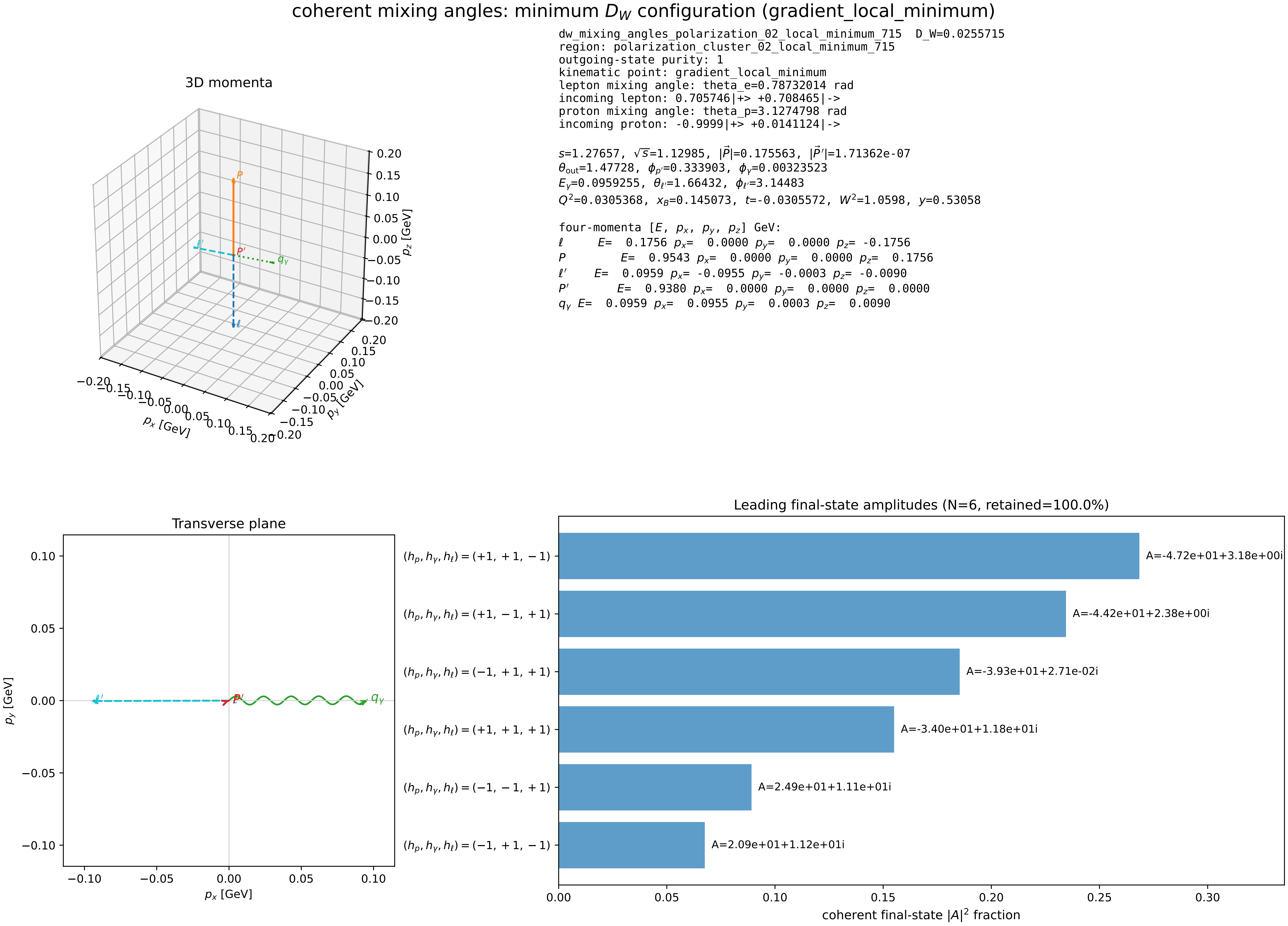
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1673 p_x= 0.0000 p_y= 0.0000 p_z= -0.1673
P E= 0.9528 p_x= 0.0000 p_y= 0.0000 p_z= 0.1673
ℓ' E= 0.0911 p_x= -0.0257 p_y= 0.0824 p_z= 0.0289
P' E= 0.9380 p_x= -0.0023 p_y= -0.0005 p_z= -0.0008
q_γ E= 0.0911 p_x= 0.0280 p_y= -0.0819 p_z= -0.0281

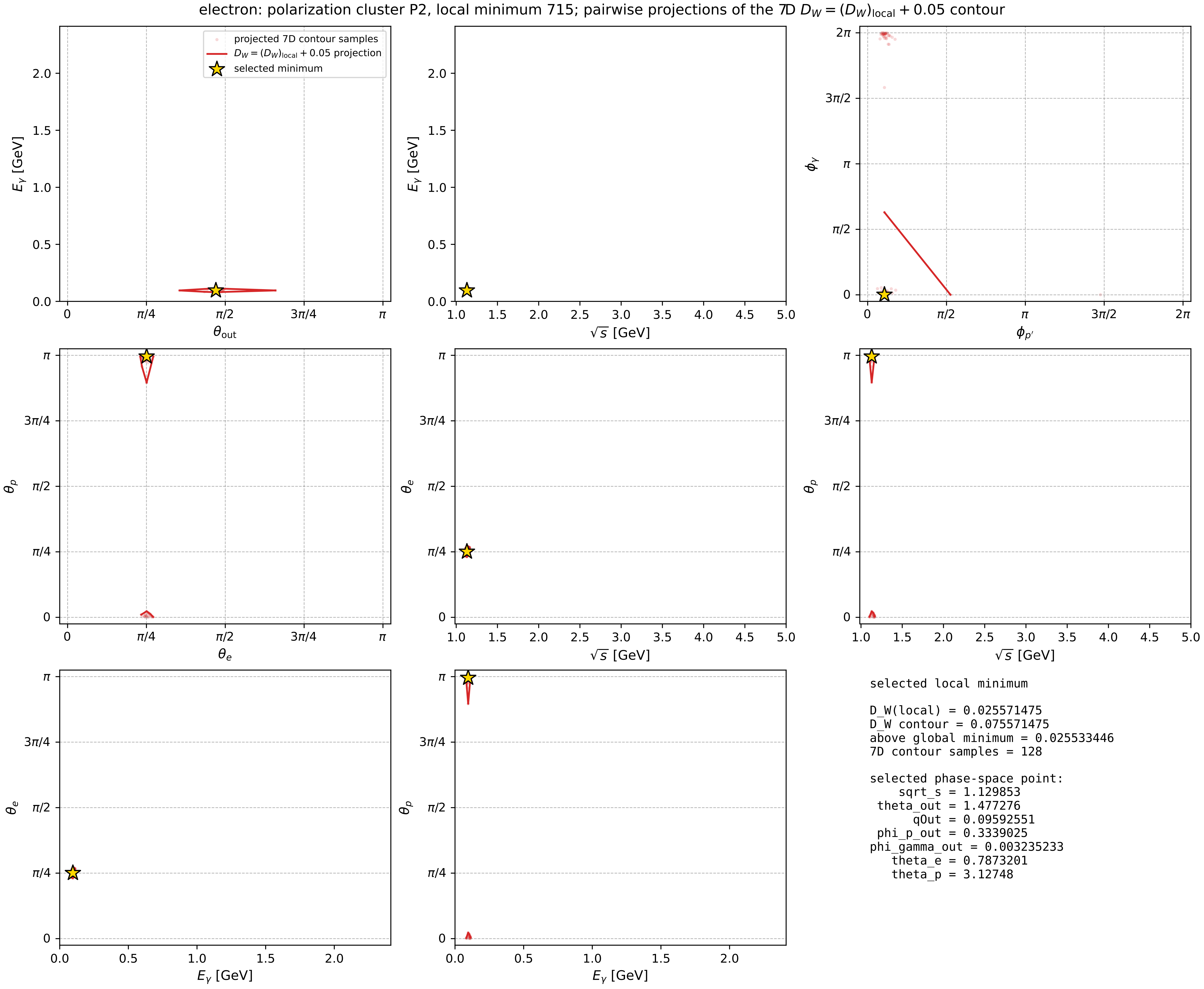




electron: polarization cluster P2, local minimum 711; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour

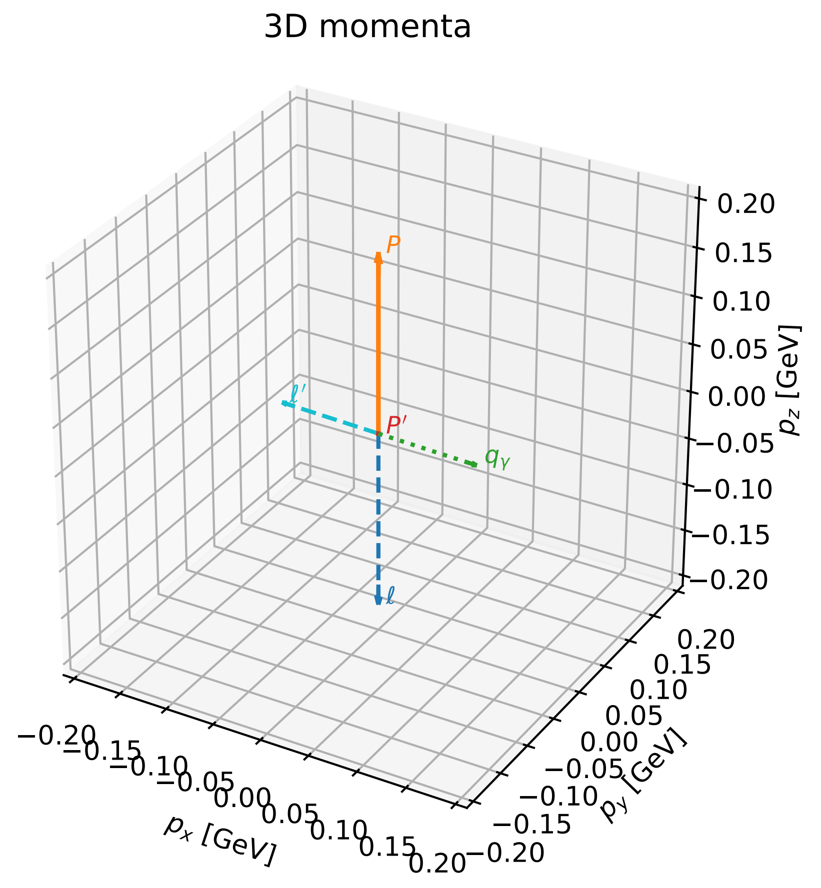



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_02_local_minimum_715 D_W=0.0255715
region: polarization_cluster_02_local_minimum_715
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.78732014 rad
incoming lepton: 0.705746|+> +0.708465|->
proton mixing angle: theta_p=3.1274798 rad
incoming proton: -0.9999|+> +0.0141124|->
s=1.27657, $\sqrt{s}$=1.12985, $|\vec{P}|$=0.175563, $|\vec{P}'|$=1.71362e-07
$\theta_{\rm out}$=1.47728, $\phi_{p'}$=0.333903, ϕ_γ=0.00323523
E_γ=0.0959255, $\theta_{\ell'}$=1.66432, $\phi_{\ell'}$=3.14483
Q^2=0.0305368, x_B=0.145073, t=-0.0305572, W^2=1.0598, y=0.53058
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1756 p_x= 0.0000 p_y= 0.0000 p_z= -0.1756
P E= 0.9543 p_x= 0.0000 p_y= 0.0000 p_z= 0.1756
ℓ' E= 0.0959 p_x= -0.0955 p_y= -0.0003 p_z= -0.0090
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= 0.0000
q_γ E= 0.0959 p_x= 0.0955 p_y= 0.0003 p_z= 0.0090
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=6, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-4.72e+01+3.18e+00i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=-4.42e+01+2.38e+00i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=-3.93e+01+2.71e-02i
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=-3.40e+01+1.18e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=2.49e+01+1.11e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=2.09e+01+1.12e+01i
coherent final-state $|A|^2$ fraction



electron: polarization cluster P2, local minimum 715; pairwise projections of the 7D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 7D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
E_γ [GeV]
θ_{out}
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
θ_p
θ_e
selected local minimum
D_W(local) = 0.025571475
D_W contour = 0.075571475
above global minimum = 0.025533446
7D contour samples = 128
selected phase-space point:
sqrt_s = 1.129853
theta_out = 1.477276
qOut = 0.09592551
phi_p_out = 0.3339025
phi_gamma_out = 0.003235233
theta_e = 0.7873201
theta_p = 3.12748

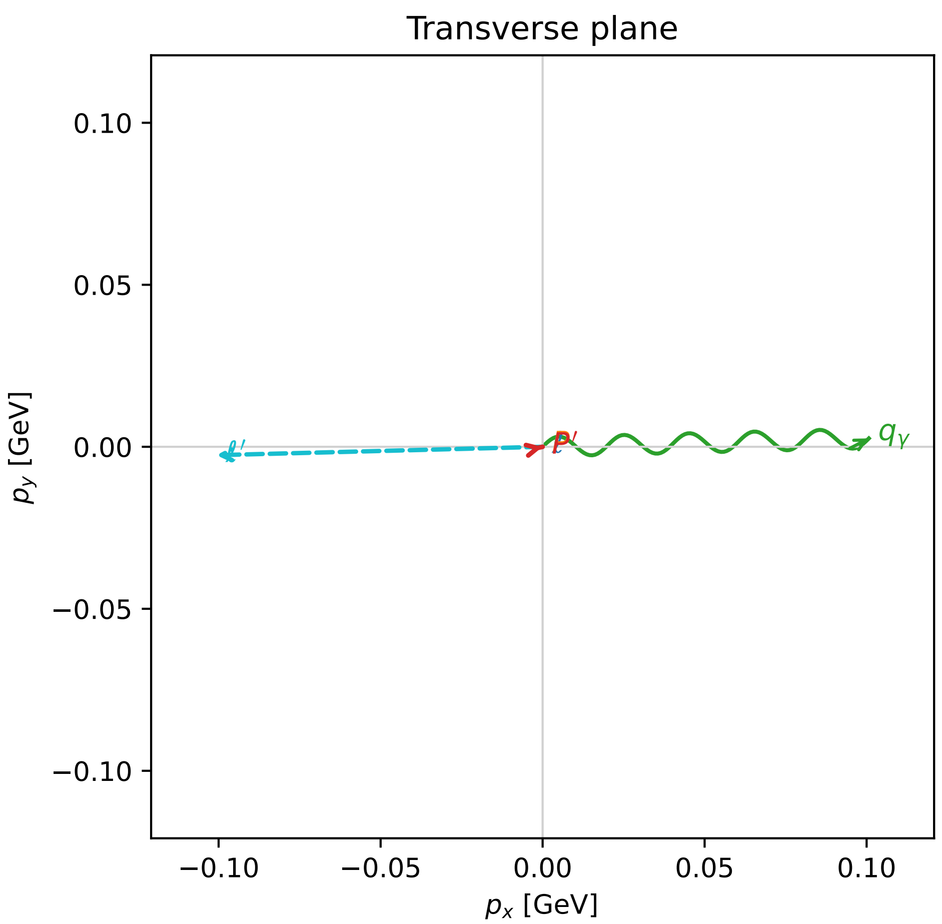
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

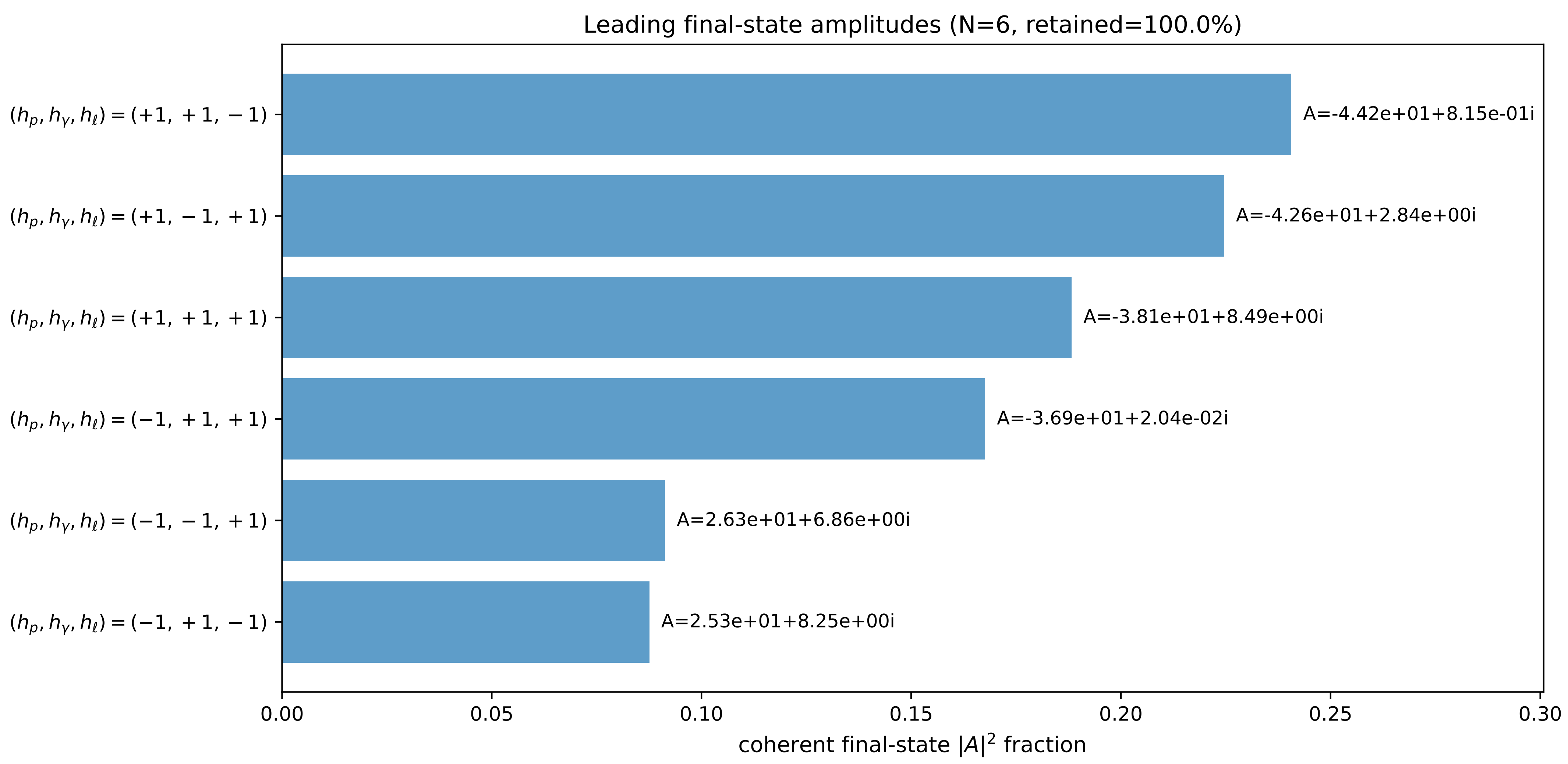


dw_mixing_angles_polarization_02_local_minimum_724 D_W=0.0303058
region: polarization_cluster_02_local_minimum_724
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: theta_e=0.79574555 rad
incoming lepton: 0.699752|+> +0.714386|->
proton mixing angle: theta_p=3.1365386 rad
incoming proton: -0.999987|+> +0.00505406|->

s=1.29899, $\sqrt{s}$=1.13973, $|\vec{P}|$=0.18388, $|\vec{P}'|$=1.04391e-06
θ_{out}=1.6239, $\phi_{p'}$=0.219641, ϕ_γ=0.025711
E_γ=0.100866, $\theta_{\ell'}$=1.5177, $\phi_{\ell'}$=3.16731
Q^2=0.0390637, x_B=0.171116, t=-0.0334931, W^2=1.06907, y=0.544645

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1839 p_x= 0.0000 p_y= 0.0000 p_z= -0.1839
P E= 0.9559 p_x= 0.0000 p_y= 0.0000 p_z= 0.1839
ℓ' E= 0.1009 p_x= -0.1007 p_y= -0.0026 p_z= 0.0054
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= -0.0000
q_γ E= 0.1009 p_x= 0.1007 p_y= 0.0026 p_z= -0.0054





electron: polarization cluster P2, local minimum 724; pairwise projections of the 7D $D_W = (D_W)_{\text{local}} + 0.05$ contour